Essays in Theoretical Microeconomics and Empirical Macroecoconomics with Implications for Social Policy All Around the World

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vorgelegt von

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Acknowledgements

I blame all of you. Writing this book has been an exercise in sustained suffering. The casual reader may, perhaps, exempt herself from excessive guilt, but for those of you who have played the larger role in prolonging my agonies with your encouragement and support, well … you know who you are, and you owe me.

—Brendan Pietsch, assistant professor of religious studies at Nazarbayev University in Astana, Kazakhstan

Source: https://www.timeshighereducation.com/blog/best-academic-acknowledgements-ever

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Ǔő R. Bèta Ȥâne-Ål
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On November 14, 1885, Senator & Mrs. Leland Stanford called together at their San Francisco mansion the 24 prominent men who had been chosen as the first trustees of The Leland Stanford Junior University. They handed to the board the Founding Grant of the University, which they had executed three days before. This document—with various amendments, legislative acts, and court decrees—remains as the University's charter. In bold, sweeping language it stipulates that the objectives of the University are "to qualify students for personal success and direct usefulness in life; and to promote the publick welfare by exercising an influence in behalf of humanity and civilization, teaching the blessings of liberty regulated by law, and inculcating love and reverence for the great principles of government as derived from the inalienable rights of man to life, liberty, and the pursuit of happiness."

¿But aren't Kafka's Schloß and Æsop's Œuvres often naïve vis-à-vis the dæmonic phœnix's official rôle in fluffy soufflés?

(¡THE DAZED BROWN FOX QUICKLY GAVE 12345–67890 JUMPS!)

Ångelå Beatrice Claire Diana Érica Françoise Ginette Hélène Iris Jackie Kāren Łauṙa María Nåťałĭe Øctave Pauline Quêneau Roxanne Sabine Tãja Uršula Vivian Wendy Xanthippe Yvønne Zäzilie

Let us cite some publications (Andersen et al. 2008; Andreoni and Sprenger 2012; Kőszegi and Szeidl 2013; Balakrishnan, Haushofer, and Jakiela 2016; Bursztyn et al. 2025). With the options set for BibLaTeX in the preamble, citations in

1. A test footnote.

the body text are automatically sorted chronologically—irrespective of the order of the "citekeys" in your input. Of course, entries are sorted alphabetically by author surname in the list of references.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

?? consists of my job market paper. I enjoyed writing that paper a lot. This also holds for the paper that makes up **??** of this dissertation. **??** includes a large variety of tests to judge the quality of the typesetting of mathematical formulas.

References

Andersen, Steffen, Glenn W. Harrison, Morten I. Lau, and E. Elisabet Rutström. 2008. "Eliciting Risk and Time Preferences." *Econometrica* 76 (3): 583–618. https://doi.org/10.1111/j.1468-0262.2008.00848.x. [1]

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Chapter 1

My Job Market Paper*

Abstract. This is where the abstract of your paper goes. The abstract is an extremely brief summary of your paper and basically follows the same structure as the paper itself: background/motivation of your study, methods, results, discussion/conclusion. Each section, however, is covered in a single sentence or maybe two sentences instead of an entire section. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

* This footnote can be used for acknowledgments. This is where you can express your gratitude to referees, editors, and colleagues for their valuable feedback and suggestions that helped improve your manuscript. Financial support by third parties can also be mentioned here.

1.1 Introduction

> *"Most people can save a few dollars a day or even $10 a day," she said. "That's doable. But if you say, 'Can you save $300 a month or a couple of thousand dollars a year?' people will say, 'Whoa.' Avoiding that 'whoa,' which is the hesitancy that can derail planning, is what consultants like Ms. Davidson are trying to do."*
>
> —*New York Times*, March 27, 2016

[Anonym 1]

This template uses the Charter typeface for the body text. Charter is a serif typeface and was designed in 1987 by Matthew Carter. By contrast, all headings, tables, and captions are set in a sans-serif typeface. The sans-serif typeface used in this document is Fira Sans, designed by Erik Spiekermann and collaborators.

The math settings are adjusted in the preamble to the effect that mathematical formulas are automatically typeset in the same font as the surrounding text. That is, math in a serif environment will be set in a serif font, while math in a sans-serif environment will use the sans-serif font. This is an aesthetic choice that may not please everyone given that a sans-serif font may be used in mathematical formulas to express a particular meaning. These cases are, however, very rare.

[Holger 1]

Ersetzt: some

[Lou E. 1]

Gelöscht: automatically

[U. R. 1]

Eingefügt

Let us cite a couple of publications (Lisi 1995; Andersen et al. 2008; Andreoni and Sprenger 2012; Balakrishnan, Haushofer, and Jakiela 2016; Bursztyn et al. 2025). With the options set for BibLaTeX in the preamble, citations in the body text are sorted chronologically—irrespective of the order of the "citekeys" in your input. In the list of references, entries are sorted alphabetically by author surname. Let's cite Andersen et al. (2008) once more.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

[Holger 2]

Eingefügt

We already included several references above.

[U. R. 2]

Check whether there are more recent publications!

Some additional references: See Sims (2003) and Gabaix (2014) for models of "rational inattention" or "goal-driven attention." See Bordalo, Gennaioli, and Shleifer (2012, 2013), Kőszegi and Szeidl (2013), Taubinsky (2014), and Bushong, Rabin, and Schwartzstein (2016) for models of "stimulus-driven attention."

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Hello, here is some text without a meaning. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.

In Section 1.3, we describe the design of our study. We present the data analysis and our results in Section 1.4. In Section 1.5, we discuss the plausibility of potential alternative explanations. Section 1.6 concludes.

[Lou E. 2]
Italics?

[Holger 3]
Gelöscht: in detail
Too wordy.

[Lou E. 3]
Ersetzt: will conclude
Let's use the present tense throughout.

1.2 Portability

1.2.1 Clean Code/Semantic Coding: Separating Content and Formatting

All written texts are structured into words and sentences. If they are long enough, texts are also structured into paragraphs. Academic texts, in addition, feature sections, subsections, and often subsubsections, as well as figures, tables, lists of references, and potentially appendices. These structural elements are associated with headings/titles and captions.

The structure of a document is conveyed to the reader through *formatting*. Effective formatting of academic documents, thus, requires that all elements of a particular type—say, section headings—be identifiable as such. Identifiability is achieved through styling that is *consistent within type* and *distinct across types*. For example, all section headings must be formatted identically, and at the same time, they must look different from subsection headings.

As a consequence, one should refrain from using manual, discretionary formatting whenever possible. In line with this principle, the source code of this template refers to the document's structure as much as possible. This is also called "semantic coding": using code that describes the *meaning* of the content and not *how* those elements should look.

- Examples for *semantic* commands are `\author`, `\title`, `\section`, `\emph`, `\url`, `\cite`, `\begin{figure}` … `\end{figure}`, `\caption`, `\begin{quote}` … `\end{quote}`.
- Examples for LaTeX commands that are *not semantic*—and should be avoided—are `\textbf`, `\textit`, `\newpage`, `\pagebreak`, `\linebreak`, `\\`, `\noindent`,

`\smallskip`, `\centering`, `\noindent`, `\hspace`, `\vspace`, `\cellcolor`, `\multirow`.

The reason why one should stick to using semantic commands is that they keep the code *portable*: Semantic code can be *reused* across document without any changes (or at least without major adjustments). This is not then case with nonsemantic code, that is, code that includes manual formatting instructions. Instructions like `\bigskip`, `\hspace`, `\\` (even worse, `\\ \\`), etc. may result in decent formatting in a particular document but are bound to lead to undesirable formatting in another document. This applies, in particular, when you face the task of combining several research papers into one, "cumulative" doctoral thesis. Moreover, manual formatting is error-prone, as it easily leads to inconsistent styling.

This entails two things: First, the formatting should be determined, as much as possible, in the *preamble* of the LaTeX document. This comes at the cost, of course, of a rather long preamble. Second, instead of defining custom commands, one should *redefine* standard LaTeX commands or use (widely adopted) packages that are available on CTAN (https://ctan.org).

This template follows the above principles, resulting in clean, portable code: The source code of the body of this template (apart from the boxes with examples) consists almost completely of semantic, basic LaTeX, with commands from popular packages on top.

1.2.2 Using BibLaTeX (Rather Than BibTeX)

This template uses the modern BibLaTeX framework (the *biblatex-chicago* package, https://ctan.org/pkg/biblatex-chicago, to be precise) instead of the vintage BibTeX framework to manage the references included in the document. The reason for this choice is that BibLaTeX is much more flexible than BibTeX and also handles multi-language references much better.

Moreover, in line with the principle describe above, BibLaTeX permits much better separation of a document's content (in this case, the *.bib* file) and its formatting (of the citations and the bibliography) than BibTeX. In particular, suppressing output of information that is present in a *.bib* file is cumbersome with BibTeX: It amounts to editing the "bibliography style" *.bst* file, and *.bst* files have a syntax that is completely different from LaTeX and difficult to learn. It is usually easier to just remove the information from the *.bib* file. This, however, impacts the portability of the *.bib* file: In other circumstances one might want that very information to be present. BibLaTeX simplifies skipping information that is present in the *.bib* file.

An example: Researchers frequently present their research in the form of posters. Space on posters is usually quite limited. Hence, one might not want authors' first names to be abbreviated on a poster. In the associated paper, however, the names should be printed in full to minimize ambiguity. With BibTeX, this requires time-consuming adjustments of the *.bst* file. With BibLaTeX, it is much easier: In the *.tex* file for the poster, one can include the command

```
\ExecuteBibliographyOptions{giveninits = true}
```

Similarly, in a paper, the list of references should include URLs or DOIs (digital object identifiers) whenever they exist, to make it easy for readers to locate a cited work. On a poster, by contrast, one might want to suppress DOIs and URLs. With BibTeX, this would require time-consuming adjustments of the *.bst* file. With BibLaTeX, one can simply do, for instance,

```
\AtEveryBibitem{%
  \ifentrytype{article}{\clearfield{doi}\clearfield{url}}{}%
}
```

Using BibLaTeX requires using the program `biber` instead of `bibtex` for compiling the bibliography. When using the Overleaf online service, using `biber` instead of `bibtex` is as simple as it gets: Overleaf automatically chooses `biber` to compile the bibliography when it encounters BibLaTeX's `backend = biber` option.[1] And when using the TeXstudio editor (https://texstudio.org), an easy way to make this happen is to include the "magic comment"

```
% !BIB program = biber
```

at the beginning of the *.tex* file.

1.2.3 Using the *tabularray* Package (Rather Than *tabularx*/*tabulary*/*tabu*, *booktabs*, *longtable*, *xltabular*, *multirow*, *makecell*, *colortbl*, *parnotes*, *threeparttable*, ...)

The principle of separating the content from its formatting as much as possible can also be applied to tables. The *tabularray* package (https://ctan.org/pkg/tabularray) allows you to do just that. *tabularray* has been around since May 2021, so it is relatively new—but it has already matured into a package that is extremely useful. *tabularray* replaces (or loads) all the other table-related LaTeX packages that you may have used so far—like *tabularx*/*tabulary*/*tabu*, *longtable*, *xltabular*, *multirow*, *makecell*, *booktabs*, *colortbl*, *parnotes*, *threeparttable*, ...

Separating the contents of a table from its formatting means that no formatting instructions whatsoever need to be placed in the table cells—neither for drawing rules between table rows nor for highlighting entries nor for increasing spacing nor for changing alignment, and so on. Instead, the placement of rules at the top and

1. This applies to the commercial version (https://www.overleaf.com) as well as to the free versions https://overleaf-students.astro.uni-bonn.de/ and https://uni-bonn.sciebo.de/apps/overleaf_nextcloud/launcher/launch.

bottom of a table or between particular rows, the specification of column widths, the specification of font sizes and cell alignment, desired highlighting via boldface or a background color, etc., can all be coded as arguments to the table environment. This is beneficial in at least three aspects:

- The source code of the table's contents is very clean. This particularly applies to animated Beamer presentations: no need to put `\pause`/`\only`/`\cellcolor`/ …in table cells.
- It makes table contents *portable*: A table's contents can be reused easily across multiple documents that are formatted in different ways.
- It also drastically simplifies the inclusion of table contents that are produced by an external program such as Python, R, or Stata.

Further additional helpful features of *tabularray* are:

- *tabularray* makes it easy to bottom-align or top-align particular table rows or individual cells entries, while keeping a different alignment for the other rows/ cells.
- Table entries that consist of multiple lines of text become easy to produce: Just enclose them in curly braces, like in this example: `... & {Line 1 \\ Line 2} & ... \\`. No need to use `\multirow` or `\makecell` any more.
- You can set document-wide formatting defaults to achieve consistent formatting of all tables. For instance, you can set defaults such that all tables feature the same top and bottom rules. Or, you could make all column heads be typeset in boldface document-wide.

When using `\UseTblrLibrary{siunitx}`, *tabularray* loads the *siunitx* package. This is another useful package which permits separating content from formatting. In particular, the `S` column type is able to round numbers to a specified precision. This means that one can include table content produced by an external program with as many decimal digits as that program produces and have LaTeX take care of the rounding.

1.3 Methods

1.3.1 Structuring Introduction and Discussion: Funnel and Reverse Funnel

A general structure that can be found in many research papers is the so-called X-shaped or "hourglass" structure.[2] This structure is brought about when the so-called

2. See https://skillsearthsciences.sites.uu.nl/writing/structure/#attachment_2387, Fig. 1, and https://pmc.ncbi.nlm.nih.gov/articles/PMC5405644/pdf/JHRS-10-3.pdf, Figure 1.

"funnel" technique is used to write the introduction,[3] and when a "reverse funnel" or "upside-down funnel" is used to write the discussion/concluding section:[4]

(1) The paper starts with a *broad* perspective at the beginning of the introduction, which becomes *narrower* with every sentence, leading to the paper's narrow research question.
(2) The *narrowest* part is when the exact research question is posed, the research methodology is described in detail, and the results are derived/presented.
(3) When the results are discussed and related to the literature in the concluding section, the perspective becomes *broad* again.

An example of the "funnel" technique can be found in Figure 1.1, and of the "reverse funnel" technique in Figure 1.2.

1.3.2 Some Questions That May Help You Guide Writing Your Paper

You may find the following list of questions helpful as a guide to making sure that your manuscript covers all important aspects. The questions are primarily intended for doctoral and advanced researchers, but they may also be helpful to students writing term papers and theses.

Your introduction might answer the following questions (not necessarily in this order, but this order will give rise to a "funnel" structure):

(1) What do we already know?
(2) What do we not know yet (the "knowledge gap")?
(3) Why is it interesting/relevant to close the gap?
(4) *What is the research question? [R]*
(5) Why is it interesting to answer the research question?
(6) How do we contribute to closing the knowledge gap: How do we answer the research question? (Which method do we use?)
(7) Why is the chosen method (more) suitable (than alternative methods) for answering the research question?

And your conclusion/discussion might answer the following questions (again, not necessarily in this order, but this order will give rise to a "reverse funnel" structure):

3. See https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1401, Figure 14.1, and https://kib.ki.se/en/write-cite/academic-writing/structure-academic-texts
4. See https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1402, Figure 14.2, and https://kib.ki.se/en/write-cite/academic-writing/writing-conclusion.

Molecular simulation methods have become an important technique in many areas of chemistry through the recent advent of effective and widespread software for molecular mechanics, molecular dynamics, and Monte Carlo simulations, and procedures for the estimation of free energy differences. In all such methods, a proper description of the electrostatic interactions within the simulated system is of key importance, … However, in most implementations, …, the simplest possible solution is used, … Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.

Unfortunately, atomic charges are not observables, i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations. Therefore, a large number of methods have been suggested for the estimation of point charges [1]. Several groups have tried to derive the charges directly from experimental quantities, … Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules. Instead, most techniques derive atomic charges from quantum chemical calculations.

The simplest way to determine quantum chemical charges is the Mulliken population analysis. … Although, Mulliken charges are known to strongly depend on the basis set [6] and to reproduce electrostatic moments poorly [1], they are still widely used due to their simplicity. Several other … methods have been devised to cure the problems of the Mulliken charges [1], …

…

In this paper, we make a critical analysis of the four most popular potential-based point-charge methods, … It is shown that these methods may give widely different results and possible explanations for this are discussed. Two alternative methods for deriving atomic charges are suggested, which avoid the arbitrariness in the selection of potential points, and their performance is judged using …

Paragraph 1
Broad introduction with general description of the research field ("many areas of chemistry"), research topic ("molecular simulation methods"), and recent development ("through the recent advent of …"). Followed by a more detailed description of crucial aspects in this context ("In all such methods, a proper description … is of key importance") and of shortcomings of existing methods.

Paragraph 2
Narrowing down the topic: Description of the state of the art and of issues and limitations observed in the literature.

Paragraph 3
Further description of the state of the art: strengths and weaknesses of existing approaches.

Paragraphs 4–6
Further description of unresolved issues.

Paragraph 7
Narrow description of the specific contribution of this paper: Identification of undesirable arbitrary variation in simulation results of existing methods, which inspired development of an "alternative methods" that avoids this issue.

Figure 1.1. Illustration of the funneling approach, using the "Introduction" section from Sigfridsson and Ryde (1998).

Note: The full text of the article by Sigfridsson and Ryde (1998) is available at https://lucris.lub.lu.se/ws/portalfiles/portal/135489685/25_charges.pdf.

Potential-based charges strongly depend on the way the potential points are selected. We have presented and tested a new method that avoids such a dependence, CHELMO. It fits the charges directly to the electrostatic moments and it performs as well, or better, as the best electrostatic potential method judging from the calculated electrostatic potential and moments. … The major disadvantage of the method is that it cannot be used for large molecules, since there is a limited number of moments. No more than 25 independent charges in a molecule can be determined if all moments up to hexadecapole moments are used. In practice it should not be used for molecules with more than about 20 atoms since otherwise the higher moments may be poorly reproduced … This could be remedied if moments higher than hexadecapole moments are calculated but such moments are not included in the output of normal quantum chemical packages.	*Paragraph 1* Narrow recap of the first contribution of the paper: brief description of the issue and how it was solved by the new method. Followed by description of the limitations of the new method. Outlook how these limitations could be overcome.
In order to get a method that can be used for larger system, we constructed the CHELP-BOW method, which estimates the charges from a Boltzmann-weighted fit to the electrostatic potential. It has the advantage over the other … methods that … The present implementation of the CHELP-BOW method is very simplified, but in a future publication we will refine and thoroughly test the method. However, the results are conclusive enough to show that the method … has the desired behaviour. … Clearly, this is the method we recommend for general use.	*Paragraph 2* Narrow recap of the central contribution of the paper and how it relates to the literature: how existing methods were further improved upon by the second method proposed in this paper. Followed by outlook on future advances and recommendation for researchers in the field.
An important advantage of the methods developed in this paper is that they are general, i.e. they can be used with any quantum chemical program and they can use experimental data (…) as well. The only thing that has to be changed is the input section of the programs. Thus any quantum chemical basis sets can be used, and any level of theory that gives a wave function may be employed. Furthermore, the methods can easily be adapted to …	*Paragraph 3* Broader implications due to the generality of the new method.
Finally, a comment on the traditional electrostatic potential methods. … Thus, we cannot see any justification to use CHELP charges except for backward comparisons.	*Paragraph 4* Other broader implications.

Figure 1.2. Illustration of the "reverse funneling" approach, using the "Concluding Remarks" subsection from Sigfridsson and Ryde (1998).

(8) What was the knowledge gap before the current study? [Similar to the introduction.]

(9) *How did we contribute to closing that gap, and what did we find? [A]* (Which method/dataset did we use, and what are the results?—Recap and take-home message.)

(10) How do our results relate to (confirm/contradict/complement/extend) the results from previous and other contemporaneous studies?

(11) What part of the gap is still open? (Phrased a bit more negatively: What are the limitations of our approach?)

(12) Next steps/avenues for future research: How could we go about closing the remaining knowledge gap (by removing the limitations of our current approach)?[5]

In her book *The Little Book of Research Writing*,[6] Varanya Chaubey proposes a simpler list of only three items—the "RAP method":

- The "R" stands for "research question."
- The "A" stands for "answer" (to the research question, of course).
- The "P" stands for "positioning statement." (How does our manuscript relate to the literature: Does it corroborate or extend or contradict others' findings?)

Relating the "RAP" approach to the the list of questions above, question 4 is the "R," the answer to question 9 is the "A," and the remaining questions spell out the "P."

1.3.3 Structuring Paragraphs in Academic Manuscripts

Obviously, there is no single "right" way of constructing paragraphs. However, a few basic rules can help you write paragraphs that are *easy to comprehend* by your readers. A common approach is to compose sentences such that they reflect the following functional sequence:[7]

(1) *topic sentence (TS)*—which, if necessary, connects to the previous paragraph;

(2) *supporting sentences (SS)*—which elaborate on the topic sentence; and

5. John Cochrane might do away with question 12. See his "Writing Tips for Ph.D. Students," https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf.

6. For a quick summary of the book, see https://mauve-porcupine-8992.squarespace.com/s/Chaubey_Research_Writing-bxdd.pdf. See also https://www.econscribe.org/about and https://arxiv.org/pdf/2012.07787.

7. See https://libguides.newcastle.edu.au/writing-paragraphs/structure, https://www.une.edu.au/library/students/academic-writing/write-paragraphs/paragraphs/Academic-paragraphs_v2.pdf, https://learningessentials.auckland.ac.nz/writing-effectively/paragraph/

(3) *concluding sentence or connecting sentence (CS)*—which refers back to the topic sentence and/or provides a link to the subsequent paragraph.

There are various ways in which the supporting sentences can be filled with content. The University of Sheffield suggests[8] that the supporting sentences could consist of

(1) "explanation or definitions (optional),"
(2) "evidence and examples," and
(3) "comment" (an interpretation and evaluation of the evidence by you).

The University of Hull and Newcastle University (UK) consider the mnemonic "PEEL" helpful, although they define its meaning in slightly different ways: "Point, Evidence, Explanation, Link"[9] versus "Point, Evidence, Evaluation, Link"[10]. One may view "Explanation"—"why the point is important and how it helps with your overall argument"—as narrower than "Evaluation"—which is a general critical reflection on the informativeness of the evidence and can include an explanation of its importance. "PEEL" relates to TS/SS/CS as follows:

(1) "P," the Point, corresponds to the topic sentence.
(2) "E," Evidence, is the first part of the supporting sentences.
(3) "E," Evaluation/Explanation, forms the second part of the supporting sentences.
(4) "L," Link, refers to the connecting sentence that links to the next paragraph.

The Wilfrid Laurier University (Canada) suggests a different mnemonic: "MEAL."[11]

(1) "M: Main Point Sentence" corresponds to the topic sentence.
(2) "E: Evidence" is the first part of the supporting sentences.
(3) "A: Analysis/Synthesis" forms the second part of the supporting sentences.
(4) "L: Linking-Back Sentence" refers to the concluding sentence that summarizes the paragraph and links back to the topic sentence or even the topic of the entire paper.

The first three components are largely identical according to these approaches. A difference exists in the meaning of "L": While "PEEL" stresses the link to the *next* step in the chain of thought, "MEAL" stresses underscoring the message of the *current* paragraph. That is, the latter advocates *closure*: summarizing the paragraph and linking *back* to the topic sentence.

8. See https://sheffield.ac.uk/study-skills/writing/academic/paragraph.
9. See https://libguides.hull.ac.uk/writing/paras.
10. See https://www.ncl.ac.uk/academic-skills-kit/writing/academic-writing/paragraphing/
11. See https://students.wlu.ca/academics/support-and-advising/student-success/assets/resources/writing/how-to-structure-an-academic-paragraph.html.

Yet slightly different is the approach put forth by Academic Writing UK:[12]

(1) Topic sentence—key topic in this paragraph.

(2) Development—the main idea/topic discussed in more detail.

(3) Example—support/evidence/data/statistics that show your development is valid/credible.

(4) Summary—overall main point summarized/evaluated.

It is perfectly fine to mix the different approaches. For some paragraphs, a closer explanation of the topic sentence ("development") may be necessary, while it is unnecessary for others. And in some paragraphs, you may want underscore the central message by reiterating it in the final sentence (concluding sentence), while for other paragraphs, you will use the final sentence to set the stage for the next step in your chain of reasoning (connecting sentence).

One thing that *all* these suggestions have in common is, however, that they start with a *topic sentence* and end with a *concluding/connecting sentence*. Hence, no matter how you fill the supporting sentences, it is probably best to start a paragraph with a topic sentence and end with a concluding/connecting sentence.

Let us illustrate the "MEAL"/"PEEL" structure—including an optional explanation or definition—using example sentences from Sigfridsson and Ryde (1998):

… Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.	**L: Concluding sentence**
Unfortunately, atomic charges are not observables,	**M/P: Topic sentence** (reference to previous paragraph)
i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations.	**Optional explanation/definition**
Therefore, a large number of methods have been suggested for the estimation of point charges [1].	**Supporting sentences:** **E: Evidence**
Several groups have tried to derive the charges directly from experimental quantities, e.g. dipole moments, electrostatic potentials, or free energy differences [2–4]. Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules.	**A/E: Analysis/Evaluation**
Instead, most techniques derive atomic charges from quantum chemical calculations.	**L: Connecting sentence** (link to next paragraph)
The simplest way to determine quantum chemical charges is the Mulliken population analysis. …	**M/P: Topic sentence** (reference to previous paragraph)

12. See https://academic-englishuk.com/paragraphing/.

The rule to finish a paragraph with a concluding sentence entails, in particular, that a theoretical or empirical finding should never be the last thing that you mention in a paragraph. An “evidence” sentence should always be followed by an interpretation in which you tell the reader what we learn from that finding. Here is an example from Sigfridsson and Ryde (1998):

… With these radii, the charges almost coincide with those obtained by the CHELPG method (within 0.02 e). Thus, we can conclude that the main difference between the Merz–Kollman and the CHELPG method is the 1.4 times larger effective radii of the former method, whereas the sampling schemes are almost equivalent.	**Evidence: Supporting sentence with empirical finding** **Concluding sentence with interpretation of that finding**

To reiterate, there is no single “right” way of constructing paragraphs. However, keeping the TS/SS/CS structure in mind as a guideline is bound to help you produce a text that enables readers to follow your chain of reasoning.

1.3.4 Additional Online Resources on Effective Writing in Economics

Additional useful recommendations and tips can be found in Plamen Nikolov’s “Writing Tips for Crafting Effective Economics Research Papers – 2023-2024 Edition” (https://docs.iza.org/dp16276.pdf) and in John H. Cochrane’s “Writing Tips for Ph.D. Students” (https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf).

In this section, we first present the design of the experiment (1.3.5) and derive behavioral predictions (1.3.6).

1.3.5 Design of the Main Experiment

1.3.5.1 General Features. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.3.5.2 More Specific Features. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Let's test the euro symbol: €, €1,234.56, €1,234.56. Let's also test text superscripts: i^{th} and text subscripts: CO_2 and H_2O. $\sigma_\epsilon, c^\alpha$. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Let's test the footnote settings.[13]

Figure 1.5 shows an exemplary decision screen with $B = €11$ and $r \approx 15\%$ for both $\text{BAL}^{\text{I}}_{1:1}$ (upper panel) and $\text{UNBAL}^{\text{I}}_{1:8}$ (lower panel). Through a slider, subjects choose their preferred $x \in \boldsymbol{X}$.[14] The slider position in Figure 1.5 indicates $x = 0.5$, i.e., the earliest payment is reduced by €5.50. Since $r \approx 15\%$ in this example, this slider position amounts to €6.30 that are paid at later payment dates. While these €6.30 are paid in a single bank transfer on the latest payment date in $\text{BAL}^{\text{I}}_{1:1}$, the amount is dispersed in equal parts over the last 8 payment dates in $\text{UNBAL}^{\text{I}}_{1:8}$—i.e., 8 consecutive payments of €0.79.[15]

1.3.5.3 Some More Details. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

13. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

14. The slider had no initial position—it appeared only after subjects first positioned the mouse cursor over the slider bar. This was done to avoid default effects.

15. We always rounded the second decimal place up so that the sum of the payments included in a dispersed payoff was always at least as great as the respective concentrated payoff.

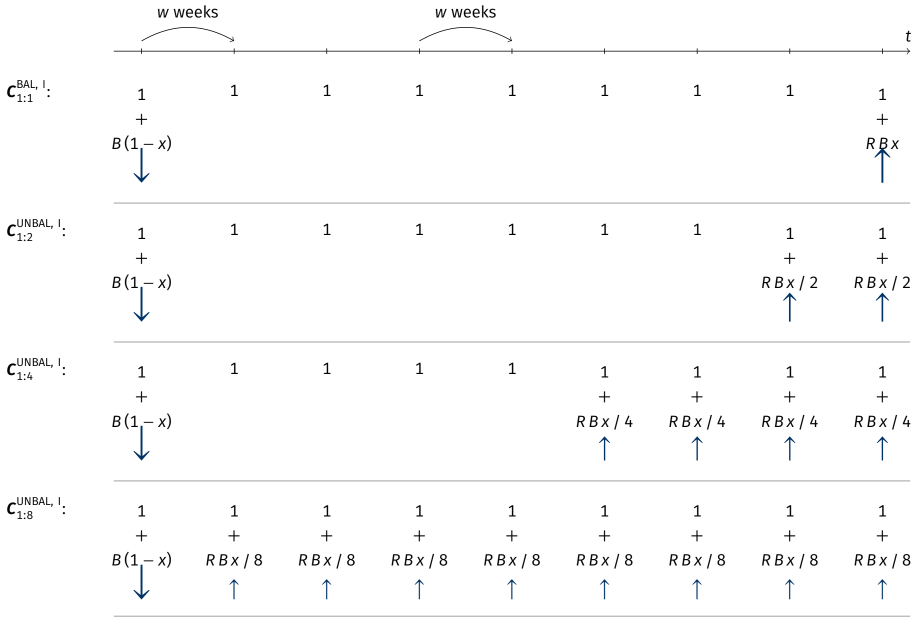

Figure 1.3. Budget Sets $C_{1:1}^{\text{BAL, I}}$ and $C_{1:n}^{\text{UNBAL, I}}$

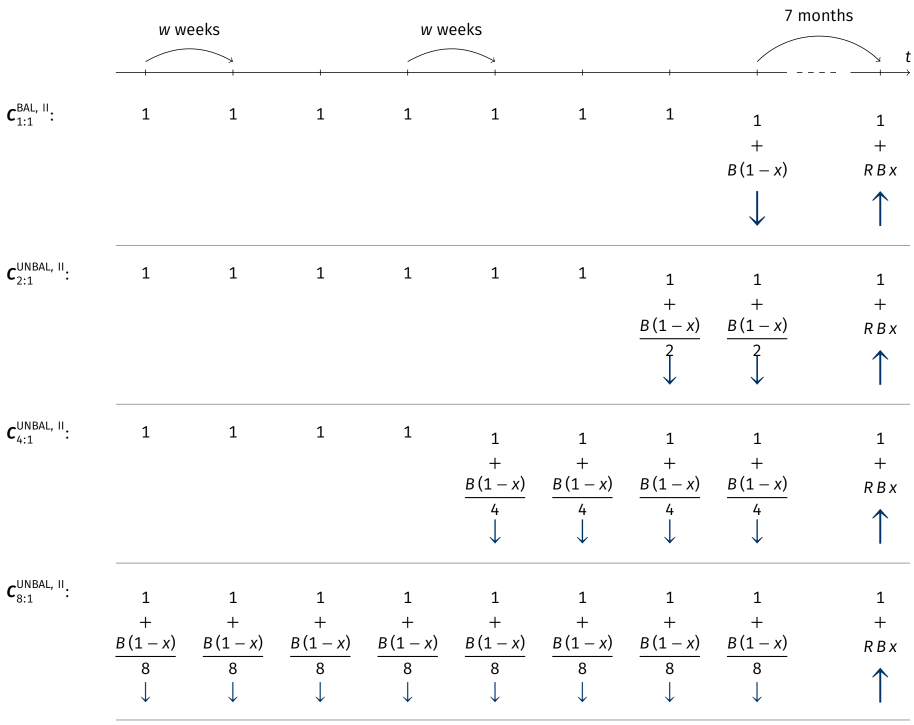

Figure 1.4. Budget Sets $C_{1:1}^{\text{BAL, II}}$ and $C_{n:1}^{\text{UNBAL, II}}$

Notes: For the values of B, R, and w that we used, see Section 1.3.5.4. The savings rate x is individuals' choice variable: they choose some $x \in \boldsymbol{X} = \{0, {}^1/_{100}, {}^2/_{100}, \ldots, 1\}$ in each trial. The arrows indicate whether and in which direction payments at the respective payment dates change if x is increased. $\sigma_\varepsilon, c^\alpha$. This figure was taken from Dertwinkel-Kalt et al. (2017).

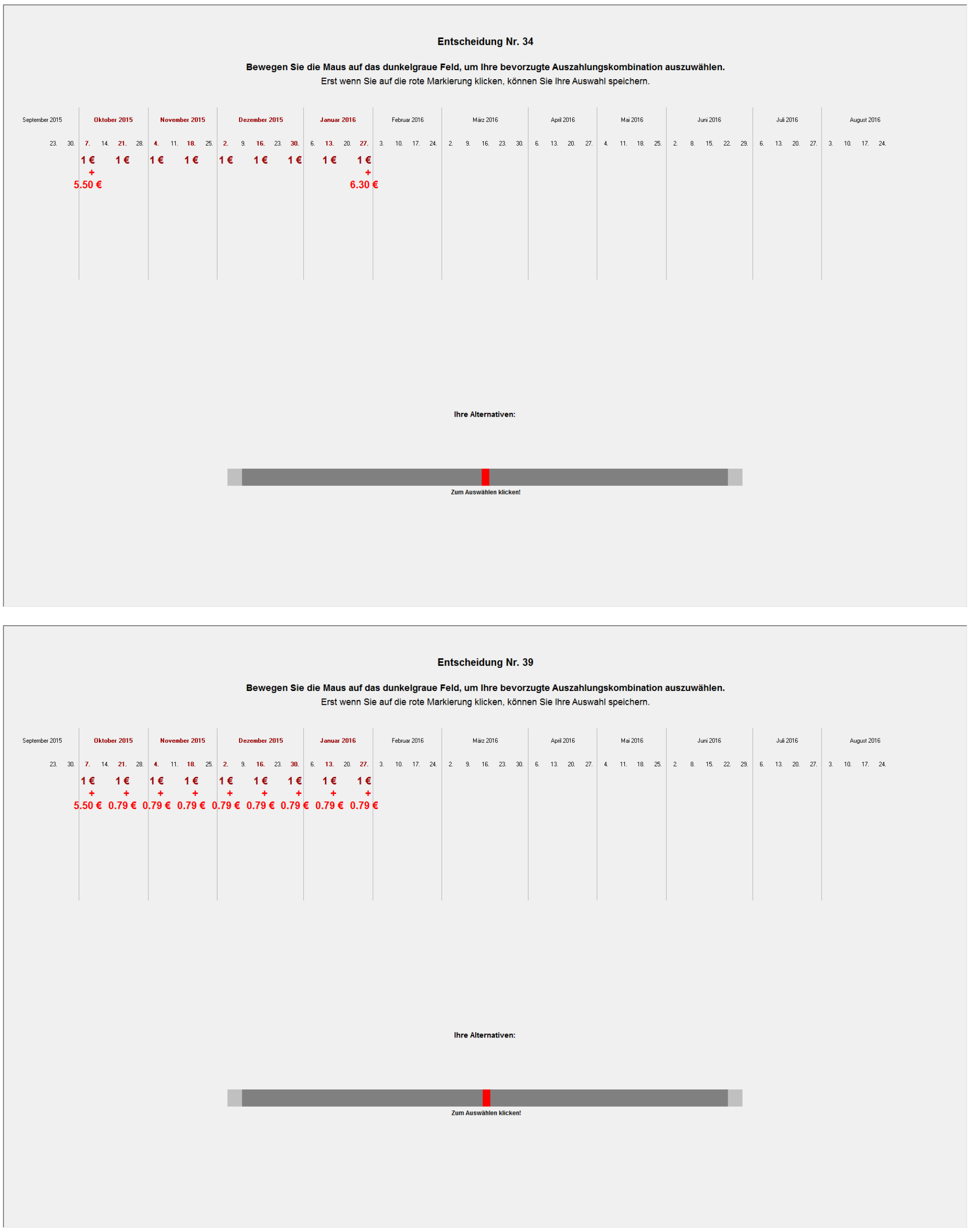

Figure 1.5. Screenshots of a $\mathrm{BAL}^{l}_{1:1}$ Decision (Top) and an $\mathrm{UNBAL}^{l}_{1:8}$ Decision (Bottom)

Note: This figure was taken from Dertwinkel-Kalt et al. (2017).

If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Here's a bulleted list:

- Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.
- Hello, here is some text without a meaning. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.
- Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

1.3.5.4 Procedure. Describe the sequence of events in your study. You could do this with the help of an enumerated list:

(1) Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look.

$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

(2) Hello, here is some text without a meaning. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.

(3) Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

1.3.6 Predictions

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages. Let's include a really, really long footnote to check how it is split across two pages.[16] Let's include a really, really long footnote to check how it is split across two

16. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information?

pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two

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Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

pages. Let's include a really, really long footnote to check how it is split across two pages.

By discounted utility we understand any intertemporal utility function that (1) is time-separable and that (2) values a payment farther in the future at most as much as an equal-sized payment closer in the future. Importantly, the predictions derived below hold for all three frequently used types of discounting—exponential, hyperbolic, and quasi-hyperbolic.

In the following, we assume that individuals base their decisions on utility derived from receiving monetary payments c_t at various dates t. This is an assumption that is frequently made in experiments on intertemporal decision making. One way to justify this assumption is that individuals anticipate to consume the payments they receive within a short period around date t. Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this assumption seems reasonable (see the arguments in favor of this view in Halevy 2014). Kőszegi and Szeidl (2013) themselves make the same assumption of "money in the utility function": "in some applications we also assume that monetary transactions induce direct utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people experience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: ... some evidence on individuals' attitudes toward money, such as narrow bracketing (...) and laboratory evidence on hyperbolic discounting (...), is difficult to explain without it." Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and monetary rewards. Additionally, we make the standard assumption that utility from money is increasing in its argument but not convex: $u'(c_t) \geq 0$ and $u''(c_t) \leq 0$.

1.3.6.1 Discounted Utility. Individuals make their allocation decisions by comparing the aggregated consumption utility of each earnings sequence $\boldsymbol{c} \in \boldsymbol{C}$. Discounted utility assumes that the utility of each period enters overall utility additively. That is, utility derived from the payment to be received at future date t can be expressed as $u_t(c_t) := D(t)\, u(c_t)$. Here, $D(t)$ denotes the individual's discount function for conversion of future utility into present utility. The discount function satisfies $0 \leq D(t)$ and $D'(t) \leq 0$, such that a payment further in the future is valued at most as much as an equal-sized payment closer in the future.[17]

The utility of earnings sequence $\boldsymbol{c}$ with payments c_t in periods $t = 1, \ldots, T$ is as follows:

17. Normalization such that $D(t) \leq 1$ is not necessary in our case. Provided that t is a metric time measure, where $t = 0$ stands for the present, examples are $D(t) := \delta^t$ with some $\delta > 0$ for exponential discounting and $D(t) := (1 + \alpha t)^{-\gamma/\alpha}$ with some $\alpha, \gamma > 0$ for generalized hyperbolic discounting.

`$$ ... $$`:

$$U(\boldsymbol{c}) = \sum_{t=1}^{T} u_t(c_t) = \sum_{t=1}^{T} D(t)\, u(c_t).$$

`\[ ... \]` with manual `\tag{...}`:

$$U(\boldsymbol{c}) \;=\; \sum_{t=1}^{T} u_t(c_t) \;=\; \sum_{t=1}^{T} D(t)\, u(c_t). \tag{II}$$

`\begin{equation} ... \end{equation}`:

$$U(\boldsymbol{c}) \;=\; \sum_{t=1}^{T} u_t(c_t) \;=\; \sum_{t=1}^{T} D(t)\, u(c_t). \tag{1.1}$$

`\begin{equation*} ... \end{equation*}`:

$$U(\boldsymbol{c}) \;=\; \sum_{t=1}^{T} u_t(c_t) \;=\; \sum_{t=1}^{T} D(t)\, u(c_t).$$

`\begin{eqnarray} ... \end{eqnarray}`:

$$U(\boldsymbol{c}) \quad = \quad \sum_{t=1}^{T} u_t(c_t) \tag{1.2}$$

$$= \quad \sum_{t=1}^{T} D(t)\, u(c_t). \tag{1.3}$$

`\begin{eqnarray*} ... \end{eqnarray*}`:

$$\begin{aligned} U(\boldsymbol{c}) \quad &= \quad \sum_{t=1}^{T} u_t(c_t) \\ &= \quad \sum_{t=1}^{T} D(t)\, u(c_t). \end{aligned}$$

`\begin{align} ... \end{align}`, equation number in the final line only:

$$\begin{aligned} U(\boldsymbol{c}) &= \sum_{t=1}^{T} u_t(c_t) \\ &= \sum_{t=1}^{T} D(t)\, u(c_t). \end{aligned} \tag{1.4}$$

`\begin{align} ... \end{align}`, equation number in each line:

$$U(\boldsymbol{c}) = \sum_{t=1}^{T} u_t(c_t) \tag{1.5}$$

$$= \sum_{t=1}^{T} D(t)\, u(c_t). \tag{1.6}$$

`\begin{align*} ... \end{align*}`:

$$\begin{aligned} U(\boldsymbol{c}) &= \sum_{t=1}^{T} u_t(c_t) \\ &= \sum_{t=1}^{T} D(t)\, u(c_t). \end{aligned}$$

`\begin{alignat}{2} ... \end{alignat}`:

$$U(\boldsymbol{c}) = \sum_{t=1}^{T} u_t(c_t) \tag{1.7}$$

$$= \sum_{t=1}^{T} D(t)\, u(c_t). \tag{1.8}$$

`\begin{alignat*}{2} ... \end{alignat*}`:

$$\begin{aligned} U(\boldsymbol{c}) &= \sum_{t=1}^{T} u_t(c_t) \\ &= \sum_{t=1}^{T} D(t)\, u(c_t). \end{aligned}$$

Individuals choose how much to allocate to the different periods by maximizing their utility over all possible earnings sequences available within a given budget set $\boldsymbol{C}$, see equations (II), (1.1), (1.2), (1.3), (1.4), (1.5), and (1.6). See also Equation 1.8. We use the superscript $^{\mathrm{DU}}$ to indicate decisions based on discounted utility.

A Subparagraph. This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$.

1.3.6.2 Focus-Weighted Utility. In this section, we extend the model of discounted utility through "focus weights," as proposed by Kőszegi and Szeidl (2013). Period-t weights g_t scale period-t consumption utility u_t. Individuals are assumed to maximize focus-weighted utility, which is defined as follows:

$$\tilde{U}(\boldsymbol{c},\boldsymbol{C}) \;:=\; \sum_{t=1}^{T} g_t(\boldsymbol{C})\,u_t(c_t). \tag{1.9}$$

In contrast to discounted utility $U(\boldsymbol{c})$, focus-weighted utility $\tilde{U}(\boldsymbol{c},\boldsymbol{C})$ has two arguments: the earnings sequence $\boldsymbol{c}$ and the choice set $\boldsymbol{C}$. The latter dependence is due to the weights g_t. These are given by a strictly increasing weighting function g that takes as its argument the difference between the maximum and the minimum attainable utility in period t over all possible earnings sequences in set $\boldsymbol{C}$:

$$g_t(\boldsymbol{C}) \;:=\; g[\Delta_t(\boldsymbol{C})] \quad \text{with} \quad \Delta_t(\boldsymbol{C}) \;:=\; \max_{\boldsymbol{c}\in\boldsymbol{C}} u_t(c_t) - \min_{\boldsymbol{c}\in\boldsymbol{C}} u_t(c_t). \tag{1.10}$$

If the underlying consumption utility function is characterized by discounted utility, then $u_t(c_t) := D(t)\,u(c_t)$. That is, focused thinkers put more weight on period t than on period t' if the discounted-utility distance between the best and worst alternative is larger for period t than for period t'.

A Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$.

Yet Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.3.6.3 Hypotheses. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. This gives rise to our first hypothesis:

Hypothesis 1.1. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Based on this, we can state our second hypothesis:

Hypothesis 1.2. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.4 Results

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. With this, we can test our hypotheses.

1.4.1 Test of Hypothesis 1.1

Our first result supports Hypothesis 1.1. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. The analysis we conducted to obtain Result 1.1 is described in detail in Table 1.1. Let's reference a section, a subsection, and a figure from the appendices: Section 1.C, Section 1.A.2, Figure 1.B.1.

Result 1.1. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place.* $\sin^2(\alpha) + \cos^2(\beta) = 1$. *If you read this text, you will get no information* $E = mc^2$. *Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text* like this *gives you information about the selected font, how the letters are written and an impression of the look.* $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. *This text should contain* all letters of the alphabet *and it should be written in of the original language.* $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. *There is no need for special contents, but the length of words should match the language.* $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Table 1.1. A *tabularray*-Based Example Table

Dependent variable	$\hat{d}$
Estimate	0.123***
	(0.011)
Observations	750
Subjects	250

Notes: Standard errors in parentheses, clustered on the subject level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.4.2 Test of Hypothesis 1.2

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. We thereby test Hypothesis 1.2.

Result 1.2. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place.* $\sin^2(\alpha) + \cos^2(\beta) = 1$. *If you read this text, you will get no information* $E = mc^2$. *Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text* like this *gives you information about the selected font, how the letters are written and an impression of the look.* $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. *This text should contain* all letters of the alphabet *and it should be written in of the original language.* $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. *There is no need for special contents, but the length of words should match the language.* $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Our second result provides evidence in support of Hypothesis 1.2. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

1.4.3 Heterogeneity

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{i=n} x_i = \frac{x_1 + x_2 + \ldots + x_n}{n}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\int_0^\infty e^{-\alpha x^2} \mathrm{d}x = \frac{1}{2}\sqrt{\int_{-\infty}^{\infty} e^{-\alpha x^2} \mathrm{d}x \int_{-\infty}^{\infty} e^{-\alpha y^2} \mathrm{d}y} = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\sum_{k=0}^{\infty} a_0 q^k = \lim_{n\to\infty} \sum_{k=0}^{n} a_0 q^k = \lim_{n\to\infty} a_0 \frac{1-q^{n+1}}{1-q} = \frac{a_0}{1-q}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

1.4.4 Structural Estimation

Inspect the variance–covariance matrix $\mathbf{\Sigma}$:

$$\mathbf{\Sigma} := \mathbf{Cov}(X) = \begin{bmatrix} \mathrm{Var}(X_1) & \cdots & \mathrm{Cov}(X_1,X_n) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n,X_1) & \cdots & \mathrm{Var}(X_n) \end{bmatrix}.$$

1.5 Discussion

1.5.1 Some Limitations

Let's reference some tables: Table 1.2 and Table 1.3. After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this*

Table 1.2. Points Awarded in Our Typeface Competition—Basic Formatting That Produces Irregular Column Widths and a Suboptimal Table Width Test Greek: $\varepsilon, \theta, \phi$

	Utopia	Computer Modern	Charter	Times Roman	Palatino
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allisőn	2	0	1	2	1
Pía	1	0	2	1	0
Đavid	1	0	2	1	1
Sum	10	5	12	8	4

gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Right?

Table 1.3. Points Awarded in Our Typeface Competition—More Sophisticated Formatting with Homogeneous Column Widths and a Table Width That Equals the Text Width

	Utopia[a]	Computer Modern[b]	Charter[c]	Times Roman[d]	Palatino[e]
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allisőn	2	0	1	2	1
Pía	1	0	2	1	0
Đavid	1	0	2	1	1
Sum	10	5	12	8	4

[a] `\usepackage{fourier}`
[b] The LaTeX standard serif font.
[c] `\usepackage[charter]{mathdesign}`
[d] `\usepackage{newtxtext, newtxmath}`
[e] `\usepackage[sc]{mathpazo}`

1.5.2 Utility from Money

In deriving our predictions (Section 1.3.6), we assume that subjects base their decisions on utility derived from receiving monetary payments c_t at various dates t. We also make the standard assumption that utility from money is increasing in its argument but not convex, i.e., $u'(c_t) > 0$ and $u''(c_t) \leq 0$. Both assumptions are frequently made in studies on intertemporal decision making.

One way to justify the assumption of utility being based on money—rather than consumption—is that individuals anticipate to consume the payments that they receive at date t within a short period around t. Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this seems reasonable (see the arguments in favor of this view in Halevy 2014).

A second justification is consistency within the discipline: Halevy (2014) points out that "in the domain of risk and uncertainty ... preferences are often defined over payments." In line with this, Kőszegi and Szeidl (2013, p. 62) make the same assumption of "money in the utility function":

> in some applications we also assume that monetary transactions induce *direct* utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people experience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: ... some evidence on individuals' attitudes toward money, such as narrow bracketing (...) and laboratory evidence on hyperbolic discounting (...), is difficult to explain without it.

Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and monetary rewards.

Let us now discuss the second assumption: that utility from money is nonconvex. We find that subjects allocate more money to the concentrated payoffs in the unbalanced than in the associated balanced budget sets—which we call concentration bias. One might argue that this relative preference for concentrated payoffs can be explained by the per-period utility function over money being convex.

Obtaining evidence on the shape of utility over money is nontrivial because it requires that at least two monetary amounts be compared with each other without the one clearly dominating the other. Thus, estimates of the curvature of the utility function over money can be obtained in two ways: the monetary amounts must be paid in different states of the world, i.e., comprise a lottery, or they have to be paid at different points in time.[18] Both methods entail particular theoretical assumptions.

Andersen et al. (2008) advocate the former approach and argue that when estimating time preference parameters, one should control for the curvature of the utility function through a measure of the curvature that is based on observed choices under risk. Their study and numerous other studies on risk attitudes consistently reveal that the vast majority of subjects is risk-averse even over small stakes. Hence, for the vast majority of subjects, utility over money is concave according to this methodology (ruling out probability weighting). Others, most notably Andreoni and Sprenger (2012), have argued that the degree of curvature measured via risky choices probably overstates the degree of curvature effective in intertemporal choices, but they also find that utility is concave (albeit close to linear). Given this unambiguous evidence from previous studies, it is implausible that our subjects exhibit convex utility over money.

1.6 Conclusion

Cite some more papers (Yaari 1965; Warner and Pleeter 2001; Davidoff, Brown, and Diamond 2005; Benartzi, Previtero, and Thaler 2011). Let's cite a book: Luce (1959). Let's cite a contribution to a collected volume: Harrison and Rutström (2008) and a collection (an edited volume) itself: Kagel and Roth (2016). Now let's cite presentations at conferences: Vosgerau et al. (2008) and Beute and Kort (2012). Attema et al. (2016) propose a method for "measuring discounting without measuring utility"[19].

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

18. As a matter of fact, the latter was the motivation behind Samuelson (1937): "Under the following four assumptions, it is believed possible to arrive theoretically at a precise measure of the marginal utility of *money income* ..." (p. 155; emphasis in the original).

19. The basic idea of their method is intriguingly simple: Imagine an individual who is indifferent between, say, Option A: \$10 today and Option B: \$10 in one year plus \$10 in two years. With a constant annual discount factor δ, this indifference translates to $u(\$10) = \delta\, u(\$10) + \delta^2\, u(\$10)$, so that $u(\$10)$ cancels out, and δ can be readily calculated as the solution to $1 = \delta + \delta^2$.

Appendix 1.A Put More Complicated Derivations and Proofs Here

1.A.1 Appendix Subsection

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$.

(1) Erster Listenpunkt, Stufe 1
   a. Erster Listenpunkt, Stufe 2
      i. Erster Listenpunkt, Stufe 3
      ii. Zweiter Listenpunkt, Stufe 3
      iii. Dritter Listenpunkt, Stufe 3
      iv. Vierter Listenpunkt, Stufe 3
   b. Zweiter Listenpunkt, Stufe 2
   c. Dritter Listenpunkt, Stufe 2
   d. Vierter Listenpunkt, Stufe 2
(2) Zweiter Listenpunkt, Stufe 1
(3) Dritter Listenpunkt, Stufe 1
(4) Vierter Listenpunkt, Stufe 1

The typeset math below follows the ISO recommendations that only variables be set in italic. Note the use of upright shapes for "d," "e," and "π." (These are entered as `\mathup{d}`, `\mathup{e}`, and `\mathup{\pi}`, respectively.)

Theorem 1.1 (Simplest form of the *Central Limit Theorem*). *Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables with mean* 0 *and variance* 1 *on a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Then*

$$\mathbb{P}\left(\frac{X_1+\cdots+X_n}{\sqrt{n}}\le y\right) \to \mathfrak{N}(y) := \int_{-\infty}^{y} \frac{\mathrm{e}^{-v^2/2}}{\sqrt{2\pi}}\,\mathrm{d}v \quad \text{as } n\to\infty,$$

or, equivalently, letting $S_n := \sum_1^n X_k$,

$$\mathbb{E}f(S_n/\sqrt{n}) \to \int_{-\infty}^{\infty} f(v)\frac{\mathrm{e}^{-v^2/2}}{\sqrt{2\pi}}\,\mathrm{d}v \quad \text{as } n\to\infty, \text{ for every } f\in \mathrm{b}\mathscr{C}(\mathbb{R}).$$

1.A.2 Salience

Salience theory (Bordalo, Gennaioli, and Shleifer 2012, 2013) represents a behavioral model according to which the most distinctive features of the available alternatives receive a particularly large share of attention and are therefore over-weighted. More precisely, a particular attribute out of all attributes of an alternative becomes the more salient, the more it differs from that attribute's average level over all available alternatives.

Formally, alternatives are assumed to be uniquely characterized by the values they take in $T \geq 1$ attributes (or, "dimensions"). Utility is assumed to be additively separable in attributes, and salience attaches a decision weight to each attribute of each good which indicates how salient the respective attribute is for that good. Suppose an agent chooses one alternative from some finite choice set $\boldsymbol{C}$. Let t index the T different attributes, and let k index the K available alternatives. Let $u_t(\cdot)$ denote the function which assigns utility to values in dimension t. Denote by a_t^k the level of attribute t of good k and define $u_t^k := u_t(a_t^k)$ as the utility that dimension t of good k yields. Let $\overline{u}_t$ be the average utility level, across all K goods, of dimension t. The salience of each dimension of good k is determined by a symmetric and continuous salience function $\sigma(\cdot,\cdot)$ that satisfies the following two properties:

(1) *Ordering.* Let $\mu := \text{sgn}(u_t^k - \overline{u}_t)$. Then for any $\varepsilon, \varepsilon' \geq 0$ with $\varepsilon + \varepsilon' > 0$, it holds that

$$\sigma(u_t^k + \mu\varepsilon, \overline{u}_t - \mu\varepsilon') > \sigma(u_t^k, \overline{u}_t). \tag{1.A.1}$$

(2) *Diminishing sensitivity.* For any $u_t^k, \overline{u}_t \geq 0$ and all $\varepsilon > 0$, it holds that

$$\sigma(u_t^k + \varepsilon, \overline{u}_t + \varepsilon) < \sigma(u_t^k, \overline{u}_t). \tag{1.A.2}$$

Following the smooth salience characterization proposed in Bordalo, Gennaioli, and Shleifer (2012, p. 1255), each dimension t of good k receives weight $\Delta^{-\sigma(u_t^k, \overline{u}_t)}$, where $\Delta \in (0,1]$ is a constant that captures an agent's susceptibility to salience. $\Delta = 1$ gives rise to a rational decision maker, and the smaller Δ, the stronger is the salience bias. We call an agent with $\Delta < 1$ a salient thinker.

A reference with a large number of authors is Henrich et al. (2005).

Appendix 1.B Additional Figures

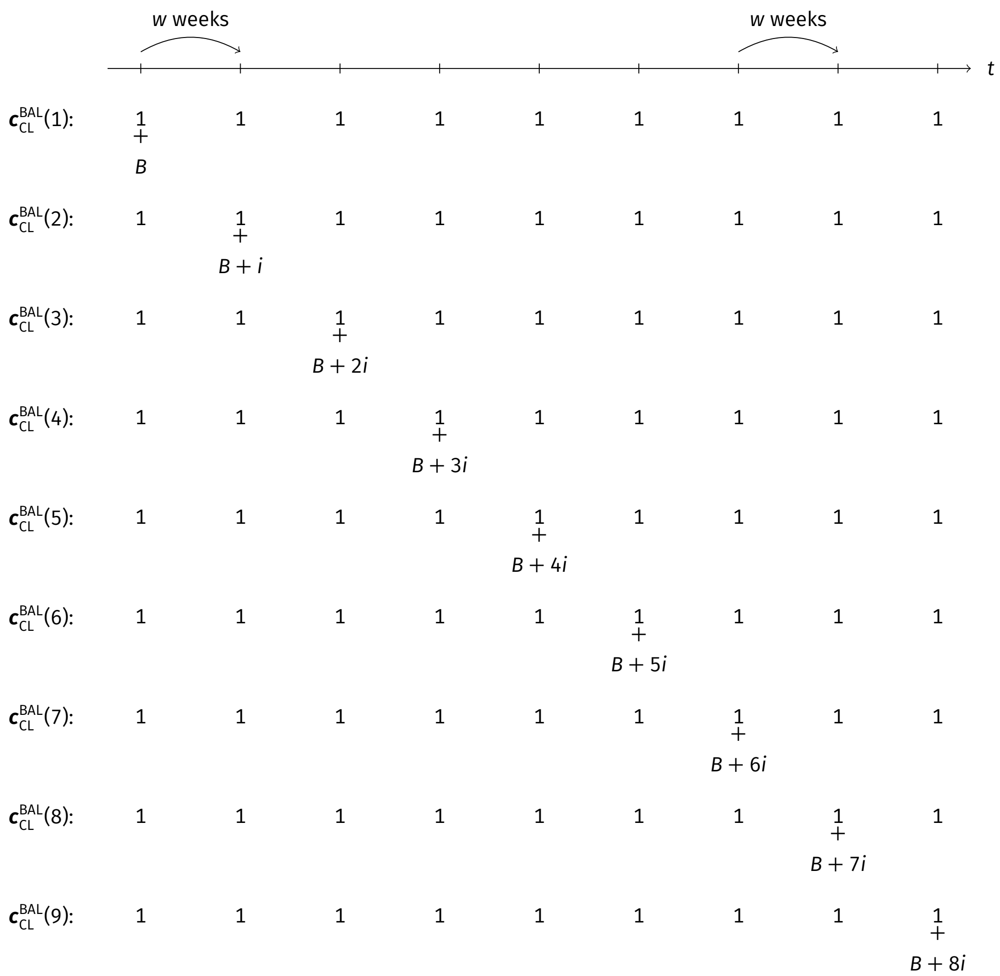

Figure 1.B.1. Earnings Sequences Included in Choice List $\boldsymbol{C}^{BAL}_{CL}$

Notes: For the values of B, i, and w that we used see Section 1.3. Figure taken from Dertwinkel-Kalt et al. (2017).

w weeks $\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad$ w weeks $\longrightarrow t$

$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(1)$:	$1+B$	1	1	1	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(2)$:	$1+\frac{B+i}{2}$	$1+\frac{B+i}{2}$	1	1	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(3)$:	$1+\frac{B+2i}{3}$	$1+\frac{B+2i}{3}$	$1+\frac{B+2i}{3}$	1	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(4)$:	$1+\frac{B+3i}{4}$	$1+\frac{B+3i}{4}$	$1+\frac{B+3i}{4}$	$1+\frac{B+3i}{4}$	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(5)$:	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(6)$:	$1+\frac{B+5i}{6}$	$1+\frac{B+5i}{6}$	$1+\frac{B+5i}{6}$	$1+\frac{B+5i}{6}$	$1+\frac{B+5i}{6}$	$1+\frac{B+5i}{6}$	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(7)$:	$1+\frac{B+6i}{7}$	$1+\frac{B+6i}{7}$	$1+\frac{B+6i}{7}$	$1+\frac{B+6i}{7}$	$1+\frac{B+6i}{7}$	$1+\frac{B+6i}{7}$	$1+\frac{B+6i}{7}$	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(8)$:	$1+\frac{B+7i}{8}$	$1+\frac{B+7i}{8}$	$1+\frac{B+7i}{8}$	$1+\frac{B+7i}{8}$	$1+\frac{B+7i}{8}$	$1+\frac{B+7i}{8}$	$1+\frac{B+7i}{8}$	$1+\frac{B+7i}{8}$	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}(9)$:	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$	$1+\frac{B+8i}{9}$

Figure 1.B.2. Earnings Sequences Included in Choice List $\boldsymbol{C}_{\mathrm{CL}}^{\mathrm{UNBAL,I}}$

Notes: For the values of B, i, and w that we used see Section 1.3. Figure taken from Dertwinkel-Kalt et al. (2017).

w weeks $\qquad\qquad\qquad\qquad$ w weeks

t

$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(1)$:	$1+\frac{B}{9}$	$1+\frac{B}{9}$	$1+\frac{B}{9}$	$1+\frac{B}{9}$	$1+\frac{B}{9}$	$1+\frac{B}{9}$	$1+\frac{B}{9}$	$1+\frac{B}{9}$	$1+\frac{B}{9}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(2)$:	1	$1+\frac{B+i}{8}$	$1+\frac{B+i}{8}$	$1+\frac{B+i}{8}$	$1+\frac{B+i}{8}$	$1+\frac{B+i}{8}$	$1+\frac{B+i}{8}$	$1+\frac{B+i}{8}$	$1+\frac{B+i}{8}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(3)$:	1	1	$1+\frac{B+2i}{7}$	$1+\frac{B+2i}{7}$	$1+\frac{B+2i}{7}$	$1+\frac{B+2i}{7}$	$1+\frac{B+2i}{7}$	$1+\frac{B+2i}{7}$	$1+\frac{B+2i}{7}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(4)$:	1	1	1	$1+\frac{B+3i}{6}$	$1+\frac{B+3i}{6}$	$1+\frac{B+3i}{6}$	$1+\frac{B+3i}{6}$	$1+\frac{B+3i}{6}$	$1+\frac{B+3i}{6}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(5)$:	1	1	1	1	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$	$1+\frac{B+4i}{5}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(6)$:	1	1	1	1	1	$1+\frac{B+5i}{4}$	$1+\frac{B+5i}{4}$	$1+\frac{B+5i}{4}$	$1+\frac{B+5i}{4}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(7)$:	1	1	1	1	1	1	$1+\frac{B+6i}{3}$	$1+\frac{B+6i}{3}$	$1+\frac{B+6i}{3}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(8)$:	1	1	1	1	1	1	1	$1+\frac{B+7i}{2}$	$1+\frac{B+7i}{2}$
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}(9)$:	1	1	1	1	1	1	1	1	$1+B+8i$

Figure 1.B.3. Earnings Sequences Included in Choice List $\boldsymbol{C}_{\mathrm{CL}}^{\mathrm{UNBAL,II}}$

Notes: For the values of B, i, and w that we used see Section 1.3. Figure taken from Dertwinkel-Kalt et al. (2017).

Appendix 1.C Additional *siunitx* and *tabularray* Example Tables

Table 1.C.1. An Example of a Regression Table (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017). Remember to Mention the Dependent Variable!

	(1)	(2)	(3)	(4)	(5)
Treatment	−0.390	−0.228	−0.729*	−0.449*	−0.453**
	(+0.352)	(−0.205)	[+0.377]	[−0.245]	{+0.204}
Female	0.948***	0.061	0.188	0.305	0.385*
	(0.354)	(0.233)	(0.372)	(0.226)	(0.222)
Female × Treatment	0.169	0.251	0.892*	0.454	0.439
	(0.514)	(0.325)	(0.533)	(0.341)	(0.307)
Final high school grade	−0.101	0.013	0.076	0.117	0.039
	(0.198)	(0.144)	(0.224)	(0.146)	(0.133)
Trait self-control	−0.016	0.002	−0.016	0.000	−0.007
	(0.016)	(0.010)	(0.015)	(0.010)	(0.009)
Constant	2.357***	1.512***	−0.322	2.158***	1.437***
	(0.239)	(0.144)	(0.265)	(0.161)	(0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	−0.221	0.023	0.163	0.004	−0.014
p_F[Treatment × (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Notes: Dependent variable: $m_{\smile}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

Table 1.C.2. Figure Grouping via `siunitx` in a Table

(1)	(2)	(3)
−0.100*	−0.100 01*	−123 456.444***
(2.871)	(2.871 23)	[+50 000.123]

Table 1.C.3. Overview of the Choice Lists Presented to Subjects (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017)

	Alternative ***A***				Alternative ***B***			
	$c_{\boldsymbol{A},1}$	$p_{\boldsymbol{A},1}$	$c_{\boldsymbol{A},2}$	$p_{\boldsymbol{A},2}$	$c_{\boldsymbol{B},1}$	$p_{\boldsymbol{B},1}$	$c_{\boldsymbol{B},2}$	$p_{\boldsymbol{B},2}$
Choice List I: risky/risky (x = €22.00, r = €7.50, k = €11.50; 25 rows)								
Top row	€ 3.00	50%	€22.00	50%	€ 3.00	50%	€ 7.00	50%
Center row	€ 3.00	50%	€22.00	50%	€ 9.00	50%	€13.00	50%
Row with $m = 0$	€ 3.00	50%	€22.00	50%	€10.50	50%	€14.50	50%
Bottom row	€ 3.00	50%	€22.00	50%	€15.00	50%	€19.00	50%
Choice List II: safe/risky (x = €16.00, r = €5.00, k = €5.00; 19 rows)								
Top row	€11.00	100%			€11.00	50%	€21.00	50%
Center row	€11.00	100%			€ 6.50	50%	€16.50	50%
Row with $m = 0$	€11.00	100%			€ 6.00	50%	€16.00	50%
Bottom row	€11.00	100%			€ 2.00	50%	€12.00	50%
Choice List III: "long shot" (x = €14.00, r = −€36.00, k = €7.00; 21 rows)								
Top row	€ 7.00	90%	€50.00	10%	€ 7.00	90%	€10.00	10%
Row with $m = 0$	€ 7.00	90%	€50.00	10%	€11.00	90%	€14.00	10%
Center row	€ 7.00	90%	€50.00	10%	€12.00	90%	€15.00	10%
Bottom row	€ 7.00	90%	€50.00	10%	€17.00	90%	€20.00	10%
Choice List IV: delayed payoffs (x = €18.00, r = €6.00, k = €8.50, paid in one week; 20 rows)								
Top row	€ 9.50	50%	€12.00	50%	€ 9.50	50%	€24.00	50%
Above-center row	€ 9.50	50%	€12.00	50%	€ 5.00	50%	€19.50	50%
Below-center row	€ 9.50	50%	€12.00	50%	€ 4.50	50%	€19.00	50%
Row with $m = 0$	€ 9.50	50%	€12.00	50%	€ 3.50	50%	€18.00	50%
Bottom row	€ 9.50	50%	€12.00	50%	€ 0.00	50%	€14.50	50%

Table 1.C.4. A Really Long Table That Spans Multiple Pages

	(1)	(2)	(3)	(4)
Row 1	0.0070	0.1356	0.1560	0.8979
Row 2	0.4223	0.7311	0.4213	0.6900
Row 3	0.0767	0.5110	0.7399	0.9491
Row 4	0.5954	0.1685	0.3778	0.9960
Row 5	0.6465	0.0524	0.8895	0.1544
Row 6	0.3838	0.7069	0.1773	0.5785
Row 7	0.1537	0.5442	0.6361	0.0327
Row 8	0.0879	0.1812	0.3082	0.2942
Row 9	0.2720	0.2565	0.6214	0.8944
Row 10	0.4873	0.3064	0.9913	0.0591
Row 11	0.8387	0.1713	0.6747	0.7455
Row 12	0.0645	0.4891	0.2892	0.1013
Row 13	0.0989	0.3798	0.5795	0.3725
Row 14	0.3256	0.7080	0.0262	0.8709
Row 15	0.7867	0.8768	0.0690	0.6081
Row 16	0.2713	0.4399	0.5838	0.6107
Row 17	0.5236	0.1527	0.4402	0.8002
Row 18	0.4851	0.4619	0.4040	0.2711
Row 19	0.1742	0.8151	0.2757	0.4184
Row 20	0.0495	0.3288	0.2759	0.1452
Row 21	0.1678	0.2403	0.1993	0.3676
Row 22	0.4977	0.9472	0.2810	0.2493
Row 23	0.6777	0.6516	0.3573	0.1413
Row 24	0.3668	0.3075	0.8724	0.3945
Row 25	0.5877	0.5670	0.0417	0.5213
Row 26	0.3599	0.5485	0.2407	0.6362
Row 27	0.1029	0.9796	0.5696	0.8696
Row 28	0.3070	0.8169	0.4015	0.4386
Row 29	0.4453	0.0670	0.3726	0.3257
Row 30	0.2648	0.9977	0.8864	0.0755
Row 31	0.4085	0.2017	0.5406	0.1333
Row 32	0.4861	0.4466	0.3472	0.2486
Row 33	0.5996	0.8639	0.1837	0.7636
Row 34	0.4446	0.3755	0.6901	0.4208
Row 35	0.9616	0.3585	0.0074	0.2867
Row 36	0.5168	0.5752	0.5778	0.0060

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 37	0.7978	0.0283	0.7998	0.9952
Row 38	0.0561	0.3133	0.1207	0.6922
Row 39	0.5237	0.1488	0.9217	0.2268
Row 40	0.0944	0.7939	0.6252	0.9836
Row 41	0.3179	0.6226	0.4493	0.4277
Row 42	0.7175	0.7267	0.8016	0.6880
Row 43	0.0192	0.4807	0.7610	0.9808
Row 44	0.9923	0.8888	0.4494	0.0645
Row 45	0.3938	0.8529	0.0496	0.0429
Row 46	0.1135	0.6166	0.5899	0.7500
Row 47	0.0654	0.1640	0.1952	0.0431
Row 48	0.8895	0.0549	0.1105	0.1284
Row 49	0.6817	0.8942	0.6597	0.3661
Row 50	0.6690	0.8817	0.2343	0.1903
Row 51	0.4091	0.0874	0.4726	0.1381
Row 52	0.9061	0.9039	0.7439	0.2061
Row 53	0.5282	0.2135	0.5223	0.7846
Row 54	0.6505	0.7404	0.8748	0.2078
Row 55	0.5824	0.8443	0.3242	0.8253
Row 56	0.0151	0.9929	0.4812	0.5010
Row 57	0.7296	0.8420	0.1535	0.4273
Row 58	0.8102	0.8068	0.1832	0.8830
Row 59	0.1650	0.5545	0.1820	0.0791
Row 60	0.5882	0.5750	0.9195	0.8993
Row 61	0.0638	0.5132	0.5994	0.0877
Row 62	0.9916	0.8032	0.0564	0.3218
Row 63	0.5555	0.4078	0.7056	0.9225
Row 64	0.8680	0.5577	0.2992	0.0941
Row 65	0.2939	0.7801	0.7039	0.7295
Row 66	0.0829	0.6756	0.5386	0.0644
Row 67	0.3868	0.4199	0.0308	0.5947
Row 68	0.0943	0.2663	0.0379	0.0887
Row 69	0.0050	0.1396	0.8348	0.2830
Row 70	0.9585	0.8018	0.4472	0.9477
Row 71	0.8153	0.2659	0.7030	0.4096
Row 72	0.7532	0.4214	0.3914	0.2360

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 73	0.6419	0.2074	0.7386	0.0653
Row 74	0.4215	0.7004	0.3193	0.9282
Row 75	0.1307	0.8242	0.1305	0.8925
Row 76	0.5812	0.6879	0.4844	0.0464
Row 77	0.1080	0.5293	0.2700	0.4844
Row 78	0.3073	0.7945	0.8300	0.3479
Row 79	0.4777	0.5842	0.2233	0.3206
Row 80	0.7218	0.7687	0.0432	0.7268
Row 81	0.1427	0.8696	0.7573	0.1263
Row 82	0.0244	0.6493	0.6750	0.9651
Row 83	0.1925	0.4131	0.3064	0.0508
Row 84	0.8678	0.3827	0.7732	0.3896
Row 85	0.6830	0.0868	0.0773	0.1712
Row 86	0.2699	0.5507	0.1200	0.4458
Row 87	0.3873	0.8615	0.0624	0.4357
Row 88	0.0610	0.0065	0.1505	0.0287
Row 89	0.3380	0.6846	0.1305	0.8998
Row 90	0.4337	0.2892	0.9326	0.7977
Row 91	0.7618	0.7254	0.6185	0.5718
Row 92	0.2404	0.2312	0.6645	0.7351
Row 93	0.8908	0.4011	0.6728	0.4192
Row 94	0.7596	0.5054	0.3343	0.1696
Row 95	0.9736	0.2894	0.8395	0.7554
Row 96	0.2555	0.3570	0.6331	0.3460
Row 97	0.5865	0.8620	0.9528	0.8383
Row 98	0.1753	0.9843	0.5822	0.7130
Row 99	0.2085	0.7513	0.4976	0.6609
Row 100	0.8550	0.6317	0.2716	0.3482
Row 101	0.0003	0.2699	0.1657	0.9740
Row 102	0.8108	0.7631	0.4779	0.7736
Row 103	0.1700	0.7518	0.6194	0.2642
Row 104	0.9089	0.7737	0.1760	0.1838
Row 105	0.2693	0.6957	0.8645	0.7214
Row 106	0.7675	0.7649	0.1831	0.5527
Row 107	0.6605	0.6763	0.6069	0.6509
Row 108	0.9355	0.8627	0.1932	0.1369

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 109	0.2459	0.2674	0.5147	0.3251
Row 110	0.1111	0.9926	0.6565	0.3905
Row 111	0.3883	0.7516	0.0597	0.2444
Row 112	0.3873	0.8884	0.8992	0.4628
Row 113	0.7374	0.3370	0.2922	0.8778
Row 114	0.9644	0.3383	0.7343	0.4642
Row 115	0.8793	0.1624	0.6602	0.6129
Row 116	0.7910	0.7928	0.9132	0.4582
Row 117	0.4158	0.6584	0.0655	0.3760
Row 118	0.6719	0.8505	0.2902	0.3726
Row 119	0.6456	0.6116	0.7580	0.3331
Row 120	0.9372	0.5338	0.9066	0.8391
Row 121	0.1427	0.6179	0.7094	0.5079
Row 122	0.1748	0.9789	0.1452	0.5829
Row 123	0.7514	0.2678	0.7714	0.1895
Row 124	0.4058	0.7714	0.4468	0.5559
Row 125	0.0799	0.6205	0.4477	0.3788
Row 126	0.3297	0.7600	0.5485	0.8005
Row 127	0.8873	0.3812	0.9346	0.4062
Row 128	0.5164	0.9326	0.8897	0.6300
Row 129	0.1876	0.8342	0.5704	0.9817
Row 130	0.3990	0.2170	0.8709	0.4717
Row 131	0.4454	0.3671	0.2185	0.9753
Row 132	0.8951	0.9321	0.3854	0.4805
Row 133	0.3442	0.8316	0.8667	0.6898
Row 134	0.0586	0.2090	0.3720	0.1668
Row 135	0.1312	0.5375	0.6314	0.2907
Row 136	0.5138	0.7588	0.2177	0.7461
Row 137	0.4966	0.1501	0.3993	0.0631
Row 138	0.7154	0.8785	0.8362	0.5782
Row 139	0.6265	0.2019	0.9703	0.2705
Row 140	0.5248	0.5235	0.5018	0.9854
Row 141	0.2711	0.5263	0.8829	0.8525
Row 142	0.1335	0.8354	0.0190	0.3996
Row 143	0.7644	0.3912	0.8849	0.7440
Row 144	0.4358	0.2065	0.4528	0.8955

Continued on next page.

Table 1.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 145	0.9038	0.0718	0.7912	0.5230
Row 146	0.1919	0.7559	0.2908	0.2352
Row 147	0.6801	0.3179	0.8315	0.7988
Row 148	0.7810	0.3397	0.5245	0.8478
Row 149	0.1458	0.1098	0.2659	0.2319
Row 150	0.7207	0.1931	0.2071	0.0241

Note: At the very end, you can add some notes to the table.

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Chapter 2

My Second Paper Has a Long Title That Spans Two Lines*

Joint with Adam Smith, Janet Smith, and Jeremiah Smith

Abstract. This is where the abstract of your paper goes. The abstract is an extremely brief summary of your paper and basically follows the same structure as the paper itself: background/motivation of your study, methods, results, discussion/conclusion. Each section, however, is covered in a single sentence or maybe two sentences instead of an entire section. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

* This footnote can be used for acknowledgments. This is where you can express your gratitude to referees, editors, and colleagues for their valuable feedback and suggestions that helped improve your manuscript. Financial support by third parties can also be mentioned here.

2.1 Introduction

> *"Most people can save a few dollars a day or even \$10 a day," she said. "That's doable. But if you say, 'Can you save \$300 a month or a couple of thousand dollars a year?' people will say, 'Whoa.' Avoiding that 'whoa,' which is the hesitancy that can derail planning, is what consultants like Ms. Davidson are trying to do."*
>
> —*New York Times*, March 27, 2016

[Anonym 2]

[Holger 4]
Ersetzt: some

[Lou E. 4]
Gelöscht: automatically

[U. R. 3]
Eingefügt

[Holger 5]
Eingefügt
We already included several references above.

[U. R. 4]
Check whether there are more recent publications!

This template uses the Charter typeface for the body text. Charter is a serif typeface and was designed in 1987 by Matthew Carter. By contrast, all headings, tables, and captions are set in a sans-serif typeface. The sans-serif typeface used in this document is Fira Sans, designed by Erik Spiekermann and collaborators.

The math settings are adjusted in the preamble to the effect that mathematical formulas are automatically typeset in the same font as the surrounding text. That is, math in a serif environment will be set in a serif font, while math in a sans-serif environment will use the sans-serif font. This is an aesthetic choice that may not please everyone given that a sans-serif font may be used in mathematical formulas to express a particular meaning. These cases are, however, very rare.

Let us cite a couple of publications (Lisi 1995; Andersen et al. 2008; Andreoni and Sprenger 2012; Balakrishnan, Haushofer, and Jakiela 2016; Bursztyn et al. 2025). With the options set for BibLaTeX in the preamble, citations in the body text are sorted chronologically—irrespective of the order of the "citekeys" in your input. In the list of references, entries are sorted alphabetically by author surname. Let's cite Andersen et al. (2008) once more.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Some additional references: See Sims (2003) and Gabaix (2014) for models of "rational inattention" or "goal-driven attention." See Bordalo, Gennaioli, and Shleifer (2012, 2013), Kőszegi and Szeidl (2013), Taubinsky (2014), and Bushong, Rabin, and Schwartzstein (2016) for models of "stimulus-driven attention."

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Hello, here is some text without a meaning. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.

In Section 2.3, we describe the design of our study. We present the data analysis and our results in Section 2.4. In Section 2.5, we discuss the plausibility of potential alternative explanations. Section 2.6 concludes.

[Lou E. 5]
Italics?

[Holger 6]
Gelöscht: in detail
Too wordy.

[Lou E. 6]
Ersetzt: will conclude
Let's use the present tense throughout.

2.2 Portability

2.2.1 Clean Code/Semantic Coding: Separating Content and Formatting

All written texts are structured into words and sentences. If they are long enough, texts are also structured into paragraphs. Academic texts, in addition, feature sections, subsections, and often subsubsections, as well as figures, tables, lists of references, and potentially appendices. These structural elements are associated with headings/titles and captions.

The structure of a document is conveyed to the reader through *formatting*. Effective formatting of academic documents, thus, requires that all elements of a particular type—say, section headings—be identifiable as such. Identifiability is achieved through styling that is *consistent within type* and *distinct across types*. For example, all section headings must be formatted identically, and at the same time, they must look different from subsection headings.

As a consequence, one should refrain from using manual, discretionary formatting whenever possible. In line with this principle, the source code of this template refers to the document's structure as much as possible. This is also called "semantic coding": using code that describes the *meaning* of the content and not *how* those elements should look.

- Examples for *semantic* commands are `\author`, `\title`, `\section`, `\emph`, `\url`, `\cite`, `\begin{figure}` … `\end{figure}`, `\caption`, `\begin{quote}` … `\end{quote}`.
- Examples for LaTeX commands that are *not semantic*—and should be avoided—are `\textbf`, `\textit`, `\newpage`, `\pagebreak`, `\linebreak`, `\\`, `\noindent`,

`\smallskip`, `\centering`, `\noindent`, `\hspace`, `\vspace`, `\cellcolor`, `\multirow`.

The reason why one should stick to using semantic commands is that they keep the code *portable*: Semantic code can be *reused* across document without any changes (or at least without major adjustments). This is not then case with nonsemantic code, that is, code that includes manual formatting instructions. Instructions like `\bigskip`, `\hspace`, `\\` (even worse, `\\ \\`), etc. may result in decent formatting in a particular document but are bound to lead to undesirable formatting in another document. This applies, in particular, when you face the task of combining several research papers into one, "cumulative" doctoral thesis. Moreover, manual formatting is error-prone, as it easily leads to inconsistent styling.

This entails two things: First, the formatting should be determined, as much as possible, in the *preamble* of the LaTeX document. This comes at the cost, of course, of a rather long preamble. Second, instead of defining custom commands, one should *redefine* standard LaTeX commands or use (widely adopted) packages that are available on CTAN (https://ctan.org).

This template follows the above principles, resulting in clean, portable code: The source code of the body of this template (apart from the boxes with examples) consists almost completely of semantic, basic LaTeX, with commands from popular packages on top.

2.2.2 Using BibLaTeX (Rather Than BibTeX)

This template uses the modern BibLaTeX framework (the *biblatex-chicago* package, https://ctan.org/pkg/biblatex-chicago, to be precise) instead of the vintage BibTeX framework to manage the references included in the document. The reason for this choice is that BibLaTeX is much more flexible than BibTeX and also handles multi-language references much better.

Moreover, in line with the principle describe above, BibLaTeX permits much better separation of a document's content (in this case, the *.bib* file) and its formatting (of the citations and the bibliography) than BibTeX. In particular, suppressing output of information that is present in a *.bib* file is cumbersome with BibTeX: It amounts to editing the "bibliography style" *.bst* file, and *.bst* files have a syntax that is completely different from LaTeX and difficult to learn. It is usually easier to just remove the information from the *.bib* file. This, however, impacts the portability of the *.bib* file: In other circumstances one might want that very information to be present. BibLaTeX simplifies skipping information that is present in the *.bib* file.

An example: Researchers frequently present their research in the form of posters. Space on posters is usually quite limited. Hence, one might not want authors' first names to be abbreviated on a poster. In the associated paper, however, the names should be printed in full to minimize ambiguity. With BibTeX, this requires time-consuming adjustments of the *.bst* file. With BibLaTeX, it is much easier: In the *.tex* file for the poster, one can include the command

```
\ExecuteBibliographyOptions{giveninits = true}
```

Similarly, in a paper, the list of references should include URLs or DOIs (digital object identifiers) whenever they exist, to make it easy for readers to locate a cited work. On a poster, by contrast, one might want to suppress DOIs and URLs. With BibTeX, this would require time-consuming adjustments of the *.bst* file. With BibLaTeX, one can simply do, for instance,

```
\AtEveryBibitem{%
  \ifentrytype{article}{\clearfield{doi}\clearfield{url}}{}%
}
```

Using BibLaTeX requires using the program `biber` instead of `bibtex` for compiling the bibliography. When using the Overleaf online service, using `biber` instead of `bibtex` is as simple as it gets: Overleaf automatically chooses `biber` to compile the bibliography when it encounters BibLaTeX's `backend = biber` option.[1] And when using the TeXstudio editor (https://texstudio.org), an easy way to make this happen is to include the "magic comment"

```
% !BIB program = biber
```

at the beginning of the *.tex* file.

2.2.3 Using the *tabularray* Package (Rather Than *tabularx*/*tabulary*/*tabu*, *booktabs*, *longtable*, *xltabular*, *multirow*, *makecell*, *colortbl*, *parnotes*, *threeparttable*, ...)

The principle of separating the content from its formatting as much as possible can also be applied to tables. The *tabularray* package (https://ctan.org/pkg/tabularray) allows you to do just that. *tabularray* has been around since May 2021, so it is relatively new—but it has already matured into a package that is extremely useful. *tabularray* replaces (or loads) all the other table-related LaTeX packages that you may have used so far—like *tabularx*/*tabulary*/*tabu*, *longtable*, *xltabular*, *multirow*, *makecell*, *booktabs*, *colortbl*, *parnotes*, *threeparttable*, ...

Separating the contents of a table from its formatting means that no formatting instructions whatsoever need to be placed in the table cells—neither for drawing rules between table rows nor for highlighting entries nor for increasing spacing nor for changing alignment, and so on. Instead, the placement of rules at the top and

1. This applies to the commercial version (https://www.overleaf.com) as well as to the free versions https://overleaf-students.astro.uni-bonn.de/ and https://uni-bonn.sciebo.de/apps/overleaf_nextcloud/launcher/launch.

bottom of a table or between particular rows, the specification of column widths, the specification of font sizes and cell alignment, desired highlighting via boldface or a background color, etc., can all be coded as arguments to the table environment. This is beneficial in at least three aspects:

- The source code of the table's contents is very clean. This particularly applies to animated Beamer presentations: no need to put `\pause`/`\only`/`\cellcolor`/ …in table cells.
- It makes table contents *portable*: A table's contents can be reused easily across multiple documents that are formatted in different ways.
- It also drastically simplifies the inclusion of table contents that are produced by an external program such as Python, R, or Stata.

Further additional helpful features of *tabularray* are:

- *tabularray* makes it easy to bottom-align or top-align particular table rows or individual cells entries, while keeping a different alignment for the other rows/ cells.
- Table entries that consist of multiple lines of text become easy to produce: Just enclose them in curly braces, like in this example: `... & {Line 1 \\ Line 2} & ... \\`. No need to use `\multirow` or `\makecell` any more.
- You can set document-wide formatting defaults to achieve consistent formatting of all tables. For instance, you can set defaults such that all tables feature the same top and bottom rules. Or, you could make all column heads be typeset in boldface document-wide.

When using `\UseTblrLibrary{siunitx}`, *tabularray* loads the *siunitx* package. This is another useful package which permits separating content from formatting. In particular, the `S` column type is able to round numbers to a specified precision. This means that one can include table content produced by an external program with as many decimal digits as that program produces and have LaTeX take care of the rounding.

2.3 Methods

2.3.1 Structuring Introduction and Discussion: Funnel and Reverse Funnel

A general structure that can be found in many research papers is the so-called X-shaped or "hourglass" structure.[2] This structure is brought about when the so-called

2. See https://skillsearthsciences.sites.uu.nl/writing/structure/#attachment_2387, Fig. 1, and https://pmc.ncbi.nlm.nih.gov/articles/PMC5405644/pdf/JHRS-10-3.pdf, Figure 1.

"funnel" technique is used to write the introduction,[3] and when a "reverse funnel" or "upside-down funnel" is used to write the discussion/concluding section:[4]

(1) The paper starts with a *broad* perspective at the beginning of the introduction, which becomes *narrower* with every sentence, leading to the paper's narrow research question.
(2) The *narrowest* part is when the exact research question is posed, the research methodology is described in detail, and the results are derived/presented.
(3) When the results are discussed and related to the literature in the concluding section, the perspective becomes *broad* again.

An example of the "funnel" technique can be found in Figure 2.1, and of the "reverse funnel" technique in Figure 2.2.

2.3.2 Some Questions That May Help You Guide Writing Your Paper

You may find the following list of questions helpful as a guide to making sure that your manuscript covers all important aspects. The questions are primarily intended for doctoral and advanced researchers, but they may also be helpful to students writing term papers and theses.

Your introduction might answer the following questions (not necessarily in this order, but this order will give rise to a "funnel" structure):

(1) What do we already know?
(2) What do we not know yet (the "knowledge gap")?
(3) Why is it interesting/relevant to close the gap?
(4) *What is the research question? [R]*
(5) Why is it interesting to answer the research question?
(6) How do we contribute to closing the knowledge gap: How do we answer the research question? (Which method do we use?)
(7) Why is the chosen method (more) suitable (than alternative methods) for answering the research question?

And your conclusion/discussion might answer the following questions (again, not necessarily in this order, but this order will give rise to a "reverse funnel" structure):

3. See https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1401, Figure 14.1, and https://kib.ki.se/en/write-cite/academic-writing/structure-academic-texts
4. See https://read.aupress.ca/read/read-think-write/section/9e945e9b-62cf-4dbb-b8cd-823549ede292#fig1402, Figure 14.2, and https://kib.ki.se/en/write-cite/academic-writing/writing-conclusion.

Molecular simulation methods have become an important technique in many areas of chemistry through the recent advent of effective and widespread software for molecular mechanics, molecular dynamics, and Monte Carlo simulations, and procedures for the estimation of free energy differences. In all such methods, a proper description of the electrostatic interactions within the simulated system is of key importance, … However, in most implementations, …, the simplest possible solution is used, … Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.	*Paragraph 1* Broad introduction with general description of the research field ("many areas of chemistry"), research topic ("molecular simulation methods"), and recent development ("through the recent advent of …"). Followed by a more detailed description of crucial aspects in this context ("In all such methods, a proper description … is of key importance") and of shortcomings of existing methods.
Unfortunately, atomic charges are not observables, i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations. Therefore, a large number of methods have been suggested for the estimation of point charges [1]. Several groups have tried to derive the charges directly from experimental quantities, … Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules. Instead, most techniques derive atomic charges from quantum chemical calculations.	*Paragraph 2* Narrowing down the topic: Description of the state of the art and of issues and limitations observed in the literature.
The simplest way to determine quantum chemical charges is the Mulliken population analysis. … Although, Mulliken charges are known to strongly depend on the basis set [6] and to reproduce electrostatic moments poorly [1], they are still widely used due to their simplicity. Several other … methods have been devised to cure the problems of the Mulliken charges [1], …	*Paragraph 3* Further description of the state of the art: strengths and weaknesses of existing approaches. *Paragraphs 4–6* Further description of unresolved issues.
…	
In this paper, we make a critical analysis of the four most popular potential-based point-charge methods, … It is shown that these methods may give widely different results and possible explanations for this are discussed. Two alternative methods for deriving atomic charges are suggested, which avoid the arbitrariness in the selection of potential points, and their performance is judged using …	*Paragraph 7* Narrow description of the specific contribution of this paper: Identification of undesirable arbitrary variation in simulation results of existing methods, which inspired development of an "alternative methods" that avoids this issue.

Figure 2.1. Illustration of the funneling approach, using the "Introduction" section from Sigfridsson and Ryde (1998).

Note: The full text of the article by Sigfridsson and Ryde (1998) is available at https://lucris.lub.lu.se/ws/portalfiles/portal/135489685/25_charges.pdf.

Potential-based charges strongly depend on the way the potential points are selected. We have presented and tested a new method that avoids such a dependence, CHELMO. It fits the charges directly to the electrostatic moments and it performs as well, or better, as the best electrostatic potential method judging from the calculated electrostatic potential and moments. … The major disadvantage of the method is that it cannot be used for large molecules, since there is a limited number of moments. No more than 25 independent charges in a molecule can be determined if all moments up to hexadecapole moments are used. In practice it should not be used for molecules with more than about 20 atoms since otherwise the higher moments may be poorly reproduced … This could be remedied if moments higher than hexadecapole moments are calculated but such moments are not included in the output of normal quantum chemical packages.

Paragraph 1
Narrow recap of the first contribution of the paper: brief description of the issue and how it was solved by the new method. Followed by description of the limitations of the new method. Outlook how these limitations could be overcome.

In order to get a method that can be used for larger system, we constructed the CHELP-BOW method, which estimates the charges from a Boltzmann-weighted fit to the electrostatic potential. It has the advantage over the other … methods that … The present implementation of the CHELP-BOW method is very simplified, but in a future publication we will refine and thoroughly test the method. However, the results are conclusive enough to show that the method … has the desired behaviour. … Clearly, this is the method we recommend for general use.

Paragraph 2
Narrow recap of the central contribution of the paper and how it relates to the literature: how existing methods were further improved upon by the second method proposed in this paper. Followed by outlook on future advances and recommendation for researchers in the field.

An important advantage of the methods developed in this paper is that they are general, i.e. they can be used with any quantum chemical program and they can use experimental data (…) as well. The only thing that has to be changed is the input section of the programs. Thus any quantum chemical basis sets can be used, and any level of theory that gives a wave function may be employed. Furthermore, the methods can easily be adapted to …

Paragraph 3
Broader implications due to the generality of the new method.

Finally, a comment on the traditional electrostatic potential methods. … Thus, we cannot see any justification to use CHELP charges except for backward comparisons.

Paragraph 4
Other broader implications.

Figure 2.2. Illustration of the “reverse funneling” approach, using the “Concluding Remarks” subsection from Sigfridsson and Ryde (1998).

(8) What was the knowledge gap before the current study? [Similar to the introduction.]

(9) *How did we contribute to closing that gap, and what did we find? [A]* (Which method/dataset did we use, and what are the results?—Recap and take-home message.)

(10) How do our results relate to (confirm/contradict/complement/extend) the results from previous and other contemporaneous studies?

(11) What part of the gap is still open? (Phrased a bit more negatively: What are the limitations of our approach?)

(12) Next steps/avenues for future research: How could we go about closing the remaining knowledge gap (by removing the limitations of our current approach)?[5]

In her book *The Little Book of Research Writing*,[6] Varanya Chaubey proposes a simpler list of only three items—the "RAP method":

- The "R" stands for "research question."
- The "A" stands for "answer" (to the research question, of course).
- The "P" stands for "positioning statement." (How does our manuscript relate to the literature: Does it corroborate or extend or contradict others' findings?)

Relating the "RAP" approach to the the list of questions above, question 4 is the "R," the answer to question 9 is the "A," and the remaining questions spell out the "P."

2.3.3 Structuring Paragraphs in Academic Manuscripts

Obviously, there is no single "right" way of constructing paragraphs. However, a few basic rules can help you write paragraphs that are *easy to comprehend* by your readers. A common approach is to compose sentences such that they reflect the following functional sequence:[7]

(1) *topic sentence (TS)*—which, if necessary, connects to the previous paragraph;

(2) *supporting sentences (SS)*—which elaborate on the topic sentence; and

5. John Cochrane might do away with question 12. See his "Writing Tips for Ph.D. Students," https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf.

6. For a quick summary of the book, see https://mauve-porcupine-8992.squarespace.com/s/Chaubey_Research_Writing-bxdd.pdf. See also https://www.econscribe.org/about and https://arxiv.org/pdf/2012.07787.

7. See https://libguides.newcastle.edu.au/writing-paragraphs/structure, https://www.une.edu.au/library/students/academic-writing/write-paragraphs/paragraphs/Academic-paragraphs_v2.pdf, https://learningessentials.auckland.ac.nz/writing-effectively/paragraph/

(3) *concluding sentence or connecting sentence (CS)*—which refers back to the topic sentence and/or provides a link to the subsequent paragraph.

There are various ways in which the supporting sentences can be filled with content. The University of Sheffield suggests[8] that the supporting sentences could consist of

(1) "explanation or definitions (optional),"
(2) "evidence and examples," and
(3) "comment" (an interpretation and evaluation of the evidence by you).

The University of Hull and Newcastle University (UK) consider the mnemonic "PEEL" helpful, although they define its meaning in slightly different ways: "Point, Evidence, Explanation, Link"[9] versus "Point, Evidence, Evaluation, Link"[10]. One may view "Explanation"—"why the point is important and how it helps with your overall argument"—as narrower than "Evaluation"—which is a general critical reflection on the informativeness of the evidence and can include an explanation of its importance. "PEEL" relates to TS/SS/CS as follows:

(1) "P," the Point, corresponds to the topic sentence.
(2) "E," Evidence, is the first part of the supporting sentences.
(3) "E," Evaluation/Explanation, forms the second part of the supporting sentences.
(4) "L," Link, refers to the connecting sentence that links to the next paragraph.

The Wilfrid Laurier University (Canada) suggests a different mnemonic: "MEAL."[11]

(1) "M: Main Point Sentence" corresponds to the topic sentence.
(2) "E: Evidence" is the first part of the supporting sentences.
(3) "A: Analysis/Synthesis" forms the second part of the supporting sentences.
(4) "L: Linking-Back Sentence" refers to the concluding sentence that summarizes the paragraph and links back to the topic sentence or even the topic of the entire paper.

The first three components are largely identical according to these approaches. A difference exists in the meaning of "L": While "PEEL" stresses the link to the *next* step in the chain of thought, "MEAL" stresses underscoring the message of the *current* paragraph. That is, the latter advocates *closure*: summarizing the paragraph and linking *back* to the topic sentence.

8. See https://sheffield.ac.uk/study-skills/writing/academic/paragraph.
9. See https://libguides.hull.ac.uk/writing/paras.
10. See https://www.ncl.ac.uk/academic-skills-kit/writing/academic-writing/paragraphing/
11. See https://students.wlu.ca/academics/support-and-advising/student-success/assets/resources/writing/how-to-structure-an-academic-paragraph.html.

Yet slightly different is the approach put forth by Academic Writing UK:[12]

(1) Topic sentence—key topic in this paragraph.
(2) Development—the main idea/topic discussed in more detail.
(3) Example—support/evidence/data/statistics that show your development is valid/credible.
(4) Summary—overall main point summarized/evaluated.

It is perfectly fine to mix the different approaches. For some paragraphs, a closer explanation of the topic sentence (“development”) may be necessary, while it is unnecessary for others. And in some paragraphs, you may want underscore the central message by reiterating it in the final sentence (concluding sentence), while for other paragraphs, you will use the final sentence to set the stage for the next step in your chain of reasoning (connecting sentence).

One thing that *all* these suggestions have in common is, however, that they start with a *topic sentence* and end with a *concluding/connecting sentence*. Hence, no matter how you fill the supporting sentences, it is probably best to start a paragraph with a topic sentence and end with a concluding/connecting sentence.

Let us illustrate the “MEAL”/“PEEL” structure—including an optional explanation or definition—using example sentences from Sigfridsson and Ryde (1998):

… Thus, most classical simulation methods need a point-charge parameterisation of the molecules of interest.	**L: Concluding sentence**
Unfortunately, atomic charges are not observables,	**M/P: Topic sentence** (reference to previous paragraph)
i.e. they cannot unambiguously be determined by experiments or quantum chemical calculations.	**Optional explanation/definition**
Therefore, a large number of methods have been suggested for the estimation of point charges [1].	**Supporting sentences:** **E: Evidence**
Several groups have tried to derive the charges directly from experimental quantities, e.g. dipole moments, electrostatic potentials, or free energy differences [2–4]. Yet, relevant data is usually missing or too scarce to allow a determination of all charges in interesting molecules.	**A/E: Analysis/Evaluation**
Instead, most techniques derive atomic charges from quantum chemical calculations.	**L: Connecting sentence** (link to next paragraph)
The simplest way to determine quantum chemical charges is the Mulliken population analysis. …	**M/P: Topic sentence** (reference to previous paragraph)

12. See https://academic-englishuk.com/paragraphing/.

The rule to finish a paragraph with a concluding sentence entails, in particular, that a theoretical or empirical finding should never be the last thing that you mention in a paragraph. An "evidence" sentence should always be followed by an interpretation in which you tell the reader what we learn from that finding. Here is an example from Sigfridsson and Ryde (1998):

> … With these radii, the charges almost coincide with those obtained by the CHELPG method (within 0.02 *e*). Thus, we can conclude that the main difference between the Merz–Kollman and the CHELPG method is the 1.4 times larger effective radii of the former method, whereas the sampling schemes are almost equivalent.

Evidence:
Supporting sentence with empirical finding

Concluding sentence with interpretation of that finding

To reiterate, there is no single "right" way of constructing paragraphs. However, keeping the TS/SS/CS structure in mind as a guideline is bound to help you produce a text that enables readers to follow your chain of reasoning.

2.3.4 Additional Online Resources on Effective Writing in Economics

Additional useful recommendations and tips can be found in Plamen Nikolov's "Writing Tips for Crafting Effective Economics Research Papers – 2023-2024 Edition" (https://docs.iza.org/dp16276.pdf) and in John H. Cochrane's "Writing Tips for Ph.D. Students" (https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf).

In this section, we first present the design of the experiment (2.3.5) and derive behavioral predictions (2.3.6).

2.3.5 Design of the Main Experiment

2.3.5.1 General Features. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.3.5.2 More Specific Features. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$.

Let’s test the euro symbol: €, €1,234.56, €1,234.56. Let’s also test text superscripts: i^{th} and text subscripts: CO_2 and H_2O. $\sigma_\epsilon, c^\alpha$. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$. Let’s test the footnote settings.[13]

Figure 2.5 shows an exemplary decision screen with $B=€11$ and $r\approx 15\%$ for both $\mathrm{BAL}^{\mathrm{I}}_{1:1}$ (upper panel) and $\mathrm{UNBAL}^{\mathrm{I}}_{1:8}$ (lower panel). Through a slider, subjects choose their preferred $x\in\boldsymbol{X}$.[14] The slider position in Figure 2.5 indicates $x=0.5$, i.e., the earliest payment is reduced by €5.50. Since $r\approx 15\%$ in this example, this slider position amounts to €6.30 that are paid at later payment dates. While these €6.30 are paid in a single bank transfer on the latest payment date in $\mathrm{BAL}^{\mathrm{I}}_{1:1}$, the amount is dispersed in equal parts over the last 8 payment dates in $\mathrm{UNBAL}^{\mathrm{I}}_{1:8}$—i.e., 8 consecutive payments of €0.79.[15]

2.3.5.3 Some More Details. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$.

13. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

14. The slider had no initial position—it appeared only after subjects first positioned the mouse cursor over the slider bar. This was done to avoid default effects.

15. We always rounded the second decimal place up so that the sum of the payments included in a dispersed payoff was always at least as great as the respective concentrated payoff.

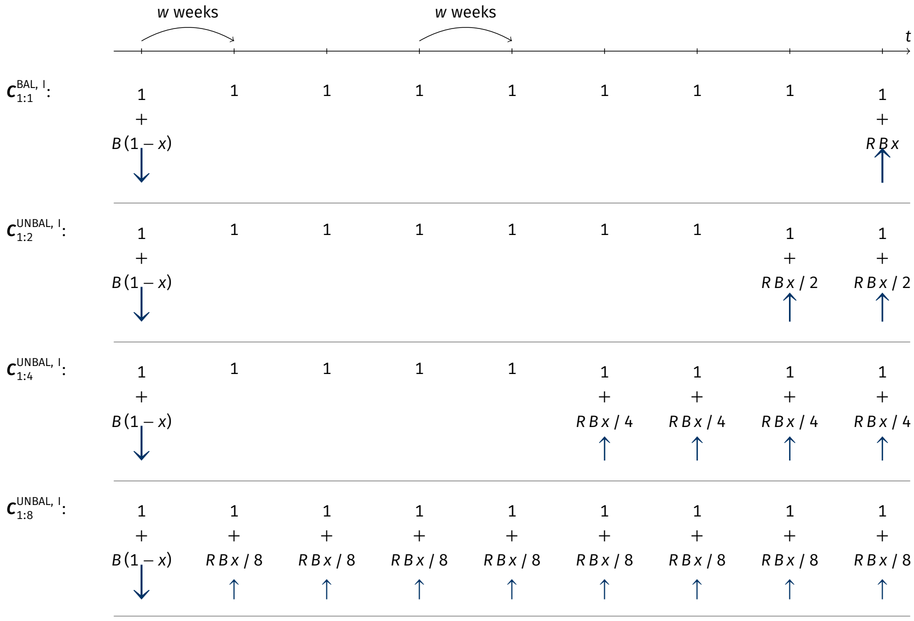

Figure 2.3. Budget Sets $\boldsymbol{C}^{\text{BAL, I}}_{1:1}$ and $\boldsymbol{C}^{\text{UNBAL, I}}_{1:n}$

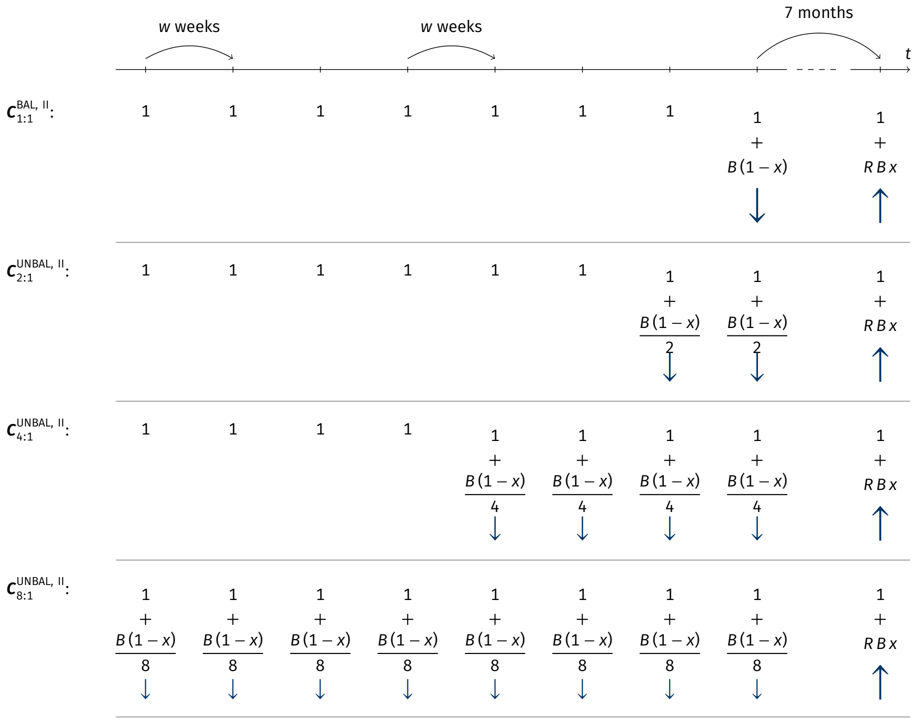

Figure 2.4. Budget Sets $\boldsymbol{C}^{\text{BAL, II}}_{1:1}$ and $\boldsymbol{C}^{\text{UNBAL, II}}_{n:1}$

Notes: For the values of B, R, and w that we used, see Section 2.3.5.4. The savings rate x is individuals' choice variable: they choose some $x \in \boldsymbol{X} = \{0, {}^{1}/_{100}, {}^{2}/_{100}, \ldots, 1\}$ in each trial. The arrows indicate whether and in which direction payments at the respective payment dates change if x is increased. σ_ε, c^α. This figure was taken from Dertwinkel-Kalt et al. (2017).

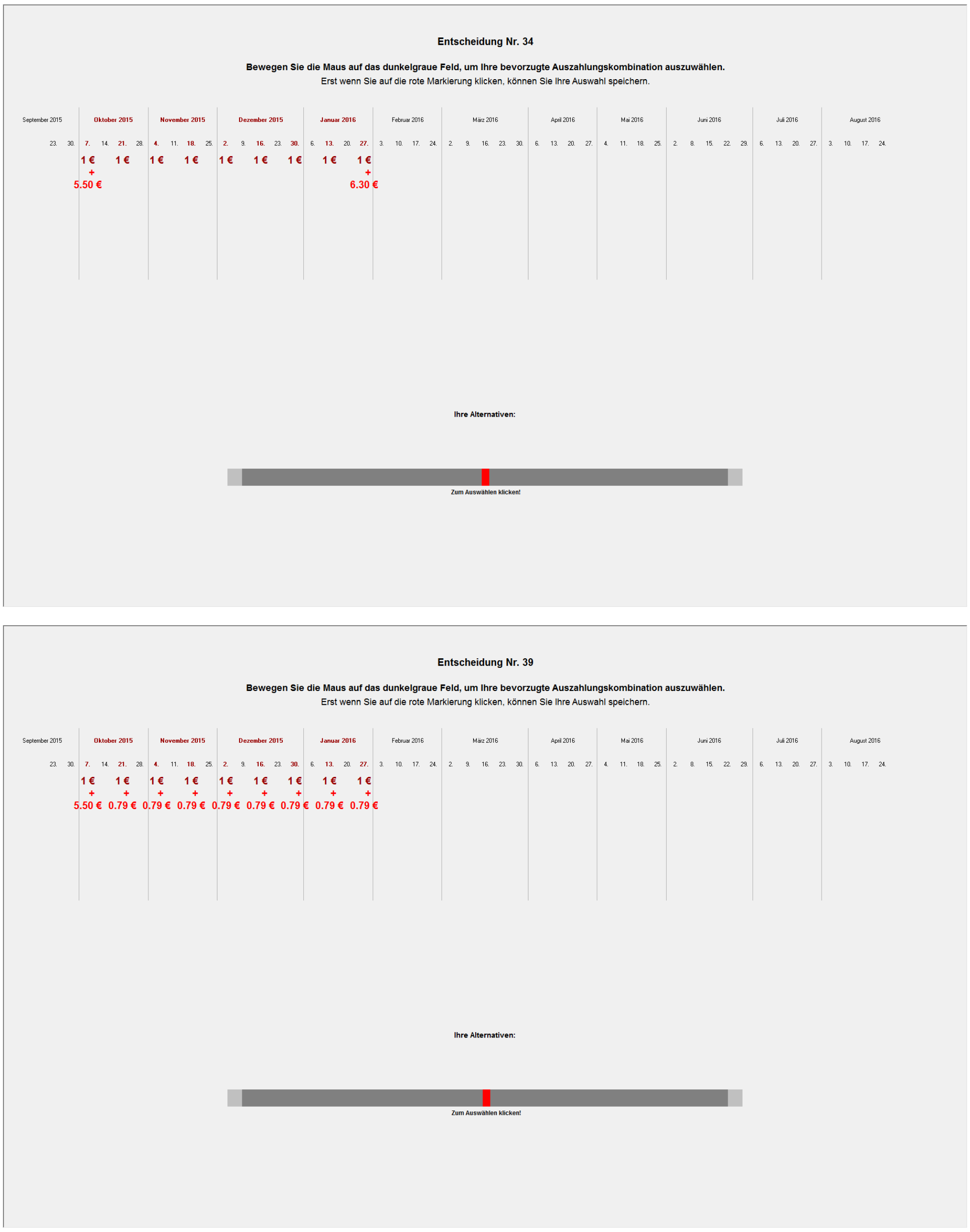

Figure 2.5. Screenshots of a $\mathrm{BAL}^{l}_{1:1}$ Decision (Top) and an $\mathrm{UNBAL}^{l}_{1:8}$ Decision (Bottom)

Note: This figure was taken from Dertwinkel-Kalt et al. (2017).

If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Here's a bulleted list:

- Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.
- Hello, here is some text without a meaning. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.
- Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

2.3.5.4 Procedure. Describe the sequence of events in your study. You could do this with the help of an enumerated list:

(1) Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look.

$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

(2) Hello, here is some text without a meaning. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha) + \cos^2(\beta) = 1$. This text should contain *all letters of the alphabet* and it should be written in of the original language $E = mc^2$. There is no need for special contents, but the length of words should match the language. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$.

(3) Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. If you read this text, you will get no information. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

2.3.6 Predictions

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages. Let’s include a really, really long footnote to check how it is split across two pages.[16] Let’s include a really, really long footnote to check how it is split across two

16. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information?

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Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

pages. Let's include a really, really long footnote to check how it is split across two pages.

By discounted utility we understand any intertemporal utility function that (1) is time-separable and that (2) values a payment farther in the future at most as much as an equal-sized payment closer in the future. Importantly, the predictions derived below hold for all three frequently used types of discounting—exponential, hyperbolic, and quasi-hyperbolic.

In the following, we assume that individuals base their decisions on utility derived from receiving monetary payments c_t at various dates t. This is an assumption that is frequently made in experiments on intertemporal decision making. One way to justify this assumption is that individuals anticipate to consume the payments they receive within a short period around date t. Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this assumption seems reasonable (see the arguments in favor of this view in Halevy 2014). Kőszegi and Szeidl (2013) themselves make the same assumption of "money in the utility function": "in some applications we also assume that monetary transactions induce direct utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people experience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: … some evidence on individuals' attitudes toward money, such as narrow bracketing (…) and laboratory evidence on hyperbolic discounting (…), is difficult to explain without it." Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and monetary rewards. Additionally, we make the standard assumption that utility from money is increasing in its argument but not convex: $u'(c_t) \geq 0$ and $u''(c_t) \leq 0$.

2.3.6.1 Discounted Utility. Individuals make their allocation decisions by comparing the aggregated consumption utility of each earnings sequence $\boldsymbol{c} \in \boldsymbol{C}$. Discounted utility assumes that the utility of each period enters overall utility additively. That is, utility derived from the payment to be received at future date t can be expressed as $u_t(c_t) := D(t)\, u(c_t)$. Here, $D(t)$ denotes the individual's discount function for conversion of future utility into present utility. The discount function satisfies $0 \leq D(t)$ and $D'(t) \leq 0$, such that a payment further in the future is valued at most as much as an equal-sized payment closer in the future.[17]

The utility of earnings sequence $\boldsymbol{c}$ with payments c_t in periods $t = 1, \ldots, T$ is as follows:

17. Normalization such that $D(t) \leq 1$ is not necessary in our case. Provided that t is a metric time measure, where $t = 0$ stands for the present, examples are $D(t) := \delta^t$ with some $\delta > 0$ for exponential discounting and $D(t) := (1 + \alpha t)^{-\gamma/\alpha}$ with some $\alpha, \gamma > 0$ for generalized hyperbolic discounting.

`$$ ... $$`:

$$U(\boldsymbol{c}) = \sum_{t=1}^{T} u_t(c_t) = \sum_{t=1}^{T} D(t)\, u(c_t).$$

`\[ ... \]` with manual `\tag{...}`:

$$U(\boldsymbol{c}) \;=\; \sum_{t=1}^{T} u_t(c_t) \;=\; \sum_{t=1}^{T} D(t)\, u(c_t). \tag{II}$$

`\begin{equation} ... \end{equation}`:

$$U(\boldsymbol{c}) \;=\; \sum_{t=1}^{T} u_t(c_t) \;=\; \sum_{t=1}^{T} D(t)\, u(c_t). \tag{2.1}$$

`\begin{equation*} ... \end{equation*}`:

$$U(\boldsymbol{c}) \;=\; \sum_{t=1}^{T} u_t(c_t) \;=\; \sum_{t=1}^{T} D(t)\, u(c_t).$$

`\begin{eqnarray} ... \end{eqnarray}`:

$$U(\boldsymbol{c}) \quad = \quad \sum_{t=1}^{T} u_t(c_t) \tag{2.2}$$

$$\quad = \quad \sum_{t=1}^{T} D(t)\, u(c_t). \tag{2.3}$$

`\begin{eqnarray*} ... \end{eqnarray*}`:

$$\begin{aligned} U(\boldsymbol{c}) \quad &= \quad \sum_{t=1}^{T} u_t(c_t) \\ &= \quad \sum_{t=1}^{T} D(t)\, u(c_t). \end{aligned}$$

`\begin{align} ... \end{align}`, equation number in the final line only:

$$\begin{aligned} U(\boldsymbol{c}) &= \sum_{t=1}^{T} u_t(c_t) \\ &= \sum_{t=1}^{T} D(t)\, u(c_t). \end{aligned} \tag{2.4}$$

`\begin{align} ... \end{align}`, equation number in each line:

$$U(\boldsymbol{c}) = \sum_{t=1}^{T} u_t(c_t) \tag{2.5}$$

$$= \sum_{t=1}^{T} D(t)\, u(c_t). \tag{2.6}$$

`\begin{align*} ... \end{align*}`:

$$
\begin{aligned}
U(\boldsymbol{c}) &= \sum_{t=1}^{T} u_t(c_t) \\
&= \sum_{t=1}^{T} D(t)\, u(c_t).
\end{aligned}
$$

`\begin{alignat}{2} ... \end{alignat}`:

$$
U(\boldsymbol{c}) = \sum_{t=1}^{T} u_t(c_t) \tag{2.7}
$$

$$
= \sum_{t=1}^{T} D(t)\, u(c_t). \tag{2.8}
$$

`\begin{alignat*}{2} ... \end{alignat*}`:

$$
\begin{aligned}
U(\boldsymbol{c}) &= \sum_{t=1}^{T} u_t(c_t) \\
&= \sum_{t=1}^{T} D(t)\, u(c_t).
\end{aligned}
$$

Individuals choose how much to allocate to the different periods by maximizing their utility over all possible earnings sequences available within a given budget set $\boldsymbol{C}$, see equations (II), (2.1), (2.2), (2.3), (2.4), (2.5), and (2.6). See also Equation 2.8. We use the superscript $^{\mathrm{DU}}$ to indicate decisions based on discounted utility.

A Subparagraph. After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$.

2.3.6.2 Focus-Weighted Utility. In this section, we extend the model of discounted utility through "focus weights," as proposed by Kőszegi and Szeidl (2013). Period-t weights g_t scale period-t consumption utility u_t. Individuals are assumed to maximize focus-weighted utility, which is defined as follows:

$$\tilde{U}(\boldsymbol{c},\boldsymbol{C}) \;:=\; \sum_{t=1}^{T} g_t(\boldsymbol{C})\,u_t(c_t). \tag{2.9}$$

In contrast to discounted utility $U(\boldsymbol{c})$, focus-weighted utility $\tilde{U}(\boldsymbol{c},\boldsymbol{C})$ has two arguments: the earnings sequence $\boldsymbol{c}$ and the choice set $\boldsymbol{C}$. The latter dependence is due to the weights g_t. These are given by a strictly increasing weighting function g that takes as its argument the difference between the maximum and the minimum attainable utility in period t over all possible earnings sequences in set $\boldsymbol{C}$:

$$g_t(\boldsymbol{C}) \;:=\; g[\Delta_t(\boldsymbol{C})] \quad \text{with} \quad \Delta_t(\boldsymbol{C}) \;:=\; \max_{\boldsymbol{c}\in\boldsymbol{C}} u_t(c_t) - \min_{\boldsymbol{c}\in\boldsymbol{C}} u_t(c_t). \tag{2.10}$$

If the underlying consumption utility function is characterized by discounted utility, then $u_t(c_t) := D(t)\,u(c_t)$. That is, focused thinkers put more weight on period t than on period t' if the discounted-utility distance between the best and worst alternative is larger for period t than for period t'.

A Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$.

Yet Another Subparagraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.3.6.3 Hypotheses. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. This gives rise to our first hypothesis:

Hypothesis 2.1. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. Based on this, we can state our second hypothesis:

Hypothesis 2.2. *This environment can be used to clearly state your hypothesis and set them apart from the body text.*

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.4 Results

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. With this, we can test our hypotheses.

2.4.1 Test of Hypothesis 2.1

Our first result supports Hypothesis 2.1. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. The analysis we conducted to obtain Result 2.1 is described in detail in Table 2.1. Let’s reference a section, a subsection, and a figure from the appendices: Section 2.C, Section 2.A.2, Figure 2.B.1.

Result 2.1. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place.* $\sin^2(\alpha) + \cos^2(\beta) = 1$. *If you read this text, you will get no information* $E = mc^2$. *Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text* like this *gives you information about the selected font, how the letters are written and an impression of the look.* $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. *This text should contain* all letters of the alphabet *and it should be written in of the original language.* $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. *There is no need for special contents, but the length of words should match the language.* $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Table 2.1. A *tabularray*-Based Example Table

Dependent variable	$\hat{d}$
Estimate	0.123***
	(0.011)
Observations	750
Subjects	250

Notes: Standard errors in parentheses, clustered on the subject level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.4.2 Test of Hypothesis 2.2

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. We thereby test Hypothesis 2.2.

Result 2.2. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place.* $\sin^2(\alpha) + \cos^2(\beta) = 1$. *If you read this text, you will get no information* $E = mc^2$. *Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text* like this *gives you information about the selected font, how the letters are written and an impression of the look.* $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. *This text should contain* all letters of the alphabet *and it should be written in of the original language.* $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. *There is no need for special contents, but the length of words should match the language.* $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

Our second result provides evidence in support of Hypothesis 2.2. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

2.4.3 Heterogeneity

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{i=n} x_i = \frac{x_1 + x_2 + \ldots + x_n}{n}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\int_0^\infty e^{-\alpha x^2} \mathrm{d}x = \frac{1}{2}\sqrt{\int_{-\infty}^{\infty} e^{-\alpha x^2} \mathrm{d}x \int_{-\infty}^{\infty} e^{-\alpha y^2} \mathrm{d}y} = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\sum_{k=0}^{\infty} a_0 q^k = \lim_{n\to\infty} \sum_{k=0}^{n} a_0 q^k = \lim_{n\to\infty} a_0 \frac{1-q^{n+1}}{1-q} = \frac{a_0}{1-q}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha) + \cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

2.4.4 Structural Estimation

Inspect the variance–covariance matrix $\mathbf{\Sigma}$:

$$\mathbf{\Sigma} := \mathbf{Cov}(X) = \begin{bmatrix} \mathrm{Var}(X_1) & \cdots & \mathrm{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n, X_1) & \cdots & \mathrm{Var}(X_n) \end{bmatrix}.$$

2.5 Discussion

2.5.1 Some Limitations

Let's reference some tables: Table 2.2 and Table 2.3. This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

Table 2.2. Points Awarded in Our Typeface Competition—Basic Formatting That Produces Irregular Column Widths and a Suboptimal Table Width Test Greek: $\varepsilon, \theta, \phi$

	Utopia	Computer Modern	Charter	Times Roman	Palatino
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allisőn	2	0	1	2	1
Pía	1	0	2	1	0
Đavid	1	0	2	1	1
Sum	10	5	12	8	4

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should

Table 2.3. Points Awarded in Our Typeface Competition—More Sophisticated Formatting with Homogeneous Column Widths and a Table Width That Equals the Text Width

	Utopia[a]	Computer Modern[b]	Charter[c]	Times Roman[d]	Palatino[e]
Yoël	1	1	2	0	1
Çelik	2	0	2	1	0
Anità	1	2	1	2	0
Uğur	1	2	0	1	0
Håkan	1	0	2	0	1
Allisőn	2	0	1	2	1
Pía	1	0	2	1	0
Đavid	1	0	2	1	1
Sum	10	5	12	8	4

[a] `\usepackage{fourier}`
[b] The LaTeX standard serif font.
[c] `\usepackage[charter]{mathdesign}`
[d] `\usepackage{newtxtext, newtxmath}`
[e] `\usepackage[sc]{mathpazo}`

be written in of the original language. There is no need for special contents, but the length of words should match the language. Right?

2.5.2 Utility from Money

In deriving our predictions (Section 2.3.6), we assume that subjects base their decisions on utility derived from receiving monetary payments c_t at various dates t. We also make the standard assumption that utility from money is increasing in its argument but not convex, i.e., $u'(c_t) > 0$ and $u''(c_t) \leq 0$. Both assumptions are frequently made in studies on intertemporal decision making.

One way to justify the assumption of utility being based on money—rather than consumption—is that individuals anticipate to consume the payments that they receive at date t within a short period around t. Given that the maximum payment was below €20 and that any two payment dates were separated by at least two weeks, this seems reasonable (see the arguments in favor of this view in Halevy 2014).

A second justification is consistency within the discipline: Halevy (2014) points out that "in the domain of risk and uncertainty … preferences are often defined over payments." In line with this, Kőszegi and Szeidl (2013, p. 62) make the same assumption of "money in the utility function":

> in some applications we also assume that monetary transactions induce *direct* utility consequences, so that for instance an agent making a payment experiences an immediate utility loss. The idea that people experience monetary transactions as immediate utility is both intuitively compelling and supported in the literature: … some evidence on individuals' attitudes toward money, such as narrow bracketing (…) and laboratory evidence on hyperbolic discounting (…), is difficult to explain without it.

Last but not least, the papers by McClure et al. (2004) and McClure et al. (2007) demonstrate that brain activation, as measured by functional magnetic resonance imaging, is similar for primary and monetary rewards.

Let us now discuss the second assumption: that utility from money is nonconvex. We find that subjects allocate more money to the concentrated payoffs in the unbalanced than in the associated balanced budget sets—which we call concentration bias. One might argue that this relative preference for concentrated payoffs can be explained by the per-period utility function over money being convex.

Obtaining evidence on the shape of utility over money is nontrivial because it requires that at least two monetary amounts be compared with each other without the one clearly dominating the other. Thus, estimates of the curvature of the utility function over money can be obtained in two ways: the monetary amounts must be paid in different states of the world, i.e., comprise a lottery, or they have to be paid at different points in time.[18] Both methods entail particular theoretical assumptions.

Andersen et al. (2008) advocate the former approach and argue that when estimating time preference parameters, one should control for the curvature of the utility function through a measure of the curvature that is based on observed choices under risk. Their study and numerous other studies on risk attitudes consistently reveal that the vast majority of subjects is risk-averse even over small stakes. Hence, for the vast majority of subjects, utility over money is concave according to this methodology (ruling out probability weighting). Others, most notably Andreoni and Sprenger (2012), have argued that the degree of curvature measured via risky choices probably overstates the degree of curvature effective in intertemporal choices, but they also find that utility is concave (albeit close to linear). Given this unambiguous evidence from previous studies, it is implausible that our subjects exhibit convex utility over money.

2.6 Conclusion

Cite some more papers (Yaari 1965; Warner and Pleeter 2001; Davidoff, Brown, and Diamond 2005; Benartzi, Previtero, and Thaler 2011). Let's cite a book: Luce (1959). Let's cite a contribution to a collected volume: Harrison and Rutström (2008) and a collection (an edited volume) itself: Kagel and Roth (2016). Now let's cite presentations at conferences: Vosgerau et al. (2008) and Beute and Kort (2012). Attema et al. (2016) propose a method for "measuring discounting without measuring utility"[19].

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. This text should contain *all letters of the alphabet* and it should be written in of the original language. There is no need for special contents, but the length of words should match the language.

18. As a matter of fact, the latter was the motivation behind Samuelson (1937): "Under the following four assumptions, it is believed possible to arrive theoretically at a precise measure of the marginal utility of *money income* ..." (p. 155; emphasis in the original).

19. The basic idea of their method is intriguingly simple: Imagine an individual who is indifferent between, say, Option A: \$10 today and Option B: \$10 in one year plus \$10 in two years. With a constant annual discount factor δ, this indifference translates to $u(\$10) = \delta\, u(\$10) + \delta^2\, u(\$10)$, so that $u(\$10)$ cancels out, and δ can be readily calculated as the solution to $1 = \delta + \delta^2$.

Appendix 2.A Put More Complicated Derivations and Proofs Here

2.A.1 Appendix Subsection

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text *like this* gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain *all letters of the alphabet* and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special contents, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^nb}$.

(1) Erster Listenpunkt, Stufe 1
   a. Erster Listenpunkt, Stufe 2
      i. Erster Listenpunkt, Stufe 3
      ii. Zweiter Listenpunkt, Stufe 3
      iii. Dritter Listenpunkt, Stufe 3
      iv. Vierter Listenpunkt, Stufe 3
   b. Zweiter Listenpunkt, Stufe 2
   c. Dritter Listenpunkt, Stufe 2
   d. Vierter Listenpunkt, Stufe 2
(2) Zweiter Listenpunkt, Stufe 1
(3) Dritter Listenpunkt, Stufe 1
(4) Vierter Listenpunkt, Stufe 1

The typeset math below follows the ISO recommendations that only variables be set in italic. Note the use of upright shapes for "d," "e," and "π." (These are entered as `\mathup{d}`, `\mathup{e}`, and `\mathup{\pi}`, respectively.)

Theorem 2.1 (Simplest form of the *Central Limit Theorem*). *Let $X_1, X_2, \ldots, X_n$ be a sequence of i.i.d. random variables with mean 0 and variance 1 on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then*

$$\mathbb{P}\left(\frac{X_1+\cdots+X_n}{\sqrt{n}}\le y\right)\to\mathfrak{N}(y):=\int_{-\infty}^{y}\frac{\mathrm{e}^{-v^2/2}}{\sqrt{2\pi}}\,\mathrm{d}v\quad\textit{as } n\to\infty,$$

or, equivalently, letting $S_n:=\sum_1^n X_k$,

$$\mathbb{E}f(S_n/\sqrt{n})\to\int_{-\infty}^{\infty}f(v)\frac{\mathrm{e}^{-v^2/2}}{\sqrt{2\pi}}\,\mathrm{d}v\quad\textit{as } n\to\infty,\textit{ for every } f\in\mathrm{b}\mathscr{C}(\mathbb{R}).$$

2.A.2 Salience

Salience theory (Bordalo, Gennaioli, and Shleifer 2012, 2013) represents a behavioral model according to which the most distinctive features of the available alternatives receive a particularly large share of attention and are therefore over-weighted. More precisely, a particular attribute out of all attributes of an alternative becomes the more salient, the more it differs from that attribute's average level over all available alternatives.

Formally, alternatives are assumed to be uniquely characterized by the values they take in $T \geq 1$ attributes (or, "dimensions"). Utility is assumed to be additively separable in attributes, and salience attaches a decision weight to each attribute of each good which indicates how salient the respective attribute is for that good. Suppose an agent chooses one alternative from some finite choice set $\boldsymbol{C}$. Let t index the T different attributes, and let k index the K available alternatives. Let $u_t(\cdot)$ denote the function which assigns utility to values in dimension t. Denote by a_t^k the level of attribute t of good k and define $u_t^k := u_t(a_t^k)$ as the utility that dimension t of good k yields. Let $\bar{u}_t$ be the average utility level, across all K goods, of dimension t. The salience of each dimension of good k is determined by a symmetric and continuous salience function $\sigma(\cdot,\cdot)$ that satisfies the following two properties:

(1) *Ordering*. Let $\mu := \operatorname{sgn}(u_t^k - \bar{u}_t)$. Then for any $\varepsilon, \varepsilon' \geq 0$ with $\varepsilon + \varepsilon' > 0$, it holds that

$$\sigma(u_t^k + \mu\,\varepsilon, \bar{u}_t - \mu\,\varepsilon') \; > \; \sigma(u_t^k, \bar{u}_t). \tag{2.A.1}$$

(2) *Diminishing sensitivity*. For any $u_t^k, \bar{u}_t \geq 0$ and all $\varepsilon > 0$, it holds that

$$\sigma(u_t^k + \varepsilon, \bar{u}_t + \varepsilon) \; < \; \sigma(u_t^k, \bar{u}_t). \tag{2.A.2}$$

Following the smooth salience characterization proposed in Bordalo, Gennaioli, and Shleifer (2012, p. 1255), each dimension t of good k receives weight $\Delta^{-\sigma(u_t^k, \bar{u}_t)}$, where $\Delta \in (0,1]$ is a constant that captures an agent's susceptibility to salience. $\Delta = 1$ gives rise to a rational decision maker, and the smaller Δ, the stronger is the salience bias. We call an agent with $\Delta < 1$ a salient thinker.

A reference with a large number of authors is Henrich et al. (2005).

Appendix 2.B Additional Figures

	w weeks						*w* weeks		t
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(1)$:	$1 + B$	1	1	1	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(2)$:	1	$1 + B + i$	1	1	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(3)$:	1	1	$1 + B + 2i$	1	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(4)$:	1	1	1	$1 + B + 3i$	1	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(5)$:	1	1	1	1	$1 + B + 4i$	1	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(6)$:	1	1	1	1	1	$1 + B + 5i$	1	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(7)$:	1	1	1	1	1	1	$1 + B + 6i$	1	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(8)$:	1	1	1	1	1	1	1	$1 + B + 7i$	1
$\boldsymbol{c}_{\mathrm{CL}}^{\mathrm{BAL}}(9)$:	1	1	1	1	1	1	1	1	$1 + B + 8i$

Figure 2.B.1. Earnings Sequences Included in Choice List $\boldsymbol{C}_{\mathrm{CL}}^{\mathrm{BAL}}$

Notes: For the values of B, i, and w that we used see Section 2.3. Figure taken from Dertwinkel-Kalt et al. (2017).

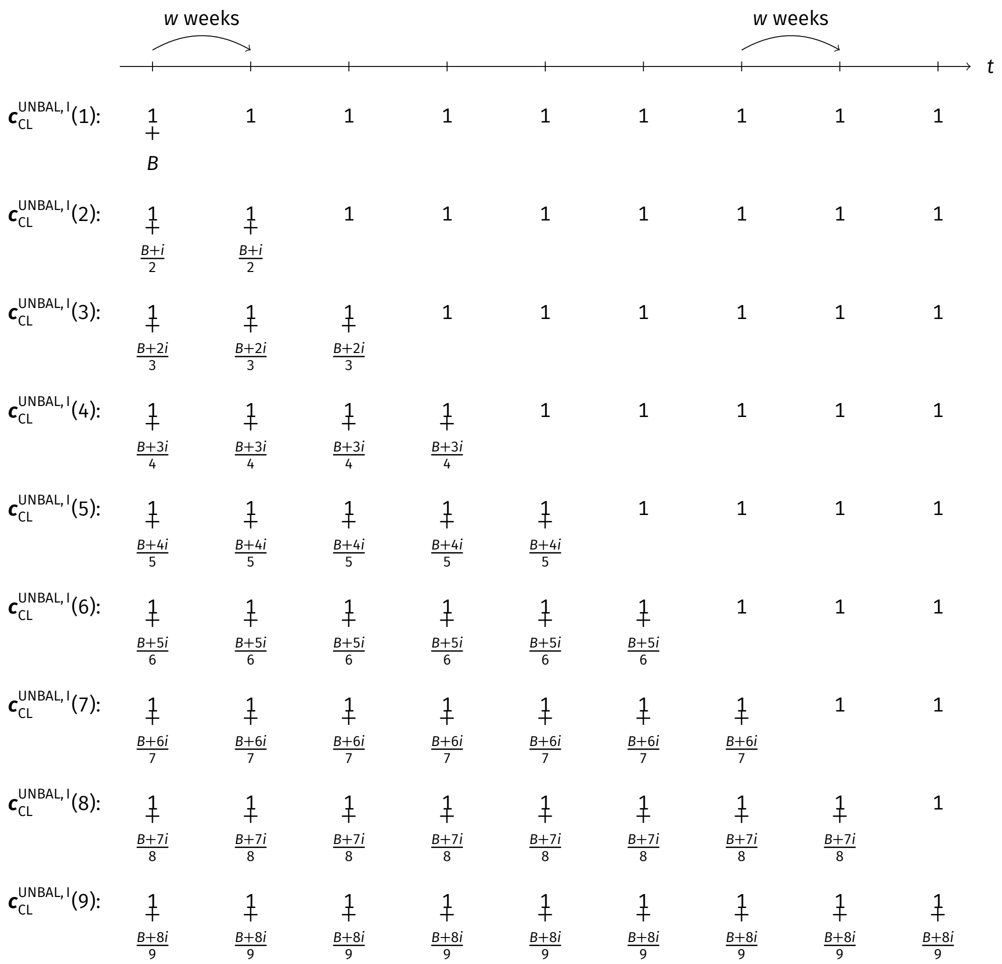

Figure 2.B.2. Earnings Sequences Included in Choice List $\boldsymbol{C}_{CL}^{UNBAL,I}$

Notes: For the values of B, i, and w that we used see Section 2.3. Figure taken from Dertwinkel-Kalt et al. (2017).

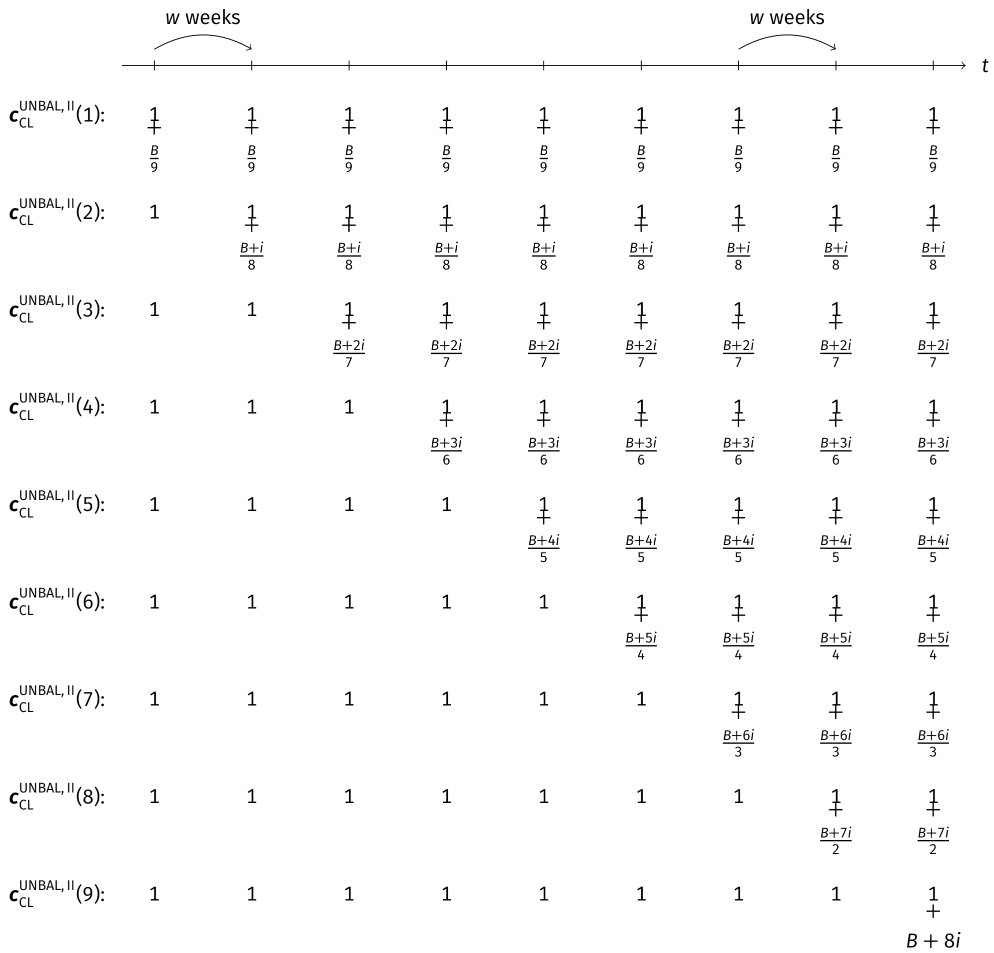

Figure 2.B.3. Earnings Sequences Included in Choice List $\boldsymbol{C}_{CL}^{UNBAL,II}$

Notes: For the values of *B*, *i*, and *w* that we used see Section 2.3. Figure taken from Dertwinkel-Kalt et al. (2017).

Appendix 2.C Additional *siunitx* and *tabularray* Example Tables

Table 2.C.1. An Example of a Regression Table (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017). Remember to Mention the Dependent Variable!

	(1)	(2)	(3)	(4)	(5)
Treatment	−0.390	−0.228	−0.729*	−0.449*	−0.453**
	(+0.352)	(−0.205)	[+0.377]	[−0.245]	{+0.204}
Female	0.948***	0.061	0.188	0.305	0.385*
	(0.354)	(0.233)	(0.372)	(0.226)	(0.222)
Female × Treatment	0.169	0.251	0.892*	0.454	0.439
	(0.514)	(0.325)	(0.533)	(0.341)	(0.307)
Final high school grade	−0.101	0.013	0.076	0.117	0.039
	(0.198)	(0.144)	(0.224)	(0.146)	(0.133)
Trait self-control	−0.016	0.002	−0.016	0.000	−0.007
	(0.016)	(0.010)	(0.015)	(0.010)	(0.009)
Constant	2.357***	1.512***	−0.322	2.158***	1.437***
	(0.239)	(0.144)	(0.265)	(0.161)	(0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	−0.221	0.023	0.163	0.004	−0.014
p_F[Treatment × (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Notes: Dependent variable: $m_{\sim}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

Table 2.C.2. Figure Grouping via `siunitx` in a Table

(1)	(2)	(3)
−0.100*	−0.100 01*	−123 456.444***
(2.871)	(2.871 23)	[+50 000.123]

Table 2.C.3. Overview of the Choice Lists Presented to Subjects (Adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017)

	Alternative ***A***				Alternative ***B***			
	$c_{\boldsymbol{A},1}$	$p_{\boldsymbol{A},1}$	$c_{\boldsymbol{A},2}$	$p_{\boldsymbol{A},2}$	$c_{\boldsymbol{B},1}$	$p_{\boldsymbol{B},1}$	$c_{\boldsymbol{B},2}$	$p_{\boldsymbol{B},2}$
Choice List I: risky/risky (x = €22.00, r = €7.50, k = €11.50; 25 rows)								
Top row	€ 3.00	50%	€22.00	50%	€ 3.00	50%	€ 7.00	50%
Center row	€ 3.00	50%	€22.00	50%	€ 9.00	50%	€13.00	50%
Row with $m = 0$	€ 3.00	50%	€22.00	50%	€10.50	50%	€14.50	50%
Bottom row	€ 3.00	50%	€22.00	50%	€15.00	50%	€19.00	50%
Choice List II: safe/risky (x = €16.00, r = €5.00, k = €5.00; 19 rows)								
Top row	€11.00	100%			€11.00	50%	€21.00	50%
Center row	€11.00	100%			€ 6.50	50%	€16.50	50%
Row with $m = 0$	€11.00	100%			€ 6.00	50%	€16.00	50%
Bottom row	€11.00	100%			€ 2.00	50%	€12.00	50%
Choice List III: “long shot” (x = €14.00, r = −€36.00, k = €7.00; 21 rows)								
Top row	€ 7.00	90%	€50.00	10%	€ 7.00	90%	€10.00	10%
Row with $m = 0$	€ 7.00	90%	€50.00	10%	€11.00	90%	€14.00	10%
Center row	€ 7.00	90%	€50.00	10%	€12.00	90%	€15.00	10%
Bottom row	€ 7.00	90%	€50.00	10%	€17.00	90%	€20.00	10%
Choice List IV: delayed payoffs (x = €18.00, r = €6.00, k = €8.50, paid in one week; 20 rows)								
Top row	€ 9.50	50%	€12.00	50%	€ 9.50	50%	€24.00	50%
Above-center row	€ 9.50	50%	€12.00	50%	€ 5.00	50%	€19.50	50%
Below-center row	€ 9.50	50%	€12.00	50%	€ 4.50	50%	€19.00	50%
Row with $m = 0$	€ 9.50	50%	€12.00	50%	€ 3.50	50%	€18.00	50%
Bottom row	€ 9.50	50%	€12.00	50%	€ 0.00	50%	€14.50	50%

Table 2.C.4. A Really Long Table That Spans Multiple Pages

	(1)	(2)	(3)	(4)
Row 1	0.0070	0.1356	0.1560	0.8979
Row 2	0.4223	0.7311	0.4213	0.6900
Row 3	0.0767	0.5110	0.7399	0.9491
Row 4	0.5954	0.1685	0.3778	0.9960
Row 5	0.6465	0.0524	0.8895	0.1544
Row 6	0.3838	0.7069	0.1773	0.5785
Row 7	0.1537	0.5442	0.6361	0.0327
Row 8	0.0879	0.1812	0.3082	0.2942
Row 9	0.2720	0.2565	0.6214	0.8944
Row 10	0.4873	0.3064	0.9913	0.0591
Row 11	0.8387	0.1713	0.6747	0.7455
Row 12	0.0645	0.4891	0.2892	0.1013
Row 13	0.0989	0.3798	0.5795	0.3725
Row 14	0.3256	0.7080	0.0262	0.8709
Row 15	0.7867	0.8768	0.0690	0.6081
Row 16	0.2713	0.4399	0.5838	0.6107
Row 17	0.5236	0.1527	0.4402	0.8002
Row 18	0.4851	0.4619	0.4040	0.2711
Row 19	0.1742	0.8151	0.2757	0.4184
Row 20	0.0495	0.3288	0.2759	0.1452
Row 21	0.1678	0.2403	0.1993	0.3676
Row 22	0.4977	0.9472	0.2810	0.2493
Row 23	0.6777	0.6516	0.3573	0.1413
Row 24	0.3668	0.3075	0.8724	0.3945
Row 25	0.5877	0.5670	0.0417	0.5213
Row 26	0.3599	0.5485	0.2407	0.6362
Row 27	0.1029	0.9796	0.5696	0.8696
Row 28	0.3070	0.8169	0.4015	0.4386
Row 29	0.4453	0.0670	0.3726	0.3257
Row 30	0.2648	0.9977	0.8864	0.0755
Row 31	0.4085	0.2017	0.5406	0.1333
Row 32	0.4861	0.4466	0.3472	0.2486
Row 33	0.5996	0.8639	0.1837	0.7636
Row 34	0.4446	0.3755	0.6901	0.4208
Row 35	0.9616	0.3585	0.0074	0.2867
Row 36	0.5168	0.5752	0.5778	0.0060

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 37	0.7978	0.0283	0.7998	0.9952
Row 38	0.0561	0.3133	0.1207	0.6922
Row 39	0.5237	0.1488	0.9217	0.2268
Row 40	0.0944	0.7939	0.6252	0.9836
Row 41	0.3179	0.6226	0.4493	0.4277
Row 42	0.7175	0.7267	0.8016	0.6880
Row 43	0.0192	0.4807	0.7610	0.9808
Row 44	0.9923	0.8888	0.4494	0.0645
Row 45	0.3938	0.8529	0.0496	0.0429
Row 46	0.1135	0.6166	0.5899	0.7500
Row 47	0.0654	0.1640	0.1952	0.0431
Row 48	0.8895	0.0549	0.1105	0.1284
Row 49	0.6817	0.8942	0.6597	0.3661
Row 50	0.6690	0.8817	0.2343	0.1903
Row 51	0.4091	0.0874	0.4726	0.1381
Row 52	0.9061	0.9039	0.7439	0.2061
Row 53	0.5282	0.2135	0.5223	0.7846
Row 54	0.6505	0.7404	0.8748	0.2078
Row 55	0.5824	0.8443	0.3242	0.8253
Row 56	0.0151	0.9929	0.4812	0.5010
Row 57	0.7296	0.8420	0.1535	0.4273
Row 58	0.8102	0.8068	0.1832	0.8830
Row 59	0.1650	0.5545	0.1820	0.0791
Row 60	0.5882	0.5750	0.9195	0.8993
Row 61	0.0638	0.5132	0.5994	0.0877
Row 62	0.9916	0.8032	0.0564	0.3218
Row 63	0.5555	0.4078	0.7056	0.9225
Row 64	0.8680	0.5577	0.2992	0.0941
Row 65	0.2939	0.7801	0.7039	0.7295
Row 66	0.0829	0.6756	0.5386	0.0644
Row 67	0.3868	0.4199	0.0308	0.5947
Row 68	0.0943	0.2663	0.0379	0.0887
Row 69	0.0050	0.1396	0.8348	0.2830
Row 70	0.9585	0.8018	0.4472	0.9477
Row 71	0.8153	0.2659	0.7030	0.4096
Row 72	0.7532	0.4214	0.3914	0.2360

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 73	0.6419	0.2074	0.7386	0.0653
Row 74	0.4215	0.7004	0.3193	0.9282
Row 75	0.1307	0.8242	0.1305	0.8925
Row 76	0.5812	0.6879	0.4844	0.0464
Row 77	0.1080	0.5293	0.2700	0.4844
Row 78	0.3073	0.7945	0.8300	0.3479
Row 79	0.4777	0.5842	0.2233	0.3206
Row 80	0.7218	0.7687	0.0432	0.7268
Row 81	0.1427	0.8696	0.7573	0.1263
Row 82	0.0244	0.6493	0.6750	0.9651
Row 83	0.1925	0.4131	0.3064	0.0508
Row 84	0.8678	0.3827	0.7732	0.3896
Row 85	0.6830	0.0868	0.0773	0.1712
Row 86	0.2699	0.5507	0.1200	0.4458
Row 87	0.3873	0.8615	0.0624	0.4357
Row 88	0.0610	0.0065	0.1505	0.0287
Row 89	0.3380	0.6846	0.1305	0.8998
Row 90	0.4337	0.2892	0.9326	0.7977
Row 91	0.7618	0.7254	0.6185	0.5718
Row 92	0.2404	0.2312	0.6645	0.7351
Row 93	0.8908	0.4011	0.6728	0.4192
Row 94	0.7596	0.5054	0.3343	0.1696
Row 95	0.9736	0.2894	0.8395	0.7554
Row 96	0.2555	0.3570	0.6331	0.3460
Row 97	0.5865	0.8620	0.9528	0.8383
Row 98	0.1753	0.9843	0.5822	0.7130
Row 99	0.2085	0.7513	0.4976	0.6609
Row 100	0.8550	0.6317	0.2716	0.3482
Row 101	0.0003	0.2699	0.1657	0.9740
Row 102	0.8108	0.7631	0.4779	0.7736
Row 103	0.1700	0.7518	0.6194	0.2642
Row 104	0.9089	0.7737	0.1760	0.1838
Row 105	0.2693	0.6957	0.8645	0.7214
Row 106	0.7675	0.7649	0.1831	0.5527
Row 107	0.6605	0.6763	0.6069	0.6509
Row 108	0.9355	0.8627	0.1932	0.1369

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 109	0.2459	0.2674	0.5147	0.3251
Row 110	0.1111	0.9926	0.6565	0.3905
Row 111	0.3883	0.7516	0.0597	0.2444
Row 112	0.3873	0.8884	0.8992	0.4628
Row 113	0.7374	0.3370	0.2922	0.8778
Row 114	0.9644	0.3383	0.7343	0.4642
Row 115	0.8793	0.1624	0.6602	0.6129
Row 116	0.7910	0.7928	0.9132	0.4582
Row 117	0.4158	0.6584	0.0655	0.3760
Row 118	0.6719	0.8505	0.2902	0.3726
Row 119	0.6456	0.6116	0.7580	0.3331
Row 120	0.9372	0.5338	0.9066	0.8391
Row 121	0.1427	0.6179	0.7094	0.5079
Row 122	0.1748	0.9789	0.1452	0.5829
Row 123	0.7514	0.2678	0.7714	0.1895
Row 124	0.4058	0.7714	0.4468	0.5559
Row 125	0.0799	0.6205	0.4477	0.3788
Row 126	0.3297	0.7600	0.5485	0.8005
Row 127	0.8873	0.3812	0.9346	0.4062
Row 128	0.5164	0.9326	0.8897	0.6300
Row 129	0.1876	0.8342	0.5704	0.9817
Row 130	0.3990	0.2170	0.8709	0.4717
Row 131	0.4454	0.3671	0.2185	0.9753
Row 132	0.8951	0.9321	0.3854	0.4805
Row 133	0.3442	0.8316	0.8667	0.6898
Row 134	0.0586	0.2090	0.3720	0.1668
Row 135	0.1312	0.5375	0.6314	0.2907
Row 136	0.5138	0.7588	0.2177	0.7461
Row 137	0.4966	0.1501	0.3993	0.0631
Row 138	0.7154	0.8785	0.8362	0.5782
Row 139	0.6265	0.2019	0.9703	0.2705
Row 140	0.5248	0.5235	0.5018	0.9854
Row 141	0.2711	0.5263	0.8829	0.8525
Row 142	0.1335	0.8354	0.0190	0.3996
Row 143	0.7644	0.3912	0.8849	0.7440
Row 144	0.4358	0.2065	0.4528	0.8955

Continued on next page.

Table 2.C.4 (continued)

	(1)	(2)	(3)	(4)
Row 145	0.9038	0.0718	0.7912	0.5230
Row 146	0.1919	0.7559	0.2908	0.2352
Row 147	0.6801	0.3179	0.8315	0.7988
Row 148	0.7810	0.3397	0.5245	0.8478
Row 149	0.1458	0.1098	0.2659	0.2319
Row 150	0.7207	0.1931	0.2071	0.0241

Note: At the very end, you can add some notes to the table.

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Chapter 3

Math Tests

Abstract. This section contains an extensive set of math “torture” tests. They are designed to check whether mathematical notation is rendered correctly across a wide range of cases. The examples include inline formulas, displayed equations, aligned equations, matrices, and other common mathematical structures. The expressions are not meant to make sense, but rather to expose potential formatting problems.

3.1 Math Test Serif

3.1.1 Overview Serif

Default: $a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma$; $\sigma_\epsilon, c^\alpha$
mathnormal: $a\alpha\alpha b\beta G\mathit{\Gamma}\mathit{\Gamma}\epsilon\varepsilon\theta\vartheta P\mathit{\Pi}\mathit{\Sigma}\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathit: $a\alpha\alpha b\beta G$``$\epsilon\varepsilon\theta\vartheta P$˝˚$\sigma$
mathbf: $\boldsymbol{\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha b\beta G}$``$\boldsymbol{\epsilon\varepsilon\theta\vartheta P}$˝˚$\boldsymbol{\sigma}$
mathbfup: $\mathbf{a\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$

Default: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$; $\boldsymbol{\sigma_\epsilon, c^\alpha}$
mathnormal: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathrm: $\mathbf{a}\boldsymbol{\alpha}\boldsymbol{\alpha}\mathbf{b}\boldsymbol{\beta}\mathbf{G\Gamma\Gamma}\boldsymbol{\epsilon\varepsilon\theta\vartheta}\mathbf{P\Pi\Sigma}\boldsymbol{\sigma}$
mathup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathit: $\boldsymbol{a\alpha\alpha b\beta G}$``$\boldsymbol{\epsilon\varepsilon\theta\vartheta P}$˝˚$\boldsymbol{\sigma}$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G}$``$\boldsymbol{\epsilon\varepsilon\theta\vartheta P}$˝˚$\boldsymbol{\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$

Default: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$; $\sigma_\varepsilon, c^\alpha$
mathnormal: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathit: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

Default: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$; $\boldsymbol{\sigma_\varepsilon, c^\alpha}$
mathnormal: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathrm: $\mathbf{a}\boldsymbol{\alpha}\boldsymbol{\alpha}\mathbf{b}\boldsymbol{\beta}\mathbf{G\Gamma\Gamma}\boldsymbol{\epsilon\varepsilon\theta\vartheta}\mathbf{P\Pi\Sigma}\boldsymbol{\sigma}$
mathup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

3.1.2 Formulas Serif

α, β, γ, δ, ϵ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, o, π, ϖ, ρ, ϱ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω,
$\digamma$, A, B, $\mathit{\Gamma}$, $\mathit{\Delta}$, E, Z, H, $\mathit{\Theta}$, I, K, $\mathit{\Lambda}$, M, N, $\mathit{\Xi}$, O, $\mathit{\Pi}$, P, $\mathit{\Sigma}$, T, $\mathit{\Upsilon}$, $\mathit{\Phi}$, X, $\mathit{\Psi}$, $\mathit{\Omega}$, F,

$\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\delta}, \boldsymbol{\epsilon}, \boldsymbol{\varepsilon}, \boldsymbol{\zeta}, \boldsymbol{\eta}, \boldsymbol{\theta}, \boldsymbol{\vartheta}, \boldsymbol{\iota}, \boldsymbol{\kappa}, \boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\nu}, \boldsymbol{\xi}, \boldsymbol{o}, \boldsymbol{\pi}, \boldsymbol{\varpi}, \boldsymbol{\rho}, \boldsymbol{\varrho}, \boldsymbol{\sigma}, \boldsymbol{\varsigma}, \boldsymbol{\tau}, \boldsymbol{\upsilon}, \boldsymbol{\phi}, \boldsymbol{\varphi}, \boldsymbol{\chi}, \boldsymbol{\psi}, \boldsymbol{\omega}, \digamma, \boldsymbol{A}, \boldsymbol{B}, \boldsymbol{\Gamma}, \boldsymbol{\Delta}, \boldsymbol{E}, \boldsymbol{Z}, \boldsymbol{H}, \boldsymbol{\Theta}, \boldsymbol{I}, \boldsymbol{K}, \boldsymbol{\Lambda}, \boldsymbol{M}, \boldsymbol{N}, \boldsymbol{\Xi}, \boldsymbol{O}, \boldsymbol{\Pi}, \boldsymbol{P}, \boldsymbol{\Sigma}, \boldsymbol{T}, \boldsymbol{\Upsilon}, \boldsymbol{\Phi}, \boldsymbol{X}, \boldsymbol{\Psi}, \boldsymbol{\Omega}, \boldsymbol{F},$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,$

$\boldsymbol{\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,}$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{\nu \to \infty} \nu(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) \, \mathrm{d}x = \binom{27}{2}$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ, ϕ ψ, – —

Latin vs. Greek: a α, d δ, e ϵ, i ι, k κ, n η, o σ, p ρ, $ß$ β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.

$\boldsymbol{\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda}$

$\boldsymbol{\lim_{\nu \to \infty} \nu(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7}$

$\boldsymbol{\hat{\beta} = (X'X)^{-1}X'y}$

$$\boldsymbol{\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)}$$

$$\boldsymbol{\int_{-\infty}^{\infty} x f(x) \, \mathrm{d}x = \binom{27}{2}}$$

Disambiguation: 0 O *O*, 1 l I | *l I* /, *i j*, *rn m*, *θ Θ*, *ϕ ψ*, – —

Latin vs. Greek: *a α*, *d δ*, *e ϵ*, *i ι*, *k κ*, *n η*, *o σ*, *p ρ*, *ß β*, *u υ*, *v ν*, *w ω*, *x χ*, *y γ*, *A Δ Λ*, *O Θ Ω*, *T Γ*, *Y ϒ*.

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{\nu \to \infty} \nu(\nu) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) \, \mathrm{d}x = \binom{27}{2}$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ, ϕ ψ, – —

Latin vs. Greek: *a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, ß β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.*

$\boldsymbol{\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda}$

$\boldsymbol{\lim_{v \to \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7}$

$\boldsymbol{\hat{\beta} = (X'X)^{-1} X'y}$

$$\boldsymbol{\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)}$$

$$\boldsymbol{\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \left(\frac{27}{2}\right)}$$

Disambiguation: 0 O *O*, 1 l I | *l I /*, *i j*, *rn m*, *θ Θ*, *ϕ ψ*, – –

Latin vs. Greek: *a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, ß β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.*

3.1.3 Math Alphabets Serif

Default

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$$
$$A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$$
$$a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$$
$$A, B, \mathit{\Gamma}, \mathit{\Delta}, E, Z, H, \mathit{\Theta}, I, K, \mathit{\Lambda}, M, N, \mathit{\Xi}, O, \mathit{\Pi}, P, \mathit{\Sigma}, T, \mathit{\Upsilon}, \mathit{\Phi}, X, \mathit{\Psi}, \mathit{\Omega},$$
$$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Normal (`\mathnormal`)

$$\mathnormal{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$$
$$\mathnormal{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$$
$$\mathnormal{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$$
$$A, B, \mathit{\Gamma}, \mathit{\Delta}, E, Z, H, \mathit{\Theta}, I, K, \mathit{\Lambda}, M, N, \mathit{\Xi}, O, \mathit{\Pi}, P, \mathit{\Sigma}, T, \mathit{\Upsilon}, \mathit{\Phi}, X, \mathit{\Psi}, \mathit{\Omega},$$
$$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Italic (`\mathit`)

$$\mathit{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$$
$$\mathit{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$$
$$\mathit{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$$
$$A, B, \grave{}, \acute{}, E, Z, H, \hat{}, I, K, \tilde{}, M, N, \ddot{}, O, ˝, P, \mathring{}, T, \check{}, \breve{}, X, \bar{}, \dot{},$$
$$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Roman (`\mathrm`)

$$\mathrm{0,1,2,3,4,5,6,7,8,9,}$$
$$\mathrm{A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z,}$$
$$\mathrm{a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z,}$$
$$\mathrm{A,B,\Gamma,\Delta,E,Z,H,\Theta,I,K,\Lambda,M,N,\Xi,O,\Pi,P,\Sigma,T,\Upsilon,\Phi,X,\Psi,\Omega,}$$
$$\alpha,\beta,\gamma,\delta,\epsilon,\zeta,\eta,\theta,\iota,\kappa,\lambda,\mu,\nu,\xi,o,\pi,\rho,\sigma,\tau,\upsilon,\phi,\chi,\psi,\omega,\varepsilon,\vartheta,\varpi,\varrho,\varsigma,\varphi,$$

Math Bold (`\mathbf`)

$$\mathbf{0,1,2,3,4,5,6,7,8,9,}$$
$$\boldsymbol{A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z,}$$
$$\boldsymbol{a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z,}$$
$$\boldsymbol{A,B,\Gamma,\Delta,E,Z,H,\Theta,I,K,\Lambda,M,N,\Xi,O,\Pi,P,\Sigma,T,\Upsilon,\Phi,X,\Psi,\Omega,}$$
$$\boldsymbol{\alpha,\beta,\gamma,\delta,\epsilon,\zeta,\eta,\theta,\iota,\kappa,\lambda,\mu,\nu,\xi,o,\pi,\rho,\sigma,\tau,\upsilon,\phi,\chi,\psi,\omega,\varepsilon,\vartheta,\varpi,\varrho,\varsigma,\varphi,}$$

Caligraphic (`\mathcal`)

$$\mathcal{A},\mathcal{B},\mathcal{C},\mathcal{D},\mathcal{E},\mathcal{F},\mathcal{G},\mathcal{H},\mathcal{I},\mathcal{J},\mathcal{K},\mathcal{L},\mathcal{M},\mathcal{N},\mathcal{O},\mathcal{P},\mathcal{Q},\mathcal{R},\mathcal{S},\mathcal{T},\mathcal{U},\mathcal{V},\mathcal{W},\mathcal{X},\mathcal{Y},\mathcal{Z},$$

Script (`\mathscr`)

$$\mathscr{A},\mathscr{B},\mathscr{C},\mathscr{D},\mathscr{E},\mathscr{F},\mathscr{G},\mathscr{H},\mathscr{I},\mathscr{J},\mathscr{K},\mathscr{L},\mathscr{M},\mathscr{N},\mathscr{O},\mathscr{P},\mathscr{Q},\mathscr{R},\mathscr{S},\mathscr{T},\mathscr{U},\mathscr{V},\mathscr{W},\mathscr{X},\mathscr{Y},\mathscr{Z},$$

Fraktur (`\mathfrak`)

$$\mathfrak{A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z,}$$
$$\mathfrak{a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z,}$$

Blackboard Bold (`\mathbb`)

$$\mathbb{A},\mathbb{B},\mathbb{C},\mathbb{D},\mathbb{E},\mathbb{F},\mathbb{G},\mathbb{H},\mathbb{I},\mathbb{J},\mathbb{K},\mathbb{L},\mathbb{M},\mathbb{N},\mathbb{O},\mathbb{P},\mathbb{Q},\mathbb{R},\mathbb{S},\mathbb{T},\mathbb{U},\mathbb{V},\mathbb{W},\mathbb{X},\mathbb{Y},\mathbb{Z},$$

3.1.4 Character Sidebearings Serif

Default

$$
\begin{gathered}
|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| + \\
|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| + \\
|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| + \\
|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| + \\
|A| + |B| + |\varGamma| + |\varDelta| + |E| + |Z| + |H| + |\varTheta| + |I| + |K| + |\varLambda| + |M| + \\
|N| + |\varXi| + |O| + |\varPi| + |P| + |\varSigma| + |T| + |\varUpsilon| + |\varPhi| + |X| + |\varPsi| + |\varOmega| + \\
|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| + \\
|\nu| + |\xi| + |o| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| + \\
|\varepsilon| + |\vartheta| + |\varpi| + |\varrho| + |\varsigma| + |\varphi| +
\end{gathered}
$$

Math Roman (`\mathrm`)

$$
\begin{gathered}
|\mathrm{A}| + |\mathrm{B}| + |\mathrm{C}| + |\mathrm{D}| + |\mathrm{E}| + |\mathrm{F}| + |\mathrm{G}| + |\mathrm{H}| + |\mathrm{I}| + |\mathrm{J}| + |\mathrm{K}| + |\mathrm{L}| + |\mathrm{M}| + \\
|\mathrm{N}| + |\mathrm{O}| + |\mathrm{P}| + |\mathrm{Q}| + |\mathrm{R}| + |\mathrm{S}| + |\mathrm{T}| + |\mathrm{U}| + |\mathrm{V}| + |\mathrm{W}| + |\mathrm{X}| + |\mathrm{Y}| + |\mathrm{Z}| + \\
|\mathrm{a}| + |\mathrm{b}| + |\mathrm{c}| + |\mathrm{d}| + |\mathrm{e}| + |\mathrm{f}| + |\mathrm{g}| + |\mathrm{h}| + |\mathrm{i}| + |\mathrm{j}| + |\mathrm{k}| + |\mathrm{l}| + |\mathrm{m}| + \\
|\mathrm{n}| + |\mathrm{o}| + |\mathrm{p}| + |\mathrm{q}| + |\mathrm{r}| + |\mathrm{s}| + |\mathrm{t}| + |\mathrm{u}| + |\mathrm{v}| + |\mathrm{w}| + |\mathrm{x}| + |\mathrm{y}| + |\mathrm{z}| + \\
|\mathrm{A}| + |\mathrm{B}| + |\Gamma| + |\Delta| + |\mathrm{E}| + |\mathrm{Z}| + |\mathrm{H}| + |\Theta| + |\mathrm{I}| + |\mathrm{K}| + |\Lambda| + |\mathrm{M}| + \\
|\mathrm{N}| + |\Xi| + |\mathrm{O}| + |\Pi| + |\mathrm{P}| + |\Sigma| + |\mathrm{T}| + |\Upsilon| + |\Phi| + |\mathrm{X}| + |\Psi| + |\Omega| +
\end{gathered}
$$

Math Bold (`\mathbf`)

$$
\begin{gathered}
|\boldsymbol{A}| + |\boldsymbol{B}| + |\boldsymbol{C}| + |\boldsymbol{D}| + |\boldsymbol{E}| + |\boldsymbol{F}| + |\boldsymbol{G}| + |\boldsymbol{H}| + |\boldsymbol{I}| + |\boldsymbol{J}| + |\boldsymbol{K}| + |\boldsymbol{L}| + |\boldsymbol{M}| + \\
|\boldsymbol{N}| + |\boldsymbol{O}| + |\boldsymbol{P}| + |\boldsymbol{Q}| + |\boldsymbol{R}| + |\boldsymbol{S}| + |\boldsymbol{T}| + |\boldsymbol{U}| + |\boldsymbol{V}| + |\boldsymbol{W}| + |\boldsymbol{X}| + |\boldsymbol{Y}| + |\boldsymbol{Z}| + \\
|\boldsymbol{a}| + |\boldsymbol{b}| + |\boldsymbol{c}| + |\boldsymbol{d}| + |\boldsymbol{e}| + |\boldsymbol{f}| + |\boldsymbol{g}| + |\boldsymbol{h}| + |\boldsymbol{i}| + |\boldsymbol{j}| + |\boldsymbol{k}| + |\boldsymbol{l}| + |\boldsymbol{m}| + \\
|\boldsymbol{n}| + |\boldsymbol{o}| + |\boldsymbol{p}| + |\boldsymbol{q}| + |\boldsymbol{r}| + |\boldsymbol{s}| + |\boldsymbol{t}| + |\boldsymbol{u}| + |\boldsymbol{v}| + |\boldsymbol{w}| + |\boldsymbol{x}| + |\boldsymbol{y}| + |\boldsymbol{z}| + \\
|\boldsymbol{A}| + |\boldsymbol{B}| + |\boldsymbol{\varGamma}| + |\boldsymbol{\varDelta}| + |\boldsymbol{E}| + |\boldsymbol{Z}| + |\boldsymbol{H}| + |\boldsymbol{\varTheta}| + |\boldsymbol{I}| + |\boldsymbol{K}| + |\boldsymbol{\varLambda}| + |\boldsymbol{M}| + \\
|\boldsymbol{N}| + |\boldsymbol{\varXi}| + |\boldsymbol{O}| + |\boldsymbol{\varPi}| + |\boldsymbol{P}| + |\boldsymbol{\varSigma}| + |\boldsymbol{T}| + |\boldsymbol{\varUpsilon}| + |\boldsymbol{\varPhi}| + |\boldsymbol{X}| + |\boldsymbol{\varPsi}| + |\boldsymbol{\varOmega}| +
\end{gathered}
$$

Math Calligraphic (`\mathcal`)

$$
\begin{gathered}
|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| + \\
|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +
\end{gathered}
$$

3.1.5 Superscript Positioning Serif

Default

$$A^2+B^2+C^2+D^2+E^2+F^2+G^2+H^2+I^2+J^2+K^2+L^2+M^2+$$
$$N^2+O^2+P^2+Q^2+R^2+S^2+T^2+U^2+V^2+W^2+X^2+Y^2+Z^2+$$
$$a^2+b^2+c^2+d^2+e^2+f^2+g^2+h^2+i^2+j^2+k^2+l^2+m^2+$$
$$n^2+o^2+p^2+q^2+r^2+s^2+t^2+u^2+v^2+w^2+x^2+y^2+z^2+$$
$$A^2+B^2+\mathit{\Gamma}^2+\mathit{\Delta}^2+E^2+Z^2+H^2+\mathit{\Theta}^2+I^2+K^2+\mathit{\Lambda}^2+M^2+$$
$$N^2+\mathit{\Xi}^2+O^2+\mathit{\Pi}^2+P^2+\mathit{\Sigma}^2+T^2+\mathit{\Upsilon}^2+\mathit{\Phi}^2+X^2+\mathit{\Psi}^2+\mathit{\Omega}^2+$$
$$\alpha^2+\beta^2+\gamma^2+\delta^2+\epsilon^2+\zeta^2+\eta^2+\theta^2+\iota^2+\kappa^2+\lambda^2+\mu^2+$$
$$\nu^2+\xi^2+o^2+\pi^2+\rho^2+\sigma^2+\tau^2+\upsilon^2+\phi^2+\chi^2+\psi^2+\omega^2+$$
$$\varepsilon^2+\vartheta^2+\varpi^2+\varrho^2+\varsigma^2+\varphi^2+$$

Math Roman (`\mathrm`)

$$\mathrm{A}^2+\mathrm{B}^2+\mathrm{C}^2+\mathrm{D}^2+\mathrm{E}^2+\mathrm{F}^2+\mathrm{G}^2+\mathrm{H}^2+\mathrm{I}^2+\mathrm{J}^2+\mathrm{K}^2+\mathrm{L}^2+\mathrm{M}^2+$$
$$\mathrm{N}^2+\mathrm{O}^2+\mathrm{P}^2+\mathrm{Q}^2+\mathrm{R}^2+\mathrm{S}^2+\mathrm{T}^2+\mathrm{U}^2+\mathrm{V}^2+\mathrm{W}^2+\mathrm{X}^2+\mathrm{Y}^2+\mathrm{Z}^2+$$
$$\mathrm{a}^2+\mathrm{b}^2+\mathrm{c}^2+\mathrm{d}^2+\mathrm{e}^2+\mathrm{f}^2+\mathrm{g}^2+\mathrm{h}^2+\mathrm{i}^2+\mathrm{j}^2+\mathrm{k}^2+\mathrm{l}^2+\mathrm{m}^2+$$
$$\mathrm{n}^2+\mathrm{o}^2+\mathrm{p}^2+\mathrm{q}^2+\mathrm{r}^2+\mathrm{s}^2+\mathrm{t}^2+\mathrm{u}^2+\mathrm{v}^2+\mathrm{w}^2+\mathrm{x}^2+\mathrm{y}^2+\mathrm{z}^2+$$
$$\mathrm{A}^2+\mathrm{B}^2+\Gamma^2+\Delta^2+\mathrm{E}^2+\mathrm{Z}^2+\mathrm{H}^2+\Theta^2+\mathrm{I}^2+\mathrm{K}^2+\Lambda^2+\mathrm{M}^2+$$
$$\mathrm{N}^2+\Xi^2+\mathrm{O}^2+\Pi^2+\mathrm{P}^2+\Sigma^2+\mathrm{T}^2+\Upsilon^2+\Phi^2+\mathrm{X}^2+\Psi^2+\Omega^2+$$

Math Bold (`\mathbf`)

$$\boldsymbol{A}^2+\boldsymbol{B}^2+\boldsymbol{C}^2+\boldsymbol{D}^2+\boldsymbol{E}^2+\boldsymbol{F}^2+\boldsymbol{G}^2+\boldsymbol{H}^2+\boldsymbol{I}^2+\boldsymbol{J}^2+\boldsymbol{K}^2+\boldsymbol{L}^2+\boldsymbol{M}^2+$$
$$\boldsymbol{N}^2+\boldsymbol{O}^2+\boldsymbol{P}^2+\boldsymbol{Q}^2+\boldsymbol{R}^2+\boldsymbol{S}^2+\boldsymbol{T}^2+\boldsymbol{U}^2+\boldsymbol{V}^2+\boldsymbol{W}^2+\boldsymbol{X}^2+\boldsymbol{Y}^2+\boldsymbol{Z}^2+$$
$$\boldsymbol{a}^2+\boldsymbol{b}^2+\boldsymbol{c}^2+\boldsymbol{d}^2+\boldsymbol{e}^2+\boldsymbol{f}^2+\boldsymbol{g}^2+\boldsymbol{h}^2+\boldsymbol{i}^2+\boldsymbol{j}^2+\boldsymbol{k}^2+\boldsymbol{l}^2+\boldsymbol{m}^2+$$
$$\boldsymbol{n}^2+\boldsymbol{o}^2+\boldsymbol{p}^2+\boldsymbol{q}^2+\boldsymbol{r}^2+\boldsymbol{s}^2+\boldsymbol{t}^2+\boldsymbol{u}^2+\boldsymbol{v}^2+\boldsymbol{w}^2+\boldsymbol{x}^2+\boldsymbol{y}^2+\boldsymbol{z}^2+$$
$$\boldsymbol{A}^2+\boldsymbol{B}^2+\boldsymbol{\mathit{\Gamma}}^2+\boldsymbol{\mathit{\Delta}}^2+\boldsymbol{E}^2+\boldsymbol{Z}^2+\boldsymbol{H}^2+\boldsymbol{\mathit{\Theta}}^2+\boldsymbol{I}^2+\boldsymbol{K}^2+\boldsymbol{\mathit{\Lambda}}^2+\boldsymbol{M}^2+$$
$$\boldsymbol{N}^2+\boldsymbol{\mathit{\Xi}}^2+\boldsymbol{O}^2+\boldsymbol{\mathit{\Pi}}^2+\boldsymbol{P}^2+\boldsymbol{\mathit{\Sigma}}^2+\boldsymbol{T}^2+\boldsymbol{\mathit{\Upsilon}}^2+\boldsymbol{\mathit{\Phi}}^2+\boldsymbol{X}^2+\boldsymbol{\mathit{\Psi}}^2+\boldsymbol{\mathit{\Omega}}^2+$$

Math Calligraphic (`\mathcal`)

$$\mathcal{A}^2+\mathcal{B}^2+\mathcal{C}^2+\mathcal{D}^2+\mathcal{E}^2+\mathcal{F}^2+\mathcal{G}^2+\mathcal{H}^2+\mathcal{I}^2+\mathcal{J}^2+\mathcal{K}^2+\mathcal{L}^2+\mathcal{M}^2+$$
$$\mathcal{N}^2+\mathcal{O}^2+\mathcal{P}^2+\mathcal{Q}^2+\mathcal{R}^2+\mathcal{S}^2+\mathcal{T}^2+\mathcal{U}^2+\mathcal{V}^2+\mathcal{W}^2+\mathcal{X}^2+\mathcal{Y}^2+\mathcal{Z}^2+$$

3.1.6 Subscript Positioning Serif

Default

$$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$$
$$N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$$
$$a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$$
$$n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$$
$$A_i + B_i + \mathit{\Gamma}_i + \mathit{\Delta}_i + E_i + Z_i + H_i + \mathit{\Theta}_i + I_i + K_i + \mathit{\Lambda}_i + M_i +$$
$$N_i + \mathit{\Xi}_i + O_i + \mathit{\Pi}_i + P_i + \mathit{\Sigma}_i + T_i + \mathit{\Upsilon}_i + \mathit{\Phi}_i + X_i + \mathit{\Psi}_i + \mathit{\Omega}_i +$$
$$\alpha_i + \beta_i + \gamma_i + \delta_i + \epsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i +$$
$$\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + \upsilon_i + \phi_i + \chi_i + \psi_i + \omega_i +$$
$$\varepsilon_i + \vartheta_i + \varpi_i + \varrho_i + \varsigma_i + \varphi_i +$$

Math Roman (\mathrm)

$$\mathrm{A}_i + \mathrm{B}_i + \mathrm{C}_i + \mathrm{D}_i + \mathrm{E}_i + \mathrm{F}_i + \mathrm{G}_i + \mathrm{H}_i + \mathrm{I}_i + \mathrm{J}_i + \mathrm{K}_i + \mathrm{L}_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \mathrm{O}_i + \mathrm{P}_i + \mathrm{Q}_i + \mathrm{R}_i + \mathrm{S}_i + \mathrm{T}_i + \mathrm{U}_i + \mathrm{V}_i + \mathrm{W}_i + \mathrm{X}_i + \mathrm{Y}_i + \mathrm{Z}_i +$$
$$\mathrm{a}_i + \mathrm{b}_i + \mathrm{c}_i + \mathrm{d}_i + \mathrm{e}_i + \mathrm{f}_i + \mathrm{g}_i + \mathrm{h}_i + \mathrm{i}_i + \mathrm{j}_i + \mathrm{k}_i + \mathrm{l}_i + \mathrm{m}_i +$$
$$\mathrm{n}_i + \mathrm{o}_i + \mathrm{p}_i + \mathrm{q}_i + \mathrm{r}_i + \mathrm{s}_i + \mathrm{t}_i + \mathrm{u}_i + \mathrm{v}_i + \mathrm{w}_i + \mathrm{x}_i + \mathrm{y}_i + \mathrm{z}_i +$$
$$\mathrm{A}_i + \mathrm{B}_i + \Gamma_i + \Delta_i + \mathrm{E}_i + \mathrm{Z}_i + \mathrm{H}_i + \Theta_i + \mathrm{I}_i + \mathrm{K}_i + \Lambda_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \Xi_i + \mathrm{O}_i + \Pi_i + \mathrm{P}_i + \Sigma_i + \mathrm{T}_i + \Upsilon_i + \Phi_i + \mathrm{X}_i + \Psi_i + \Omega_i +$$

Math Bold (\mathbf)

$$\boldsymbol{A}_i + \boldsymbol{B}_i + \boldsymbol{C}_i + \boldsymbol{D}_i + \boldsymbol{E}_i + \boldsymbol{F}_i + \boldsymbol{G}_i + \boldsymbol{H}_i + \boldsymbol{I}_i + \boldsymbol{J}_i + \boldsymbol{K}_i + \boldsymbol{L}_i + \boldsymbol{M}_i +$$
$$\boldsymbol{N}_i + \boldsymbol{O}_i + \boldsymbol{P}_i + \boldsymbol{Q}_i + \boldsymbol{R}_i + \boldsymbol{S}_i + \boldsymbol{T}_i + \boldsymbol{U}_i + \boldsymbol{V}_i + \boldsymbol{W}_i + \boldsymbol{X}_i + \boldsymbol{Y}_i + \boldsymbol{Z}_i +$$
$$\boldsymbol{a}_i + \boldsymbol{b}_i + \boldsymbol{c}_i + \boldsymbol{d}_i + \boldsymbol{e}_i + \boldsymbol{f}_i + \boldsymbol{g}_i + \boldsymbol{h}_i + \boldsymbol{i}_i + \boldsymbol{j}_i + \boldsymbol{k}_i + \boldsymbol{l}_i + \boldsymbol{m}_i +$$
$$\boldsymbol{n}_i + \boldsymbol{o}_i + \boldsymbol{p}_i + \boldsymbol{q}_i + \boldsymbol{r}_i + \boldsymbol{s}_i + \boldsymbol{t}_i + \boldsymbol{u}_i + \boldsymbol{v}_i + \boldsymbol{w}_i + \boldsymbol{x}_i + \boldsymbol{y}_i + \boldsymbol{z}_i +$$
$$\boldsymbol{A}_i + \boldsymbol{B}_i + \boldsymbol{\Gamma}_i + \boldsymbol{\Delta}_i + \boldsymbol{E}_i + \boldsymbol{Z}_i + \boldsymbol{H}_i + \boldsymbol{\Theta}_i + \boldsymbol{I}_i + \boldsymbol{K}_i + \boldsymbol{\Lambda}_i + \boldsymbol{M}_i +$$
$$\boldsymbol{N}_i + \boldsymbol{\Xi}_i + \boldsymbol{O}_i + \boldsymbol{\Pi}_i + \boldsymbol{P}_i + \boldsymbol{\Sigma}_i + \boldsymbol{T}_i + \boldsymbol{\Upsilon}_i + \boldsymbol{\Phi}_i + \boldsymbol{X}_i + \boldsymbol{\Psi}_i + \boldsymbol{\Omega}_i +$$

Math Calligraphic (\mathcal)

$$\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i +$$
$$\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +$$

3.1.7 Accent Positioning Serif

Default

$$\hat{0}+\hat{1}+\hat{2}+\hat{3}+\hat{4}+\hat{5}+\hat{6}+\hat{7}+\hat{8}+\hat{9}+$$
$$\hat{A}+\hat{B}+\hat{C}+\hat{D}+\hat{E}+\hat{F}+\hat{G}+\hat{H}+\hat{I}+\hat{J}+\hat{K}+\hat{L}+\hat{M}+$$
$$\hat{N}+\hat{O}+\hat{P}+\hat{Q}+\hat{R}+\hat{S}+\hat{T}+\hat{U}+\hat{V}+\hat{W}+\hat{X}+\hat{Y}+\hat{Z}+$$
$$\hat{a}+\hat{b}+\hat{c}+\hat{d}+\hat{e}+\hat{f}+\hat{g}+\hat{h}+\hat{\imath}+\hat{\jmath}+\hat{k}+\hat{l}+\hat{m}+$$
$$\hat{n}+\hat{o}+\hat{p}+\hat{q}+\hat{r}+\hat{s}+\hat{t}+\hat{u}+\hat{v}+\hat{w}+\hat{x}+\hat{y}+\hat{z}+$$
$$\hat{\mathit{A}}+\hat{\mathit{B}}+\hat{\mathit{\Gamma}}+\hat{\mathit{\Delta}}+\hat{\mathit{E}}+\hat{\mathit{Z}}+\hat{\mathit{H}}+\hat{\mathit{\Theta}}+\hat{\mathit{I}}+\hat{\mathit{K}}+\hat{\mathit{\Lambda}}+\hat{\mathit{M}}+$$
$$\hat{\mathit{N}}+\hat{\mathit{\Xi}}+\hat{\mathit{O}}+\hat{\mathit{\Pi}}+\hat{\mathit{P}}+\hat{\mathit{\Sigma}}+\hat{\mathit{T}}+\hat{\mathit{\Upsilon}}+\hat{\mathit{\Phi}}+\hat{\mathit{X}}+\hat{\mathit{\Psi}}+\hat{\mathit{\Omega}}+$$
$$\hat{\alpha}+\hat{\beta}+\hat{\gamma}+\hat{\delta}+\hat{\epsilon}+\hat{\zeta}+\hat{\eta}+\hat{\theta}+\hat{\iota}+\hat{\kappa}+\hat{\lambda}+\hat{\mu}+$$
$$\hat{\nu}+\hat{\xi}+\hat{o}+\hat{\pi}+\hat{\rho}+\hat{\sigma}+\hat{\tau}+\hat{\upsilon}+\hat{\phi}+\hat{\chi}+\hat{\psi}+\hat{\omega}+$$
$$\hat{\varepsilon}+\hat{\vartheta}+\hat{\varpi}+\hat{\varrho}+\hat{\varsigma}+\hat{\varphi}+$$

Math Italic (`\mathit`)

$$\hat{\mathit{0}}+\hat{\mathit{1}}+\hat{\mathit{2}}+\hat{\mathit{3}}+\hat{\mathit{4}}+\hat{\mathit{5}}+\hat{\mathit{6}}+\hat{\mathit{7}}+\hat{\mathit{8}}+\hat{\mathit{9}}+$$
$$\hat{\mathit{A}}+\hat{\mathit{B}}+\hat{\mathit{C}}+\hat{\mathit{D}}+\hat{\mathit{E}}+\hat{\mathit{F}}+\hat{\mathit{G}}+\hat{\mathit{H}}+\hat{\mathit{I}}+\hat{\mathit{J}}+\hat{\mathit{K}}+\hat{\mathit{L}}+\hat{\mathit{M}}+$$
$$\hat{\mathit{N}}+\hat{\mathit{O}}+\hat{\mathit{P}}+\hat{\mathit{Q}}+\hat{\mathit{R}}+\hat{\mathit{S}}+\hat{\mathit{T}}+\hat{\mathit{U}}+\hat{\mathit{V}}+\hat{\mathit{W}}+\hat{\mathit{X}}+\hat{\mathit{Y}}+\hat{\mathit{Z}}+$$
$$\hat{\mathit{a}}+\hat{\mathit{b}}+\hat{\mathit{c}}+\hat{\mathit{d}}+\hat{\mathit{e}}+\hat{\mathit{f}}+\hat{\mathit{g}}+\hat{\mathit{h}}+\hat{\mathit{i}}+\hat{\mathit{j}}+\hat{\mathit{k}}+\hat{\mathit{l}}+\hat{\mathit{m}}+\hat{\ell}+\hat{\wp}+\hat{\imath}+\hat{\jmath}+\tilde{\mathit{i}}$$
$$\hat{\mathit{n}}+\hat{\mathit{o}}+\hat{\mathit{p}}+\hat{\mathit{q}}+\hat{\mathit{r}}+\hat{\mathit{s}}+\hat{\mathit{t}}+\hat{\mathit{u}}+\hat{\mathit{v}}+\hat{\mathit{w}}+\hat{\mathit{x}}+\hat{\mathit{y}}+\hat{\mathit{z}}+$$
$$\hat{\mathit{A}}+\hat{\mathit{B}}+\hat{\grave{}}+\hat{\acute{}}+\hat{\mathit{E}}+\hat{\mathit{Z}}+\hat{\mathit{H}}+\hat{\hat{}}+\hat{\mathit{I}}+\hat{\mathit{K}}+\hat{\tilde{}}+\hat{\mathit{M}}+$$
$$\hat{\mathit{N}}+\hat{\ddot{}}+\hat{\mathit{O}}+\hat{\text{\H{}}}+\hat{\mathit{P}}+\hat{\mathring{}}+\hat{\mathit{T}}+\hat{\check{}}+\hat{\breve{}}+\hat{\mathit{X}}+\hat{\bar{}}+\hat{\dot{}}+$$
$$\hat{\alpha}+\hat{\beta}+\hat{\gamma}+\hat{\delta}+\hat{\epsilon}+\hat{\zeta}+\hat{\eta}+\hat{\theta}+\hat{\iota}+\hat{\kappa}+\hat{\lambda}+\hat{\mu}+$$
$$\hat{\nu}+\hat{\xi}+\hat{o}+\hat{\pi}+\hat{\rho}+\hat{\sigma}+\hat{\tau}+\hat{\upsilon}+\hat{\phi}+\hat{\chi}+\hat{\psi}+\hat{\omega}+$$
$$\hat{\varepsilon}+\hat{\vartheta}+\hat{\varpi}+\hat{\varrho}+\hat{\varsigma}+\hat{\varphi}+$$

Math Roman (`\mathrm`)

$$\hat{\mathrm{0}}+\hat{\mathrm{1}}+\hat{\mathrm{2}}+\hat{\mathrm{3}}+\hat{\mathrm{4}}+\hat{\mathrm{5}}+\hat{\mathrm{6}}+\hat{\mathrm{7}}+\hat{\mathrm{8}}+\hat{\mathrm{9}}+$$
$$\hat{\mathrm{A}}+\hat{\mathrm{B}}+\hat{\mathrm{C}}+\hat{\mathrm{D}}+\hat{\mathrm{E}}+\hat{\mathrm{F}}+\hat{\mathrm{G}}+\hat{\mathrm{H}}+\hat{\mathrm{I}}+\hat{\mathrm{J}}+\hat{\mathrm{K}}+\hat{\mathrm{L}}+\hat{\mathrm{M}}+$$
$$\hat{\mathrm{N}}+\hat{\mathrm{O}}+\hat{\mathrm{P}}+\hat{\mathrm{Q}}+\hat{\mathrm{R}}+\hat{\mathrm{S}}+\hat{\mathrm{T}}+\hat{\mathrm{U}}+\hat{\mathrm{V}}+\hat{\mathrm{W}}+\hat{\mathrm{X}}+\hat{\mathrm{Y}}+\hat{\mathrm{Z}}+$$
$$\hat{\mathrm{a}}+\hat{\mathrm{b}}+\hat{\mathrm{c}}+\hat{\mathrm{d}}+\hat{\mathrm{e}}+\hat{\mathrm{f}}+\hat{\mathrm{g}}+\hat{\mathrm{h}}+\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}+\hat{\mathrm{l}}+\hat{\mathrm{m}}+$$
$$\hat{\mathrm{n}}+\hat{\mathrm{o}}+\hat{\mathrm{p}}+\hat{\mathrm{q}}+\hat{\mathrm{r}}+\hat{\mathrm{s}}+\hat{\mathrm{t}}+\hat{\mathrm{u}}+\hat{\mathrm{v}}+\hat{\mathrm{w}}+\hat{\mathrm{x}}+\hat{\mathrm{y}}+\hat{\mathrm{z}}+$$
$$\hat{\mathrm{A}}+\hat{\mathrm{B}}+\hat{\Gamma}+\hat{\Delta}+\hat{\mathrm{E}}+\hat{\mathrm{Z}}+\hat{\mathrm{H}}+\hat{\Theta}+\hat{\mathrm{I}}+\hat{\mathrm{K}}+\hat{\Lambda}+\hat{\mathrm{M}}+$$
$$\hat{\mathrm{N}}+\hat{\Xi}+\hat{\mathrm{O}}+\hat{\Pi}+\hat{\mathrm{P}}+\hat{\Sigma}+\hat{\mathrm{T}}+\hat{\Upsilon}+\hat{\Phi}+\hat{\mathrm{X}}+\hat{\Psi}+\hat{\Omega}+$$

Math Bold (`\mathbf`)

$$
\begin{gathered}
\hat{\mathbf{0}} + \hat{\mathbf{1}} + \hat{\mathbf{2}} + \hat{\mathbf{3}} + \hat{\mathbf{4}} + \hat{\mathbf{5}} + \hat{\mathbf{6}} + \hat{\mathbf{7}} + \hat{\mathbf{8}} + \hat{\mathbf{9}} + \\
\hat{\boldsymbol{A}} + \hat{\boldsymbol{B}} + \hat{\boldsymbol{C}} + \hat{\boldsymbol{D}} + \hat{\boldsymbol{E}} + \hat{\boldsymbol{F}} + \hat{\boldsymbol{G}} + \hat{\boldsymbol{H}} + \hat{\boldsymbol{I}} + \hat{\boldsymbol{J}} + \hat{\boldsymbol{K}} + \hat{\boldsymbol{L}} + \hat{\boldsymbol{M}} + \\
\hat{\boldsymbol{N}} + \hat{\boldsymbol{O}} + \hat{\boldsymbol{P}} + \hat{\boldsymbol{Q}} + \hat{\boldsymbol{R}} + \hat{\boldsymbol{S}} + \hat{\boldsymbol{T}} + \hat{\boldsymbol{U}} + \hat{\boldsymbol{V}} + \hat{\boldsymbol{W}} + \hat{\boldsymbol{X}} + \hat{\boldsymbol{Y}} + \hat{\boldsymbol{Z}} + \\
\hat{\boldsymbol{a}} + \hat{\boldsymbol{b}} + \hat{\boldsymbol{c}} + \hat{\boldsymbol{d}} + \hat{\boldsymbol{e}} + \hat{\boldsymbol{f}} + \hat{\boldsymbol{g}} + \hat{\boldsymbol{h}} + \hat{\boldsymbol{i}} + \hat{\boldsymbol{j}} + \hat{\boldsymbol{k}} + \hat{\boldsymbol{l}} + \hat{\boldsymbol{m}} + \\
\hat{\boldsymbol{n}} + \hat{\boldsymbol{o}} + \hat{\boldsymbol{p}} + \hat{\boldsymbol{q}} + \hat{\boldsymbol{r}} + \hat{\boldsymbol{s}} + \hat{\boldsymbol{t}} + \hat{\boldsymbol{u}} + \hat{\boldsymbol{v}} + \hat{\boldsymbol{w}} + \hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} + \hat{\boldsymbol{z}} + \\
\hat{\boldsymbol{A}} + \hat{\boldsymbol{B}} + \hat{\boldsymbol{\mathit{\Gamma}}} + \hat{\boldsymbol{\Delta}} + \hat{\boldsymbol{E}} + \hat{\boldsymbol{Z}} + \hat{\boldsymbol{H}} + \hat{\boldsymbol{\mathit{\Theta}}} + \hat{\boldsymbol{I}} + \hat{\boldsymbol{K}} + \hat{\boldsymbol{\mathit{\Lambda}}} + \hat{\boldsymbol{M}} + \\
\hat{\boldsymbol{N}} + \hat{\boldsymbol{\mathit{\Xi}}} + \hat{\boldsymbol{O}} + \hat{\boldsymbol{\mathit{\Pi}}} + \hat{\boldsymbol{P}} + \hat{\boldsymbol{\mathit{\Sigma}}} + \hat{\boldsymbol{T}} + \hat{\boldsymbol{\mathit{\Upsilon}}} + \hat{\boldsymbol{\mathit{\Phi}}} + \hat{\boldsymbol{X}} + \hat{\boldsymbol{\mathit{\Psi}}} + \hat{\boldsymbol{\mathit{\Omega}}} +
\end{gathered}
$$

Math Calligraphic (`\mathcal`)

$$
\begin{gathered}
\hat{\mathcal{A}} + \hat{\mathcal{B}} + \hat{\mathcal{C}} + \hat{\mathcal{D}} + \hat{\mathcal{E}} + \hat{\mathcal{F}} + \hat{\mathcal{G}} + \hat{\mathcal{H}} + \hat{\mathcal{I}} + \hat{\mathcal{J}} + \hat{\mathcal{K}} + \hat{\mathcal{L}} + \hat{\mathcal{M}} + \\
\hat{\mathcal{N}} + \hat{\mathcal{O}} + \hat{\mathcal{P}} + \hat{\mathcal{Q}} + \hat{\mathcal{R}} + \hat{\mathcal{S}} + \hat{\mathcal{T}} + \hat{\mathcal{U}} + \hat{\mathcal{V}} + \hat{\mathcal{W}} + \hat{\mathcal{X}} + \hat{\mathcal{Y}} + \hat{\mathcal{Z}} +
\end{gathered}
$$

3.1.8 Differentials Serif

$$
\begin{aligned}
& dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM + \\
& dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ + \\
& da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm + \\
& dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz + \\
& dA + dB + d\mathit{\Gamma} + d\Delta + dE + dZ + dH + d\mathit{\Theta} + dI + dK + d\mathit{\Lambda} + dM + \\
& dN + d\mathit{\Xi} + dO + d\mathit{\Pi} + dP + d\mathit{\Sigma} + dT + d\mathit{\Upsilon} + d\mathit{\Phi} + dX + d\mathit{\Psi} + d\mathit{\Omega} + \\
& d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu + \\
& d\nu + d\xi + do + d\pi + d\rho + d\sigma + d\tau + d\upsilon + d\phi + d\chi + d\psi + d\omega + \\
& d\varepsilon + d\vartheta + d\varpi + d\varrho + d\varsigma + d\varphi + \\
& d\mathrm{A} + d\mathrm{B} + d\Gamma + d\Delta + d\mathrm{E} + d\mathrm{Z} + d\mathrm{H} + d\Theta + d\mathrm{I} + d\mathrm{K} + d\Lambda + d\mathrm{M} + \\
& d\mathrm{N} + d\Xi + d\mathrm{O} + d\Pi + d\mathrm{P} + d\Sigma + d\mathrm{T} + d\Upsilon + d\Phi + d\mathrm{X} + d\Psi + d\Omega +
\end{aligned}
$$

$$
\begin{gathered}
\mathrm{d}A+\mathrm{d}B+\mathrm{d}C+\mathrm{d}D+\mathrm{d}E+\mathrm{d}F+\mathrm{d}G+\mathrm{d}H+\mathrm{d}I+\mathrm{d}J+\mathrm{d}K+\mathrm{d}L+\mathrm{d}M+\\
\mathrm{d}N+\mathrm{d}O+\mathrm{d}P+\mathrm{d}Q+\mathrm{d}R+\mathrm{d}S+\mathrm{d}T+\mathrm{d}U+\mathrm{d}V+\mathrm{d}W+\mathrm{d}X+\mathrm{d}Y+\mathrm{d}Z+\\
\mathrm{d}a+\mathrm{d}b+\mathrm{d}c+\mathrm{d}d+\mathrm{d}e+\mathrm{d}f+\mathrm{d}g+\mathrm{d}h+\mathrm{d}i+\mathrm{d}j+\mathrm{d}k+\mathrm{d}l+\mathrm{d}m+\\
\mathrm{d}n+\mathrm{d}o+\mathrm{d}p+\mathrm{d}q+\mathrm{d}r+\mathrm{d}s+\mathrm{d}t+\mathrm{d}u+\mathrm{d}v+\mathrm{d}w+\mathrm{d}x+\mathrm{d}y+\mathrm{d}z+\\
\mathrm{d}A+\mathrm{d}B+\mathrm{d}\varGamma+\mathrm{d}\varDelta+\mathrm{d}E+\mathrm{d}Z+\mathrm{d}H+\mathrm{d}\varTheta+\mathrm{d}I+\mathrm{d}K+\mathrm{d}\varLambda+\mathrm{d}M+\\
\mathrm{d}N+\mathrm{d}\varXi+\mathrm{d}O+\mathrm{d}\varPi+\mathrm{d}P+\mathrm{d}\varSigma+\mathrm{d}T+\mathrm{d}\varUpsilon+\mathrm{d}\varPhi+\mathrm{d}X+\mathrm{d}\varPsi+\mathrm{d}\varOmega+\\
\mathrm{d}\alpha+\mathrm{d}\beta+\mathrm{d}\gamma+\mathrm{d}\delta+\mathrm{d}\epsilon+\mathrm{d}\zeta+\mathrm{d}\eta+\mathrm{d}\theta+\mathrm{d}\iota+\mathrm{d}\kappa+\mathrm{d}\lambda+\mathrm{d}\mu+\\
\mathrm{d}\nu+\mathrm{d}\xi+\mathrm{d}o+\mathrm{d}\pi+\mathrm{d}\rho+\mathrm{d}\sigma+\mathrm{d}\tau+\mathrm{d}\upsilon+\mathrm{d}\phi+\mathrm{d}\chi+\mathrm{d}\psi+\mathrm{d}\omega+\\
\mathrm{d}\varepsilon+\mathrm{d}\vartheta+\mathrm{d}\varpi+\mathrm{d}\varrho+\mathrm{d}\varsigma+\mathrm{d}\varphi+\\
\mathrm{d}\mathrm{A}+\mathrm{d}\mathrm{B}+\mathrm{d}\Gamma+\mathrm{d}\Delta+\mathrm{d}\mathrm{E}+\mathrm{d}\mathrm{Z}+\mathrm{d}\mathrm{H}+\mathrm{d}\Theta+\mathrm{d}\mathrm{I}+\mathrm{d}\mathrm{K}+\mathrm{d}\Lambda+\mathrm{d}\mathrm{M}+\\
\mathrm{d}\mathrm{N}+\mathrm{d}\Xi+\mathrm{d}\mathrm{O}+\mathrm{d}\Pi+\mathrm{d}\mathrm{P}+\mathrm{d}\Sigma+\mathrm{d}\mathrm{T}+\mathrm{d}\Upsilon+\mathrm{d}\Phi+\mathrm{d}\mathrm{X}+\mathrm{d}\Psi+\mathrm{d}\Omega+
\end{gathered}
$$

$$
\begin{gathered}
\partial A+\partial B+\partial C+\partial D+\partial E+\partial F+\partial G+\partial H+\partial I+\partial J+\partial K+\partial L+\partial M+\\
\partial N+\partial O+\partial P+\partial Q+\partial R+\partial S+\partial T+\partial U+\partial V+\partial W+\partial X+\partial Y+\partial Z+\\
\partial a+\partial b+\partial c+\partial d+\partial e+\partial f+\partial g+\partial h+\partial i+\partial j+\partial k+\partial l+\partial m+\\
\partial n+\partial o+\partial p+\partial q+\partial r+\partial s+\partial t+\partial u+\partial v+\partial w+\partial x+\partial y+\partial z+\\
\partial A+\partial B+\partial\varGamma+\partial\varDelta+\partial E+\partial Z+\partial H+\partial\varTheta+\partial I+\partial K+\partial\varLambda+\partial M+\\
\partial N+\partial\varXi+\partial O+\partial\varPi+\partial P+\partial\varSigma+\partial T+\partial\varUpsilon+\partial\varPhi+\partial X+\partial\varPsi+\partial\varOmega+\\
\partial\alpha+\partial\beta+\partial\gamma+\partial\delta+\partial\epsilon+\partial\zeta+\partial\eta+\partial\theta+\partial\iota+\partial\kappa+\partial\lambda+\partial\mu+\\
\partial\nu+\partial\xi+\partial o+\partial\pi+\partial\rho+\partial\sigma+\partial\tau+\partial\upsilon+\partial\phi+\partial\chi+\partial\psi+\partial\omega+\\
\partial\varepsilon+\partial\vartheta+\partial\varpi+\partial\varrho+\partial\varsigma+\partial\varphi+\\
\partial\mathrm{A}+\partial\mathrm{B}+\partial\Gamma+\partial\Delta+\partial\mathrm{E}+\partial\mathrm{Z}+\partial\mathrm{H}+\partial\Theta+\partial\mathrm{I}+\partial\mathrm{K}+\partial\Lambda+\partial\mathrm{M}+\\
\partial\mathrm{N}+\partial\Xi+\partial\mathrm{O}+\partial\Pi+\partial\mathrm{P}+\partial\Sigma+\partial\mathrm{T}+\partial\Upsilon+\partial\Phi+\partial\mathrm{X}+\partial\Psi+\partial\Omega+
\end{gathered}
$$

3.1.9 Slash Kerning Serif

$$
\begin{gathered}
1/A+1/B+1/C+1/D+1/E+1/F+1/G+1/H+1/I+1/J+1/K+1/L+1/M+\\
1/N+1/O+1/P+1/Q+1/R+1/S+1/T+1/U+1/V+1/W+1/X+1/Y+1/Z+\\
1/a+1/b+1/c+1/d+1/e+1/f+1/g+1/h+1/i+1/j+1/k+1/l+1/m+\\
1/n+1/o+1/p+1/q+1/r+1/s+1/t+1/u+1/v+1/w+1/x+1/y+1/z+\\
1/A+1/B+1/\varGamma+1/\varDelta+1/E+1/Z+1/H+1/\varTheta+1/I+1/K+1/\varLambda+1/M+\\
1/N+1/\varXi+1/O+1/\varPi+1/P+1/\varSigma+1/T+1/\varUpsilon+1/\varPhi+1/X+1/\varPsi+1/\varOmega+\\
1/\alpha+1/\beta+1/\gamma+1/\delta+1/\epsilon+1/\zeta+1/\eta+1/\theta+1/\iota+1/\kappa+1/\lambda+1/\mu+\\
1/\nu+1/\xi+1/o+1/\pi+1/\rho+1/\sigma+1/\tau+1/\upsilon+1/\phi+1/\chi+1/\psi+1/\omega+\\
1/\varepsilon+1/\vartheta+1/\varpi+1/\varrho+1/\varsigma+1/\varphi+
\end{gathered}
$$

$$A/2+B/2+C/2+D/2+E/2+F/2+G/2+H/2+I/2+J/2+K/2+L/2+M/2+$$
$$N/2+O/2+P/2+Q/2+R/2+S/2+T/2+U/2+V/2+W/2+X/2+Y/2+Z/2+$$
$$a/2+b/2+c/2+d/2+e/2+f/2+g/2+h/2+i/2+j/2+k/2+l/2+m/2+$$
$$n/2+o/2+p/2+q/2+r/2+s/2+t/2+u/2+v/2+w/2+x/2+y/2+z/2+$$
$$A/2+B/2+\Gamma/2+\Delta/2+E/2+Z/2+H/2+\Theta/2+I/2+K/2+\Lambda/2+M/2+$$
$$N/2+\Xi/2+O/2+\Pi/2+P/2+\Sigma/2+T/2+\Upsilon/2+\Phi/2+X/2+\Psi/2+\Omega/2+$$
$$\alpha/2+\beta/2+\gamma/2+\delta/2+\epsilon/2+\zeta/2+\eta/2+\theta/2+\iota/2+\kappa/2+\lambda/2+\mu/2+$$
$$\nu/2+\xi/2+o/2+\pi/2+\rho/2+\sigma/2+\tau/2+\upsilon/2+\phi/2+\chi/2+\psi/2+\omega/2+$$
$$\varepsilon/2+\vartheta/2+\varpi/2+\varrho/2+\varsigma/2+\varphi/2+$$

3.1.10 (Big) Operators Serif

$\sum_{i=1}^n x^n$ $\prod_{i=1}^n x^n$ $\coprod_{i=1}^n x^n$ $\int_{i=1}^n x^n$ $\oint_{i=1}^n x^n$
$\bigotimes_{i=1}^n x^n$ $\bigoplus_{i=1}^n x^n$ $\bigodot_{i=1}^n x^n$ $\bigwedge_{i=1}^n x^n$ $\bigvee_{i=1}^n x^n$ $\biguplus_{i=1}^n x^n$ $\bigcup_{i=1}^n x^n$ $\bigcap_{i=1}^n x^n$ $\bigsqcup_{i=1}^n x^n$

$$\sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n$$

$$\bigotimes_{i=1}^n x^n \quad \bigoplus_{i=1}^n x^n \quad \bigodot_{i=1}^n x^n \quad \bigwedge_{i=1}^n x^n \quad \bigvee_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n$$

3.1.11 Radicals Serif

$$\sqrt{x+y} \quad \sqrt{x^2+y^2} \quad \sqrt{x_i^2+y_j^2} \quad \sqrt{\left(\frac{\cos x}{2}\right)} \quad \sqrt{\left(\frac{\sin x}{2}\right)}$$

$$\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}$$

3.1.12 Over- and Underbraces Serif

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.1.13 Normal and Wide Accents Serif

$$\dot{x} \quad \ddot{x} \quad \vec{x} \quad \bar{x} \quad \overline{x} \quad \overline{xx} \quad \tilde{x} \quad \widetilde{x} \quad \widetilde{xx} \quad \widetilde{xxx} \quad \hat{x} \quad \widehat{x} \quad \widehat{xx} \quad \widehat{xxx}$$

$$\hat{x} \quad \check{x} \quad \tilde{x} \quad \acute{x} \quad \grave{x} \quad \dot{x} \quad \ddot{x} \quad \breve{x} \quad \bar{x} \quad \vec{x}$$

3.1.14 Long Arrows Serif

$$\leftarrow\!\!-\!\!-\!\!\rightarrow \quad \leftrightarrow \quad \longleftarrow \quad \longrightarrow \quad \longleftrightarrow \quad \Longleftarrow\!\!\!=\!\!\!\Longrightarrow \quad \Leftrightarrow \quad \Longleftarrow \quad \Longrightarrow \quad \Longleftrightarrow$$

3.1.15 Left and Right Delimiters Serif

$$-(f)--[f]--\lfloor f\rfloor--\lceil f\rceil--\langle f\rangle--\{f\}-$$

Using `\left` and `\right`.

$$-\left(f\right)--\left[f\right]--\left\lfloor f\right\rfloor--\left\lceil f\right\rceil--\left\langle f\right\rangle--\left\{f\right\}-$$

$$-\left)f\right(--\left]f\right[--\left/f\right/--\left\backslash f\right\backslash--\left/f\right\backslash--\left\backslash f\right/-$$

3.1.16 Big-g-g Delimiters Serif

$$-\Biggl\lfloor\biggl\lfloor\Bigl\lfloor\bigl\lfloor\lfloor-\rfloor\bigr\rfloor\Bigr\rfloor\biggr\rfloor\Biggr\rfloor- \qquad -\Biggl(\biggl(\Bigl(\bigl((-)\bigr)\Bigr)\biggr)\Biggr)-$$

$$-\Biggl\lceil\biggl\lceil\Bigl\lceil\bigl\lceil\lceil-\rceil\bigr\rceil\Bigr\rceil\biggr\rceil\Biggr\rceil- \qquad -\Biggl\{\biggl\{\Bigl\{\bigl\{\{-\}\bigr\}\Bigr\}\biggr\}\Biggr\}-$$

$$-\Biggl[\biggl[\Bigl[\bigl[[-]\bigr]\Bigr]\biggr]\Biggr]- \qquad -\Biggl(\biggl(\Bigl(\bigl((-)\bigr)\Bigr)\biggr)\Biggr)-$$

$$-\Biggl\langle\biggl\langle\Bigl\langle\bigl\langle\langle-\rangle\bigr\rangle\Bigr\rangle\biggr\rangle\Biggr\rangle- \qquad -\Bigg\int\bigg\int\Big\int\big\int\int-\int\big\int\Big\int\bigg\int\Bigg\int-$$

$$-\Bigg\uparrow\bigg\uparrow\Big\uparrow\big\uparrow\uparrow-\downarrow\big\downarrow\Big\downarrow\bigg\downarrow\Bigg\downarrow- \qquad -\Bigg\Uparrow\bigg\Uparrow\Big\Uparrow\big\Uparrow\Uparrow-\Downarrow\big\Downarrow\Big\Downarrow\bigg\Downarrow\Bigg\Downarrow-$$

3.1.17 Binary Operators Serif

$x \pm y$	\pm	$x \cap y$	\cap	$x \diamond y$	\diamond	$x \oplus y$	\oplus
$x \mp y$	\mp	$x \cup y$	\cup	$x \bigtriangleup y$	\bigtriangleup	$x \ominus y$	\ominus
$x \times y$	\times	$x \uplus y$	\uplus	$x \bigtriangledown y$	\bigtriangledown	$x \otimes y$	\otimes
$x \div y$	\div	$x \sqcap y$	\sqcap	$x \triangleleft y$	\triangleleft	$x \oslash y$	\oslash
$x \ast y$	\ast	$x \sqcup y$	\sqcup	$x \triangleright y$	\triangleright	$x \odot y$	\odot
$x \star y$	\star	$x \vee y$	\vee	$x \lhd y$	\lhd	$x \bigcirc y$	\bigcirc
$x \circ y$	\circ	$x \wedge y$	\wedge	$x \rhd y$	\rhd	$x \dagger y$	\dagger
$x \bullet y$	\bullet	$x \setminus y$	\setminus	$x \unlhd y$	\unlhd	$x \ddagger y$	\ddagger
$x \cdot y$	\cdot	$x \wr y$	\wr	$x \unrhd y$	\unrhd	$x \S y$	\S
$x + y$	+	$x - y$	-	$x \amalg y$	\amalg	$x \P y$	\P

3.1.18 Relations Serif

$x \leq y$	\leq	$x \geq y$	\geq	$x \equiv y$	\equiv	$x \models y$	\models
$x \prec y$	\prec	$x \succ y$	\succ	$x \sim y$	\sim	$x \perp y$	\perp
$x \preceq y$	\preceq	$x \succeq y$	\succeq	$x \simeq y$	\simeq	$x \mid y$	\mid
$x \ll y$	\ll	$x \gg y$	\gg	$x \asymp y$	\asymp	$x \parallel y$	\parallel
$x \subset y$	\subset	$x \supset y$	\supset	$x \approx y$	\approx	$x \bowtie y$	\bowtie
$x \subseteq y$	\subseteq	$x \supseteq y$	\supseteq	$x \cong y$	\cong	$x \Join y$	\Join
$x \sqsubset y$	\sqsubset	$x \sqsupset y$	\sqsupset	$x \neq y$	\neq	$x \smile y$	\smile
$x \sqsubseteq y$	\sqsubseteq	$x \sqsupseteq y$	\sqsupseteq	$x \doteq y$	\doteq	$x \frown y$	\frown
$x \in y$	\in	$x \ni y$	\ni	$x \propto y$	\propto	$x = y$	=
$x \vdash y$	\vdash	$x \dashv y$	\dashv	$x < y$	<	$x > y$	>
$x : y$	:						

3.1.19 Punctuation Serif

x,y	,	$x;y$	;	$x\colon y$	\colon	$x.y$	\ldotp	$x\cdotp y$	\cdotp

3.1.20 Arrows Serif

$x \leftarrow y$	`\leftarrow`	$x \longleftarrow y$	`\longleftarrow`	$x \uparrow y$	`\uparrow`
$x \Leftarrow y$	`\Leftarrow`	$x \Longleftarrow y$	`\Longleftarrow`	$x \Uparrow y$	`\Uparrow`
$x \rightarrow y$	`\rightarrow`	$x \longrightarrow y$	`\longrightarrow`	$x \downarrow y$	`\downarrow`
$x \Rightarrow y$	`\Rightarrow`	$x \Longrightarrow y$	`\Longrightarrow`	$x \Downarrow y$	`\Downarrow`
$x \leftrightarrow y$	`\leftrightarrow`	$x \longleftrightarrow y$	`\longleftrightarrow`	$x \updownarrow y$	`\updownarrow`
$x \Leftrightarrow y$	`\Leftrightarrow`	$x \Longleftrightarrow y$	`\Longleftrightarrow`	$x \Updownarrow y$	`\Updownarrow`
$x \mapsto y$	`\mapsto`	$x \longmapsto y$	`\longmapsto`	$x \nearrow y$	`\nearrow`
$x \hookleftarrow y$	`\hookleftarrow`	$x \hookrightarrow y$	`\hookrightarrow`	$x \searrow y$	`\searrow`
$x \leftharpoonup y$	`\leftharpoonup`	$x \rightharpoonup y$	`\rightharpoonup`	$x \swarrow y$	`\swarrow`
$x \leftharpoondown y$	`\leftharpoondown`	$x \rightharpoondown y$	`\rightharpoondown`	$x \nwarrow y$	`\nwarrow`
$x \rightleftharpoons y$	`\rightleftharpoons`	$x \leadsto y$	`\leadsto`		

3.1.21 Miscellaneous Symbols Serif

<table>
<tr><td>$x \ldots y$</td><td><code>\ldots</code></td><td>$x \cdots y$</td><td><code>\cdots</code></td><td>$x \vdots y$</td><td><code>\vdots</code></td><td>$x \ddots y$</td><td><code>\ddots</code></td></tr>
<tr><td>$x \aleph y$</td><td><code>\aleph</code></td><td>$x \prime y$</td><td><code>\prime</code></td><td>$x \forall y$</td><td><code>\forall</code></td><td>$x \infty y$</td><td><code>\infty</code></td></tr>
<tr><td>$x \hbar y$</td><td><code>\hbar</code></td><td>$x \emptyset y$</td><td><code>\emptyset</code></td><td>$x \exists y$</td><td><code>\exists</code></td><td>$x \Box y$</td><td><code>\Box</code></td></tr>
<tr><td>$x \imath y$</td><td><code>\imath</code></td><td>$x \nabla y$</td><td><code>\nabla</code></td><td>$x \neg y$</td><td><code>\neg</code></td><td>$x \Diamond y$</td><td><code>\Diamond</code></td></tr>
<tr><td>$x \jmath y$</td><td><code>\jmath</code></td><td>$x \surd y$</td><td><code>\surd</code></td><td>$x \flat y$</td><td><code>\flat</code></td><td>$x \triangle y$</td><td><code>\triangle</code></td></tr>
<tr><td>$x \ell y$</td><td><code>\ell</code></td><td>$x \top y$</td><td><code>\top</code></td><td>$x \natural y$</td><td><code>\natural</code></td><td>$x \clubsuit y$</td><td><code>\clubsuit</code></td></tr>
<tr><td>$x \wp y$</td><td><code>\wp</code></td><td>$x \bot y$</td><td><code>\bot</code></td><td>$x \sharp y$</td><td><code>\sharp</code></td><td>$x \diamondsuit y$</td><td><code>\diamondsuit</code></td></tr>
<tr><td>$x \Re y$</td><td><code>\Re</code></td><td>$x \| y$</td><td><code>\|</code></td><td>$x \backslash y$</td><td><code>\backslash</code></td><td>$x \heartsuit y$</td><td><code>\heartsuit</code></td></tr>
<tr><td>$x \Im y$</td><td><code>\Im</code></td><td>$x \angle y$</td><td><code>\angle</code></td><td>$x \partial y$</td><td><code>\partial</code></td><td>$x \spadesuit y$</td><td><code>\spadesuit</code></td></tr>
<tr><td>$x \mho y$</td><td><code>\mho</code></td><td>$x . y$</td><td><code>.</code></td><td>$x | y$</td><td><code>|</code></td><td>$x ! y$</td><td><code>!</code></td></tr>
</table>

3.1.22 Variable-Sized Operators Serif

$x \sum y$	`\sum`	$x \bigcap y$	`\bigcap`	$x \bigodot y$	`\bigodot`
$x \prod y$	`\prod`	$x \bigcup y$	`\bigcup`	$x \bigotimes y$	`\bigotimes`
$x \coprod y$	`\coprod`	$x \bigsqcup y$	`\bigsqcup`	$x \bigoplus y$	`\bigoplus`
$x \int y$	`\int`	$x \bigvee y$	`\bigvee`	$x \biguplus y$	`\biguplus`
$x \oint y$	`\oint`	$x \bigwedge y$	`\bigwedge`		

3.1.23 Log-Like Operators Serif

$x \arccos y$	$x \cos y$	$x \csc y$	$x \exp y$	$x \ker y$	$x \limsup y$	$x \min y$	$x \sinh y$
$x \arcsin y$	$x \cosh y$	$x \deg y$	$x \gcd y$	$x \lg y$	$x \ln y$	$x \Pr y$	$x \sup y$
$x \arctan y$	$x \cot y$	$x \det y$	$x \hom y$	$x \lim y$	$x \log y$	$x \sec y$	$x \tan y$
$x \arg y$	$x \coth y$	$x \dim y$	$x \inf y$	$x \liminf y$	$x \max y$	$x \sin y$	$x \tanh y$

3.1.24 Delimiters Serif

$x(y$	`(`	$x)y$	`)`	$x\uparrow y$	`\uparrow`	$x\Uparrow y$	`\Uparrow`
$x[y$	`[`	$x]y$	`]`	$x\downarrow y$	`\downarrow`	$x\Downarrow y$	`\Downarrow`
$x\{y$	`\{`	$x\}y$	`\}`	$x\updownarrow y$	`\updownarrow`	$x\Updownarrow y$	`\Updownarrow`
$x\lfloor y$	`\lfloor`	$x\rfloor y$	`\rfloor`	$x\lceil y$	`\lceil`	$x\rceil y$	`\rceil`
$x\langle y$	`\langle`	$x\rangle y$	`\rangle`	x/y	`/`	$x\backslash y$	`\backslash`
$x\vert y$	`|`	$x\Vert y$	`\\|`				

3.1.25 Large Delimiters Serif

$\rmoustache$	`\rmoustache`	$\lmoustache$	`\lmoustache`	$\rgroup$	`\rgroup`	$\lgroup$	`\lgroup`
$\arrowvert$	`\arrowvert`	$\Arrowvert$	`\Arrowvert`	$\bracevert$	`\bracevert`		

3.1.26 Math Mode Accents Serif

$\hat{a}$	`\hat{a}`	$\acute{a}$	`\acute{a}`	$\bar{a}$	`\bar{a}`	$\dot{a}$	`\dot{a}`	$\breve{a}$	`\breve{a}`
$\check{a}$	`\check{a}`	$\grave{a}$	`\grave{a}`	$\vec{a}$	`\vec{a}`	$\ddot{a}$	`\ddot{a}`	$\tilde{a}$	`\tilde{a}`

3.1.27 Miscellaneous Constructions Serif

$\widetilde{abc}$	`\widetilde{abc}`	$\widehat{abc}$	`\widehat{abc}`
$\overleftarrow{abc}$	`\overleftarrow{abc}`	$\overrightarrow{abc}$	`\overrightarrow{abc}`
$\overline{abc}$	`\overline{abc}`	$\underline{abc}$	`\underline{abc}`
$\overbrace{abc}$	`\overbrace{abc}`	$\underbrace{abc}$	`\underbrace{abc}`
$\sqrt{abc}$	`\sqrt{abc}`	$\sqrt[n]{abc}$	`\sqrt[n]{abc}`
f'	`f'`	$\frac{abc}{xyz}$	`\frac{abc}{xyz}`

3.1.28 AMS Delimiters Serif

$x\ulcorner y$	`\ulcorner`	$x\urcorner y$	`\urcorner`	$x\llcorner y$	`\llcorner`	$x\lrcorner y$	`\lrcorner`

3.1.29 AMS Arrows Serif

$x \dashrightarrow y$	\dashrightarrow	$x \dashleftarrow y$	\dashleftarrow
$x \leftleftarrows y$	\leftleftarrows	$x \leftrightarrows y$	\leftrightarrows
$x \Lleftarrow y$	\Lleftarrow	$x \twoheadleftarrow y$	\twoheadleftarrow
$x \leftarrowtail y$	\leftarrowtail	$x \looparrowleft y$	\looparrowleft
$x \leftrightharpoons y$	\leftrightharpoons	$x \curvearrowleft y$	\curvearrowleft
$x \circlearrowleft y$	\circlearrowleft	$x \Lsh y$	\Lsh
$x \upuparrows y$	\upuparrows	$x \upharpoonleft y$	\upharpoonleft
$x \downharpoonleft y$	\downharpoonleft	$x \multimap y$	\multimap
$x \leftrightsquigarrow y$	\leftrightsquigarrow	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \twoheadrightarrow y$	\twoheadrightarrow
$x \rightarrowtail y$	\rightarrowtail	$x \looparrowright y$	\looparrowright
$x \rightleftharpoons y$	\rightleftharpoons	$x \curvearrowright y$	\curvearrowright
$x \circlearrowright y$	\circlearrowright	$x \Rsh y$	\Rsh
$x \downdownarrows y$	\downdownarrows	$x \upharpoonright y$	\upharpoonright
$x \downharpoonright y$	\downharpoonright	$x \rightsquigarrow y$	\rightsquigarrow

3.1.30 AMS Negated Arrows Serif

$x \nleftarrow y$	\nleftarrow	$x \nrightarrow y$	\nrightarrow
$x \nLeftarrow y$	\nLeftarrow	$x \nRightarrow y$	\nRightarrow
$x \nleftrightarrow y$	\nleftrightarrow	$x \nLeftrightarrow y$	\nLeftrightarrow

3.1.31 AMS Greek Serif

$x\digamma y$	\digamma	$x\varkappa y$	\varkappa

3.1.32 AMS Hebrew Serif

$x\beth y$	\beth	$x\daleth y$	\daleth	$x\gimel y$	\gimel

3.1.33 AMS Miscellaneous Serif

$x\hbar y$	`\hbar`	$x\hslash y$	`\hslash`
$x \vartriangle y$	`\vartriangle`	$x\triangledown y$	`\triangledown`
$x\square y$	`\square`	$x\lozenge y$	`\lozenge`
$x\circledS y$	`\circledS`	$x\angle y$	`\angle`
$x\measuredangle y$	`\measuredangle`	$x\nexists y$	`\nexists`
$x\mho y$	`\mho`	$x\Finv y$	`\Finv`^{u}
$x\Game y$	`\Game`^{u}	$x\Bbbk y$	`\Bbbk`^{u}
$x\backprime y$	`\backprime`	$x\varnothing y$	`\varnothing`
$x\blacktriangle y$	`\blacktriangle`	$x\blacktriangledown y$	`\blacktriangledown`
$x\blacksquare y$	`\blacksquare`	$x\blacklozenge y$	`\blacklozenge`
$x\bigstar y$	`\bigstar`	$x\sphericalangle y$	`\sphericalangle`
$x\complement y$	`\complement`	$x\eth y$	`\eth`
$x\diagup y$	`\diagup`^{u}	$x\diagdown y$	`\diagdown`^{u}

^{u} Not defined in `amssymb.sty`, define using the `\newsymbol` command.

3.1.34 AMS Binary Operators Serif

$x \dotplus y$	`\dotplus`	$x \smallsetminus y$	`\smallsetminus`
$x \Cap y$	`\Cap`	$x \Cup y$	`\Cup`
$x \barwedge y$	`\barwedge`	$x \veebar y$	`\veebar`
$x \doublebarwedge y$	`\doublebarwedge`	$x \boxminus y$	`\boxminus`
$x \boxtimes y$	`\boxtimes`	$x \boxdot y$	`\boxdot`
$x \boxplus y$	`\boxplus`	$x \divideontimes y$	`\divideontimes`
$x \ltimes y$	`\ltimes`	$x \rtimes y$	`\rtimes`
$x \leftthreetimes y$	`\leftthreetimes`	$x \rightthreetimes y$	`\rightthreetimes`
$x \curlywedge y$	`\curlywedge`	$x \curlyvee y$	`\curlyvee`
$x \circleddash y$	`\circleddash`	$x \circledast y$	`\circledast`
$x \circledcirc y$	`\circledcirc`	$x \centerdot y$	`\centerdot`
$x \intercal y$	`\intercal`		

3.1.35 AMS Relations Serif

$x \leqslant y$	`\leqslant`
$x \lesssim y$	`\lesssim`
$x \approxeq y$	`\approxeq`
$x \lll y$	`\lll`
$x \lesseqgtr y$	`\lesseqgtr`
$x \doteqdot y$	`\doteqdot`
$x \fallingdotseq y$	`\fallingdotseq`
$x \backsimeq y$	`\backsimeq`
$x \Subset y$	`\Subset`
$x \preccurlyeq y$	`\preccurlyeq`
$x \precsim y$	`\precsim`
$x \vartriangleleft y$	`\vartriangleleft`
$x \vDash y$	`\vDash`
$x \smallsmile y$	`\smallsmile`
$x \bumpeq y$	`\bumpeq`
$x \geqq y$	`\geqq`
$x \eqslantgtr y$	`\eqslantgtr`
$x \gtrapprox y$	`\gtrapprox`
$x \ggg y$	`\ggg`
$x \gtreqless y$	`\gtreqless`
$x \eqcirc y$	`\eqcirc`
$x \triangleq y$	`\triangleq`
$x \thickapprox y$	`\thickapprox`
$x \Supset y$	`\Supset`
$x \succcurlyeq y$	`\succcurlyeq`
$x \succsim y$	`\succsim`
$x \vartriangleright y$	`\vartriangleright`
$x \Vdash y$	`\Vdash`
$x \shortparallel y$	`\shortparallel`
$x \pitchfork y$	`\pitchfork`
$x \blacktriangleleft y$	`\blacktriangleleft`
$x \backepsilon y$	`\backepsilon`
$x \because y$	`\because`

3.1.36 AMS Negated Relations Serif

$x \nless y$	`\nless`	$x \nleq y$	`\nleq`
$x \nleqslant y$	`\nleqslant`	$x \nleqq y$	`\nleqq`
$x \lneq y$	`\lneq`	$x \lneqq y$	`\lneqq`
$x \lvertneqq y$	`\lvertneqq`	$x \lnsim y$	`\lnsim`
$x \lnapprox y$	`\lnapprox`	$x \nprec y$	`\nprec`
$x \npreceq y$	`\npreceq`	$x \precnsim y$	`\precnsim`
$x \precnapprox y$	`\precnapprox`	$x \nsim y$	`\nsim`
$x \nshortmid y$	`\nshortmid`	$x \nmid y$	`\nmid`
$x \nvdash y$	`\nvdash`	$x \nvDash y$	`\nvDash`
$x \ntriangleleft y$	`\ntriangleleft`	$x \ntrianglelefteq y$	`\ntrianglelefteq`
$x \nsubseteq y$	`\nsubseteq`	$x \subsetneq y$	`\subsetneq`
$x \varsubsetneq y$	`\varsubsetneq`	$x \subsetneqq y$	`\subsetneqq`
$x \varsubsetneqq y$	`\varsubsetneqq`	$x \ngtr y$	`\ngtr`
$x \ngeq y$	`\ngeq`	$x \ngeqslant y$	`\ngeqslant`
$x \ngeqq y$	`\ngeqq`	$x \gneq y$	`\gneq`
$x \gneqq y$	`\gneqq`	$x \gvertneqq y$	`\gvertneqq`
$x \gnsim y$	`\gnsim`	$x \gnapprox y$	`\gnapprox`
$x \nsucc y$	`\nsucc`	$x \nsucceq y$	`\nsucceq`
$x \nsucceqq y$	`\nsucceqq`	$x \succnsim y$	`\succnsim`
$x \succnapprox y$	`\succnapprox`	$x \ncong y$	`\ncong`
$x \nshortparallel y$	`\nshortparallel`	$x \nparallel y$	`\nparallel`
$x \nvDash y$	`\nvDash`	$x \nVDash y$	`\nVDash`
$x \ntriangleright y$	`\ntriangleright`	$x \ntrianglerighteq y$	`\ntrianglerighteq`
$x \nsupseteq y$	`\nsupseteq`	$x \nsupseteqq y$	`\nsupseteqq`
$x \supsetneq y$	`\supsetneq`	$x \varsupsetneq y$	`\varsupsetneq`
$x \supsetneqq y$	`\supsetneqq`	$x \varsupsetneqq y$	`\varsupsetneqq`

3.1.37 Math “Torture” Test Serif

Most of the following examples are taken from *The TEXbook* (Knuth 1984, see https://ctan.org/pkg/texbook) and were adapted for LATEX from Karl Berry’s torture test for plain TEX math fonts.

$x+y-z,\quad x+y*z,\quad z*y/z,\quad (x+y)(x-y)=x^2-y^2,$
$x\times y\cdot z=[x\,y\,z],\quad x\circ y\bullet z,\quad x\cup y\cap z,\quad x\sqcup y\sqcap z,$
$x\vee y\wedge z,\quad x\pm y\mp z,\quad x=y/z,\quad x:=y,\quad x\le y\ne z,\quad x\sim y\simeq z\; x\equiv y\not\equiv z,\quad x\subset y\subseteq z$
$\sin 2\theta=2\sin\theta\cos\theta,\quad \mathrm{O}(n\log n\log n),\quad \Pr(X>x)=\exp(-x/\mu),$
$\bigl(x\in A(n)\bigm| x\in B(n)\bigr),\quad \bigcup_n X_n \parallel \bigcap_n Y_n$
In-text matrices $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}a&b&c\\l&m&n\end{smallmatrix}\right)$.

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x} \frac{1}{1-x^2} = \frac{a+1}{b} \bigg/ \frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |\varphi(x+\mathrm{i}y)|^2 = 0$$

$$\pi(n) = \sum_{m=2}^{n} \left\lfloor \left(\sum_{k=1}^{m-1} \lfloor (m/k)/\lceil m/k \rceil \rfloor \right)^{-1} \right\rfloor.$$

$$\int_0^\infty \frac{t - \mathrm{i}b}{t^2+b^2} e^{\mathrm{i}at} \,\mathrm{d}t = e^{ab} E_1(ab), \quad a, b > 0.$$

$$\boldsymbol{A} := \begin{pmatrix} x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda \end{pmatrix}.$$

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \begin{pmatrix} u & x \\ v & y \\ w & z \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

$$\boldsymbol{M} = \begin{array}{c} \\ C \\ I \\ C' \end{array} \!\!\begin{array}{c} \begin{array}{ccc} C & I & C' \end{array} \\ \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{array}$$

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{converges if} \quad |z| < \left(\limsup_{n\to\infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x+\Delta x)-f(x)}{\Delta x} \to f'(x) \qquad \text{as } \Delta x \to 0.$$

$$\|u_i\| = 1, \qquad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of} \quad \left\{ \begin{array}{c} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{array} \right\} \quad \text{is} \quad \left\{ \begin{array}{c} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{array} \right\}.$$

$$\begin{aligned} T(n) \;\le\; T(2^{\lceil \lg n \rceil}) &\le c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\ &< 3c \cdot 3^{\lg n} \\ &= 3c\, n^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2 - xy + yx - y^2 \\ &= x^2 - y^2 \\ (x+y)^2 &= x^2 + 2xy + y^2. \end{aligned}$$

$$\begin{aligned} \left(\int\limits_{-\infty}^{\infty} \mathrm{e}^{-x^2}\, \mathrm{d}x \right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)}\, \mathrm{d}x\, \mathrm{d}y \\ &= \int_0^{2\pi} \int_0^{\infty} \mathrm{e}^{-r^2}\, \mathrm{d}r\, \mathrm{d}\theta \\ &= \int_0^{2\pi} \left(\mathrm{e}^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) \mathrm{d}\theta \\ &= \pi. \end{aligned}$$

$$\prod_{k\ge 0} \frac{1}{(1-q^k z)} = \sum_{n\ge 0} z^n \Bigg/ \prod_{1\le k\le n} (1-q^k).$$

$$\sum_{\substack{0<i\le m \\ 0<j\le n}} p(i,j) \neq \sum_{i=1}^{p} \sum_{j=1}^{q} \sum_{k=1}^{r} a_{ij} b_{jk} c_{ki} \neq \sum_{\substack{1\le i\le p \\ 1\le j\le q \\ 1\le k\le r}} a_{ij} b_{jk} c_{ki}$$

$$\max_{1\le n\le m} \log_2 P_n \quad \text{and} \quad \lim_{x\to 0} \frac{\sin x}{x} = 1$$

Inline math: $\max_{1\le n\le m} \log_2 P_n$ and $\lim_{x\to 0} \frac{\sin x}{x} = 1$

$$p_1(n) = \lim_{m\to\infty} \sum_{\nu=0}^{\infty} \big(1 - \cos^{2m}(\nu!^n \pi/n)\big)$$

Inline math: $p_1(n) = \lim_{m\to\infty} \sum_{\nu=0}^{\infty} \big(1 - \cos^{2m}(\nu!^n \pi/n)\big)$

3.2 Math Test Serif Bold

3.2.1 Overview Serif Bold

Default: $\alpha a\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma;\ \sigma_\epsilon, c^\alpha$
mathnormal: $\alpha a\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathup: aααbβGΓΓϵεθϑPΠΣσ
mathit: $\mathit{a\alpha\alpha b\beta G``\epsilon\varepsilon\theta\vartheta P˝˚\sigma}$
mathbf: $\boldsymbol{\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{\alpha b\beta G``\epsilon\varepsilon\theta\vartheta P˝˚\sigma}$
mathbfup: $\mathbf{a\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$

Default: $\alpha a\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma;\ \sigma_\epsilon, c^\alpha$
mathnormal: $\alpha a\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathup: aααbβGΓΓϵεθϑPΠΣσ
mathit: $\mathit{a\alpha\alpha b\beta G``\epsilon\varepsilon\theta\vartheta P˝˚\sigma}$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G``\epsilon\varepsilon\theta\vartheta P˝˚\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$

Default: $\alpha a\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma;\ \sigma_\varepsilon, c^\alpha$
mathnormal: $\alpha a\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathup: aααbβGΓΓεεθθPΠΣσ
mathit: $\mathit{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

Default: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma;\ \sigma_\varepsilon, c^\alpha}$
mathnormal: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathrm: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\epsilon\varepsilon\theta\vartheta P\Pi\Sigma\sigma}$
mathup: **aααbβGΓΓεεθθPΠΣσ**
mathit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

3.2.2 Formulas Serif Bold

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$
$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,$
$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,$
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{\nu \to \infty} \nu(\nu) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$
$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x = \binom{27}{2}$$

Disambiguation: 0 O *O*, 1 l I | *l I* /, *i j*, *rn m*, θ Θ, ϕ ψ, – —
Latin vs. Greek: $a\ \alpha$, $d\ \delta$, $e\ \epsilon$, $i\ \iota$, $k\ \kappa$, $n\ \eta$, $o\ \sigma$, $p\ \rho$, ß β, $u\ \upsilon$, $v\ \nu$, $w\ \omega$, $x\ \chi$, $y\ \gamma$, $A\ \Delta\ \Lambda$, $O\ \Theta\ \Omega$, $T\ \Gamma$, $Y\ \Upsilon$.
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{\nu \to \infty} \nu(\nu) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$
$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x = \binom{27}{2}$$

Disambiguation: 0 O *O*, 1 l I | *l I* /, *i j*, *rn m*, θ Θ, ϕ ψ, – —
Latin vs. Greek: $a\ \alpha$, $d\ \delta$, $e\ \epsilon$, $i\ \iota$, $k\ \kappa$, $n\ \eta$, $o\ \sigma$, $p\ \rho$, ß β, $u\ \upsilon$, $v\ \nu$, $w\ \omega$, $x\ \chi$, $y\ \gamma$, $A\ \Delta\ \Lambda$, $O\ \Theta\ \Omega$, $T\ \Gamma$, $Y\ \Upsilon$.
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{v \to \infty} v(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$
$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x = \binom{27}{2}$$

Disambiguation: 0 O *O*, 1 l I | *l I* /, *i j*, *rn m*, θ Θ, ϕ ψ, – –

Latin vs. Greek: $a\ \alpha, d\ \delta, e\ \varepsilon, i\ \iota, k\ \kappa, n\ \eta, o\ \sigma, p\ \rho$, ß $\beta, u\ \upsilon, v\ \nu, w\ \omega, x\ \chi, y\ \gamma, A\ \Delta\ \Lambda, O\ \Theta\ \Omega, T\ \Gamma, Y\ \Upsilon$.
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{\nu \to \infty} \nu(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$
$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \left(\frac{27}{2}\right)$$

Disambiguation: 0 O O, 1 l I | l I $/$, i j, rn m, θ Θ, ϕ ψ, − –
Latin vs. Greek: $a\ \alpha, d\ \delta, e\ \varepsilon, i\ \iota, k\ \kappa, n\ \eta, o\ \sigma, p\ \rho$, ß $\beta, u\ \upsilon, v\ \nu, w\ \omega, x\ \chi, y\ \gamma, A\ \Delta\ \Lambda, O\ \Theta\ \Omega, T\ \Gamma, Y\ \Upsilon$.

3.2.3 Math Alphabets Serif Bold

Default

$$0,1,2,3,4,5,6,7,8,9,$$
$$A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z,$$
$$a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z,$$
$$A,B,\Gamma,\Delta,E,Z,H,\Theta,I,K,\Lambda,M,N,\Xi,O,\Pi,P,\Sigma,T,\Upsilon,\Phi,X,\Psi,\Omega,$$
$$\alpha,\beta,\gamma,\delta,\epsilon,\zeta,\eta,\theta,\iota,\kappa,\lambda,\mu,\nu,\xi,o,\pi,\rho,\sigma,\tau,\upsilon,\phi,\chi,\psi,\omega,\varepsilon,\vartheta,\varpi,\varrho,\varsigma,\varphi,$$

Math Normal (`\mathnormal`)

$$0,1,2,3,4,5,6,7,8,9,$$
$$A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z,$$
$$a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z,$$
$$A,B,\Gamma,\Delta,E,Z,H,\Theta,I,K,\Lambda,M,N,\Xi,O,\Pi,P,\Sigma,T,\Upsilon,\Phi,X,\Psi,\Omega,$$
$$\alpha,\beta,\gamma,\delta,\epsilon,\zeta,\eta,\theta,\iota,\kappa,\lambda,\mu,\nu,\xi,o,\pi,\rho,\sigma,\tau,\upsilon,\phi,\chi,\psi,\omega,\varepsilon,\vartheta,\varpi,\varrho,\varsigma,\varphi,$$

Math Italic (`\mathit`)

$$0,1,2,3,4,5,6,7,8,9,$$
$$A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z,$$
$$a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z,$$
A, B, `, ´, E, Z, H, ˆ, I, K, ˜, M, N, ¨, O, ˝, P, ˚, T, ˇ, ˘, X, ¯, ˙,
$$\alpha,\beta,\gamma,\delta,\epsilon,\zeta,\eta,\theta,\iota,\kappa,\lambda,\mu,\nu,\xi,o,\pi,\rho,\sigma,\tau,\upsilon,\phi,\chi,\psi,\omega,\varepsilon,\vartheta,\varpi,\varrho,\varsigma,\varphi,$$

Math Roman (`\mathrm`)

$$\mathrm{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$$
$$\mathrm{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$$
$$\mathrm{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$$
$$\mathrm{A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,}$$
$$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Bold (`\mathbf`)

$$\mathbf{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$$
$$\mathbf{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$$
$$\mathbf{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$$
$$\mathbf{A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,}$$
$$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Caligraphic (`\mathcal`)

$$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{K}, \mathcal{L}, \mathcal{M}, \mathcal{N}, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{S}, \mathcal{T}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z},$$

Script (`\mathscr`)

$$\mathscr{A}, \mathscr{B}, \mathscr{C}, \mathscr{D}, \mathscr{E}, \mathscr{F}, \mathscr{G}, \mathscr{H}, \mathscr{I}, \mathscr{J}, \mathscr{K}, \mathscr{L}, \mathscr{M}, \mathscr{N}, \mathscr{O}, \mathscr{P}, \mathscr{Q}, \mathscr{R}, \mathscr{S}, \mathscr{T}, \mathscr{U}, \mathscr{V}, \mathscr{W}, \mathscr{X}, \mathscr{Y}, \mathscr{Z},$$

Fraktur (`\mathfrak`)

$$\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{I}, \mathfrak{J}, \mathfrak{K}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N}, \mathfrak{O}, \mathfrak{P}, \mathfrak{Q}, \mathfrak{R}, \mathfrak{S}, \mathfrak{T}, \mathfrak{U}, \mathfrak{V}, \mathfrak{W}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z},$$
$$\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{e}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n}, \mathfrak{o}, \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}, \mathfrak{t}, \mathfrak{u}, \mathfrak{v}, \mathfrak{w}, \mathfrak{x}, \mathfrak{y}, \mathfrak{z},$$

Blackboard Bold (`\mathbb`)

$$\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}, \mathbb{E}, \mathbb{F}, \mathbb{G}, \mathbb{H}, \mathbb{I}, \mathbb{J}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \mathbb{O}, \mathbb{P}, \mathbb{Q}, \mathbb{R}, \mathbb{S}, \mathbb{T}, \mathbb{U}, \mathbb{V}, \mathbb{W}, \mathbb{X}, \mathbb{Y}, \mathbb{Z},$$

3.2.4 Character Sidebearings Serif Bold

Default

$$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$$
$$|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$$
$$|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$$
$$|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$$
$$|A| + |B| + |\mathit{\Gamma}| + |\mathit{\Delta}| + |E| + |Z| + |H| + |\mathit{\Theta}| + |I| + |K| + |\mathit{\Lambda}| + |M| +$$
$$|N| + |\mathit{\Xi}| + |O| + |\mathit{\Pi}| + |P| + |\mathit{\Sigma}| + |T| + |\mathit{\Upsilon}| + |\mathit{\Phi}| + |X| + |\mathit{\Psi}| + |\mathit{\Omega}| +$$
$$|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| +$$
$$|\nu| + |\xi| + |o| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| +$$
$$|\varepsilon| + |\vartheta| + |\varpi| + |\varrho| + |\varsigma| + |\varphi| +$$

Math Roman (`\mathrm`)

$$|\mathrm{A}| + |\mathrm{B}| + |\mathrm{C}| + |\mathrm{D}| + |\mathrm{E}| + |\mathrm{F}| + |\mathrm{G}| + |\mathrm{H}| + |\mathrm{I}| + |\mathrm{J}| + |\mathrm{K}| + |\mathrm{L}| + |\mathrm{M}| +$$
$$|\mathrm{N}| + |\mathrm{O}| + |\mathrm{P}| + |\mathrm{Q}| + |\mathrm{R}| + |\mathrm{S}| + |\mathrm{T}| + |\mathrm{U}| + |\mathrm{V}| + |\mathrm{W}| + |\mathrm{X}| + |\mathrm{Y}| + |\mathrm{Z}| +$$
$$|\mathrm{a}| + |\mathrm{b}| + |\mathrm{c}| + |\mathrm{d}| + |\mathrm{e}| + |\mathrm{f}| + |\mathrm{g}| + |\mathrm{h}| + |\mathrm{i}| + |\mathrm{j}| + |\mathrm{k}| + |\mathrm{l}| + |\mathrm{m}| +$$
$$|\mathrm{n}| + |\mathrm{o}| + |\mathrm{p}| + |\mathrm{q}| + |\mathrm{r}| + |\mathrm{s}| + |\mathrm{t}| + |\mathrm{u}| + |\mathrm{v}| + |\mathrm{w}| + |\mathrm{x}| + |\mathrm{y}| + |\mathrm{z}| +$$
$$|\mathrm{A}| + |\mathrm{B}| + |\Gamma| + |\Delta| + |\mathrm{E}| + |\mathrm{Z}| + |\mathrm{H}| + |\Theta| + |\mathrm{I}| + |\mathrm{K}| + |\Lambda| + |\mathrm{M}| +$$
$$|\mathrm{N}| + |\Xi| + |\mathrm{O}| + |\Pi| + |\mathrm{P}| + |\Sigma| + |\mathrm{T}| + |\Upsilon| + |\Phi| + |\mathrm{X}| + |\Psi| + |\Omega| +$$

Math Bold (`\mathbf`)

$$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$$
$$|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$$
$$|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$$
$$|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$$
$$|A| + |B| + |\mathit{\Gamma}| + |\mathit{\Delta}| + |E| + |Z| + |H| + |\mathit{\Theta}| + |I| + |K| + |\mathit{\Lambda}| + |M| +$$
$$|N| + |\mathit{\Xi}| + |O| + |\mathit{\Pi}| + |P| + |\mathit{\Sigma}| + |T| + |\mathit{\Upsilon}| + |\mathit{\Phi}| + |X| + |\mathit{\Psi}| + |\mathit{\Omega}| +$$

Math Calligraphic (`\mathcal`)

$$|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| +$$
$$|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +$$

3.2.5 Superscript Positioning Serif Bold

Default

$$\begin{aligned}
&A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 + \\
&N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 + \\
&a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 + \\
&n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 + \\
&A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 + \\
&N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 + \\
&\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \epsilon^2 + \zeta^2 + \eta^2 + \theta^2 + \iota^2 + \kappa^2 + \lambda^2 + \mu^2 + \\
&\nu^2 + \xi^2 + o^2 + \pi^2 + \rho^2 + \sigma^2 + \tau^2 + \upsilon^2 + \phi^2 + \chi^2 + \psi^2 + \omega^2 + \\
&\varepsilon^2 + \vartheta^2 + \varpi^2 + \varrho^2 + \varsigma^2 + \varphi^2 +
\end{aligned}$$

Math Roman (`\mathrm`)

$$\begin{aligned}
&\mathrm{A}^2 + \mathrm{B}^2 + \mathrm{C}^2 + \mathrm{D}^2 + \mathrm{E}^2 + \mathrm{F}^2 + \mathrm{G}^2 + \mathrm{H}^2 + \mathrm{I}^2 + \mathrm{J}^2 + \mathrm{K}^2 + \mathrm{L}^2 + \mathrm{M}^2 + \\
&\mathrm{N}^2 + \mathrm{O}^2 + \mathrm{P}^2 + \mathrm{Q}^2 + \mathrm{R}^2 + \mathrm{S}^2 + \mathrm{T}^2 + \mathrm{U}^2 + \mathrm{V}^2 + \mathrm{W}^2 + \mathrm{X}^2 + \mathrm{Y}^2 + \mathrm{Z}^2 + \\
&\mathrm{a}^2 + \mathrm{b}^2 + \mathrm{c}^2 + \mathrm{d}^2 + \mathrm{e}^2 + \mathrm{f}^2 + \mathrm{g}^2 + \mathrm{h}^2 + \mathrm{i}^2 + \mathrm{j}^2 + \mathrm{k}^2 + \mathrm{l}^2 + \mathrm{m}^2 + \\
&\mathrm{n}^2 + \mathrm{o}^2 + \mathrm{p}^2 + \mathrm{q}^2 + \mathrm{r}^2 + \mathrm{s}^2 + \mathrm{t}^2 + \mathrm{u}^2 + \mathrm{v}^2 + \mathrm{w}^2 + \mathrm{x}^2 + \mathrm{y}^2 + \mathrm{z}^2 + \\
&\mathrm{A}^2 + \mathrm{B}^2 + \mathrm{\Gamma}^2 + \mathrm{\Delta}^2 + \mathrm{E}^2 + \mathrm{Z}^2 + \mathrm{H}^2 + \mathrm{\Theta}^2 + \mathrm{I}^2 + \mathrm{K}^2 + \mathrm{\Lambda}^2 + \mathrm{M}^2 + \\
&\mathrm{N}^2 + \mathrm{\Xi}^2 + \mathrm{O}^2 + \mathrm{\Pi}^2 + \mathrm{P}^2 + \mathrm{\Sigma}^2 + \mathrm{T}^2 + \mathrm{\Upsilon}^2 + \mathrm{\Phi}^2 + \mathrm{X}^2 + \mathrm{\Psi}^2 + \mathrm{\Omega}^2 +
\end{aligned}$$

Math Bold (`\mathbf`)

$$\begin{aligned}
&\mathbf{A}^2 + \mathbf{B}^2 + \mathbf{C}^2 + \mathbf{D}^2 + \mathbf{E}^2 + \mathbf{F}^2 + \mathbf{G}^2 + \mathbf{H}^2 + \mathbf{I}^2 + \mathbf{J}^2 + \mathbf{K}^2 + \mathbf{L}^2 + \mathbf{M}^2 + \\
&\mathbf{N}^2 + \mathbf{O}^2 + \mathbf{P}^2 + \mathbf{Q}^2 + \mathbf{R}^2 + \mathbf{S}^2 + \mathbf{T}^2 + \mathbf{U}^2 + \mathbf{V}^2 + \mathbf{W}^2 + \mathbf{X}^2 + \mathbf{Y}^2 + \mathbf{Z}^2 + \\
&\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2 + \mathbf{d}^2 + \mathbf{e}^2 + \mathbf{f}^2 + \mathbf{g}^2 + \mathbf{h}^2 + \mathbf{i}^2 + \mathbf{j}^2 + \mathbf{k}^2 + \mathbf{l}^2 + \mathbf{m}^2 + \\
&\mathbf{n}^2 + \mathbf{o}^2 + \mathbf{p}^2 + \mathbf{q}^2 + \mathbf{r}^2 + \mathbf{s}^2 + \mathbf{t}^2 + \mathbf{u}^2 + \mathbf{v}^2 + \mathbf{w}^2 + \mathbf{x}^2 + \mathbf{y}^2 + \mathbf{z}^2 + \\
&\mathbf{A}^2 + \mathbf{B}^2 + \mathbf{\Gamma}^2 + \mathbf{\Delta}^2 + \mathbf{E}^2 + \mathbf{Z}^2 + \mathbf{H}^2 + \mathbf{\Theta}^2 + \mathbf{I}^2 + \mathbf{K}^2 + \mathbf{\Lambda}^2 + \mathbf{M}^2 + \\
&\mathbf{N}^2 + \mathbf{\Xi}^2 + \mathbf{O}^2 + \mathbf{\Pi}^2 + \mathbf{P}^2 + \mathbf{\Sigma}^2 + \mathbf{T}^2 + \mathbf{\Upsilon}^2 + \mathbf{\Phi}^2 + \mathbf{X}^2 + \mathbf{\Psi}^2 + \mathbf{\Omega}^2 +
\end{aligned}$$

Math Calligraphic (`\mathcal`)

$$\begin{aligned}
&\mathcal{A}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{D}^2 + \mathcal{E}^2 + \mathcal{F}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{L}^2 + \mathcal{M}^2 + \\
&\mathcal{N}^2 + \mathcal{O}^2 + \mathcal{P}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{S}^2 + \mathcal{T}^2 + \mathcal{U}^2 + \mathcal{V}^2 + \mathcal{W}^2 + \mathcal{X}^2 + \mathcal{Y}^2 + \mathcal{Z}^2 +
\end{aligned}$$

3.2.6 Subscript Positioning Serif Bold

Default

$$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$$
$$N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$$
$$a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$$
$$n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$$
$$A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +$$
$$N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i +$$
$$\alpha_i + \beta_i + \gamma_i + \delta_i + \epsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i +$$
$$\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + \upsilon_i + \phi_i + \chi_i + \psi_i + \omega_i +$$
$$\varepsilon_i + \vartheta_i + \varpi_i + \varrho_i + \varsigma_i + \varphi_i +$$

Math Roman (`\mathrm`)

$$\mathrm{A}_i + \mathrm{B}_i + \mathrm{C}_i + \mathrm{D}_i + \mathrm{E}_i + \mathrm{F}_i + \mathrm{G}_i + \mathrm{H}_i + \mathrm{I}_i + \mathrm{J}_i + \mathrm{K}_i + \mathrm{L}_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \mathrm{O}_i + \mathrm{P}_i + \mathrm{Q}_i + \mathrm{R}_i + \mathrm{S}_i + \mathrm{T}_i + \mathrm{U}_i + \mathrm{V}_i + \mathrm{W}_i + \mathrm{X}_i + \mathrm{Y}_i + \mathrm{Z}_i +$$
$$\mathrm{a}_i + \mathrm{b}_i + \mathrm{c}_i + \mathrm{d}_i + \mathrm{e}_i + \mathrm{f}_i + \mathrm{g}_i + \mathrm{h}_i + \mathrm{i}_i + \mathrm{j}_i + \mathrm{k}_i + \mathrm{l}_i + \mathrm{m}_i +$$
$$\mathrm{n}_i + \mathrm{o}_i + \mathrm{p}_i + \mathrm{q}_i + \mathrm{r}_i + \mathrm{s}_i + \mathrm{t}_i + \mathrm{u}_i + \mathrm{v}_i + \mathrm{w}_i + \mathrm{x}_i + \mathrm{y}_i + \mathrm{z}_i +$$
$$\mathrm{A}_i + \mathrm{B}_i + \mathrm{\Gamma}_i + \mathrm{\Delta}_i + \mathrm{E}_i + \mathrm{Z}_i + \mathrm{H}_i + \mathrm{\Theta}_i + \mathrm{I}_i + \mathrm{K}_i + \mathrm{\Lambda}_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \mathrm{\Xi}_i + \mathrm{O}_i + \mathrm{\Pi}_i + \mathrm{P}_i + \mathrm{\Sigma}_i + \mathrm{T}_i + \mathrm{\Upsilon}_i + \mathrm{\Phi}_i + \mathrm{X}_i + \mathrm{\Psi}_i + \mathrm{\Omega}_i +$$

Math Bold (`\mathbf`)

$$\mathbf{A}_i + \mathbf{B}_i + \mathbf{C}_i + \mathbf{D}_i + \mathbf{E}_i + \mathbf{F}_i + \mathbf{G}_i + \mathbf{H}_i + \mathbf{I}_i + \mathbf{J}_i + \mathbf{K}_i + \mathbf{L}_i + \mathbf{M}_i +$$
$$\mathbf{N}_i + \mathbf{O}_i + \mathbf{P}_i + \mathbf{Q}_i + \mathbf{R}_i + \mathbf{S}_i + \mathbf{T}_i + \mathbf{U}_i + \mathbf{V}_i + \mathbf{W}_i + \mathbf{X}_i + \mathbf{Y}_i + \mathbf{Z}_i +$$
$$\mathbf{a}_i + \mathbf{b}_i + \mathbf{c}_i + \mathbf{d}_i + \mathbf{e}_i + \mathbf{f}_i + \mathbf{g}_i + \mathbf{h}_i + \mathbf{i}_i + \mathbf{j}_i + \mathbf{k}_i + \mathbf{l}_i + \mathbf{m}_i +$$
$$\mathbf{n}_i + \mathbf{o}_i + \mathbf{p}_i + \mathbf{q}_i + \mathbf{r}_i + \mathbf{s}_i + \mathbf{t}_i + \mathbf{u}_i + \mathbf{v}_i + \mathbf{w}_i + \mathbf{x}_i + \mathbf{y}_i + \mathbf{z}_i +$$
$$\mathbf{A}_i + \mathbf{B}_i + \mathbf{\Gamma}_i + \mathbf{\Delta}_i + \mathbf{E}_i + \mathbf{Z}_i + \mathbf{H}_i + \mathbf{\Theta}_i + \mathbf{I}_i + \mathbf{K}_i + \mathbf{\Lambda}_i + \mathbf{M}_i +$$
$$\mathbf{N}_i + \mathbf{\Xi}_i + \mathbf{O}_i + \mathbf{\Pi}_i + \mathbf{P}_i + \mathbf{\Sigma}_i + \mathbf{T}_i + \mathbf{\Upsilon}_i + \mathbf{\Phi}_i + \mathbf{X}_i + \mathbf{\Psi}_i + \mathbf{\Omega}_i +$$

Math Calligraphic (`\mathcal`)

$$\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i +$$
$$\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +$$

3.2.7 Accent Positioning Serif Bold

Default

$$
\begin{gathered}
\hat{0}+\hat{1}+\hat{2}+\hat{3}+\hat{4}+\hat{5}+\hat{6}+\hat{7}+\hat{8}+\hat{9}+\\
\hat{A}+\hat{B}+\hat{C}+\hat{D}+\hat{E}+\hat{F}+\hat{G}+\hat{H}+\hat{I}+\hat{J}+\hat{K}+\hat{L}+\hat{M}+\\
\hat{N}+\hat{O}+\hat{P}+\hat{Q}+\hat{R}+\hat{S}+\hat{T}+\hat{U}+\hat{V}+\hat{W}+\hat{X}+\hat{Y}+\hat{Z}+\\
\hat{a}+\hat{b}+\hat{c}+\hat{d}+\hat{e}+\hat{f}+\hat{g}+\hat{h}+\hat{i}+\hat{j}+\hat{k}+\hat{l}+\hat{m}+\\
\hat{n}+\hat{o}+\hat{p}+\hat{q}+\hat{r}+\hat{s}+\hat{t}+\hat{u}+\hat{v}+\hat{w}+\hat{x}+\hat{y}+\hat{z}+\\
\hat{A}+\hat{B}+\hat{\mathit{\Gamma}}+\hat{\mathit{\Delta}}+\hat{E}+\hat{Z}+\hat{H}+\hat{\mathit{\Theta}}+\hat{I}+\hat{K}+\hat{\mathit{\Lambda}}+\hat{M}+\\
\hat{N}+\hat{\mathit{\Xi}}+\hat{O}+\hat{\mathit{\Pi}}+\hat{P}+\hat{\mathit{\Sigma}}+\hat{T}+\hat{\mathit{\Upsilon}}+\hat{\mathit{\Phi}}+\hat{X}+\hat{\mathit{\Psi}}+\hat{\mathit{\Omega}}+\\
\hat{\alpha}+\hat{\beta}+\hat{\gamma}+\hat{\delta}+\hat{\epsilon}+\hat{\zeta}+\hat{\eta}+\hat{\theta}+\hat{\iota}+\hat{\kappa}+\hat{\lambda}+\hat{\mu}+\\
\hat{\nu}+\hat{\xi}+\hat{o}+\hat{\pi}+\hat{\rho}+\hat{\sigma}+\hat{\tau}+\hat{\upsilon}+\hat{\phi}+\hat{\chi}+\hat{\psi}+\hat{\omega}+\\
\hat{\varepsilon}+\hat{\vartheta}+\hat{\varpi}+\hat{\varrho}+\hat{\varsigma}+\hat{\varphi}+
\end{gathered}
$$

Math Italic (`\mathit`)

$$
\begin{gathered}
\hat{\mathit{0}}+\hat{\mathit{1}}+\hat{\mathit{2}}+\hat{\mathit{3}}+\hat{\mathit{4}}+\hat{\mathit{5}}+\hat{\mathit{6}}+\hat{\mathit{7}}+\hat{\mathit{8}}+\hat{\mathit{9}}+\\
\hat{A}+\hat{B}+\hat{C}+\hat{D}+\hat{E}+\hat{F}+\hat{G}+\hat{H}+\hat{I}+\hat{J}+\hat{K}+\hat{L}+\hat{M}+\\
\hat{N}+\hat{O}+\hat{P}+\hat{Q}+\hat{R}+\hat{S}+\hat{T}+\hat{U}+\hat{V}+\hat{W}+\hat{X}+\hat{Y}+\hat{Z}+\\
\hat{a}+\hat{b}+\hat{c}+\hat{d}+\hat{e}+\hat{f}+\hat{g}+\hat{h}+\hat{i}+\hat{j}+\hat{k}+\hat{l}+\hat{m}+\hat{\ell}+\hat{\wp}+\hat{\imath}+\hat{\jmath}+\tilde{i}\\
\hat{n}+\hat{o}+\hat{p}+\hat{q}+\hat{r}+\hat{s}+\hat{t}+\hat{u}+\hat{v}+\hat{w}+\hat{x}+\hat{y}+\hat{z}+\\
\hat{A}+\hat{B}+\hat{}+\hat{}+\hat{E}+\hat{Z}+\hat{H}+\hat{}+\hat{I}+\hat{K}+\hat{}+\hat{M}+\\
\hat{N}+\hat{}+\hat{O}+\hat{}+\hat{P}+\hat{}+\hat{T}+\hat{}+\hat{}+\hat{X}+\hat{}+\hat{}+\\
\hat{\alpha}+\hat{\beta}+\hat{\gamma}+\hat{\delta}+\hat{\epsilon}+\hat{\zeta}+\hat{\eta}+\hat{\theta}+\hat{\iota}+\hat{\kappa}+\hat{\lambda}+\hat{\mu}+\\
\hat{\nu}+\hat{\xi}+\hat{o}+\hat{\pi}+\hat{\rho}+\hat{\sigma}+\hat{\tau}+\hat{\upsilon}+\hat{\phi}+\hat{\chi}+\hat{\psi}+\hat{\omega}+\\
\hat{\varepsilon}+\hat{\vartheta}+\hat{\varpi}+\hat{\varrho}+\hat{\varsigma}+\hat{\varphi}+
\end{gathered}
$$

Math Roman (`\mathrm`)

$$
\begin{gathered}
\hat{0}+\hat{1}+\hat{2}+\hat{3}+\hat{4}+\hat{5}+\hat{6}+\hat{7}+\hat{8}+\hat{9}+\\
\hat{\mathrm{A}}+\hat{\mathrm{B}}+\hat{\mathrm{C}}+\hat{\mathrm{D}}+\hat{\mathrm{E}}+\hat{\mathrm{F}}+\hat{\mathrm{G}}+\hat{\mathrm{H}}+\hat{\mathrm{I}}+\hat{\mathrm{J}}+\hat{\mathrm{K}}+\hat{\mathrm{L}}+\hat{\mathrm{M}}+\\
\hat{\mathrm{N}}+\hat{\mathrm{O}}+\hat{\mathrm{P}}+\hat{\mathrm{Q}}+\hat{\mathrm{R}}+\hat{\mathrm{S}}+\hat{\mathrm{T}}+\hat{\mathrm{U}}+\hat{\mathrm{V}}+\hat{\mathrm{W}}+\hat{\mathrm{X}}+\hat{\mathrm{Y}}+\hat{\mathrm{Z}}+\\
\hat{\mathrm{a}}+\hat{\mathrm{b}}+\hat{\mathrm{c}}+\hat{\mathrm{d}}+\hat{\mathrm{e}}+\hat{\mathrm{f}}+\hat{\mathrm{g}}+\hat{\mathrm{h}}+\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}+\hat{\mathrm{l}}+\hat{\mathrm{m}}+\\
\hat{\mathrm{n}}+\hat{\mathrm{o}}+\hat{\mathrm{p}}+\hat{\mathrm{q}}+\hat{\mathrm{r}}+\hat{\mathrm{s}}+\hat{\mathrm{t}}+\hat{\mathrm{u}}+\hat{\mathrm{v}}+\hat{\mathrm{w}}+\hat{\mathrm{x}}+\hat{\mathrm{y}}+\hat{\mathrm{z}}+\\
\hat{\mathrm{A}}+\hat{\mathrm{B}}+\hat{\Gamma}+\hat{\Delta}+\hat{\mathrm{E}}+\hat{\mathrm{Z}}+\hat{\mathrm{H}}+\hat{\Theta}+\hat{\mathrm{I}}+\hat{\mathrm{K}}+\hat{\Lambda}+\hat{\mathrm{M}}+\\
\hat{\mathrm{N}}+\hat{\Xi}+\hat{\mathrm{O}}+\hat{\Pi}+\hat{\mathrm{P}}+\hat{\Sigma}+\hat{\mathrm{T}}+\hat{\Upsilon}+\hat{\Phi}+\hat{\mathrm{X}}+\hat{\Psi}+\hat{\Omega}+
\end{gathered}
$$

Math Bold (\mathbf)

$$\hat{\mathbf{0}} + \hat{\mathbf{1}} + \hat{\mathbf{2}} + \hat{\mathbf{3}} + \hat{\mathbf{4}} + \hat{\mathbf{5}} + \hat{\mathbf{6}} + \hat{\mathbf{7}} + \hat{\mathbf{8}} + \hat{\mathbf{9}} +$$
$$\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$$
$$\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$$
$$\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$$
$$\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$$
$$\hat{A} + \hat{B} + \hat{\Gamma} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$$
$$\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$$

Math Calligraphic (\mathcal)

$$\hat{\mathcal{A}} + \hat{\mathcal{B}} + \hat{\mathcal{C}} + \hat{\mathcal{D}} + \hat{\mathcal{E}} + \hat{\mathcal{F}} + \hat{\mathcal{G}} + \hat{\mathcal{H}} + \hat{\mathcal{I}} + \hat{\mathcal{J}} + \hat{\mathcal{K}} + \hat{\mathcal{L}} + \hat{\mathcal{M}} +$$
$$\hat{\mathcal{N}} + \hat{\mathcal{O}} + \hat{\mathcal{P}} + \hat{\mathcal{Q}} + \hat{\mathcal{R}} + \hat{\mathcal{S}} + \hat{\mathcal{T}} + \hat{\mathcal{U}} + \hat{\mathcal{V}} + \hat{\mathcal{W}} + \hat{\mathcal{X}} + \hat{\mathcal{Y}} + \hat{\mathcal{Z}} +$$

3.2.8 Differentials Serif Bold

$$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$$
$$dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$$
$$da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$$
$$dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$$
$$dA + dB + d\Gamma + d\Delta + dE + dZ + dH + d\Theta + dI + dK + d\Lambda + dM +$$
$$dN + d\Xi + dO + d\Pi + dP + d\Sigma + dT + d\Upsilon + d\Phi + dX + d\Psi + d\Omega +$$
$$d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$$
$$d\nu + d\xi + do + d\pi + d\rho + d\sigma + d\tau + d\upsilon + d\phi + d\chi + d\psi + d\omega +$$
$$d\varepsilon + d\vartheta + d\varpi + d\varrho + d\varsigma + d\varphi +$$
$$d\mathrm{A} + d\mathrm{B} + d\Gamma + d\Delta + d\mathrm{E} + d\mathrm{Z} + d\mathrm{H} + d\Theta + d\mathrm{I} + d\mathrm{K} + d\Lambda + d\mathrm{M} +$$
$$d\mathrm{N} + d\Xi + d\mathrm{O} + d\Pi + d\mathrm{P} + d\Sigma + d\mathrm{T} + d\Upsilon + d\Phi + d\mathrm{X} + d\Psi + d\Omega +$$

$$\mathrm{d}A + \mathrm{d}B + \mathrm{d}C + \mathrm{d}D + \mathrm{d}E + \mathrm{d}F + \mathrm{d}G + \mathrm{d}H + \mathrm{d}I + \mathrm{d}J + \mathrm{d}K + \mathrm{d}L + \mathrm{d}M +$$
$$\mathrm{d}N + \mathrm{d}O + \mathrm{d}P + \mathrm{d}Q + \mathrm{d}R + \mathrm{d}S + \mathrm{d}T + \mathrm{d}U + \mathrm{d}V + \mathrm{d}W + \mathrm{d}X + \mathrm{d}Y + \mathrm{d}Z +$$
$$\mathrm{d}a + \mathrm{d}b + \mathrm{d}c + \mathrm{d}d + \mathrm{d}e + \mathrm{d}f + \mathrm{d}g + \mathrm{d}h + \mathrm{d}i + \mathrm{d}j + \mathrm{d}k + \mathrm{d}l + \mathrm{d}m +$$
$$\mathrm{d}n + \mathrm{d}o + \mathrm{d}p + \mathrm{d}q + \mathrm{d}r + \mathrm{d}s + \mathrm{d}t + \mathrm{d}u + \mathrm{d}v + \mathrm{d}w + \mathrm{d}x + \mathrm{d}y + \mathrm{d}z +$$
$$\mathrm{d}A + \mathrm{d}B + \mathrm{d}\varGamma + \mathrm{d}\varDelta + \mathrm{d}E + \mathrm{d}Z + \mathrm{d}H + \mathrm{d}\varTheta + \mathrm{d}I + \mathrm{d}K + \mathrm{d}\varLambda + \mathrm{d}M +$$
$$\mathrm{d}N + \mathrm{d}\varXi + \mathrm{d}O + \mathrm{d}\varPi + \mathrm{d}P + \mathrm{d}\varSigma + \mathrm{d}T + \mathrm{d}\varUpsilon + \mathrm{d}\varPhi + \mathrm{d}X + \mathrm{d}\varPsi + \mathrm{d}\varOmega +$$
$$\mathrm{d}\alpha + \mathrm{d}\beta + \mathrm{d}\gamma + \mathrm{d}\delta + \mathrm{d}\epsilon + \mathrm{d}\zeta + \mathrm{d}\eta + \mathrm{d}\theta + \mathrm{d}\iota + \mathrm{d}\kappa + \mathrm{d}\lambda + \mathrm{d}\mu +$$
$$\mathrm{d}\nu + \mathrm{d}\xi + \mathrm{d}o + \mathrm{d}\pi + \mathrm{d}\rho + \mathrm{d}\sigma + \mathrm{d}\tau + \mathrm{d}\upsilon + \mathrm{d}\phi + \mathrm{d}\chi + \mathrm{d}\psi + \mathrm{d}\omega +$$
$$\mathrm{d}\varepsilon + \mathrm{d}\vartheta + \mathrm{d}\varpi + \mathrm{d}\varrho + \mathrm{d}\varsigma + \mathrm{d}\varphi +$$
$$\mathrm{d}\mathrm{A} + \mathrm{d}\mathrm{B} + \mathrm{d}\Gamma + \mathrm{d}\Delta + \mathrm{d}\mathrm{E} + \mathrm{d}\mathrm{Z} + \mathrm{d}\mathrm{H} + \mathrm{d}\Theta + \mathrm{d}\mathrm{I} + \mathrm{d}\mathrm{K} + \mathrm{d}\Lambda + \mathrm{d}\mathrm{M} +$$
$$\mathrm{d}\mathrm{N} + \mathrm{d}\Xi + \mathrm{d}\mathrm{O} + \mathrm{d}\Pi + \mathrm{d}\mathrm{P} + \mathrm{d}\Sigma + \mathrm{d}\mathrm{T} + \mathrm{d}\Upsilon + \mathrm{d}\Phi + \mathrm{d}\mathrm{X} + \mathrm{d}\Psi + \mathrm{d}\Omega +$$

$$\partial A + \partial B + \partial C + \partial D + \partial E + \partial F + \partial G + \partial H + \partial I + \partial J + \partial K + \partial L + \partial M +$$
$$\partial N + \partial O + \partial P + \partial Q + \partial R + \partial S + \partial T + \partial U + \partial V + \partial W + \partial X + \partial Y + \partial Z +$$
$$\partial a + \partial b + \partial c + \partial d + \partial e + \partial f + \partial g + \partial h + \partial i + \partial j + \partial k + \partial l + \partial m +$$
$$\partial n + \partial o + \partial p + \partial q + \partial r + \partial s + \partial t + \partial u + \partial v + \partial w + \partial x + \partial y + \partial z +$$
$$\partial A + \partial B + \partial \varGamma + \partial \varDelta + \partial E + \partial Z + \partial H + \partial \varTheta + \partial I + \partial K + \partial \varLambda + \partial M +$$
$$\partial N + \partial \varXi + \partial O + \partial \varPi + \partial P + \partial \varSigma + \partial T + \partial \varUpsilon + \partial \varPhi + \partial X + \partial \varPsi + \partial \varOmega +$$
$$\partial \alpha + \partial \beta + \partial \gamma + \partial \delta + \partial \epsilon + \partial \zeta + \partial \eta + \partial \theta + \partial \iota + \partial \kappa + \partial \lambda + \partial \mu +$$
$$\partial \nu + \partial \xi + \partial o + \partial \pi + \partial \rho + \partial \sigma + \partial \tau + \partial \upsilon + \partial \phi + \partial \chi + \partial \psi + \partial \omega +$$
$$\partial \varepsilon + \partial \vartheta + \partial \varpi + \partial \varrho + \partial \varsigma + \partial \varphi +$$
$$\partial \mathrm{A} + \partial \mathrm{B} + \partial \Gamma + \partial \Delta + \partial \mathrm{E} + \partial \mathrm{Z} + \partial \mathrm{H} + \partial \Theta + \partial \mathrm{I} + \partial \mathrm{K} + \partial \Lambda + \partial \mathrm{M} +$$
$$\partial \mathrm{N} + \partial \Xi + \partial \mathrm{O} + \partial \Pi + \partial \mathrm{P} + \partial \Sigma + \partial \mathrm{T} + \partial \Upsilon + \partial \Phi + \partial \mathrm{X} + \partial \Psi + \partial \Omega +$$

3.2.9 Slash Kerning Serif Bold

$$1/A + 1/B + 1/C + 1/D + 1/E + 1/F + 1/G + 1/H + 1/I + 1/J + 1/K + 1/L + 1/M +$$
$$1/N + 1/O + 1/P + 1/Q + 1/R + 1/S + 1/T + 1/U + 1/V + 1/W + 1/X + 1/Y + 1/Z +$$
$$1/a + 1/b + 1/c + 1/d + 1/e + 1/f + 1/g + 1/h + 1/i + 1/j + 1/k + 1/l + 1/m +$$
$$1/n + 1/o + 1/p + 1/q + 1/r + 1/s + 1/t + 1/u + 1/v + 1/w + 1/x + 1/y + 1/z +$$
$$1/A + 1/B + 1/\varGamma + 1/\varDelta + 1/E + 1/Z + 1/H + 1/\varTheta + 1/I + 1/K + 1/\varLambda + 1/M +$$
$$1/N + 1/\varXi + 1/O + 1/\varPi + 1/P + 1/\varSigma + 1/T + 1/\varUpsilon + 1/\varPhi + 1/X + 1/\varPsi + 1/\varOmega +$$
$$1/\alpha + 1/\beta + 1/\gamma + 1/\delta + 1/\epsilon + 1/\zeta + 1/\eta + 1/\theta + 1/\iota + 1/\kappa + 1/\lambda + 1/\mu +$$
$$1/\nu + 1/\xi + 1/o + 1/\pi + 1/\rho + 1/\sigma + 1/\tau + 1/\upsilon + 1/\phi + 1/\chi + 1/\psi + 1/\omega +$$
$$1/\varepsilon + 1/\vartheta + 1/\varpi + 1/\varrho + 1/\varsigma + 1/\varphi +$$

$$A/2+B/2+C/2+D/2+E/2+F/2+G/2+H/2+I/2+J/2+K/2+L/2+M/2+ N/2+O/2+P/2+Q/2+R/2+S/2+T/2+U/2+V/2+W/2+X/2+Y/2+Z/2+ a/2+b/2+c/2+d/2+e/2+f/2+g/2+h/2+i/2+j/2+k/2+l/2+m/2+ n/2+o/2+p/2+q/2+r/2+s/2+t/2+u/2+v/2+w/2+x/2+y/2+z/2+ A/2+B/2+\Gamma/2+\Delta/2+E/2+Z/2+H/2+\Theta/2+I/2+K/2+\Lambda/2+M/2+ N/2+\Xi/2+O/2+\Pi/2+P/2+\Sigma/2+T/2+\Upsilon/2+\Phi/2+X/2+\Psi/2+\Omega/2+ \alpha/2+\beta/2+\gamma/2+\delta/2+\epsilon/2+\zeta/2+\eta/2+\theta/2+\iota/2+\kappa/2+\lambda/2+\mu/2+ \nu/2+\xi/2+o/2+\pi/2+\rho/2+\sigma/2+\tau/2+\upsilon/2+\phi/2+\chi/2+\psi/2+\omega/2+ \varepsilon/2+\vartheta/2+\varpi/2+\varrho/2+\varsigma/2+\varphi/2+$$

3.2.10 (Big) Operators Serif Bold

$\sum_{i=1}^n x^n$ $\prod_{i=1}^n x^n$ $\coprod_{i=1}^n x^n$ $\int_{i=1}^n x^n$ $\oint_{i=1}^n x^n$
$\bigotimes_{i=1}^n x^n$ $\bigoplus_{i=1}^n x^n$ $\bigodot_{i=1}^n x^n$ $\bigwedge_{i=1}^n x^n$ $\bigvee_{i=1}^n x^n$ $\biguplus_{i=1}^n x^n$ $\bigcup_{i=1}^n x^n$ $\bigcap_{i=1}^n x^n$ $\bigsqcup_{i=1}^n x^n$

$$\sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n$$

$$\bigotimes_{i=1}^n x^n \quad \bigoplus_{i=1}^n x^n \quad \bigodot_{i=1}^n x^n \quad \bigwedge_{i=1}^n x^n \quad \bigvee_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n$$

3.2.11 Radicals Serif Bold

$$\sqrt{x+y} \qquad \sqrt{x^2+y^2} \qquad \sqrt{x_i^2+y_j^2} \qquad \sqrt{\left(\frac{\cos x}{2}\right)} \qquad \sqrt{\left(\frac{\sin x}{2}\right)}$$

$$\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}$$

3.2.12 Over- and Underbraces Serif Bold

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.2.13 Normal and Wide Accents Serif Bold

$$\dot{x} \quad \ddot{x} \quad \vec{x} \quad \bar{x} \quad \overline{x} \quad \overline{xx} \quad \tilde{x} \quad \widetilde{x} \quad \widetilde{xx} \quad \widetilde{xxx} \quad \hat{x} \quad \widehat{x} \quad \widehat{xx} \quad \widehat{xxx}$$

$$\hat{x} \quad \check{x} \quad \tilde{x} \quad \acute{x} \quad \grave{x} \quad \dot{x} \quad \ddot{x} \quad \breve{x} \quad \bar{x} \quad \vec{x}$$

3.2.14 Long Arrows Serif Bold

$$\longleftrightarrow \quad \leftrightarrow \quad \longleftarrow \quad \longrightarrow \quad \longleftrightarrow \quad \Longleftrightarrow \quad \Leftrightarrow \quad \Longleftarrow \quad \Longrightarrow \quad \Longleftrightarrow$$

3.2.15 Left and Right Delimiters Serif Bold

$$-(f)--[f]--\lfloor f\rfloor--\lceil f\rceil--\langle f\rangle--\{f\}-$$

Using `\left` **and** `\right`.

$$-\left(f\right)--\left[f\right]--\left\lfloor f\right\rfloor--\left\lceil f\right\rceil--\left\langle f\right\rangle--\left\{f\right\}-$$

$$-\left)f\right(--\left]f\right[--\left/f\right/--\left\backslash f\right\backslash--\left/f\right\backslash--\left\backslash f\right/-$$

3.2.16 Big-g-g Delimiters Serif Bold

$$-\Biggl\lfloor\biggl\lfloor\Bigl\lfloor\bigl\lfloor\lfloor-\rfloor\bigr\rfloor\Bigr\rfloor\biggr\rfloor\Biggr\rfloor- \qquad -\Biggl(\biggl(\Bigl(\bigl((-)\bigr)\Bigr)\biggr)\Biggr)-$$

$$-\Biggl\lceil\biggl\lceil\Bigl\lceil\bigl\lceil\lceil-\rceil\bigr\rceil\Bigr\rceil\biggr\rceil\Biggr\rceil- \qquad -\Biggl\{\biggl\{\Bigl\{\bigl\{\{-\}\bigr\}\Bigr\}\biggr\}\Biggr\}-$$

$$-\Biggl[\biggl[\Bigl[\bigl[[-]\bigr]\Bigr]\biggr]\Biggr]- \qquad -\Biggl(\biggl(\Bigl(\bigl((-)\bigr)\Bigr)\biggr)\Biggr)-$$

$$-\Biggl\langle\biggl\langle\Bigl\langle\bigl\langle\langle-\rangle\bigr\rangle\Bigr\rangle\biggr\rangle\Biggr\rangle- \qquad -\Biggl\lmoustache\biggl\lmoustache\Bigl\lmoustache\bigl\lmoustache\lmoustache-\rmoustache\bigr\rmoustache\Bigr\rmoustache\biggr\rmoustache\Biggr\rmoustache-$$

$$-\Biggl\uparrow\biggl\uparrow\Bigl\uparrow\bigl\uparrow\uparrow-\downarrow\bigr\downarrow\Bigr\downarrow\biggr\downarrow\Biggr\downarrow- \qquad -\Biggl\Uparrow\biggl\Uparrow\Bigl\Uparrow\bigl\Uparrow\Uparrow-\Downarrow\bigr\Downarrow\Bigr\Downarrow\biggr\Downarrow\Biggr\Downarrow-$$

3.2.17 Binary Operators Serif Bold

$x \pm y$	\pm	$x \cap y$	\cap	$x \diamond y$	\diamond	$x \oplus y$	\oplus
$x \mp y$	\mp	$x \cup y$	\cup	$x \bigtriangleup y$	\bigtriangleup	$x \ominus y$	\ominus
$x \times y$	\times	$x \uplus y$	\uplus	$x \bigtriangledown y$	\bigtriangledown	$x \otimes y$	\otimes
$x \div y$	\div	$x \sqcap y$	\sqcap	$x \triangleleft y$	\triangleleft	$x \oslash y$	\oslash
$x \ast y$	\ast	$x \sqcup y$	\sqcup	$x \triangleright y$	\triangleright	$x \odot y$	\odot
$x \star y$	\star	$x \vee y$	\vee	$x \lhd y$	\lhd	$x \bigcirc y$	\bigcirc
$x \circ y$	\circ	$x \wedge y$	\wedge	$x \rhd y$	\rhd	$x \dagger y$	\dagger
$x \bullet y$	\bullet	$x \setminus y$	\setminus	$x \unlhd y$	\unlhd	$x \ddagger y$	\ddagger
$x \cdot y$	\cdot	$x \wr y$	\wr	$x \unrhd y$	\unrhd	$x \S y$	\S
$x + y$	+	$x - y$	-	$x \amalg y$	\amalg	$x \P y$	\P

3.2.18 Relations Serif Bold

$x \leq y$	\leq	$x \geq y$	\geq	$x \equiv y$	\equiv	$x \models y$	\models
$x \prec y$	\prec	$x \succ y$	\succ	$x \sim y$	\sim	$x \perp y$	\perp
$x \preceq y$	\preceq	$x \succeq y$	\succeq	$x \simeq y$	\simeq	$x \mid y$	\mid
$x \ll y$	\ll	$x \gg y$	\gg	$x \asymp y$	\asymp	$x \parallel y$	\parallel
$x \subset y$	\subset	$x \supset y$	\supset	$x \approx y$	\approx	$x \bowtie y$	\bowtie
$x \subseteq y$	\subseteq	$x \supseteq y$	\supseteq	$x \cong y$	\cong	$x \Join y$	\Join
$x \sqsubset y$	\sqsubset	$x \sqsupset y$	\sqsupset	$x \neq y$	\neq	$x \smile y$	\smile
$x \sqsubseteq y$	\sqsubseteq	$x \sqsupseteq y$	\sqsupseteq	$x \doteq y$	\doteq	$x \frown y$	\frown
$x \in y$	\in	$x \ni y$	\ni	$x \propto y$	\propto	$x = y$	=
$x \vdash y$	\vdash	$x \dashv y$	\dashv	$x < y$	<	$x > y$	>
$x : y$	:						

3.2.19 Punctuation Serif Bold

x, y	,	$x; y$	;	$x \colon y$	\colon	$x \ldotp y$	\ldotp	$x \cdotp y$	\cdotp

3.2.20 Arrows Serif Bold

$x \leftarrow y$	\leftarrow	$x \longleftarrow y$	\longleftarrow	$x \uparrow y$	\uparrow
$x \Leftarrow y$	\Leftarrow	$x \Longleftarrow y$	\Longleftarrow	$x \Uparrow y$	\Uparrow
$x \rightarrow y$	\rightarrow	$x \longrightarrow y$	\longrightarrow	$x \downarrow y$	\downarrow
$x \Rightarrow y$	\Rightarrow	$x \Longrightarrow y$	\Longrightarrow	$x \Downarrow y$	\Downarrow
$x \leftrightarrow y$	\leftrightarrow	$x \longleftrightarrow y$	\longleftrightarrow	$x \updownarrow y$	\updownarrow
$x \Leftrightarrow y$	\Leftrightarrow	$x \Longleftrightarrow y$	\Longleftrightarrow	$x \Updownarrow y$	\Updownarrow
$x \mapsto y$	\mapsto	$x \longmapsto y$	\longmapsto	$x \nearrow y$	\nearrow
$x \hookleftarrow y$	\hookleftarrow	$x \hookrightarrow y$	\hookrightarrow	$x \searrow y$	\searrow
$x \leftharpoonup y$	\leftharpoonup	$x \rightharpoonup y$	\rightharpoonup	$x \swarrow y$	\swarrow
$x \leftharpoondown y$	\leftharpoondown	$x \rightharpoondown y$	\rightharpoondown	$x \nwarrow y$	\nwarrow
$x \rightleftharpoons y$	\rightleftharpoons	$x \leadsto y$	\leadsto		

3.2.21 Miscellaneous Symbols Serif Bold

$x \ldots y$	\ldots	$x \cdots y$	\cdots	$x \vdots y$	\vdots	$x \ddots y$	\ddots
$x \aleph y$	\aleph	$x \prime y$	\prime	$x \forall y$	\forall	$x \infty y$	\infty
$x \hbar y$	\hbar	$x \emptyset y$	\emptyset	$x \exists y$	\exists	$x \Box y$	\Box
$x \imath y$	\imath	$x \nabla y$	\nabla	$x \neg y$	\neg	$x \Diamond y$	\Diamond
$x \jmath y$	\jmath	$x \surd y$	\surd	$x \flat y$	\flat	$x \triangle y$	\triangle
$x \ell y$	\ell	$x \top y$	\top	$x \natural y$	\natural	$x \clubsuit y$	\clubsuit
$x \wp y$	\wp	$x \bot y$	\bot	$x \sharp y$	\sharp	$x \diamondsuit y$	\diamondsuit
$x \Re y$	\Re	$x \| y$	\\|	$x \backslash y$	\backslash	$x \heartsuit y$	\heartsuit
$x \Im y$	\Im	$x \angle y$	\angle	$x \partial y$	\partial	$x \spadesuit y$	\spadesuit
$x \mho y$	\mho	$x.y$	.	$x\vert y$	|	$x!y$	!

3.2.22 Variable-Sized Operators Serif Bold

$x \sum y$	\sum	$x \bigcap y$	\bigcap	$x \bigodot y$	\bigodot
$x \prod y$	\prod	$x \bigcup y$	\bigcup	$x \bigotimes y$	\bigotimes
$x \coprod y$	\coprod	$x \bigsqcup y$	\bigsqcup	$x \bigoplus y$	\bigoplus
$x \int y$	\int	$x \bigvee y$	\bigvee	$x \biguplus y$	\biguplus
$x \oint y$	\oint	$x \bigwedge y$	\bigwedge		

3.2.23 Log-Like Operators Serif Bold

$x \arccos y$	$x \cos y$	$x \csc y$	$x \exp y$	$x \ker y$	$x \limsup y$	$x \min y$	$x \sinh y$
$x \arcsin y$	$x \cosh y$	$x \deg y$	$x \gcd y$	$x \lg y$	$x \ln y$	$x \Pr y$	$x \sup y$
$x \arctan y$	$x \cot y$	$x \det y$	$x \hom y$	$x \lim y$	$x \log y$	$x \sec y$	$x \tan y$
$x \arg y$	$x \coth y$	$x \dim y$	$x \inf y$	$x \liminf y$	$x \max y$	$x \sin y$	$x \tanh y$

3.2.24 Delimiters Serif Bold

$x(y$	(	$x)y$	)	$x\uparrow y$	\uparrow	$x\Uparrow y$	\Uparrow
$x[y$	[	$x]y$	]	$x\downarrow y$	\downarrow	$x\Downarrow y$	\Downarrow
$x\{y$	\{	$x\}y$	\}	$x\updownarrow y$	\updownarrow	$x\Updownarrow y$	\Updownarrow
$x\lfloor y$	\lfloor	$x\rfloor y$	\rfloor	$x\lceil y$	\lceil	$x\rceil y$	\rceil
$x\langle y$	\langle	$x\rangle y$	\rangle	x/y	/	$x\backslash y$	\backslash
$x\vert y$	|	$x\Vert y$	\\\|				

3.2.25 Large Delimiters Serif Bold

$\rmoustache$	\rmoustache	$\lmoustache$	\lmoustache	$\rgroup$	\rgroup	$\lgroup$	\lgroup
$\arrowvert$	\arrowvert	$\Arrowvert$	\Arrowvert	$\bracevert$	\bracevert		

3.2.26 Math Mode Accents Serif Bold

$\hat{a}$	\hat{a}	$\acute{a}$	\acute{a}	$\bar{a}$	\bar{a}	$\dot{a}$	\dot{a}	$\breve{a}$	\breve{a}
$\check{a}$	\check{a}	$\grave{a}$	\grave{a}	$\vec{a}$	\vec{a}	$\ddot{a}$	\ddot{a}	$\tilde{a}$	\tilde{a}

3.2.27 Miscellaneous Constructions Serif Bold

$\widetilde{abc}$	\widetilde{abc}	$\widehat{abc}$	\widehat{abc}
$\overleftarrow{abc}$	\overleftarrow{abc}	$\overrightarrow{abc}$	\overrightarrow{abc}
$\overline{abc}$	\overline{abc}	$\underline{abc}$	\underline{abc}
$\overbrace{abc}$	\overbrace{abc}	$\underbrace{abc}$	\underbrace{abc}
$\sqrt{abc}$	\sqrt{abc}	$\sqrt[n]{abc}$	\sqrt[n]{abc}
f'	f'	$\frac{abc}{xyz}$	\frac{abc}{xyz}

3.2.28 AMS Delimiters Serif Bold

$x\ulcorner y$	\ulcorner	$x\urcorner y$	\urcorner	$x\llcorner y$	\llcorner	$x\lrcorner y$	\lrcorner

3.2.29 AMS Arrows Serif Bold

$x \dashrightarrow y$	\dashrightarrow	$x \dashleftarrow y$	\dashleftarrow
$x \leftleftarrows y$	\leftleftarrows	$x \leftrightarrows y$	\leftrightarrows
$x \Lleftarrow y$	\Lleftarrow	$x \twoheadleftarrow y$	\twoheadleftarrow
$x \leftarrowtail y$	\leftarrowtail	$x \looparrowleft y$	\looparrowleft
$x \leftrightharpoons y$	\leftrightharpoons	$x \curvearrowleft y$	\curvearrowleft
$x \circlearrowleft y$	\circlearrowleft	$x \Lsh y$	\Lsh
$x \upuparrows y$	\upuparrows	$x \upharpoonleft y$	\upharpoonleft
$x \downharpoonleft y$	\downharpoonleft	$x \multimap y$	\multimap
$x \leftrightsquigarrow y$	\leftrightsquigarrow	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \twoheadrightarrow y$	\twoheadrightarrow
$x \rightarrowtail y$	\rightarrowtail	$x \looparrowright y$	\looparrowright
$x \rightleftharpoons y$	\rightleftharpoons	$x \curvearrowright y$	\curvearrowright
$x \circlearrowright y$	\circlearrowright	$x \Rsh y$	\Rsh
$x \downdownarrows y$	\downdownarrows	$x \upharpoonright y$	\upharpoonright
$x \downharpoonright y$	\downharpoonright	$x \rightsquigarrow y$	\rightsquigarrow

3.2.30 AMS Negated Arrows Serif Bold

$x \nleftarrow y$	\nleftarrow	$x \nrightarrow y$	\nrightarrow
$x \nLeftarrow y$	\nLeftarrow	$x \nRightarrow y$	\nRightarrow
$x \nleftrightarrow y$	\nleftrightarrow	$x \nLeftrightarrow y$	\nLeftrightarrow

3.2.31 AMS Greek Serif Bold

$x \digamma y$	\digamma	$x \varkappa y$	\varkappa

3.2.32 AMS Hebrew Serif Bold

$x \beth y$	\beth	$x \daleth y$	\daleth	$x \gimel y$	\gimel

3.2.33 AMS Miscellaneous Serif Bold

$x\hbar y$	`\hbar`	$x\hslash y$	`\hslash`
$x\vartriangle y$	`\vartriangle`	$x\triangledown y$	`\triangledown`
$x\square y$	`\square`	$x\lozenge y$	`\lozenge`
$x\circledS y$	`\circledS`	$x\angle y$	`\angle`
$x\measuredangle y$	`\measuredangle`	$x\nexists y$	`\nexists`
$x\mho y$	`\mho`	$x\Finv y$	`\Finv`^{u}
$x\Game y$	`\Game`^{u}	$x\Bbbk y$	`\Bbbk`^{u}
$x\backprime y$	`\backprime`	$x\varnothing y$	`\varnothing`
$x\blacktriangle y$	`\blacktriangle`	$x\blacktriangledown y$	`\blacktriangledown`
$x\blacksquare y$	`\blacksquare`	$x\blacklozenge y$	`\blacklozenge`
$x\bigstar y$	`\bigstar`	$x\sphericalangle y$	`\sphericalangle`
$x\complement y$	`\complement`	$x\eth y$	`\eth`
$x\diagup y$	`\diagup`^{u}	$x\diagdown y$	`\diagdown`^{u}

^{u} **Not defined in** `amssymb.sty`**, define using the** `\newsymbol` **command.**

3.2.34 AMS Binary Operators Serif Bold

$x\dotplus y$	`\dotplus`	$x\smallsetminus y$	`\smallsetminus`
$x\Cap y$	`\Cap`	$x\Cup y$	`\Cup`
$x\barwedge y$	`\barwedge`	$x\veebar y$	`\veebar`
$x\doublebarwedge y$	`\doublebarwedge`	$x\boxminus y$	`\boxminus`
$x\boxtimes y$	`\boxtimes`	$x\boxdot y$	`\boxdot`
$x\boxplus y$	`\boxplus`	$x\divideontimes y$	`\divideontimes`
$x\ltimes y$	`\ltimes`	$x\rtimes y$	`\rtimes`
$x\leftthreetimes y$	`\leftthreetimes`	$x\rightthreetimes y$	`\rightthreetimes`
$x\curlywedge y$	`\curlywedge`	$x\curlyvee y$	`\curlyvee`
$x\circleddash y$	`\circleddash`	$x\circledast y$	`\circledast`
$x\circledcirc y$	`\circledcirc`	$x\centerdot y$	`\centerdot`
$x\intercal y$	`\intercal`		

3.2.35 AMS Relations Serif Bold

$x \leqslant y$	`\leqslant`
$x \lesssim y$	`\lesssim`
$x \approxeq y$	`\approxeq`
$x \lll y$	`\lll`
$x \lesseqgtr y$	`\lesseqgtr`
$x \doteqdot y$	`\doteqdot`
$x \fallingdotseq y$	`\fallingdotseq`
$x \backsimeq y$	`\backsimeq`
$x \Subset y$	`\Subset`
$x \preccurlyeq y$	`\preccurlyeq`
$x \precsim y$	`\precsim`
$x \vartriangleleft y$	`\vartriangleleft`
$x \vDash y$	`\vDash`
$x \smallsmile y$	`\smallsmile`
$x \bumpeq y$	`\bumpeq`
$x \geqq y$	`\geqq`
$x \eqslantgtr y$	`\eqslantgtr`
$x \gtrapprox y$	`\gtrapprox`
$x \ggg y$	`\ggg`
$x \gtreqless y$	`\gtreqless`
$x \eqcirc y$	`\eqcirc`
$x \triangleq y$	`\triangleq`
$x \thickapprox y$	`\thickapprox`
$x \Supset y$	`\Supset`
$x \succcurlyeq y$	`\succcurlyeq`
$x \succsim y$	`\succsim`
$x \vartriangleright y$	`\vartriangleright`
$x \Vdash y$	`\Vdash`
$x \shortparallel y$	`\shortparallel`
$x \pitchfork y$	`\pitchfork`
$x \blacktriangleleft y$	`\blacktriangleleft`
$x \backepsilon y$	`\backepsilon`
$x \because y$	`\because`

3.2.36 AMS Negated Relations Serif Bold

$x \nless y$	`\nless`	$x \nleq y$	`\nleq`
$x \nleqslant y$	`\nleqslant`	$x \nleqq y$	`\nleqq`
$x \lneq y$	`\lneq`	$x \lneqq y$	`\lneqq`
$x \lvertneqq y$	`\lvertneqq`	$x \lnsim y$	`\lnsim`
$x \lnapprox y$	`\lnapprox`	$x \nprec y$	`\nprec`
$x \npreceq y$	`\npreceq`	$x \precnsim y$	`\precnsim`
$x \precnapprox y$	`\precnapprox`	$x \nsim y$	`\nsim`
$x \nshortmid y$	`\nshortmid`	$x \nmid y$	`\nmid`
$x \nvdash y$	`\nvdash`	$x \nvDash y$	`\nvDash`
$x \ntriangleleft y$	`\ntriangleleft`	$x \ntrianglelefteq y$	`\ntrianglelefteq`
$x \nsubseteq y$	`\nsubseteq`	$x \subsetneq y$	`\subsetneq`
$x \varsubsetneq y$	`\varsubsetneq`	$x \subsetneqq y$	`\subsetneqq`
$x \varsubsetneqq y$	`\varsubsetneqq`	$x \ngtr y$	`\ngtr`
$x \ngeq y$	`\ngeq`	$x \ngeqslant y$	`\ngeqslant`
$x \ngeqq y$	`\ngeqq`	$x \gneq y$	`\gneq`
$x \gneqq y$	`\gneqq`	$x \gvertneqq y$	`\gvertneqq`
$x \gnsim y$	`\gnsim`	$x \gnapprox y$	`\gnapprox`
$x \nsucc y$	`\nsucc`	$x \nsucceq y$	`\nsucceq`
$x \nsucceqq y$	`\nsucceqq`	$x \succnsim y$	`\succnsim`
$x \succnapprox y$	`\succnapprox`	$x \ncong y$	`\ncong`
$x \nshortparallel y$	`\nshortparallel`	$x \nparallel y$	`\nparallel`
$x \nvDash y$	`\nvDash`	$x \nVDash y$	`\nVDash`
$x \ntriangleright y$	`\ntriangleright`	$x \ntrianglerighteq y$	`\ntrianglerighteq`
$x \nsupseteq y$	`\nsupseteq`	$x \nsupseteqq y$	`\nsupseteqq`
$x \supsetneq y$	`\supsetneq`	$x \varsupsetneq y$	`\varsupsetneq`
$x \supsetneqq y$	`\supsetneqq`	$x \varsupsetneqq y$	`\varsupsetneqq`

3.2.37 Math “Torture” Test Serif Bold

Most of the following examples are taken from *The TEXbook* (Knuth 1984, see https://ctan.org/pkg/texbook) and were adapted for LaTeX from Karl Berry’s torture test for plain TEX math fonts.

$x+y-z$, $x+y*z$, $z*y/z$, $(x+y)(x-y)=x^2-y^2$,
$x\times y\cdot z=[x\,y\,z]$, $x\circ y\bullet z$, $x\cup y\cap z$, $x\sqcup y\sqcap z$,
$x\vee y\wedge z$, $x\pm y\mp z$, $x=y/z$, $x:=y$, $x\le y\ne z$, $x\sim y\simeq z\, x\equiv y\not\equiv z$, $x\subset y\subseteq z$
$\sin 2\theta = 2\sin\theta\cos\theta$, $\mathrm{O}(n\log n\log n)$, $\Pr(X>x)=\exp(-x/\mu)$,
$\bigl(x\in A(n)\bigm| x\in B(n)\bigr)$, $\bigcup_n X_n \parallel \bigcap_n Y_n$
In-text matrices $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}a&b&c\\1&m&n\end{smallmatrix}\right)$.

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x}\frac{1}{1-x^2} = \frac{a+1}{b} \bigg/ \frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |\varphi(x+\mathrm{i}y)|^2 = 0$$

$$\pi(n) = \sum_{m=2}^{n} \left\lfloor \left(\sum_{k=1}^{m-1} \lfloor (m/k)/\lceil m/k \rceil \rfloor \right)^{-1} \right\rfloor.$$

$$\int_0^\infty \frac{t - \mathrm{i}b}{t^2 + b^2} e^{\mathrm{i}at}\,\mathrm{d}t = e^{ab} E_1(ab), \quad a, b > 0.$$

$$A := \begin{pmatrix} x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda \end{pmatrix}.$$

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \begin{pmatrix} u & x \\ v & y \\ w & z \end{pmatrix}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

$$M = \begin{array}{c} \\ C \\ I \\ C' \end{array} \!\!\begin{array}{c} \begin{array}{ccc} C & I & C' \end{array} \\ \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{array}$$

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{converges if} \quad |z| < \left(\limsup_{n\to\infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x+\Delta x)-f(x)}{\Delta x}\to f'(x) \qquad \text{as } \Delta x\to 0.$$

$$\|u_i\|=1, \qquad u_i\cdot u_j=0 \quad \text{if } i\neq j.$$

$$\text{The confluent image of} \begin{Bmatrix}\text{an arc}\\ \text{a circle}\\ \text{a fan}\end{Bmatrix} \text{ is } \begin{Bmatrix}\text{an arc}\\ \text{an arc or a circle}\\ \text{a fan or an arc}\end{Bmatrix}.$$

$$\begin{aligned} T(n) \le T(2^{\lceil \lg n\rceil}) &\le c(3^{\lceil \lg n\rceil}-2^{\lceil \lg n\rceil})\\ &< 3c\cdot 3^{\lg n}\\ &= 3c\,n^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2-xy+yx-y^2\\ &= x^2-y^2\\ (x+y)^2 &= x^2+2xy+y^2. \end{aligned}$$

$$\begin{aligned} \left(\int_{-\infty}^{\infty} \mathrm{e}^{-x^2}\,\mathrm{d}x\right)^2 &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y\\ &= \int_0^{2\pi}\int_0^{\infty} \mathrm{e}^{-r^2}\,\mathrm{d}r\,\mathrm{d}\theta\\ &= \int_0^{2\pi}\left(\mathrm{e}^{-\frac{r^2}{2}}\Big|_{r=0}^{r=\infty}\right)\mathrm{d}\theta\\ &= \pi. \end{aligned}$$

$$\prod_{k\ge 0}\frac{1}{(1-q^k z)} = \sum_{n\ge 0} z^n \Big/ \prod_{1\le k\le n}(1-q^k).$$

$$\sum_{\substack{0<i\le m\\ 0<j\le n}} p(i,j) \neq \sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r} a_{ij}b_{jk}c_{ki} \neq \sum_{\substack{1\le i\le p\\ 1\le j\le q\\ 1\le k\le r}} a_{ij}b_{jk}c_{ki}$$

$$\max_{1\le n\le m}\log_2 P_n \quad \text{and} \quad \lim_{x\to 0}\frac{\sin x}{x}=1$$

Inline math: $\max_{1\le n\le m}\log_2 P_n$ **and** $\lim_{x\to 0}\frac{\sin x}{x}=1$

$$p_1(n)=\lim_{m\to\infty}\sum_{\nu=0}^{\infty}\bigl(1-\cos^{2m}(\nu!^n\pi/n)\bigr)$$

Inline math: $p_1(n)=\lim_{m\to\infty}\sum_{\nu=0}^{\infty}\bigl(1-\cos^{2m}(\nu!^n\pi/n)\bigr)$

3.3 Math Test Sans Serif

3.3.1 Overview Sans Serif

Default: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma; \sigma_{\varepsilon}, c^{\alpha}$
mathnormal: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathit: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathbf: $\boldsymbol{a\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

Default: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma; \sigma_{\varepsilon}, c^{\alpha}}$
mathnormal: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathrm: $\boldsymbol{\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma}$
mathup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

Default: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma; \sigma_{\varepsilon}, c^{\alpha}$
mathnormal: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: $\mathrm{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathit: $a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

Default: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma; \sigma_{\varepsilon}, c^{\alpha}}$
mathnormal: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathrm: $\boldsymbol{\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma}$
mathup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbf: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: $\mathbf{a\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$

3.3.2 Formulas Sans Serif

α, β, γ, δ, ϵ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, o, π, ϖ, ρ, ϱ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, $\digamma$, A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Υ, Φ, X, Ψ, Ω, F,

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,$

$\mathrm{\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,}$

$\boldsymbol{\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, Y, \Phi, X, \Psi, \Omega, F,}$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \to \infty} v(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \left(\frac{27}{2}\right)$$

Disambiguation: 0 O O, 1 l I | l I $/$, i j, rn m, θ Θ, ϕ ψ, $-$ –

Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, $ß$ β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Y.

$\boldsymbol{\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda}$

$\boldsymbol{\lim_{v \to \infty} v(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7}$

$\boldsymbol{\hat{\beta} = (X'X)^{-1}X'y}$

$$\boldsymbol{\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)}$$

$$\boldsymbol{\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \left(\frac{27}{2}\right)}$$

Disambiguation: 0 O *O*, 1 l I | *l I /*, *i j*, *rn m*, $\boldsymbol{\theta}$ $\boldsymbol{\Theta}$, $\boldsymbol{\phi}$ $\boldsymbol{\psi}$, – –

Latin vs. Greek: $\boldsymbol{a\ \alpha, d\ \delta, e\ \varepsilon, i\ \iota, k\ \kappa, n\ \eta, o\ \sigma, p\ \rho, ß\ \beta, u\ \upsilon, v\ \nu, w\ \omega, x\ \chi, y\ \gamma, A\ \Delta\ \Lambda, O\ \Theta\ \Omega, T\ \Gamma, Y\ Y.}$

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$

$\lim_{v \to \infty} v(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$

$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \left(\frac{27}{2}\right)$$

Disambiguation: 0 O O, 1 l I | l I $/$, i j, rn m, θ Θ, ϕ ψ, $-$ –

Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, $ß$ β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Y.

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{v \to \infty} v(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$
$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x)\, dx = \binom{27}{2}$$

Disambiguation: 0 O O, 1 l I | l I $/$, i j, rn m, θ Θ, ϕ ψ, - –
Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, $ß$ β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.

3.3.3 Math Alphabets Sans Serif

Default

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$$
$$A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$$
$$a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$$
$$A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,$$
$$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \epsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Normal (`\mathnormal`)

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$$
$$A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$$
$$a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$$
$$A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,$$
$$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \epsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Italic (`\mathit`)

$$\mathit{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$$
$$\mathit{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$$
$$\mathit{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$$
$$A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,$$
$$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \epsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Roman (`\mathrm`)

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$$
$$\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}, \mathrm{H}, \mathrm{I}, \mathrm{J}, \mathrm{K}, \mathrm{L}, \mathrm{M}, \mathrm{N}, \mathrm{O}, \mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}, \mathrm{T}, \mathrm{U}, \mathrm{V}, \mathrm{W}, \mathrm{X}, \mathrm{Y}, \mathrm{Z},$$
$$\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}, \mathrm{f}, \mathrm{g}, \mathrm{h}, \mathrm{i}, \mathrm{j}, \mathrm{k}, \mathrm{l}, \mathrm{m}, \mathrm{n}, \mathrm{o}, \mathrm{p}, \mathrm{q}, \mathrm{r}, \mathrm{s}, \mathrm{t}, \mathrm{u}, \mathrm{v}, \mathrm{w}, \mathrm{x}, \mathrm{y}, \mathrm{z},$$
$$\mathrm{A}, \mathrm{B}, \Gamma, \Delta, \mathrm{E}, \mathrm{Z}, \mathrm{H}, \Theta, \mathrm{I}, \mathrm{K}, \Lambda, \mathrm{M}, \mathrm{N}, \Xi, \mathrm{O}, \Pi, \mathrm{P}, \Sigma, \mathrm{T}, \Upsilon, \Phi, \mathrm{X}, \Psi, \Omega,$$
$$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Bold (`\mathbf`)

$$\mathbf{0}, \mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{5}, \mathbf{6}, \mathbf{7}, \mathbf{8}, \mathbf{9},$$
$$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}, \boldsymbol{E}, \boldsymbol{F}, \boldsymbol{G}, \boldsymbol{H}, \boldsymbol{I}, \boldsymbol{J}, \boldsymbol{K}, \boldsymbol{L}, \boldsymbol{M}, \boldsymbol{N}, \boldsymbol{O}, \boldsymbol{P}, \boldsymbol{Q}, \boldsymbol{R}, \boldsymbol{S}, \boldsymbol{T}, \boldsymbol{U}, \boldsymbol{V}, \boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z},$$
$$\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}, \boldsymbol{f}, \boldsymbol{g}, \boldsymbol{h}, \boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}, \boldsymbol{l}, \boldsymbol{m}, \boldsymbol{n}, \boldsymbol{o}, \boldsymbol{p}, \boldsymbol{q}, \boldsymbol{r}, \boldsymbol{s}, \boldsymbol{t}, \boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}, \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z},$$
$$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{\Gamma}, \boldsymbol{\Delta}, \boldsymbol{E}, \boldsymbol{Z}, \boldsymbol{H}, \boldsymbol{\Theta}, \boldsymbol{I}, \boldsymbol{K}, \boldsymbol{\Lambda}, \boldsymbol{M}, \boldsymbol{N}, \boldsymbol{\Xi}, \boldsymbol{O}, \boldsymbol{\Pi}, \boldsymbol{P}, \boldsymbol{\Sigma}, \boldsymbol{T}, \boldsymbol{\Upsilon}, \boldsymbol{\Phi}, \boldsymbol{X}, \boldsymbol{\Psi}, \boldsymbol{\Omega},$$
$$\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\delta}, \boldsymbol{\varepsilon}, \boldsymbol{\zeta}, \boldsymbol{\eta}, \boldsymbol{\theta}, \boldsymbol{\iota}, \boldsymbol{\kappa}, \boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\nu}, \boldsymbol{\xi}, \boldsymbol{o}, \boldsymbol{\pi}, \boldsymbol{\rho}, \boldsymbol{\sigma}, \boldsymbol{\tau}, \boldsymbol{\upsilon}, \boldsymbol{\phi}, \boldsymbol{\chi}, \boldsymbol{\psi}, \boldsymbol{\omega}, \boldsymbol{\varepsilon}, \boldsymbol{\theta}, \boldsymbol{\pi}, \boldsymbol{\rho}, \boldsymbol{\varsigma}, \boldsymbol{\phi},$$

Caligraphic (`\mathcal`)

$$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{K}, \mathcal{L}, \mathcal{M}, \mathcal{N}, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{S}, \mathcal{T}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z},$$

Script (`\mathscr`)

$$\mathscr{A}, \mathscr{B}, \mathscr{C}, \mathscr{D}, \mathscr{E}, \mathscr{F}, \mathscr{G}, \mathscr{H}, \mathscr{I}, \mathscr{J}, \mathscr{K}, \mathscr{L}, \mathscr{M}, \mathscr{N}, \mathscr{O}, \mathscr{P}, \mathscr{Q}, \mathscr{R}, \mathscr{S}, \mathscr{T}, \mathscr{U}, \mathscr{V}, \mathscr{W}, \mathscr{X}, \mathscr{Y}, \mathscr{Z},$$

Fraktur (`\mathfrak`)

$$\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{I}, \mathfrak{J}, \mathfrak{K}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N}, \mathfrak{O}, \mathfrak{P}, \mathfrak{Q}, \mathfrak{R}, \mathfrak{S}, \mathfrak{T}, \mathfrak{U}, \mathfrak{V}, \mathfrak{W}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z},$$
$$\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{e}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n}, \mathfrak{o}, \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}, \mathfrak{t}, \mathfrak{u}, \mathfrak{v}, \mathfrak{w}, \mathfrak{x}, \mathfrak{y}, \mathfrak{z},$$

Blackboard Bold (`\mathbb`)

$$\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}, \mathbb{E}, \mathbb{F}, \mathbb{G}, \mathbb{H}, \mathbb{I}, \mathbb{J}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \mathbb{O}, \mathbb{P}, \mathbb{Q}, \mathbb{R}, \mathbb{S}, \mathbb{T}, \mathbb{U}, \mathbb{V}, \mathbb{W}, \mathbb{X}, \mathbb{Y}, \mathbb{Z},$$

3.3.4 Character Sidebearings Sans Serif

Default

$$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$$
$$|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$$
$$|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$$
$$|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$$
$$|A| + |B| + |\mathit{\Gamma}| + |\mathit{\Delta}| + |E| + |Z| + |H| + |\mathit{\Theta}| + |I| + |K| + |\mathit{\Lambda}| + |M| +$$
$$|N| + |\mathit{\Xi}| + |O| + |\mathit{\Pi}| + |P| + |\mathit{\Sigma}| + |T| + |Y| + |\mathit{\Phi}| + |X| + |\mathit{\Psi}| + |\mathit{\Omega}| +$$
$$|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| +$$
$$|\nu| + |\xi| + |o| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| +$$
$$|\varepsilon| + |\vartheta| + |\varpi| + |\varrho| + |\varsigma| + |\varphi| +$$

Math Roman (`\mathrm`)

$$|\mathrm{A}| + |\mathrm{B}| + |\mathrm{C}| + |\mathrm{D}| + |\mathrm{E}| + |\mathrm{F}| + |\mathrm{G}| + |\mathrm{H}| + |\mathrm{I}| + |\mathrm{J}| + |\mathrm{K}| + |\mathrm{L}| + |\mathrm{M}| +$$
$$|\mathrm{N}| + |\mathrm{O}| + |\mathrm{P}| + |\mathrm{Q}| + |\mathrm{R}| + |\mathrm{S}| + |\mathrm{T}| + |\mathrm{U}| + |\mathrm{V}| + |\mathrm{W}| + |\mathrm{X}| + |\mathrm{Y}| + |\mathrm{Z}| +$$
$$|\mathrm{a}| + |\mathrm{b}| + |\mathrm{c}| + |\mathrm{d}| + |\mathrm{e}| + |\mathrm{f}| + |\mathrm{g}| + |\mathrm{h}| + |\mathrm{i}| + |\mathrm{j}| + |\mathrm{k}| + |\mathrm{l}| + |\mathrm{m}| +$$
$$|\mathrm{n}| + |\mathrm{o}| + |\mathrm{p}| + |\mathrm{q}| + |\mathrm{r}| + |\mathrm{s}| + |\mathrm{t}| + |\mathrm{u}| + |\mathrm{v}| + |\mathrm{w}| + |\mathrm{x}| + |\mathrm{y}| + |\mathrm{z}| +$$
$$|\mathrm{A}| + |\mathrm{B}| + |\Gamma| + |\Delta| + |\mathrm{E}| + |\mathrm{Z}| + |\mathrm{H}| + |\Theta| + |\mathrm{I}| + |\mathrm{K}| + |\Lambda| + |\mathrm{M}| +$$
$$|\mathrm{N}| + |\Xi| + |\mathrm{O}| + |\Pi| + |\mathrm{P}| + |\Sigma| + |\mathrm{T}| + |\Upsilon| + |\Phi| + |\mathrm{X}| + |\Psi| + |\Omega| +$$

Math Bold (`\mathbf`)

$$|\mathbf{A}| + |\mathbf{B}| + |\mathbf{C}| + |\mathbf{D}| + |\mathbf{E}| + |\mathbf{F}| + |\mathbf{G}| + |\mathbf{H}| + |\mathbf{I}| + |\mathbf{J}| + |\mathbf{K}| + |\mathbf{L}| + |\mathbf{M}| +$$
$$|\mathbf{N}| + |\mathbf{O}| + |\mathbf{P}| + |\mathbf{Q}| + |\mathbf{R}| + |\mathbf{S}| + |\mathbf{T}| + |\mathbf{U}| + |\mathbf{V}| + |\mathbf{W}| + |\mathbf{X}| + |\mathbf{Y}| + |\mathbf{Z}| +$$
$$|\mathbf{a}| + |\mathbf{b}| + |\mathbf{c}| + |\mathbf{d}| + |\mathbf{e}| + |\mathbf{f}| + |\mathbf{g}| + |\mathbf{h}| + |\mathbf{i}| + |\mathbf{j}| + |\mathbf{k}| + |\mathbf{l}| + |\mathbf{m}| +$$
$$|\mathbf{n}| + |\mathbf{o}| + |\mathbf{p}| + |\mathbf{q}| + |\mathbf{r}| + |\mathbf{s}| + |\mathbf{t}| + |\mathbf{u}| + |\mathbf{v}| + |\mathbf{w}| + |\mathbf{x}| + |\mathbf{y}| + |\mathbf{z}| +$$
$$|\mathbf{A}| + |\mathbf{B}| + |\mathbf{\Gamma}| + |\mathbf{\Delta}| + |\mathbf{E}| + |\mathbf{Z}| + |\mathbf{H}| + |\mathbf{\Theta}| + |\mathbf{I}| + |\mathbf{K}| + |\mathbf{\Lambda}| + |\mathbf{M}| +$$
$$|\mathbf{N}| + |\mathbf{\Xi}| + |\mathbf{O}| + |\mathbf{\Pi}| + |\mathbf{P}| + |\mathbf{\Sigma}| + |\mathbf{T}| + |\mathbf{Y}| + |\mathbf{\Phi}| + |\mathbf{X}| + |\mathbf{\Psi}| + |\mathbf{\Omega}| +$$

Math Calligraphic (`\mathcal`)

$$|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| +$$
$$|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +$$

3.3.5 Superscript Positioning Sans Serif

Default

$$A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 +$$
$$N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 +$$
$$a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 +$$
$$n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 +$$
$$A^2 + B^2 + \mathit{\Gamma}^2 + \mathit{\Delta}^2 + E^2 + Z^2 + H^2 + \mathit{\Theta}^2 + I^2 + K^2 + \mathit{\Lambda}^2 + M^2 +$$
$$N^2 + \mathit{\Xi}^2 + O^2 + \mathit{\Pi}^2 + P^2 + \mathit{\Sigma}^2 + T^2 + \mathit{\Upsilon}^2 + \mathit{\Phi}^2 + X^2 + \mathit{\Psi}^2 + \mathit{\Omega}^2 +$$
$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \epsilon^2 + \zeta^2 + \eta^2 + \theta^2 + \iota^2 + \kappa^2 + \lambda^2 + \mu^2 +$$
$$\nu^2 + \xi^2 + o^2 + \pi^2 + \rho^2 + \sigma^2 + \tau^2 + \upsilon^2 + \phi^2 + \chi^2 + \psi^2 + \omega^2 +$$
$$\varepsilon^2 + \vartheta^2 + \varpi^2 + \varrho^2 + \varsigma^2 + \varphi^2 +$$

Math Roman (\mathrm)

$$\mathrm{A}^2 + \mathrm{B}^2 + \mathrm{C}^2 + \mathrm{D}^2 + \mathrm{E}^2 + \mathrm{F}^2 + \mathrm{G}^2 + \mathrm{H}^2 + \mathrm{I}^2 + \mathrm{J}^2 + \mathrm{K}^2 + \mathrm{L}^2 + \mathrm{M}^2 +$$
$$\mathrm{N}^2 + \mathrm{O}^2 + \mathrm{P}^2 + \mathrm{Q}^2 + \mathrm{R}^2 + \mathrm{S}^2 + \mathrm{T}^2 + \mathrm{U}^2 + \mathrm{V}^2 + \mathrm{W}^2 + \mathrm{X}^2 + \mathrm{Y}^2 + \mathrm{Z}^2 +$$
$$\mathrm{a}^2 + \mathrm{b}^2 + \mathrm{c}^2 + \mathrm{d}^2 + \mathrm{e}^2 + \mathrm{f}^2 + \mathrm{g}^2 + \mathrm{h}^2 + \mathrm{i}^2 + \mathrm{j}^2 + \mathrm{k}^2 + \mathrm{l}^2 + \mathrm{m}^2 +$$
$$\mathrm{n}^2 + \mathrm{o}^2 + \mathrm{p}^2 + \mathrm{q}^2 + \mathrm{r}^2 + \mathrm{s}^2 + \mathrm{t}^2 + \mathrm{u}^2 + \mathrm{v}^2 + \mathrm{w}^2 + \mathrm{x}^2 + \mathrm{y}^2 + \mathrm{z}^2 +$$
$$\mathrm{A}^2 + \mathrm{B}^2 + \Gamma^2 + \Delta^2 + \mathrm{E}^2 + \mathrm{Z}^2 + \mathrm{H}^2 + \Theta^2 + \mathrm{I}^2 + \mathrm{K}^2 + \Lambda^2 + \mathrm{M}^2 +$$
$$\mathrm{N}^2 + \Xi^2 + \mathrm{O}^2 + \Pi^2 + \mathrm{P}^2 + \Sigma^2 + \mathrm{T}^2 + \Upsilon^2 + \Phi^2 + \mathrm{X}^2 + \Psi^2 + \Omega^2 +$$

Math Bold (\mathbf)

$$\boldsymbol{A}^2 + \boldsymbol{B}^2 + \boldsymbol{C}^2 + \boldsymbol{D}^2 + \boldsymbol{E}^2 + \boldsymbol{F}^2 + \boldsymbol{G}^2 + \boldsymbol{H}^2 + \boldsymbol{I}^2 + \boldsymbol{J}^2 + \boldsymbol{K}^2 + \boldsymbol{L}^2 + \boldsymbol{M}^2 +$$
$$\boldsymbol{N}^2 + \boldsymbol{O}^2 + \boldsymbol{P}^2 + \boldsymbol{Q}^2 + \boldsymbol{R}^2 + \boldsymbol{S}^2 + \boldsymbol{T}^2 + \boldsymbol{U}^2 + \boldsymbol{V}^2 + \boldsymbol{W}^2 + \boldsymbol{X}^2 + \boldsymbol{Y}^2 + \boldsymbol{Z}^2 +$$
$$\boldsymbol{a}^2 + \boldsymbol{b}^2 + \boldsymbol{c}^2 + \boldsymbol{d}^2 + \boldsymbol{e}^2 + \boldsymbol{f}^2 + \boldsymbol{g}^2 + \boldsymbol{h}^2 + \boldsymbol{i}^2 + \boldsymbol{j}^2 + \boldsymbol{k}^2 + \boldsymbol{l}^2 + \boldsymbol{m}^2 +$$
$$\boldsymbol{n}^2 + \boldsymbol{o}^2 + \boldsymbol{p}^2 + \boldsymbol{q}^2 + \boldsymbol{r}^2 + \boldsymbol{s}^2 + \boldsymbol{t}^2 + \boldsymbol{u}^2 + \boldsymbol{v}^2 + \boldsymbol{w}^2 + \boldsymbol{x}^2 + \boldsymbol{y}^2 + \boldsymbol{z}^2 +$$
$$\boldsymbol{A}^2 + \boldsymbol{B}^2 + \boldsymbol{\mathit{\Gamma}}^2 + \boldsymbol{\mathit{\Delta}}^2 + \boldsymbol{E}^2 + \boldsymbol{Z}^2 + \boldsymbol{H}^2 + \boldsymbol{\mathit{\Theta}}^2 + \boldsymbol{I}^2 + \boldsymbol{K}^2 + \boldsymbol{\mathit{\Lambda}}^2 + \boldsymbol{M}^2 +$$
$$\boldsymbol{N}^2 + \boldsymbol{\mathit{\Xi}}^2 + \boldsymbol{O}^2 + \boldsymbol{\mathit{\Pi}}^2 + \boldsymbol{P}^2 + \boldsymbol{\mathit{\Sigma}}^2 + \boldsymbol{T}^2 + \boldsymbol{\mathit{\Upsilon}}^2 + \boldsymbol{\mathit{\Phi}}^2 + \boldsymbol{X}^2 + \boldsymbol{\mathit{\Psi}}^2 + \boldsymbol{\mathit{\Omega}}^2 +$$

Math Calligraphic (\mathcal)

$$\mathcal{A}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{D}^2 + \mathcal{E}^2 + \mathcal{F}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{L}^2 + \mathcal{M}^2 +$$
$$\mathcal{N}^2 + \mathcal{O}^2 + \mathcal{P}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{S}^2 + \mathcal{T}^2 + \mathcal{U}^2 + \mathcal{V}^2 + \mathcal{W}^2 + \mathcal{X}^2 + \mathcal{Y}^2 + \mathcal{Z}^2 +$$

3.3.6 Subscript Positioning Sans Serif

Default

$$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$$
$$N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$$
$$a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$$
$$n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$$
$$A_i + B_i + \mathit{\Gamma}_i + \mathit{\Delta}_i + E_i + Z_i + H_i + \mathit{\Theta}_i + I_i + K_i + \mathit{\Lambda}_i + M_i +$$
$$N_i + \mathit{\Xi}_i + O_i + \mathit{\Pi}_i + P_i + \mathit{\Sigma}_i + T_i + Y_i + \mathit{\Phi}_i + X_i + \mathit{\Psi}_i + \mathit{\Omega}_i +$$
$$\alpha_i + \beta_i + \gamma_i + \delta_i + \epsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i +$$
$$\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + \upsilon_i + \phi_i + \chi_i + \psi_i + \omega_i +$$
$$\varepsilon_i + \vartheta_i + \varpi_i + \varrho_i + \varsigma_i + \varphi_i +$$

Math Roman (`\mathrm`)

$$\mathrm{A}_i + \mathrm{B}_i + \mathrm{C}_i + \mathrm{D}_i + \mathrm{E}_i + \mathrm{F}_i + \mathrm{G}_i + \mathrm{H}_i + \mathrm{I}_i + \mathrm{J}_i + \mathrm{K}_i + \mathrm{L}_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \mathrm{O}_i + \mathrm{P}_i + \mathrm{Q}_i + \mathrm{R}_i + \mathrm{S}_i + \mathrm{T}_i + \mathrm{U}_i + \mathrm{V}_i + \mathrm{W}_i + \mathrm{X}_i + \mathrm{Y}_i + \mathrm{Z}_i +$$
$$\mathrm{a}_i + \mathrm{b}_i + \mathrm{c}_i + \mathrm{d}_i + \mathrm{e}_i + \mathrm{f}_i + \mathrm{g}_i + \mathrm{h}_i + \mathrm{i}_i + \mathrm{j}_i + \mathrm{k}_i + \mathrm{l}_i + \mathrm{m}_i +$$
$$\mathrm{n}_i + \mathrm{o}_i + \mathrm{p}_i + \mathrm{q}_i + \mathrm{r}_i + \mathrm{s}_i + \mathrm{t}_i + \mathrm{u}_i + \mathrm{v}_i + \mathrm{w}_i + \mathrm{x}_i + \mathrm{y}_i + \mathrm{z}_i +$$
$$\mathrm{A}_i + \mathrm{B}_i + \Gamma_i + \Delta_i + \mathrm{E}_i + \mathrm{Z}_i + \mathrm{H}_i + \Theta_i + \mathrm{I}_i + \mathrm{K}_i + \Lambda_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \Xi_i + \mathrm{O}_i + \Pi_i + \mathrm{P}_i + \Sigma_i + \mathrm{T}_i + \Upsilon_i + \Phi_i + \mathrm{X}_i + \Psi_i + \Omega_i +$$

Math Bold (`\mathbf`)

$$\mathbf{A}_i + \mathbf{B}_i + \mathbf{C}_i + \mathbf{D}_i + \mathbf{E}_i + \mathbf{F}_i + \mathbf{G}_i + \mathbf{H}_i + \mathbf{I}_i + \mathbf{J}_i + \mathbf{K}_i + \mathbf{L}_i + \mathbf{M}_i +$$
$$\mathbf{N}_i + \mathbf{O}_i + \mathbf{P}_i + \mathbf{Q}_i + \mathbf{R}_i + \mathbf{S}_i + \mathbf{T}_i + \mathbf{U}_i + \mathbf{V}_i + \mathbf{W}_i + \mathbf{X}_i + \mathbf{Y}_i + \mathbf{Z}_i +$$
$$\mathbf{a}_i + \mathbf{b}_i + \mathbf{c}_i + \mathbf{d}_i + \mathbf{e}_i + \mathbf{f}_i + \mathbf{g}_i + \mathbf{h}_i + \mathbf{i}_i + \mathbf{j}_i + \mathbf{k}_i + \mathbf{l}_i + \mathbf{m}_i +$$
$$\mathbf{n}_i + \mathbf{o}_i + \mathbf{p}_i + \mathbf{q}_i + \mathbf{r}_i + \mathbf{s}_i + \mathbf{t}_i + \mathbf{u}_i + \mathbf{v}_i + \mathbf{w}_i + \mathbf{x}_i + \mathbf{y}_i + \mathbf{z}_i +$$
$$\mathbf{A}_i + \mathbf{B}_i + \mathbf{\Gamma}_i + \mathbf{\Delta}_i + \mathbf{E}_i + \mathbf{Z}_i + \mathbf{H}_i + \mathbf{\Theta}_i + \mathbf{I}_i + \mathbf{K}_i + \mathbf{\Lambda}_i + \mathbf{M}_i +$$
$$\mathbf{N}_i + \mathbf{\Xi}_i + \mathbf{O}_i + \mathbf{\Pi}_i + \mathbf{P}_i + \mathbf{\Sigma}_i + \mathbf{T}_i + \mathbf{Y}_i + \mathbf{\Phi}_i + \mathbf{X}_i + \mathbf{\Psi}_i + \mathbf{\Omega}_i +$$

Math Calligraphic (`\mathcal`)

$$\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i +$$
$$\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +$$

3.3.7 Accent Positioning Sans Serif

Default

$$\hat{0}+\hat{1}+\hat{2}+\hat{3}+\hat{4}+\hat{5}+\hat{6}+\hat{7}+\hat{8}+\hat{9}+$$
$$\hat{A}+\hat{B}+\hat{C}+\hat{D}+\hat{E}+\hat{F}+\hat{G}+\hat{H}+\hat{I}+\hat{J}+\hat{K}+\hat{L}+\hat{M}+$$
$$\hat{N}+\hat{O}+\hat{P}+\hat{Q}+\hat{R}+\hat{S}+\hat{T}+\hat{U}+\hat{V}+\hat{W}+\hat{X}+\hat{Y}+\hat{Z}+$$
$$\hat{a}+\hat{b}+\hat{c}+\hat{d}+\hat{e}+\hat{f}+\hat{g}+\hat{h}+\hat{i}+\hat{j}+\hat{k}+\hat{l}+\hat{m}+$$
$$\hat{n}+\hat{o}+\hat{p}+\hat{q}+\hat{r}+\hat{s}+\hat{t}+\hat{u}+\hat{v}+\hat{w}+\hat{x}+\hat{y}+\hat{z}+$$
$$\hat{A}+\hat{B}+\hat{\Gamma}+\hat{\Delta}+\hat{E}+\hat{Z}+\hat{H}+\hat{\Theta}+\hat{I}+\hat{K}+\hat{\Lambda}+\hat{M}+$$
$$\hat{N}+\hat{\Xi}+\hat{O}+\hat{\Pi}+\hat{P}+\hat{\Sigma}+\hat{T}+\hat{\Upsilon}+\hat{\Phi}+\hat{X}+\hat{\Psi}+\hat{\Omega}+$$
$$\hat{\alpha}+\hat{\beta}+\hat{\gamma}+\hat{\delta}+\hat{\varepsilon}+\hat{\zeta}+\hat{\eta}+\hat{\theta}+\hat{\iota}+\hat{\kappa}+\hat{\lambda}+\hat{\mu}+$$
$$\hat{\nu}+\hat{\xi}+\hat{o}+\hat{\pi}+\hat{\rho}+\hat{\sigma}+\hat{\tau}+\hat{\upsilon}+\hat{\phi}+\hat{\chi}+\hat{\psi}+\hat{\omega}+$$
$$\hat{\epsilon}+\hat{\vartheta}+\hat{\varpi}+\hat{\varrho}+\hat{\varsigma}+\hat{\varphi}+$$

Math Italic (\mathit)

$$\hat{\mathit{0}}+\hat{\mathit{1}}+\hat{\mathit{2}}+\hat{\mathit{3}}+\hat{\mathit{4}}+\hat{\mathit{5}}+\hat{\mathit{6}}+\hat{\mathit{7}}+\hat{\mathit{8}}+\hat{\mathit{9}}+$$
$$\hat{A}+\hat{B}+\hat{C}+\hat{D}+\hat{E}+\hat{F}+\hat{G}+\hat{H}+\hat{I}+\hat{J}+\hat{K}+\hat{L}+\hat{M}+$$
$$\hat{N}+\hat{O}+\hat{P}+\hat{Q}+\hat{R}+\hat{S}+\hat{T}+\hat{U}+\hat{V}+\hat{W}+\hat{X}+\hat{Y}+\hat{Z}+$$
$$\hat{a}+\hat{b}+\hat{c}+\hat{d}+\hat{e}+\hat{f}+\hat{g}+\hat{h}+\hat{i}+\hat{j}+\hat{k}+\hat{l}+\hat{m}+\hat{\ell}+\hat{\wp}+\hat{\imath}+\hat{\jmath}+\tilde{\imath}$$
$$\hat{n}+\hat{o}+\hat{p}+\hat{q}+\hat{r}+\hat{s}+\hat{t}+\hat{u}+\hat{v}+\hat{w}+\hat{x}+\hat{y}+\hat{z}+$$
$$\hat{A}+\hat{B}+\hat{\Gamma}+\hat{\Delta}+\hat{E}+\hat{Z}+\hat{H}+\hat{\Theta}+\hat{I}+\hat{K}+\hat{\Lambda}+\hat{M}+$$
$$\hat{N}+\hat{\Xi}+\hat{O}+\hat{\Pi}+\hat{P}+\hat{\Sigma}+\hat{T}+\hat{\Upsilon}+\hat{\Phi}+\hat{X}+\hat{\Psi}+\hat{\Omega}+$$
$$\hat{\alpha}+\hat{\beta}+\hat{\gamma}+\hat{\delta}+\hat{\varepsilon}+\hat{\zeta}+\hat{\eta}+\hat{\theta}+\hat{\iota}+\hat{\kappa}+\hat{\lambda}+\hat{\mu}+$$
$$\hat{\nu}+\hat{\xi}+\hat{o}+\hat{\pi}+\hat{\rho}+\hat{\sigma}+\hat{\tau}+\hat{\upsilon}+\hat{\phi}+\hat{\chi}+\hat{\psi}+\hat{\omega}+$$
$$\hat{\epsilon}+\hat{\vartheta}+\hat{\varpi}+\hat{\varrho}+\hat{\varsigma}+\hat{\varphi}+$$

Math Roman (\mathrm)

$$\hat{\mathrm{0}}+\hat{\mathrm{1}}+\hat{\mathrm{2}}+\hat{\mathrm{3}}+\hat{\mathrm{4}}+\hat{\mathrm{5}}+\hat{\mathrm{6}}+\hat{\mathrm{7}}+\hat{\mathrm{8}}+\hat{\mathrm{9}}+$$
$$\hat{\mathrm{A}}+\hat{\mathrm{B}}+\hat{\mathrm{C}}+\hat{\mathrm{D}}+\hat{\mathrm{E}}+\hat{\mathrm{F}}+\hat{\mathrm{G}}+\hat{\mathrm{H}}+\hat{\mathrm{I}}+\hat{\mathrm{J}}+\hat{\mathrm{K}}+\hat{\mathrm{L}}+\hat{\mathrm{M}}+$$
$$\hat{\mathrm{N}}+\hat{\mathrm{O}}+\hat{\mathrm{P}}+\hat{\mathrm{Q}}+\hat{\mathrm{R}}+\hat{\mathrm{S}}+\hat{\mathrm{T}}+\hat{\mathrm{U}}+\hat{\mathrm{V}}+\hat{\mathrm{W}}+\hat{\mathrm{X}}+\hat{\mathrm{Y}}+\hat{\mathrm{Z}}+$$
$$\hat{\mathrm{a}}+\hat{\mathrm{b}}+\hat{\mathrm{c}}+\hat{\mathrm{d}}+\hat{\mathrm{e}}+\hat{\mathrm{f}}+\hat{\mathrm{g}}+\hat{\mathrm{h}}+\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}+\hat{\mathrm{l}}+\hat{\mathrm{m}}+$$
$$\hat{\mathrm{n}}+\hat{\mathrm{o}}+\hat{\mathrm{p}}+\hat{\mathrm{q}}+\hat{\mathrm{r}}+\hat{\mathrm{s}}+\hat{\mathrm{t}}+\hat{\mathrm{u}}+\hat{\mathrm{v}}+\hat{\mathrm{w}}+\hat{\mathrm{x}}+\hat{\mathrm{y}}+\hat{\mathrm{z}}+$$
$$\hat{\mathrm{A}}+\hat{\mathrm{B}}+\hat{\mathrm{\Gamma}}+\hat{\mathrm{\Delta}}+\hat{\mathrm{E}}+\hat{\mathrm{Z}}+\hat{\mathrm{H}}+\hat{\mathrm{\Theta}}+\hat{\mathrm{I}}+\hat{\mathrm{K}}+\hat{\mathrm{\Lambda}}+\hat{\mathrm{M}}+$$
$$\hat{\mathrm{N}}+\hat{\mathrm{\Xi}}+\hat{\mathrm{O}}+\hat{\mathrm{\Pi}}+\hat{\mathrm{P}}+\hat{\mathrm{\Sigma}}+\hat{\mathrm{T}}+\hat{\mathrm{\Upsilon}}+\hat{\mathrm{\Phi}}+\hat{\mathrm{X}}+\hat{\mathrm{\Psi}}+\hat{\mathrm{\Omega}}+$$

Math Bold (\mathbf)

$$\hat{\mathbf{0}} + \hat{\mathbf{1}} + \hat{\mathbf{2}} + \hat{\mathbf{3}} + \hat{\mathbf{4}} + \hat{\mathbf{5}} + \hat{\mathbf{6}} + \hat{\mathbf{7}} + \hat{\mathbf{8}} + \hat{\mathbf{9}} +$$
$$\hat{\boldsymbol{A}} + \hat{\boldsymbol{B}} + \hat{\boldsymbol{C}} + \hat{\boldsymbol{D}} + \hat{\boldsymbol{E}} + \hat{\boldsymbol{F}} + \hat{\boldsymbol{G}} + \hat{\boldsymbol{H}} + \hat{\boldsymbol{I}} + \hat{\boldsymbol{J}} + \hat{\boldsymbol{K}} + \hat{\boldsymbol{L}} + \hat{\boldsymbol{M}} +$$
$$\hat{\boldsymbol{N}} + \hat{\boldsymbol{O}} + \hat{\boldsymbol{P}} + \hat{\boldsymbol{Q}} + \hat{\boldsymbol{R}} + \hat{\boldsymbol{S}} + \hat{\boldsymbol{T}} + \hat{\boldsymbol{U}} + \hat{\boldsymbol{V}} + \hat{\boldsymbol{W}} + \hat{\boldsymbol{X}} + \hat{\boldsymbol{Y}} + \hat{\boldsymbol{Z}} +$$
$$\hat{\boldsymbol{a}} + \hat{\boldsymbol{b}} + \hat{\boldsymbol{c}} + \hat{\boldsymbol{d}} + \hat{\boldsymbol{e}} + \hat{\boldsymbol{f}} + \hat{\boldsymbol{g}} + \hat{\boldsymbol{h}} + \hat{\boldsymbol{i}} + \hat{\boldsymbol{j}} + \hat{\boldsymbol{k}} + \hat{\boldsymbol{l}} + \hat{\boldsymbol{m}} +$$
$$\hat{\boldsymbol{n}} + \hat{\boldsymbol{o}} + \hat{\boldsymbol{p}} + \hat{\boldsymbol{q}} + \hat{\boldsymbol{r}} + \hat{\boldsymbol{s}} + \hat{\boldsymbol{t}} + \hat{\boldsymbol{u}} + \hat{\boldsymbol{v}} + \hat{\boldsymbol{w}} + \hat{\boldsymbol{x}} + \hat{\boldsymbol{y}} + \hat{\boldsymbol{z}} +$$
$$\hat{\boldsymbol{A}} + \hat{\boldsymbol{B}} + \hat{\boldsymbol{\Gamma}} + \hat{\boldsymbol{\Delta}} + \hat{\boldsymbol{E}} + \hat{\boldsymbol{Z}} + \hat{\boldsymbol{H}} + \hat{\boldsymbol{\Theta}} + \hat{\boldsymbol{I}} + \hat{\boldsymbol{K}} + \hat{\boldsymbol{\Lambda}} + \hat{\boldsymbol{M}} +$$
$$\hat{\boldsymbol{N}} + \hat{\boldsymbol{\Xi}} + \hat{\boldsymbol{O}} + \hat{\boldsymbol{\Pi}} + \hat{\boldsymbol{P}} + \hat{\boldsymbol{\Sigma}} + \hat{\boldsymbol{T}} + \hat{\boldsymbol{Y}} + \hat{\boldsymbol{\Phi}} + \hat{\boldsymbol{X}} + \hat{\boldsymbol{\Psi}} + \hat{\boldsymbol{\Omega}} +$$

Math Calligraphic (\mathcal)

$$\hat{\mathcal{A}} + \hat{\mathcal{B}} + \hat{\mathcal{C}} + \hat{\mathcal{D}} + \hat{\mathcal{E}} + \hat{\mathcal{F}} + \hat{\mathcal{G}} + \hat{\mathcal{H}} + \hat{\mathcal{I}} + \hat{\mathcal{J}} + \hat{\mathcal{K}} + \hat{\mathcal{L}} + \hat{\mathcal{M}} +$$
$$\hat{\mathcal{N}} + \hat{\mathcal{O}} + \hat{\mathcal{P}} + \hat{\mathcal{Q}} + \hat{\mathcal{R}} + \hat{\mathcal{S}} + \hat{\mathcal{T}} + \hat{\mathcal{U}} + \hat{\mathcal{V}} + \hat{\mathcal{W}} + \hat{\mathcal{X}} + \hat{\mathcal{Y}} + \hat{\mathcal{Z}} +$$

3.3.8 Differentials Sans Serif

$$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$$
$$dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$$
$$da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$$
$$dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$$
$$dA + dB + d\mathit{\Gamma} + d\mathit{\Delta} + dE + dZ + dH + d\mathit{\Theta} + dI + dK + d\mathit{\Lambda} + dM +$$
$$dN + d\mathit{\Xi} + dO + d\mathit{\Pi} + dP + d\mathit{\Sigma} + dT + dY + d\mathit{\Phi} + dX + d\mathit{\Psi} + d\mathit{\Omega} +$$
$$d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$$
$$d\nu + d\xi + do + d\pi + d\rho + d\sigma + d\tau + d\upsilon + d\phi + d\chi + d\psi + d\omega +$$
$$d\varepsilon + d\vartheta + d\varpi + d\varrho + d\varsigma + d\varphi +$$
$$d\mathrm{A} + d\mathrm{B} + d\Gamma + d\Delta + d\mathrm{E} + d\mathrm{Z} + d\mathrm{H} + d\Theta + d\mathrm{I} + d\mathrm{K} + d\Lambda + d\mathrm{M} +$$
$$d\mathrm{N} + d\Xi + d\mathrm{O} + d\Pi + d\mathrm{P} + d\Sigma + d\mathrm{T} + d\Upsilon + d\Phi + d\mathrm{X} + d\Psi + d\Omega +$$

$\mathrm{d}A + \mathrm{d}B + \mathrm{d}C + \mathrm{d}D + \mathrm{d}E + \mathrm{d}F + \mathrm{d}G + \mathrm{d}H + \mathrm{d}I + \mathrm{d}J + \mathrm{d}K + \mathrm{d}L + \mathrm{d}M +$
$\mathrm{d}N + \mathrm{d}O + \mathrm{d}P + \mathrm{d}Q + \mathrm{d}R + \mathrm{d}S + \mathrm{d}T + \mathrm{d}U + \mathrm{d}V + \mathrm{d}W + \mathrm{d}X + \mathrm{d}Y + \mathrm{d}Z +$
$\mathrm{d}a + \mathrm{d}b + \mathrm{d}c + \mathrm{d}d + \mathrm{d}e + \mathrm{d}f + \mathrm{d}g + \mathrm{d}h + \mathrm{d}i + \mathrm{d}j + \mathrm{d}k + \mathrm{d}l + \mathrm{d}m +$
$\mathrm{d}n + \mathrm{d}o + \mathrm{d}p + \mathrm{d}q + \mathrm{d}r + \mathrm{d}s + \mathrm{d}t + \mathrm{d}u + \mathrm{d}v + \mathrm{d}w + \mathrm{d}x + \mathrm{d}y + \mathrm{d}z +$
$\mathrm{d}A + \mathrm{d}B + \mathrm{d}\mathit{\Gamma} + \mathrm{d}\mathit{\Delta} + \mathrm{d}E + \mathrm{d}Z + \mathrm{d}H + \mathrm{d}\mathit{\Theta} + \mathrm{d}I + \mathrm{d}K + \mathrm{d}\mathit{\Lambda} + \mathrm{d}M +$
$\mathrm{d}N + \mathrm{d}\mathit{\Xi} + \mathrm{d}O + \mathrm{d}\mathit{\Pi} + \mathrm{d}P + \mathrm{d}\mathit{\Sigma} + \mathrm{d}T + \mathrm{d}\mathit{\Upsilon} + \mathrm{d}\mathit{\Phi} + \mathrm{d}X + \mathrm{d}\mathit{\Psi} + \mathrm{d}\mathit{\Omega} +$
$\mathrm{d}\alpha + \mathrm{d}\beta + \mathrm{d}\gamma + \mathrm{d}\delta + \mathrm{d}\epsilon + \mathrm{d}\zeta + \mathrm{d}\eta + \mathrm{d}\theta + \mathrm{d}\iota + \mathrm{d}\kappa + \mathrm{d}\lambda + \mathrm{d}\mu +$
$\mathrm{d}\nu + \mathrm{d}\xi + \mathrm{d}o + \mathrm{d}\pi + \mathrm{d}\rho + \mathrm{d}\sigma + \mathrm{d}\tau + \mathrm{d}\upsilon + \mathrm{d}\phi + \mathrm{d}\chi + \mathrm{d}\psi + \mathrm{d}\omega +$
$\mathrm{d}\varepsilon + \mathrm{d}\vartheta + \mathrm{d}\varpi + \mathrm{d}\varrho + \mathrm{d}\varsigma + \mathrm{d}\varphi +$
$\mathrm{d}\mathrm{A} + \mathrm{d}\mathrm{B} + \mathrm{d}\Gamma + \mathrm{d}\Delta + \mathrm{d}\mathrm{E} + \mathrm{d}\mathrm{Z} + \mathrm{d}\mathrm{H} + \mathrm{d}\Theta + \mathrm{d}\mathrm{I} + \mathrm{d}\mathrm{K} + \mathrm{d}\Lambda + \mathrm{d}\mathrm{M} +$
$\mathrm{d}\mathrm{N} + \mathrm{d}\Xi + \mathrm{d}\mathrm{O} + \mathrm{d}\Pi + \mathrm{d}\mathrm{P} + \mathrm{d}\Sigma + \mathrm{d}\mathrm{T} + \mathrm{d}\Upsilon + \mathrm{d}\Phi + \mathrm{d}\mathrm{X} + \mathrm{d}\Psi + \mathrm{d}\Omega +$

$\partial A + \partial B + \partial C + \partial D + \partial E + \partial F + \partial G + \partial H + \partial I + \partial J + \partial K + \partial L + \partial M +$
$\partial N + \partial O + \partial P + \partial Q + \partial R + \partial S + \partial T + \partial U + \partial V + \partial W + \partial X + \partial Y + \partial Z +$
$\partial a + \partial b + \partial c + \partial d + \partial e + \partial f + \partial g + \partial h + \partial i + \partial j + \partial k + \partial l + \partial m +$
$\partial n + \partial o + \partial p + \partial q + \partial r + \partial s + \partial t + \partial u + \partial v + \partial w + \partial x + \partial y + \partial z +$
$\partial A + \partial B + \partial \mathit{\Gamma} + \partial \mathit{\Delta} + \partial E + \partial Z + \partial H + \partial \mathit{\Theta} + \partial I + \partial K + \partial \mathit{\Lambda} + \partial M +$
$\partial N + \partial \mathit{\Xi} + \partial O + \partial \mathit{\Pi} + \partial P + \partial \mathit{\Sigma} + \partial T + \partial \mathit{\Upsilon} + \partial \mathit{\Phi} + \partial X + \partial \mathit{\Psi} + \partial \mathit{\Omega} +$
$\partial \alpha + \partial \beta + \partial \gamma + \partial \delta + \partial \epsilon + \partial \zeta + \partial \eta + \partial \theta + \partial \iota + \partial \kappa + \partial \lambda + \partial \mu +$
$\partial \nu + \partial \xi + \partial o + \partial \pi + \partial \rho + \partial \sigma + \partial \tau + \partial \upsilon + \partial \phi + \partial \chi + \partial \psi + \partial \omega +$
$\partial \varepsilon + \partial \vartheta + \partial \varpi + \partial \varrho + \partial \varsigma + \partial \varphi +$
$\partial \mathrm{A} + \partial \mathrm{B} + \partial \Gamma + \partial \Delta + \partial \mathrm{E} + \partial \mathrm{Z} + \partial \mathrm{H} + \partial \Theta + \partial \mathrm{I} + \partial \mathrm{K} + \partial \Lambda + \partial \mathrm{M} +$
$\partial \mathrm{N} + \partial \Xi + \partial \mathrm{O} + \partial \Pi + \partial \mathrm{P} + \partial \Sigma + \partial \mathrm{T} + \partial \Upsilon + \partial \Phi + \partial \mathrm{X} + \partial \Psi + \partial \Omega +$

3.3.9 Slash Kerning Sans Serif

$1/A + 1/B + 1/C + 1/D + 1/E + 1/F + 1/G + 1/H + 1/I + 1/J + 1/K + 1/L + 1/M +$
$1/N + 1/O + 1/P + 1/Q + 1/R + 1/S + 1/T + 1/U + 1/V + 1/W + 1/X + 1/Y + 1/Z +$
$1/a + 1/b + 1/c + 1/d + 1/e + 1/f + 1/g + 1/h + 1/i + 1/j + 1/k + 1/l + 1/m +$
$1/n + 1/o + 1/p + 1/q + 1/r + 1/s + 1/t + 1/u + 1/v + 1/w + 1/x + 1/y + 1/z +$
$1/A + 1/B + 1/\mathit{\Gamma} + 1/\mathit{\Delta} + 1/E + 1/Z + 1/H + 1/\mathit{\Theta} + 1/I + 1/K + 1/\mathit{\Lambda} + 1/M +$
$1/N + 1/\mathit{\Xi} + 1/O + 1/\mathit{\Pi} + 1/P + 1/\mathit{\Sigma} + 1/T + 1/\mathit{\Upsilon} + 1/\mathit{\Phi} + 1/X + 1/\mathit{\Psi} + 1/\mathit{\Omega} +$
$1/\alpha + 1/\beta + 1/\gamma + 1/\delta + 1/\epsilon + 1/\zeta + 1/\eta + 1/\theta + 1/\iota + 1/\kappa + 1/\lambda + 1/\mu +$
$1/\nu + 1/\xi + 1/o + 1/\pi + 1/\rho + 1/\sigma + 1/\tau + 1/\upsilon + 1/\phi + 1/\chi + 1/\psi + 1/\omega +$
$1/\varepsilon + 1/\vartheta + 1/\varpi + 1/\varrho + 1/\varsigma + 1/\varphi +$

$$A/2+B/2+C/2+D/2+E/2+F/2+G/2+H/2+I/2+J/2+K/2+L/2+M/2+N/2+O/2+P/2+Q/2+R/2+S/2+T/2+U/2+V/2+W/2+X/2+Y/2+Z/2+a/2+b/2+c/2+d/2+e/2+f/2+g/2+h/2+i/2+j/2+k/2+l/2+m/2+n/2+o/2+p/2+q/2+r/2+s/2+t/2+u/2+v/2+w/2+x/2+y/2+z/2+\mathrm{A}/2+\mathrm{B}/2+\Gamma/2+\Delta/2+\mathrm{E}/2+\mathrm{Z}/2+\mathrm{H}/2+\Theta/2+\mathrm{I}/2+\mathrm{K}/2+\Lambda/2+\mathrm{M}/2+\mathrm{N}/2+\Xi/2+\mathrm{O}/2+\Pi/2+\mathrm{P}/2+\Sigma/2+\mathrm{T}/2+\Upsilon/2+\Phi/2+\mathrm{X}/2+\Psi/2+\Omega/2+\alpha/2+\beta/2+\gamma/2+\delta/2+\epsilon/2+\zeta/2+\eta/2+\theta/2+\iota/2+\kappa/2+\lambda/2+\mu/2+\nu/2+\xi/2+o/2+\pi/2+\rho/2+\sigma/2+\tau/2+\upsilon/2+\phi/2+\chi/2+\psi/2+\omega/2+\varepsilon/2+\vartheta/2+\varpi/2+\varrho/2+\varsigma/2+\varphi/2+$$

3.3.10 (Big) Operators Sans Serif

$\sum_{i=1}^n x^n$ $\prod_{i=1}^n x^n$ $\coprod_{i=1}^n x^n$ $\int_{i=1}^n x^n$ $\oint_{i=1}^n x^n$ $\bigotimes_{i=1}^n x^n$ $\bigoplus_{i=1}^n x^n$ $\bigodot_{i=1}^n x^n$ $\bigwedge_{i=1}^n x^n$ $\bigvee_{i=1}^n x^n$ $\biguplus_{i=1}^n x^n$ $\bigcup_{i=1}^n x^n$ $\bigcap_{i=1}^n x^n$ $\bigsqcup_{i=1}^n x^n$

$$\sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n$$

$$\bigotimes_{i=1}^n x^n \quad \bigoplus_{i=1}^n x^n \quad \bigodot_{i=1}^n x^n \quad \bigwedge_{i=1}^n x^n \quad \bigvee_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n$$

3.3.11 Radicals Sans Serif

$$\sqrt{x+y} \quad \sqrt{x^2+y^2} \quad \sqrt{x_i^2+y_j^2} \quad \sqrt{\left(\frac{\cos x}{2}\right)} \quad \sqrt{\left(\frac{\sin x}{2}\right)}$$

$$\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}$$

3.3.12 Over- and Underbraces Sans Serif

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.3.13 Normal and Wide Accents Sans Serif

$$\dot{x} \quad \ddot{x} \quad \vec{x} \quad \bar{x} \quad \overline{x} \quad \overline{xx} \quad \tilde{x} \quad \widetilde{x} \quad \widetilde{xx} \quad \widetilde{xxx} \quad \hat{x} \quad \widehat{x} \quad \widehat{xx} \quad \widehat{xxx}$$

$$\hat{x} \quad \check{x} \quad \tilde{x} \quad \acute{x} \quad \grave{x} \quad \dot{x} \quad \ddot{x} \quad \breve{x} \quad \bar{x} \quad \vec{x}$$

3.3.14 Long Arrows Sans Serif

$$\longleftarrow\!\!\!\longrightarrow \quad \leftrightarrow \quad \longleftarrow \quad \longrightarrow \quad \longleftrightarrow \quad \Longleftarrow\!\!\!\Longrightarrow \quad \Leftrightarrow \quad \Longleftarrow \quad \Longrightarrow \quad \Longleftrightarrow$$

3.3.15 Left and Right Delimiters Sans Serif

$$-(f)- -[f]- -\lfloor f\rfloor- -\lceil f\rceil- -\langle f\rangle- -\{f\}-$$

Using \left and \right.

$$-\left(f\right)- -\left[f\right]- -\left\lfloor f\right\rfloor- -\left\lceil f\right\rceil- -\left\langle f\right\rangle- -\left\{f\right\}-$$

$$-\left)f\right(- -\left]f\right[- -\left/f\right/- -\left\backslash f\right\backslash- -\left/f\right\backslash- -\left\backslash f\right/-$$

3.3.16 Big-g-g Delimiters Sans Serif

$$-\Bigg\lfloor\bigg\lfloor\Big\lfloor\big\lfloor\lfloor-\rfloor\big\rfloor\Big\rfloor\bigg\rfloor\Bigg\rfloor- \qquad -\Bigg(\bigg(\Big(\big((-)\big)\Big)\bigg)\Bigg)-$$

$$-\Bigg\lceil\bigg\lceil\Big\lceil\big\lceil\lceil-\rceil\big\rceil\Big\rceil\bigg\rceil\Bigg\rceil- \qquad -\Bigg\{\bigg\{\Big\{\big\{\{-\}\big\}\Big\}\bigg\}\Bigg\}-$$

$$-\Bigg[\bigg[\Big[\big[[-]\big]\Big]\bigg]\Bigg]- \qquad -\Bigg(\bigg(\Big(\big((-)\big)\Big)\bigg)\Bigg)-$$

$$-\Bigg\langle\bigg\langle\Big\langle\big\langle\langle-\rangle\big\rangle\Big\rangle\bigg\rangle\Bigg\rangle- \qquad -\Bigg\lgroup\bigg\lgroup\Big\lgroup\big\lgroup\lgroup-\rgroup\big\rgroup\Big\rgroup\bigg\rgroup\Bigg\rgroup-$$

$$-\Bigg\uparrow\bigg\uparrow\Big\uparrow\big\uparrow\uparrow-\downarrow\big\downarrow\Big\downarrow\bigg\downarrow\Bigg\downarrow- \qquad -\Bigg\Uparrow\bigg\Uparrow\Big\Uparrow\big\Uparrow\Uparrow-\Downarrow\big\Downarrow\Big\Downarrow\bigg\Downarrow\Bigg\Downarrow-$$

3.3.17 Binary Operators Sans Serif

$x \pm y$	\pm	$x \cap y$	\cap	$x \diamond y$	\diamond	$x \oplus y$	\oplus
$x \mp y$	\mp	$x \cup y$	\cup	$x \bigtriangleup y$	\bigtriangleup	$x \ominus y$	\ominus
$x \times y$	\times	$x \uplus y$	\uplus	$x \bigtriangledown y$	\bigtriangledown	$x \otimes y$	\otimes
$x \div y$	\div	$x \sqcap y$	\sqcap	$x \triangleleft y$	\triangleleft	$x \oslash y$	\oslash
$x \ast y$	\ast	$x \sqcup y$	\sqcup	$x \triangleright y$	\triangleright	$x \odot y$	\odot
$x \star y$	\star	$x \vee y$	\vee	$x \lhd y$	\lhd	$x \bigcirc y$	\bigcirc
$x \circ y$	\circ	$x \wedge y$	\wedge	$x \rhd y$	\rhd	$x \dagger y$	\dagger
$x \bullet y$	\bullet	$x \setminus y$	\setminus	$x \unlhd y$	\unlhd	$x \ddagger y$	\ddagger
$x \cdot y$	\cdot	$x \wr y$	\wr	$x \unrhd y$	\unrhd	$x \S y$	\S
$x + y$	+	$x - y$	-	$x \amalg y$	\amalg	$x \P y$	\P

3.3.18 Relations Sans Serif

$x \leq y$	\leq	$x \geq y$	\geq	$x \equiv y$	\equiv	$x \models y$	\models
$x \prec y$	\prec	$x \succ y$	\succ	$x \sim y$	\sim	$x \perp y$	\perp
$x \preceq y$	\preceq	$x \succeq y$	\succeq	$x \simeq y$	\simeq	$x \mid y$	\mid
$x \ll y$	\ll	$x \gg y$	\gg	$x \asymp y$	\asymp	$x \parallel y$	\parallel
$x \subset y$	\subset	$x \supset y$	\supset	$x \approx y$	\approx	$x \bowtie y$	\bowtie
$x \subseteq y$	\subseteq	$x \supseteq y$	\supseteq	$x \cong y$	\cong	$x \Join y$	\Join
$x \sqsubset y$	\sqsubset	$x \sqsupset y$	\sqsupset	$x \neq y$	\neq	$x \smile y$	\smile
$x \sqsubseteq y$	\sqsubseteq	$x \sqsupseteq y$	\sqsupseteq	$x \doteq y$	\doteq	$x \frown y$	\frown
$x \in y$	\in	$x \ni y$	\ni	$x \propto y$	\propto	$x = y$	=
$x \vdash y$	\vdash	$x \dashv y$	\dashv	$x < y$	<	$x > y$	>
$x : y$	:						

3.3.19 Punctuation Sans Serif

x, y	,	$x; y$	;	$x \colon y$	\colon	$x \ldotp y$	\ldotp	$x \cdotp y$	\cdotp

3.3.20 Arrows Sans Serif

$x \leftarrow y$	\leftarrow	$x \longleftarrow y$	\longleftarrow	$x \uparrow y$	\uparrow
$x \Leftarrow y$	\Leftarrow	$x \Longleftarrow y$	\Longleftarrow	$x \Uparrow y$	\Uparrow
$x \rightarrow y$	\rightarrow	$x \longrightarrow y$	\longrightarrow	$x \downarrow y$	\downarrow
$x \Rightarrow y$	\Rightarrow	$x \Longrightarrow y$	\Longrightarrow	$x \Downarrow y$	\Downarrow
$x \leftrightarrow y$	\leftrightarrow	$x \longleftrightarrow y$	\longleftrightarrow	$x \updownarrow y$	\updownarrow
$x \Leftrightarrow y$	\Leftrightarrow	$x \Longleftrightarrow y$	\Longleftrightarrow	$x \Updownarrow y$	\Updownarrow
$x \mapsto y$	\mapsto	$x \longmapsto y$	\longmapsto	$x \nearrow y$	\nearrow
$x \hookleftarrow y$	\hookleftarrow	$x \hookrightarrow y$	\hookrightarrow	$x \searrow y$	\searrow
$x \leftharpoonup y$	\leftharpoonup	$x \rightharpoonup y$	\rightharpoonup	$x \swarrow y$	\swarrow
$x \leftharpoondown y$	\leftharpoondown	$x \rightharpoondown y$	\rightharpoondown	$x \nwarrow y$	\nwarrow
$x \rightleftharpoons y$	\rightleftharpoons	$x \leadsto y$	\leadsto		

3.3.21 Miscellaneous Symbols Sans Serif

<table>
<tr><td>$x \ldots y$</td><td>\ldots</td><td>$x \cdots y$</td><td>\cdots</td><td>$x \vdots y$</td><td>\vdots</td><td>$x \ddots y$</td><td>\ddots</td></tr>
<tr><td>$x\aleph y$</td><td>\aleph</td><td>$x\prime y$</td><td>\prime</td><td>$x\forall y$</td><td>\forall</td><td>$x\infty y$</td><td>\infty</td></tr>
<tr><td>$x\hbar y$</td><td>\hbar</td><td>$x\emptyset y$</td><td>\emptyset</td><td>$x\exists y$</td><td>\exists</td><td>$x\Box y$</td><td>\Box</td></tr>
<tr><td>$x\imath y$</td><td>\imath</td><td>$x\nabla y$</td><td>\nabla</td><td>$x\neg y$</td><td>\neg</td><td>$x\Diamond y$</td><td>\Diamond</td></tr>
<tr><td>$x\jmath y$</td><td>\jmath</td><td>$x\surd y$</td><td>\surd</td><td>$x\flat y$</td><td>\flat</td><td>$x\triangle y$</td><td>\triangle</td></tr>
<tr><td>$x\ell y$</td><td>\ell</td><td>$x\top y$</td><td>\top</td><td>$x\natural y$</td><td>\natural</td><td>$x\clubsuit y$</td><td>\clubsuit</td></tr>
<tr><td>$x\wp y$</td><td>\wp</td><td>$x\bot y$</td><td>\bot</td><td>$x\sharp y$</td><td>\sharp</td><td>$x\diamondsuit y$</td><td>\diamondsuit</td></tr>
<tr><td>$x\Re y$</td><td>\Re</td><td>$x\| y$</td><td>\|</td><td>$x\backslash y$</td><td>\backslash</td><td>$x\heartsuit y$</td><td>\heartsuit</td></tr>
<tr><td>$x\Im y$</td><td>\Im</td><td>$x\angle y$</td><td>\angle</td><td>$x\partial y$</td><td>\partial</td><td>$x\spadesuit y$</td><td>\spadesuit</td></tr>
<tr><td>$x\mho y$</td><td>\mho</td><td>$x.y$</td><td>.</td><td>$x|y$</td><td>|</td><td>$x!y$</td><td>!</td></tr>
</table>

3.3.22 Variable-Sized Operators Sans Serif

$x \sum y$	\sum	$x \bigcap y$	\bigcap	$x \bigodot y$	\bigodot
$x \prod y$	\prod	$x \bigcup y$	\bigcup	$x \bigotimes y$	\bigotimes
$x \coprod y$	\coprod	$x \bigsqcup y$	\bigsqcup	$x \bigoplus y$	\bigoplus
$x \int y$	\int	$x \bigvee y$	\bigvee	$x \biguplus y$	\biguplus
$x \oint y$	\oint	$x \bigwedge y$	\bigwedge		

3.3.23 Log-Like Operators Sans Serif

$x \arccos y$	$x \cos y$	$x \csc y$	$x \exp y$	$x \ker y$	$x \limsup y$	$x \min y$	$x \sinh y$
$x \arcsin y$	$x \cosh y$	$x \deg y$	$x \gcd y$	$x \lg y$	$x \ln y$	$x \Pr y$	$x \sup y$
$x \arctan y$	$x \cot y$	$x \det y$	$x \hom y$	$x \lim y$	$x \log y$	$x \sec y$	$x \tan y$
$x \arg y$	$x \coth y$	$x \dim y$	$x \inf y$	$x \liminf y$	$x \max y$	$x \sin y$	$x \tanh y$

3.3.24 Delimiters Sans Serif

$x(y$	(	$x)y$	)	$x\uparrow y$	\uparrow	$x\Uparrow y$	\Uparrow
$x[y$	[	$x]y$	]	$x\downarrow y$	\downarrow	$x\Downarrow y$	\Downarrow
$x\{y$	\{	$x\}y$	\}	$x\updownarrow y$	\updownarrow	$x\Updownarrow y$	\Updownarrow
$x\lfloor y$	\lfloor	$x\rfloor y$	\rfloor	$x\lceil y$	\lceil	$x\rceil y$	\rceil
$x\langle y$	\langle	$x\rangle y$	\rangle	x/y	/	$x\backslash y$	\backslash
$x\vert y$	|	$x\Vert y$	\\|				

3.3.25 Large Delimiters Sans Serif

$\rmoustache$	\rmoustache	$\lmoustache$	\lmoustache	$\rgroup$	\rgroup	$\lgroup$	\lgroup
$\arrowvert$	\arrowvert	$\Arrowvert$	\Arrowvert	$\bracevert$	\bracevert		

3.3.26 Math Mode Accents Sans Serif

$\hat{a}$	\hat{a}	$\acute{a}$	\acute{a}	$\bar{a}$	\bar{a}	$\dot{a}$	\dot{a}	$\breve{a}$	\breve{a}
$\check{a}$	\check{a}	$\grave{a}$	\grave{a}	$\vec{a}$	\vec{a}	$\ddot{a}$	\ddot{a}	$\tilde{a}$	\tilde{a}

3.3.27 Miscellaneous Constructions Sans Serif

$\widetilde{abc}$	\widetilde{abc}	$\widehat{abc}$	\widehat{abc}
$\overleftarrow{abc}$	\overleftarrow{abc}	$\overrightarrow{abc}$	\overrightarrow{abc}
$\overline{abc}$	\overline{abc}	$\underline{abc}$	\underline{abc}
$\overbrace{abc}$	\overbrace{abc}	$\underbrace{abc}$	\underbrace{abc}
$\sqrt{abc}$	\sqrt{abc}	$\sqrt[n]{abc}$	\sqrt[n]{abc}
f'	f'	$\frac{abc}{xyz}$	\frac{abc}{xyz}

3.3.28 AMS Delimiters Sans Serif

$x\ulcorner y$	\ulcorner	$x\urcorner y$	\urcorner	$x\llcorner y$	\llcorner	$x\lrcorner y$	\lrcorner

3.3.29 AMS Arrows Sans Serif

$x \dashrightarrow y$	\dashrightarrow	$x \dashleftarrow y$	\dashleftarrow
$x \leftleftarrows y$	\leftleftarrows	$x \leftrightarrows y$	\leftrightarrows
$x \Lleftarrow y$	\Lleftarrow	$x \twoheadleftarrow y$	\twoheadleftarrow
$x \leftarrowtail y$	\leftarrowtail	$x \looparrowleft y$	\looparrowleft
$x \leftrightharpoons y$	\leftrightharpoons	$x \curvearrowleft y$	\curvearrowleft
$x \circlearrowleft y$	\circlearrowleft	$x \Lsh y$	\Lsh
$x \upuparrows y$	\upuparrows	$x \upharpoonleft y$	\upharpoonleft
$x \downharpoonleft y$	\downharpoonleft	$x \multimap y$	\multimap
$x \leftrightsquigarrow y$	\leftrightsquigarrow	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \twoheadrightarrow y$	\twoheadrightarrow
$x \rightarrowtail y$	\rightarrowtail	$x \looparrowright y$	\looparrowright
$x \rightleftharpoons y$	\rightleftharpoons	$x \curvearrowright y$	\curvearrowright
$x \circlearrowright y$	\circlearrowright	$x \Rsh y$	\Rsh
$x \downdownarrows y$	\downdownarrows	$x \upharpoonright y$	\upharpoonright
$x \downharpoonright y$	\downharpoonright	$x \rightsquigarrow y$	\rightsquigarrow

3.3.30 AMS Negated Arrows Sans Serif

$x \nleftarrow y$	\nleftarrow	$x \nrightarrow y$	\nrightarrow
$x \nLeftarrow y$	\nLeftarrow	$x \nRightarrow y$	\nRightarrow
$x \nleftrightarrow y$	\nleftrightarrow	$x \nLeftrightarrow y$	\nLeftrightarrow

3.3.31 AMS Greek Sans Serif

$x\digamma y$	\digamma	$x\varkappa y$	\varkappa

3.3.32 AMS Hebrew Sans Serif

$x\beth y$	\beth	$x\daleth y$	\daleth	$x\gimel y$	\gimel

3.3.33 AMS Miscellaneous Sans Serif

$x\hbar y$	\hbar	$x\hslash y$	\hslash
$x \vartriangle y$	\vartriangle	$x\triangledown y$	\triangledown
$x\square y$	\square	$x\lozenge y$	\lozenge
$x\circledS y$	\circledS	$x\angle y$	\angle
$x\measuredangle y$	\measuredangle	$x\nexists y$	\nexists
$x\mho y$	\mho	$x\Finv y$	\Finv[u]
$x\Game y$	\Game[u]	$x\Bbbk y$	\Bbbk[u]
$x\backprime y$	\backprime	$x\varnothing y$	\varnothing
$x\blacktriangle y$	\blacktriangle	$x\blacktriangledown y$	\blacktriangledown
$x\blacksquare y$	\blacksquare	$x\blacklozenge y$	\blacklozenge
$x\bigstar y$	\bigstar	$x\sphericalangle y$	\sphericalangle
$x\complement y$	\complement	$x\eth y$	\eth
$x\diagup y$	\diagup[u]	$x\diagdown y$	\diagdown[u]

[u] Not defined in amssymb.sty, define using the \newsymbol command.

3.3.34 AMS Binary Operators Sans Serif

$x \dotplus y$	\dotplus	$x \smallsetminus y$	\smallsetminus
$x \Cap y$	\Cap	$x \Cup y$	\Cup
$x \barwedge y$	\barwedge	$x \veebar y$	\veebar
$x \doublebarwedge y$	\doublebarwedge	$x \boxminus y$	\boxminus
$x \boxtimes y$	\boxtimes	$x \boxdot y$	\boxdot
$x \boxplus y$	\boxplus	$x \divideontimes y$	\divideontimes
$x \ltimes y$	\ltimes	$x \rtimes y$	\rtimes
$x \leftthreetimes y$	\leftthreetimes	$x \rightthreetimes y$	\rightthreetimes
$x \curlywedge y$	\curlywedge	$x \curlyvee y$	\curlyvee
$x \circleddash y$	\circleddash	$x \circledast y$	\circledast
$x \circledcirc y$	\circledcirc	$x \centerdot y$	\centerdot
$x \intercal y$	\intercal		

3.3.35 AMS Relations Sans Serif

$x \leqslant y$	\leqslant
$x \lesssim y$	\lesssim
$x \approxeq y$	\approxeq
$x \lll y$	\lll
$x \lesseqgtr y$	\lesseqgtr
$x \doteqdot y$	\doteqdot
$x \fallingdotseq y$	\fallingdotseq
$x \backsimeq y$	\backsimeq
$x \Subset y$	\Subset
$x \preccurlyeq y$	\preccurlyeq
$x \precsim y$	\precsim
$x \vartriangleleft y$	\vartriangleleft
$x \vDash y$	\vDash
$x \smallsmile y$	\smallsmile
$x \bumpeq y$	\bumpeq
$x \geqq y$	\geqq
$x \eqslantgtr y$	\eqslantgtr
$x \gtrapprox y$	\gtrapprox
$x \ggg y$	\ggg
$x \gtreqless y$	\gtreqless
$x \eqcirc y$	\eqcirc
$x \triangleq y$	\triangleq
$x \thickapprox y$	\thickapprox
$x \Supset y$	\Supset
$x \succcurlyeq y$	\succcurlyeq
$x \succsim y$	\succsim
$x \vartriangleright y$	\vartriangleright
$x \Vdash y$	\Vdash
$x \shortparallel y$	\shortparallel
$x \pitchfork y$	\pitchfork
$x \blacktriangleleft y$	\blacktriangleleft
$x \backepsilon y$	\backepsilon
$x \because y$	\because

3.3.36 AMS Negated Relations Sans Serif

$x \nless y$	\nless	$x \nleq y$	\nleq
$x \nleqslant y$	\nleqslant	$x \nleqq y$	\nleqq
$x \lneq y$	\lneq	$x \lneqq y$	\lneqq
$x \lvertneqq y$	\lvertneqq	$x \lnsim y$	\lnsim
$x \lnapprox y$	\lnapprox	$x \nprec y$	\nprec
$x \npreceq y$	\npreceq	$x \precnsim y$	\precnsim
$x \precnapprox y$	\precnapprox	$x \nsim y$	\nsim
$x \nshortmid y$	\nshortmid	$x \nmid y$	\nmid
$x \nvdash y$	\nvdash	$x \nvDash y$	\nvDash
$x \ntriangleleft y$	\ntriangleleft	$x \ntrianglelefteq y$	\ntrianglelefteq
$x \nsubseteq y$	\nsubseteq	$x \subsetneq y$	\subsetneq
$x \varsubsetneq y$	\varsubsetneq	$x \subsetneqq y$	\subsetneqq
$x \varsubsetneqq y$	\varsubsetneqq	$x \ngtr y$	\ngtr
$x \ngeq y$	\ngeq	$x \ngeqslant y$	\ngeqslant
$x \ngeqq y$	\ngeqq	$x \gneq y$	\gneq
$x \gneqq y$	\gneqq	$x \gvertneqq y$	\gvertneqq
$x \gnsim y$	\gnsim	$x \gnapprox y$	\gnapprox
$x \nsucc y$	\nsucc	$x \nsucceq y$	\nsucceq
$x \nsucceq y$	\nsucceqq	$x \succnsim y$	\succnsim
$x \succnapprox y$	\succnapprox	$x \ncong y$	\ncong
$x \nshortparallel y$	\nshortparallel	$x \nparallel y$	\nparallel
$x \nvDash y$	\nvDash	$x \nVDash y$	\nVDash
$x \ntriangleright y$	\ntriangleright	$x \ntrianglerighteq y$	\ntrianglerighteq
$x \nsupseteq y$	\nsupseteq	$x \nsupseteqq y$	\nsupseteqq
$x \supsetneq y$	\supsetneq	$x \varsupsetneq y$	\varsupsetneq
$x \supsetneqq y$	\supsetneqq	$x \varsupsetneqq y$	\varsupsetneqq

3.3.37 Math “Torture” Test Sans Serif

Most of the following examples are taken from *The TEXbook* (Knuth 1984, see https://ctan.org/pkg/texbook) and were adapted for LATEX from Karl Berry’s torture test for plain TEX math fonts.

$x + y - z, \quad x + y * z, \quad z * y / z, \quad (x + y)(x - y) = x^2 - y^2,$

$x \times y \cdot z = [x\, y\, z], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$

$x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y/z, \quad x := y, \quad x \le y \ne z, \quad x \sim y \simeq z\, x \equiv y \not\equiv z, \quad x \subset y \subseteq z$

$\sin 2\theta = 2 \sin\theta \cos\theta, \quad \mathrm{O}(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),$

$\bigl(x \in A(n) \bigm| x \in B(n)\bigr), \quad \bigcup_n X_n \bigm\| \bigcap_n Y_n$

In-text matrices $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}a&b&c\\1&m&n\end{smallmatrix}\right)$.

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x} \frac{1}{1-x^2} = \frac{a+1}{b} \bigg/ \frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \left|\phi(x+\mathrm{i}y)\right|^2 = 0$$

$$\pi(n) = \sum_{m=2}^{n} \left\lfloor \left(\sum_{k=1}^{m-1} \lfloor (m/k) / \lceil m/k \rceil \rfloor \right)^{-1} \right\rfloor.$$

$$\int_0^\infty \frac{t - \mathrm{i}b}{t^2 + b^2} e^{\mathrm{i}at} \,\mathrm{d}t = e^{ab} E_1(ab), \quad a, b > 0.$$

$$\boldsymbol{A} := \begin{pmatrix} x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda \end{pmatrix}.$$

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \begin{pmatrix} u & x \\ v & y \\ w & z \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

$$\boldsymbol{M} = \begin{array}{c c} & \begin{array}{c c c} C & I & C' \end{array} \\ \begin{array}{c} C \\ I \\ C' \end{array} & \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{array}$$

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{converges if} \quad |z| < \left(\limsup_{n\to\infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x+\Delta x)-f(x)}{\Delta x} \to f'(x) \qquad \text{as } \Delta x \to 0.$$

$$\|u_i\| = 1, \qquad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of} \quad \left\{\begin{array}{c}\text{an arc}\\ \text{a circle}\\ \text{a fan}\end{array}\right\} \quad \text{is} \quad \left\{\begin{array}{c}\text{an arc}\\ \text{an arc or a circle}\\ \text{a fan or an arc}\end{array}\right\}.$$

$$\begin{aligned} T(n) \le T(2^{\lceil \lg n \rceil}) &\le c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\ &< 3c \cdot 3^{\lg n} \\ &= 3c\, n^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2 - xy + yx - y^2 \\ &= x^2 - y^2 \\ (x+y)^2 &= x^2 + 2xy + y^2. \end{aligned}$$

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2}\, dx\right)^2 &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\, dx\, dy \\ &= \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\, dr\, d\theta \\ &= \int_0^{2\pi} \left(e^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) d\theta \\ &= \pi. \end{aligned}$$

$$\prod_{k\ge 0} \frac{1}{(1-q^k z)} = \sum_{n\ge 0} z^n \Bigg/ \prod_{1\le k\le n} (1-q^k).$$

$$\sum_{\substack{0<i\le m\\ 0<j\le n}} p(i,j) \neq \sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r} a_{ij}b_{jk}c_{ki} \neq \sum_{\substack{1\le i\le p\\ 1\le j\le q\\ 1\le k\le r}} a_{ij}b_{jk}c_{ki}$$

$$\max_{1\le n\le m} \log_2 P_n \quad \text{and} \quad \lim_{x\to 0} \frac{\sin x}{x} = 1$$

Inline math: $\max_{1\le n\le m} \log_2 P_n$ and $\lim_{x\to 0} \frac{\sin x}{x} = 1$

$$p_1(n) = \lim_{m\to\infty} \sum_{\nu=0}^{\infty} \bigl(1 - \cos^{2m}(\nu!^n \pi/n)\bigr)$$

Inline math: $p_1(n) = \lim_{m\to\infty} \sum_{\nu=0}^{\infty} \bigl(1 - \cos^{2m}(\nu!^n \pi/n)\bigr)$

3.4 Math Test Sans Serif Bold

3.4.1 Overview Sans Serif Bold

Default: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$; $\sigma_\varepsilon, c^\alpha$
mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: αααbβGΓΓεεθθPΠΣσ
mathit: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathbf: $\boldsymbol{\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: **ααbβGΓΓεεθθPΠΣσ**

Default: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$; $\sigma_\varepsilon, c^\alpha$
mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: αααbβGΓΓεεθθPΠΣσ
mathit: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathbf: $\boldsymbol{\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: **αααbβGΓΓεεθθPΠΣσ**

Default: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$; $\sigma_\varepsilon, c^\alpha$
mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: αααbβGΓΓεεθθPΠΣσ
mathit: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathbf: $\boldsymbol{\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: **αααbβGΓΓεεθθPΠΣσ**

Default: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$; $\sigma_\varepsilon, c^\alpha$
mathnormal: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathrm: $\mathrm{a}\alpha\alpha\mathrm{b}\beta\mathrm{G}\Gamma\Gamma\epsilon\varepsilon\theta\vartheta\mathrm{P}\Pi\Sigma\sigma$
mathup: αααbβGΓΓεεθθPΠΣσ
mathit: $\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma$
mathbf: $\boldsymbol{\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfit: $\boldsymbol{\alpha\alpha\alpha b\beta G\Gamma\Gamma\varepsilon\varepsilon\theta\theta P\Pi\Sigma\sigma}$
mathbfup: **αααbβGΓΓεεθθPΠΣσ**

3.4.2 Formulas Sans Serif Bold

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \digamma, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F,$

α, β, γ, δ, ε, ε, ζ, η, θ, θ, ι, κ, λ, μ, ν, ξ, ο, π, π, ρ, ρ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, ϝ, A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Y, Φ, X, Ψ, Ω, F,
α, β, γ, δ, ε, ε, ζ, η, θ, θ, ι, κ, λ, μ, ν, ξ, ο, π, π, ρ, ρ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, ϝ, A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Y, Φ, X, Ψ, Ω, F,
α, β, γ, δ, ε, ε, ζ, η, θ, θ, ι, κ, λ, μ, ν, ξ, ο, π, π, ρ, ρ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, ϝ, A, B, Γ, Δ, E, Z, H, Θ, I, K, Λ, M, N, Ξ, O, Π, P, Σ, T, Y, Φ, X, Ψ, Ω, F,
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{v \to \infty} v(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$
$\hat{\beta} = (X'X)^{-1}X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \left(\frac{27}{2}\right)$$

Disambiguation: 0 O *O*, 1 l I | *l I /*, *i j*, *rn m*, *θ Θ*, *ϕ ψ*, – —
Latin vs. Greek: *a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, ß β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.*
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{v \to \infty} v(v) = \max_{s \in S}\{s \pm 3\gamma + y - 1\} = 4 \times 7$
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Disambiguation: 0 O *O*, 1 l I | *l I /*, *i j*, *rn m*, *θ Θ*, *ϕ ψ*, – —
Latin vs. Greek: *a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, ß β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.*
$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
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$$\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \left(\frac{27}{2}\right)$$

Disambiguation: 0 O *O*, 1 l I | *l I /*, *i j*, *rn m*, *θ Θ*, *ϕ ψ*, – —
Latin vs. Greek: *a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, ß β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.*

$\alpha a > 0, \beta b + (3 \times 27), \Gamma G = 7 < 8, \lambda$
$\lim_{v \to \infty} v(v) = \max_{s \in S} \{s \pm 3\gamma + y - 1\} = 4 \times 7$
$\hat{\beta} = (X'X)^{-1} X'y$

$$\lim_{N \to \infty} \sum_{i=0}^{N} x^i = \min_{x \in \mathbb{R}} S(x)$$

$$\int_{-\infty}^{\infty} x f(x) \, dx = \binom{27}{2}$$

Disambiguation: 0 O O, 1 l I | l I /, i j, rn m, θ Θ, ϕ ψ, − –
Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, β β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.

3.4.3 Math Alphabets Sans Serif Bold

Default

$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$
$A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$
$a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$
$A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,$
$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \epsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Normal (`\mathnormal`)

$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$
$A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$
$a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$
$A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,$
$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \epsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Italic (`\mathit`)

$0, 1, 2, 3, 4, 5, 6, 7, 8, 9,$
$A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,$
$a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,$
$A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,$
$\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \epsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$

Math Roman (`\mathrm`)

$$\mathrm{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$$
$$\mathrm{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$$
$$\mathrm{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$$
$$\mathrm{A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,}$$
$$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \mathrm{o}, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi,$$

Math Bold (`\mathbf`)

$$\mathbf{0, 1, 2, 3, 4, 5, 6, 7, 8, 9,}$$
$$\boldsymbol{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z,}$$
$$\boldsymbol{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,}$$
$$\boldsymbol{A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega,}$$
$$\boldsymbol{\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega, \varepsilon, \theta, \pi, \rho, \varsigma, \phi,}$$

Caligraphic (`\mathcal`)

$$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{K}, \mathcal{L}, \mathcal{M}, \mathcal{N}, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{S}, \mathcal{T}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z},$$

Script (`\mathscr`)

$$\mathscr{A}, \mathscr{B}, \mathscr{C}, \mathscr{D}, \mathscr{E}, \mathscr{F}, \mathscr{G}, \mathscr{H}, \mathscr{I}, \mathscr{J}, \mathscr{K}, \mathscr{L}, \mathscr{M}, \mathscr{N}, \mathscr{O}, \mathscr{P}, \mathscr{Q}, \mathscr{R}, \mathscr{S}, \mathscr{T}, \mathscr{U}, \mathscr{V}, \mathscr{W}, \mathscr{X}, \mathscr{Y}, \mathscr{Z},$$

Fraktur (`\mathfrak`)

$$\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{I}, \mathfrak{J}, \mathfrak{K}, \mathfrak{L}, \mathfrak{M}, \mathfrak{N}, \mathfrak{O}, \mathfrak{P}, \mathfrak{Q}, \mathfrak{R}, \mathfrak{S}, \mathfrak{T}, \mathfrak{U}, \mathfrak{V}, \mathfrak{W}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z},$$
$$\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{e}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n}, \mathfrak{o}, \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}, \mathfrak{t}, \mathfrak{u}, \mathfrak{v}, \mathfrak{w}, \mathfrak{x}, \mathfrak{y}, \mathfrak{z},$$

Blackboard Bold (`\mathbb`)

$$\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}, \mathbb{E}, \mathbb{F}, \mathbb{G}, \mathbb{H}, \mathbb{I}, \mathbb{J}, \mathbb{K}, \mathbb{L}, \mathbb{M}, \mathbb{N}, \mathbb{O}, \mathbb{P}, \mathbb{Q}, \mathbb{R}, \mathbb{S}, \mathbb{T}, \mathbb{U}, \mathbb{V}, \mathbb{W}, \mathbb{X}, \mathbb{Y}, \mathbb{Z},$$

3.4.4 Character Sidebearings Sans Serif Bold

Default

$$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$$
$$|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$$
$$|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$$
$$|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$$
$$|A| + |B| + |\mathit{\Gamma}| + |\mathit{\Delta}| + |E| + |Z| + |H| + |\mathit{\Theta}| + |I| + |K| + |\mathit{\Lambda}| + |M| +$$
$$|N| + |\mathit{\Xi}| + |O| + |\mathit{\Pi}| + |P| + |\mathit{\Sigma}| + |T| + |\mathit{\Upsilon}| + |\mathit{\Phi}| + |X| + |\mathit{\Psi}| + |\mathit{\Omega}| +$$
$$|\alpha| + |\beta| + |\gamma| + |\delta| + |\epsilon| + |\zeta| + |\eta| + |\theta| + |\iota| + |\kappa| + |\lambda| + |\mu| +$$
$$|\nu| + |\xi| + |o| + |\pi| + |\rho| + |\sigma| + |\tau| + |\upsilon| + |\phi| + |\chi| + |\psi| + |\omega| +$$
$$|\varepsilon| + |\vartheta| + |\varpi| + |\varrho| + |\varsigma| + |\varphi| +$$

Math Roman (`\mathrm`)

$$|\mathrm{A}| + |\mathrm{B}| + |\mathrm{C}| + |\mathrm{D}| + |\mathrm{E}| + |\mathrm{F}| + |\mathrm{G}| + |\mathrm{H}| + |\mathrm{I}| + |\mathrm{J}| + |\mathrm{K}| + |\mathrm{L}| + |\mathrm{M}| +$$
$$|\mathrm{N}| + |\mathrm{O}| + |\mathrm{P}| + |\mathrm{Q}| + |\mathrm{R}| + |\mathrm{S}| + |\mathrm{T}| + |\mathrm{U}| + |\mathrm{V}| + |\mathrm{W}| + |\mathrm{X}| + |\mathrm{Y}| + |\mathrm{Z}| +$$
$$|\mathrm{a}| + |\mathrm{b}| + |\mathrm{c}| + |\mathrm{d}| + |\mathrm{e}| + |\mathrm{f}| + |\mathrm{g}| + |\mathrm{h}| + |\mathrm{i}| + |\mathrm{j}| + |\mathrm{k}| + |\mathrm{l}| + |\mathrm{m}| +$$
$$|\mathrm{n}| + |\mathrm{o}| + |\mathrm{p}| + |\mathrm{q}| + |\mathrm{r}| + |\mathrm{s}| + |\mathrm{t}| + |\mathrm{u}| + |\mathrm{v}| + |\mathrm{w}| + |\mathrm{x}| + |\mathrm{y}| + |\mathrm{z}| +$$
$$|\mathrm{A}| + |\mathrm{B}| + |\Gamma| + |\Delta| + |\mathrm{E}| + |\mathrm{Z}| + |\mathrm{H}| + |\Theta| + |\mathrm{I}| + |\mathrm{K}| + |\Lambda| + |\mathrm{M}| +$$
$$|\mathrm{N}| + |\Xi| + |\mathrm{O}| + |\Pi| + |\mathrm{P}| + |\Sigma| + |\mathrm{T}| + |\Upsilon| + |\Phi| + |\mathrm{X}| + |\Psi| + |\Omega| +$$

Math Bold (`\mathbf`)

$$|A| + |B| + |C| + |D| + |E| + |F| + |G| + |H| + |I| + |J| + |K| + |L| + |M| +$$
$$|N| + |O| + |P| + |Q| + |R| + |S| + |T| + |U| + |V| + |W| + |X| + |Y| + |Z| +$$
$$|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i| + |j| + |k| + |l| + |m| +$$
$$|n| + |o| + |p| + |q| + |r| + |s| + |t| + |u| + |v| + |w| + |x| + |y| + |z| +$$
$$|A| + |B| + |\mathit{\Gamma}| + |\mathit{\Delta}| + |E| + |Z| + |H| + |\mathit{\Theta}| + |I| + |K| + |\mathit{\Lambda}| + |M| +$$
$$|N| + |\mathit{\Xi}| + |O| + |\mathit{\Pi}| + |P| + |\mathit{\Sigma}| + |T| + |\mathit{\Upsilon}| + |\mathit{\Phi}| + |X| + |\mathit{\Psi}| + |\mathit{\Omega}| +$$

Math Calligraphic (`\mathcal`)

$$|\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{D}| + |\mathcal{E}| + |\mathcal{F}| + |\mathcal{G}| + |\mathcal{H}| + |\mathcal{I}| + |\mathcal{J}| + |\mathcal{K}| + |\mathcal{L}| + |\mathcal{M}| +$$
$$|\mathcal{N}| + |\mathcal{O}| + |\mathcal{P}| + |\mathcal{Q}| + |\mathcal{R}| + |\mathcal{S}| + |\mathcal{T}| + |\mathcal{U}| + |\mathcal{V}| + |\mathcal{W}| + |\mathcal{X}| + |\mathcal{Y}| + |\mathcal{Z}| +$$

3.4.5 Superscript Positioning Sans Serif Bold

Default

$$A^2 + B^2 + C^2 + D^2 + E^2 + F^2 + G^2 + H^2 + I^2 + J^2 + K^2 + L^2 + M^2 +$$
$$N^2 + O^2 + P^2 + Q^2 + R^2 + S^2 + T^2 + U^2 + V^2 + W^2 + X^2 + Y^2 + Z^2 +$$
$$a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 + j^2 + k^2 + l^2 + m^2 +$$
$$n^2 + o^2 + p^2 + q^2 + r^2 + s^2 + t^2 + u^2 + v^2 + w^2 + x^2 + y^2 + z^2 +$$
$$A^2 + B^2 + \Gamma^2 + \Delta^2 + E^2 + Z^2 + H^2 + \Theta^2 + I^2 + K^2 + \Lambda^2 + M^2 +$$
$$N^2 + \Xi^2 + O^2 + \Pi^2 + P^2 + \Sigma^2 + T^2 + \Upsilon^2 + \Phi^2 + X^2 + \Psi^2 + \Omega^2 +$$
$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \epsilon^2 + \zeta^2 + \eta^2 + \theta^2 + \iota^2 + \kappa^2 + \lambda^2 + \mu^2 +$$
$$\nu^2 + \xi^2 + o^2 + \pi^2 + \rho^2 + \sigma^2 + \tau^2 + \upsilon^2 + \phi^2 + \chi^2 + \psi^2 + \omega^2 +$$
$$\varepsilon^2 + \vartheta^2 + \varpi^2 + \varrho^2 + \varsigma^2 + \varphi^2 +$$

Math Roman (`\mathrm`)

$$\mathrm{A}^2 + \mathrm{B}^2 + \mathrm{C}^2 + \mathrm{D}^2 + \mathrm{E}^2 + \mathrm{F}^2 + \mathrm{G}^2 + \mathrm{H}^2 + \mathrm{I}^2 + \mathrm{J}^2 + \mathrm{K}^2 + \mathrm{L}^2 + \mathrm{M}^2 +$$
$$\mathrm{N}^2 + \mathrm{O}^2 + \mathrm{P}^2 + \mathrm{Q}^2 + \mathrm{R}^2 + \mathrm{S}^2 + \mathrm{T}^2 + \mathrm{U}^2 + \mathrm{V}^2 + \mathrm{W}^2 + \mathrm{X}^2 + \mathrm{Y}^2 + \mathrm{Z}^2 +$$
$$\mathrm{a}^2 + \mathrm{b}^2 + \mathrm{c}^2 + \mathrm{d}^2 + \mathrm{e}^2 + \mathrm{f}^2 + \mathrm{g}^2 + \mathrm{h}^2 + \mathrm{i}^2 + \mathrm{j}^2 + \mathrm{k}^2 + \mathrm{l}^2 + \mathrm{m}^2 +$$
$$\mathrm{n}^2 + \mathrm{o}^2 + \mathrm{p}^2 + \mathrm{q}^2 + \mathrm{r}^2 + \mathrm{s}^2 + \mathrm{t}^2 + \mathrm{u}^2 + \mathrm{v}^2 + \mathrm{w}^2 + \mathrm{x}^2 + \mathrm{y}^2 + \mathrm{z}^2 +$$
$$\mathrm{A}^2 + \mathrm{B}^2 + \Gamma^2 + \Delta^2 + \mathrm{E}^2 + \mathrm{Z}^2 + \mathrm{H}^2 + \Theta^2 + \mathrm{I}^2 + \mathrm{K}^2 + \Lambda^2 + \mathrm{M}^2 +$$
$$\mathrm{N}^2 + \Xi^2 + \mathrm{O}^2 + \Pi^2 + \mathrm{P}^2 + \Sigma^2 + \mathrm{T}^2 + \Upsilon^2 + \Phi^2 + \mathrm{X}^2 + \Psi^2 + \Omega^2 +$$

Math Bold (`\mathbf`)

$$\mathbf{A}^2 + \mathbf{B}^2 + \mathbf{C}^2 + \mathbf{D}^2 + \mathbf{E}^2 + \mathbf{F}^2 + \mathbf{G}^2 + \mathbf{H}^2 + \mathbf{I}^2 + \mathbf{J}^2 + \mathbf{K}^2 + \mathbf{L}^2 + \mathbf{M}^2 +$$
$$\mathbf{N}^2 + \mathbf{O}^2 + \mathbf{P}^2 + \mathbf{Q}^2 + \mathbf{R}^2 + \mathbf{S}^2 + \mathbf{T}^2 + \mathbf{U}^2 + \mathbf{V}^2 + \mathbf{W}^2 + \mathbf{X}^2 + \mathbf{Y}^2 + \mathbf{Z}^2 +$$
$$\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2 + \mathbf{d}^2 + \mathbf{e}^2 + \mathbf{f}^2 + \mathbf{g}^2 + \mathbf{h}^2 + \mathbf{i}^2 + \mathbf{j}^2 + \mathbf{k}^2 + \mathbf{l}^2 + \mathbf{m}^2 +$$
$$\mathbf{n}^2 + \mathbf{o}^2 + \mathbf{p}^2 + \mathbf{q}^2 + \mathbf{r}^2 + \mathbf{s}^2 + \mathbf{t}^2 + \mathbf{u}^2 + \mathbf{v}^2 + \mathbf{w}^2 + \mathbf{x}^2 + \mathbf{y}^2 + \mathbf{z}^2 +$$
$$\mathbf{A}^2 + \mathbf{B}^2 + \boldsymbol{\Gamma}^2 + \boldsymbol{\Delta}^2 + \mathbf{E}^2 + \mathbf{Z}^2 + \mathbf{H}^2 + \boldsymbol{\Theta}^2 + \mathbf{I}^2 + \mathbf{K}^2 + \boldsymbol{\Lambda}^2 + \mathbf{M}^2 +$$
$$\mathbf{N}^2 + \boldsymbol{\Xi}^2 + \mathbf{O}^2 + \boldsymbol{\Pi}^2 + \mathbf{P}^2 + \boldsymbol{\Sigma}^2 + \mathbf{T}^2 + \boldsymbol{\Upsilon}^2 + \boldsymbol{\Phi}^2 + \mathbf{X}^2 + \boldsymbol{\Psi}^2 + \boldsymbol{\Omega}^2 +$$

Math Calligraphic (`\mathcal`)

$$\mathcal{A}^2 + \mathcal{B}^2 + \mathcal{C}^2 + \mathcal{D}^2 + \mathcal{E}^2 + \mathcal{F}^2 + \mathcal{G}^2 + \mathcal{H}^2 + \mathcal{I}^2 + \mathcal{J}^2 + \mathcal{K}^2 + \mathcal{L}^2 + \mathcal{M}^2 +$$
$$\mathcal{N}^2 + \mathcal{O}^2 + \mathcal{P}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{S}^2 + \mathcal{T}^2 + \mathcal{U}^2 + \mathcal{V}^2 + \mathcal{W}^2 + \mathcal{X}^2 + \mathcal{Y}^2 + \mathcal{Z}^2 +$$

3.4.6 Subscript Positioning Sans Serif Bold

Default

$$A_i + B_i + C_i + D_i + E_i + F_i + G_i + H_i + I_i + J_i + K_i + L_i + M_i +$$
$$N_i + O_i + P_i + Q_i + R_i + S_i + T_i + U_i + V_i + W_i + X_i + Y_i + Z_i +$$
$$a_i + b_i + c_i + d_i + e_i + f_i + g_i + h_i + i_i + j_i + k_i + l_i + m_i +$$
$$n_i + o_i + p_i + q_i + r_i + s_i + t_i + u_i + v_i + w_i + x_i + y_i + z_i +$$
$$A_i + B_i + \Gamma_i + \Delta_i + E_i + Z_i + H_i + \Theta_i + I_i + K_i + \Lambda_i + M_i +$$
$$N_i + \Xi_i + O_i + \Pi_i + P_i + \Sigma_i + T_i + \Upsilon_i + \Phi_i + X_i + \Psi_i + \Omega_i +$$
$$\alpha_i + \beta_i + \gamma_i + \delta_i + \epsilon_i + \zeta_i + \eta_i + \theta_i + \iota_i + \kappa_i + \lambda_i + \mu_i +$$
$$\nu_i + \xi_i + o_i + \pi_i + \rho_i + \sigma_i + \tau_i + \upsilon_i + \phi_i + \chi_i + \psi_i + \omega_i +$$
$$\varepsilon_i + \vartheta_i + \varpi_i + \varrho_i + \varsigma_i + \varphi_i +$$

Math Roman (`\mathrm`)

$$\mathrm{A}_i + \mathrm{B}_i + \mathrm{C}_i + \mathrm{D}_i + \mathrm{E}_i + \mathrm{F}_i + \mathrm{G}_i + \mathrm{H}_i + \mathrm{I}_i + \mathrm{J}_i + \mathrm{K}_i + \mathrm{L}_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \mathrm{O}_i + \mathrm{P}_i + \mathrm{Q}_i + \mathrm{R}_i + \mathrm{S}_i + \mathrm{T}_i + \mathrm{U}_i + \mathrm{V}_i + \mathrm{W}_i + \mathrm{X}_i + \mathrm{Y}_i + \mathrm{Z}_i +$$
$$\mathrm{a}_i + \mathrm{b}_i + \mathrm{c}_i + \mathrm{d}_i + \mathrm{e}_i + \mathrm{f}_i + \mathrm{g}_i + \mathrm{h}_i + \mathrm{i}_i + \mathrm{j}_i + \mathrm{k}_i + \mathrm{l}_i + \mathrm{m}_i +$$
$$\mathrm{n}_i + \mathrm{o}_i + \mathrm{p}_i + \mathrm{q}_i + \mathrm{r}_i + \mathrm{s}_i + \mathrm{t}_i + \mathrm{u}_i + \mathrm{v}_i + \mathrm{w}_i + \mathrm{x}_i + \mathrm{y}_i + \mathrm{z}_i +$$
$$\mathrm{A}_i + \mathrm{B}_i + \Gamma_i + \Delta_i + \mathrm{E}_i + \mathrm{Z}_i + \mathrm{H}_i + \Theta_i + \mathrm{I}_i + \mathrm{K}_i + \Lambda_i + \mathrm{M}_i +$$
$$\mathrm{N}_i + \Xi_i + \mathrm{O}_i + \Pi_i + \mathrm{P}_i + \Sigma_i + \mathrm{T}_i + \Upsilon_i + \Phi_i + \mathrm{X}_i + \Psi_i + \Omega_i +$$

Math Bold (`\mathbf`)

$$\mathbf{A}_i + \mathbf{B}_i + \mathbf{C}_i + \mathbf{D}_i + \mathbf{E}_i + \mathbf{F}_i + \mathbf{G}_i + \mathbf{H}_i + \mathbf{I}_i + \mathbf{J}_i + \mathbf{K}_i + \mathbf{L}_i + \mathbf{M}_i +$$
$$\mathbf{N}_i + \mathbf{O}_i + \mathbf{P}_i + \mathbf{Q}_i + \mathbf{R}_i + \mathbf{S}_i + \mathbf{T}_i + \mathbf{U}_i + \mathbf{V}_i + \mathbf{W}_i + \mathbf{X}_i + \mathbf{Y}_i + \mathbf{Z}_i +$$
$$\mathbf{a}_i + \mathbf{b}_i + \mathbf{c}_i + \mathbf{d}_i + \mathbf{e}_i + \mathbf{f}_i + \mathbf{g}_i + \mathbf{h}_i + \mathbf{i}_i + \mathbf{j}_i + \mathbf{k}_i + \mathbf{l}_i + \mathbf{m}_i +$$
$$\mathbf{n}_i + \mathbf{o}_i + \mathbf{p}_i + \mathbf{q}_i + \mathbf{r}_i + \mathbf{s}_i + \mathbf{t}_i + \mathbf{u}_i + \mathbf{v}_i + \mathbf{w}_i + \mathbf{x}_i + \mathbf{y}_i + \mathbf{z}_i +$$
$$\mathbf{A}_i + \mathbf{B}_i + \mathbf{\Gamma}_i + \mathbf{\Delta}_i + \mathbf{E}_i + \mathbf{Z}_i + \mathbf{H}_i + \mathbf{\Theta}_i + \mathbf{I}_i + \mathbf{K}_i + \mathbf{\Lambda}_i + \mathbf{M}_i +$$
$$\mathbf{N}_i + \mathbf{\Xi}_i + \mathbf{O}_i + \mathbf{\Pi}_i + \mathbf{P}_i + \mathbf{\Sigma}_i + \mathbf{T}_i + \mathbf{\Upsilon}_i + \mathbf{\Phi}_i + \mathbf{X}_i + \mathbf{\Psi}_i + \mathbf{\Omega}_i +$$

Math Calligraphic (`\mathcal`)

$$\mathcal{A}_i + \mathcal{B}_i + \mathcal{C}_i + \mathcal{D}_i + \mathcal{E}_i + \mathcal{F}_i + \mathcal{G}_i + \mathcal{H}_i + \mathcal{I}_i + \mathcal{J}_i + \mathcal{K}_i + \mathcal{L}_i + \mathcal{M}_i +$$
$$\mathcal{N}_i + \mathcal{O}_i + \mathcal{P}_i + \mathcal{Q}_i + \mathcal{R}_i + \mathcal{S}_i + \mathcal{T}_i + \mathcal{U}_i + \mathcal{V}_i + \mathcal{W}_i + \mathcal{X}_i + \mathcal{Y}_i + \mathcal{Z}_i +$$

3.4.7 Accent Positioning Sans Serif Bold

Default

$$\hat{0} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$$
$$\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$$
$$\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$$
$$\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} +$$
$$\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$$
$$\hat{A} + \hat{B} + \hat{\Gamma} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$$
$$\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$$
$$\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{\iota} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$$
$$\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{\upsilon} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$$
$$\hat{\varepsilon} + \hat{\vartheta} + \hat{\varpi} + \hat{\varrho} + \hat{\varsigma} + \hat{\varphi} +$$

Math Italic (`\mathit`)

$$\hat{0} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$$
$$\hat{A} + \hat{B} + \hat{C} + \hat{D} + \hat{E} + \hat{F} + \hat{G} + \hat{H} + \hat{I} + \hat{J} + \hat{K} + \hat{L} + \hat{M} +$$
$$\hat{N} + \hat{O} + \hat{P} + \hat{Q} + \hat{R} + \hat{S} + \hat{T} + \hat{U} + \hat{V} + \hat{W} + \hat{X} + \hat{Y} + \hat{Z} +$$
$$\hat{a} + \hat{b} + \hat{c} + \hat{d} + \hat{e} + \hat{f} + \hat{g} + \hat{h} + \hat{i} + \hat{j} + \hat{k} + \hat{l} + \hat{m} + \hat{\ell} + \hat{\wp} + \hat{\imath} + \hat{\jmath} + \tilde{i}$$
$$\hat{n} + \hat{o} + \hat{p} + \hat{q} + \hat{r} + \hat{s} + \hat{t} + \hat{u} + \hat{v} + \hat{w} + \hat{x} + \hat{y} + \hat{z} +$$
$$\hat{A} + \hat{B} + \hat{\Gamma} + \hat{\Delta} + \hat{E} + \hat{Z} + \hat{H} + \hat{\Theta} + \hat{I} + \hat{K} + \hat{\Lambda} + \hat{M} +$$
$$\hat{N} + \hat{\Xi} + \hat{O} + \hat{\Pi} + \hat{P} + \hat{\Sigma} + \hat{T} + \hat{\Upsilon} + \hat{\Phi} + \hat{X} + \hat{\Psi} + \hat{\Omega} +$$
$$\hat{\alpha} + \hat{\beta} + \hat{\gamma} + \hat{\delta} + \hat{\epsilon} + \hat{\zeta} + \hat{\eta} + \hat{\theta} + \hat{\iota} + \hat{\kappa} + \hat{\lambda} + \hat{\mu} +$$
$$\hat{\nu} + \hat{\xi} + \hat{o} + \hat{\pi} + \hat{\rho} + \hat{\sigma} + \hat{\tau} + \hat{\upsilon} + \hat{\phi} + \hat{\chi} + \hat{\psi} + \hat{\omega} +$$
$$\hat{\varepsilon} + \hat{\vartheta} + \hat{\varpi} + \hat{\varrho} + \hat{\varsigma} + \hat{\varphi} +$$

Math Roman (`\mathrm`)

$$\hat{0} + \hat{1} + \hat{2} + \hat{3} + \hat{4} + \hat{5} + \hat{6} + \hat{7} + \hat{8} + \hat{9} +$$
$$\hat{\mathrm{A}} + \hat{\mathrm{B}} + \hat{\mathrm{C}} + \hat{\mathrm{D}} + \hat{\mathrm{E}} + \hat{\mathrm{F}} + \hat{\mathrm{G}} + \hat{\mathrm{H}} + \hat{\mathrm{I}} + \hat{\mathrm{J}} + \hat{\mathrm{K}} + \hat{\mathrm{L}} + \hat{\mathrm{M}} +$$
$$\hat{\mathrm{N}} + \hat{\mathrm{O}} + \hat{\mathrm{P}} + \hat{\mathrm{Q}} + \hat{\mathrm{R}} + \hat{\mathrm{S}} + \hat{\mathrm{T}} + \hat{\mathrm{U}} + \hat{\mathrm{V}} + \hat{\mathrm{W}} + \hat{\mathrm{X}} + \hat{\mathrm{Y}} + \hat{\mathrm{Z}} +$$
$$\hat{\mathrm{a}} + \hat{\mathrm{b}} + \hat{\mathrm{c}} + \hat{\mathrm{d}} + \hat{\mathrm{e}} + \hat{\mathrm{f}} + \hat{\mathrm{g}} + \hat{\mathrm{h}} + \hat{\mathrm{i}} + \hat{\mathrm{j}} + \hat{\mathrm{k}} + \hat{\mathrm{l}} + \hat{\mathrm{m}} +$$
$$\hat{\mathrm{n}} + \hat{\mathrm{o}} + \hat{\mathrm{p}} + \hat{\mathrm{q}} + \hat{\mathrm{r}} + \hat{\mathrm{s}} + \hat{\mathrm{t}} + \hat{\mathrm{u}} + \hat{\mathrm{v}} + \hat{\mathrm{w}} + \hat{\mathrm{x}} + \hat{\mathrm{y}} + \hat{\mathrm{z}} +$$
$$\hat{\mathrm{A}} + \hat{\mathrm{B}} + \hat{\Gamma} + \hat{\Delta} + \hat{\mathrm{E}} + \hat{\mathrm{Z}} + \hat{\mathrm{H}} + \hat{\Theta} + \hat{\mathrm{I}} + \hat{\mathrm{K}} + \hat{\Lambda} + \hat{\mathrm{M}} +$$
$$\hat{\mathrm{N}} + \hat{\Xi} + \hat{\mathrm{O}} + \hat{\Pi} + \hat{\mathrm{P}} + \hat{\Sigma} + \hat{\mathrm{T}} + \hat{\Upsilon} + \hat{\Phi} + \hat{\mathrm{X}} + \hat{\Psi} + \hat{\Omega} +$$

Math Bold (\mathbf)

$$\hat{\mathbf{0}} + \hat{\mathbf{1}} + \hat{\mathbf{2}} + \hat{\mathbf{3}} + \hat{\mathbf{4}} + \hat{\mathbf{5}} + \hat{\mathbf{6}} + \hat{\mathbf{7}} + \hat{\mathbf{8}} + \hat{\mathbf{9}} +$$
$$\hat{\mathbf{A}} + \hat{\mathbf{B}} + \hat{\mathbf{C}} + \hat{\mathbf{D}} + \hat{\mathbf{E}} + \hat{\mathbf{F}} + \hat{\mathbf{G}} + \hat{\mathbf{H}} + \hat{\mathbf{I}} + \hat{\mathbf{J}} + \hat{\mathbf{K}} + \hat{\mathbf{L}} + \hat{\mathbf{M}} +$$
$$\hat{\mathbf{N}} + \hat{\mathbf{O}} + \hat{\mathbf{P}} + \hat{\mathbf{Q}} + \hat{\mathbf{R}} + \hat{\mathbf{S}} + \hat{\mathbf{T}} + \hat{\mathbf{U}} + \hat{\mathbf{V}} + \hat{\mathbf{W}} + \hat{\mathbf{X}} + \hat{\mathbf{Y}} + \hat{\mathbf{Z}} +$$
$$\hat{\mathbf{a}} + \hat{\mathbf{b}} + \hat{\mathbf{c}} + \hat{\mathbf{d}} + \hat{\mathbf{e}} + \hat{\mathbf{f}} + \hat{\mathbf{g}} + \hat{\mathbf{h}} + \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}} + \hat{\mathbf{l}} + \hat{\mathbf{m}} +$$
$$\hat{\mathbf{n}} + \hat{\mathbf{o}} + \hat{\mathbf{p}} + \hat{\mathbf{q}} + \hat{\mathbf{r}} + \hat{\mathbf{s}} + \hat{\mathbf{t}} + \hat{\mathbf{u}} + \hat{\mathbf{v}} + \hat{\mathbf{w}} + \hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}} +$$
$$\hat{\mathbf{A}} + \hat{\mathbf{B}} + \hat{\mathbf{\Gamma}} + \hat{\mathbf{\Delta}} + \hat{\mathbf{E}} + \hat{\mathbf{Z}} + \hat{\mathbf{H}} + \hat{\mathbf{\Theta}} + \hat{\mathbf{I}} + \hat{\mathbf{K}} + \hat{\mathbf{\Lambda}} + \hat{\mathbf{M}} +$$
$$\hat{\mathbf{N}} + \hat{\mathbf{\Xi}} + \hat{\mathbf{O}} + \hat{\mathbf{\Pi}} + \hat{\mathbf{P}} + \hat{\mathbf{\Sigma}} + \hat{\mathbf{T}} + \hat{\mathbf{\Upsilon}} + \hat{\mathbf{\Phi}} + \hat{\mathbf{X}} + \hat{\mathbf{\Psi}} + \hat{\mathbf{\Omega}} +$$

Math Calligraphic (\mathcal)

$$\hat{\mathcal{A}} + \hat{\mathcal{B}} + \hat{\mathcal{C}} + \hat{\mathcal{D}} + \hat{\mathcal{E}} + \hat{\mathcal{F}} + \hat{\mathcal{G}} + \hat{\mathcal{H}} + \hat{\mathcal{I}} + \hat{\mathcal{J}} + \hat{\mathcal{K}} + \hat{\mathcal{L}} + \hat{\mathcal{M}} +$$
$$\hat{\mathcal{N}} + \hat{\mathcal{O}} + \hat{\mathcal{P}} + \hat{\mathcal{Q}} + \hat{\mathcal{R}} + \hat{\mathcal{S}} + \hat{\mathcal{T}} + \hat{\mathcal{U}} + \hat{\mathcal{V}} + \hat{\mathcal{W}} + \hat{\mathcal{X}} + \hat{\mathcal{Y}} + \hat{\mathcal{Z}} +$$

3.4.8 Differentials Sans Serif Bold

$$dA + dB + dC + dD + dE + dF + dG + dH + dI + dJ + dK + dL + dM +$$
$$dN + dO + dP + dQ + dR + dS + dT + dU + dV + dW + dX + dY + dZ +$$
$$da + db + dc + dd + de + df + dg + dh + di + dj + dk + dl + dm +$$
$$dn + do + dp + dq + dr + ds + dt + du + dv + dw + dx + dy + dz +$$
$$dA + dB + d\mathit{\Gamma} + d\mathit{\Delta} + dE + dZ + dH + d\mathit{\Theta} + dI + dK + d\mathit{\Lambda} + dM +$$
$$dN + d\mathit{\Xi} + dO + d\mathit{\Pi} + dP + d\mathit{\Sigma} + dT + d\mathit{\Upsilon} + d\mathit{\Phi} + dX + d\mathit{\Psi} + d\mathit{\Omega} +$$
$$d\alpha + d\beta + d\gamma + d\delta + d\epsilon + d\zeta + d\eta + d\theta + d\iota + d\kappa + d\lambda + d\mu +$$
$$d\nu + d\xi + do + d\pi + d\rho + d\sigma + d\tau + d\upsilon + d\phi + d\chi + d\psi + d\omega +$$
$$d\varepsilon + d\vartheta + d\varpi + d\varrho + d\varsigma + d\varphi +$$
$$d\mathrm{A} + d\mathrm{B} + d\Gamma + d\Delta + d\mathrm{E} + d\mathrm{Z} + d\mathrm{H} + d\Theta + d\mathrm{I} + d\mathrm{K} + d\Lambda + d\mathrm{M} +$$
$$d\mathrm{N} + d\Xi + d\mathrm{O} + d\Pi + d\mathrm{P} + d\Sigma + d\mathrm{T} + d\Upsilon + d\Phi + d\mathrm{X} + d\Psi + d\Omega +$$

$$\mathbf{d}A + \mathbf{d}B + \mathbf{d}C + \mathbf{d}D + \mathbf{d}E + \mathbf{d}F + \mathbf{d}G + \mathbf{d}H + \mathbf{d}I + \mathbf{d}J + \mathbf{d}K + \mathbf{d}L + \mathbf{d}M +$$
$$\mathbf{d}N + \mathbf{d}O + \mathbf{d}P + \mathbf{d}Q + \mathbf{d}R + \mathbf{d}S + \mathbf{d}T + \mathbf{d}U + \mathbf{d}V + \mathbf{d}W + \mathbf{d}X + \mathbf{d}Y + \mathbf{d}Z +$$
$$\mathbf{d}a + \mathbf{d}b + \mathbf{d}c + \mathbf{d}d + \mathbf{d}e + \mathbf{d}f + \mathbf{d}g + \mathbf{d}h + \mathbf{d}i + \mathbf{d}j + \mathbf{d}k + \mathbf{d}l + \mathbf{d}m +$$
$$\mathbf{d}n + \mathbf{d}o + \mathbf{d}p + \mathbf{d}q + \mathbf{d}r + \mathbf{d}s + \mathbf{d}t + \mathbf{d}u + \mathbf{d}v + \mathbf{d}w + \mathbf{d}x + \mathbf{d}y + \mathbf{d}z +$$
$$\mathbf{d}A + \mathbf{d}B + \mathbf{d}\varGamma + \mathbf{d}\varDelta + \mathbf{d}E + \mathbf{d}Z + \mathbf{d}H + \mathbf{d}\varTheta + \mathbf{d}I + \mathbf{d}K + \mathbf{d}\varLambda + \mathbf{d}M +$$
$$\mathbf{d}N + \mathbf{d}\varXi + \mathbf{d}O + \mathbf{d}\varPi + \mathbf{d}P + \mathbf{d}\varSigma + \mathbf{d}T + \mathbf{d}\varUpsilon + \mathbf{d}\varPhi + \mathbf{d}X + \mathbf{d}\varPsi + \mathbf{d}\varOmega +$$
$$\mathbf{d}\alpha + \mathbf{d}\beta + \mathbf{d}\gamma + \mathbf{d}\delta + \mathbf{d}\epsilon + \mathbf{d}\zeta + \mathbf{d}\eta + \mathbf{d}\theta + \mathbf{d}\iota + \mathbf{d}\kappa + \mathbf{d}\lambda + \mathbf{d}\mu +$$
$$\mathbf{d}\nu + \mathbf{d}\xi + \mathbf{d}o + \mathbf{d}\pi + \mathbf{d}\rho + \mathbf{d}\sigma + \mathbf{d}\tau + \mathbf{d}\upsilon + \mathbf{d}\phi + \mathbf{d}\chi + \mathbf{d}\psi + \mathbf{d}\omega +$$
$$\mathbf{d}\varepsilon + \mathbf{d}\vartheta + \mathbf{d}\varpi + \mathbf{d}\varrho + \mathbf{d}\varsigma + \mathbf{d}\varphi +$$
$$\mathbf{d}\mathrm{A} + \mathbf{d}\mathrm{B} + \mathbf{d}\Gamma + \mathbf{d}\Delta + \mathbf{d}\mathrm{E} + \mathbf{d}\mathrm{Z} + \mathbf{d}\mathrm{H} + \mathbf{d}\Theta + \mathbf{d}\mathrm{I} + \mathbf{d}\mathrm{K} + \mathbf{d}\Lambda + \mathbf{d}\mathrm{M} +$$
$$\mathbf{d}\mathrm{N} + \mathbf{d}\Xi + \mathbf{d}\mathrm{O} + \mathbf{d}\Pi + \mathbf{d}\mathrm{P} + \mathbf{d}\Sigma + \mathbf{d}\mathrm{T} + \mathbf{d}\Upsilon + \mathbf{d}\Phi + \mathbf{d}\mathrm{X} + \mathbf{d}\Psi + \mathbf{d}\Omega +$$

$$\partial A + \partial B + \partial C + \partial D + \partial E + \partial F + \partial G + \partial H + \partial I + \partial J + \partial K + \partial L + \partial M +$$
$$\partial N + \partial O + \partial P + \partial Q + \partial R + \partial S + \partial T + \partial U + \partial V + \partial W + \partial X + \partial Y + \partial Z +$$
$$\partial a + \partial b + \partial c + \partial d + \partial e + \partial f + \partial g + \partial h + \partial i + \partial j + \partial k + \partial l + \partial m +$$
$$\partial n + \partial o + \partial p + \partial q + \partial r + \partial s + \partial t + \partial u + \partial v + \partial w + \partial x + \partial y + \partial z +$$
$$\partial A + \partial B + \partial \varGamma + \partial \varDelta + \partial E + \partial Z + \partial H + \partial \varTheta + \partial I + \partial K + \partial \varLambda + \partial M +$$
$$\partial N + \partial \varXi + \partial O + \partial \varPi + \partial P + \partial \varSigma + \partial T + \partial \varUpsilon + \partial \varPhi + \partial X + \partial \varPsi + \partial \varOmega +$$
$$\partial \alpha + \partial \beta + \partial \gamma + \partial \delta + \partial \epsilon + \partial \zeta + \partial \eta + \partial \theta + \partial \iota + \partial \kappa + \partial \lambda + \partial \mu +$$
$$\partial \nu + \partial \xi + \partial o + \partial \pi + \partial \rho + \partial \sigma + \partial \tau + \partial \upsilon + \partial \phi + \partial \chi + \partial \psi + \partial \omega +$$
$$\partial \varepsilon + \partial \vartheta + \partial \varpi + \partial \varrho + \partial \varsigma + \partial \varphi +$$
$$\partial \mathrm{A} + \partial \mathrm{B} + \partial \Gamma + \partial \Delta + \partial \mathrm{E} + \partial \mathrm{Z} + \partial \mathrm{H} + \partial \Theta + \partial \mathrm{I} + \partial \mathrm{K} + \partial \Lambda + \partial \mathrm{M} +$$
$$\partial \mathrm{N} + \partial \Xi + \partial \mathrm{O} + \partial \Pi + \partial \mathrm{P} + \partial \Sigma + \partial \mathrm{T} + \partial \Upsilon + \partial \Phi + \partial \mathrm{X} + \partial \Psi + \partial \Omega +$$

3.4.9 Slash Kerning Sans Serif Bold

$$1/A + 1/B + 1/C + 1/D + 1/E + 1/F + 1/G + 1/H + 1/I + 1/J + 1/K + 1/L + 1/M +$$
$$1/N + 1/O + 1/P + 1/Q + 1/R + 1/S + 1/T + 1/U + 1/V + 1/W + 1/X + 1/Y + 1/Z +$$
$$1/a + 1/b + 1/c + 1/d + 1/e + 1/f + 1/g + 1/h + 1/i + 1/j + 1/k + 1/l + 1/m +$$
$$1/n + 1/o + 1/p + 1/q + 1/r + 1/s + 1/t + 1/u + 1/v + 1/w + 1/x + 1/y + 1/z +$$
$$1/A + 1/B + 1/\varGamma + 1/\varDelta + 1/E + 1/Z + 1/H + 1/\varTheta + 1/I + 1/K + 1/\varLambda + 1/M +$$
$$1/N + 1/\varXi + 1/O + 1/\varPi + 1/P + 1/\varSigma + 1/T + 1/\varUpsilon + 1/\varPhi + 1/X + 1/\varPsi + 1/\varOmega +$$
$$1/\alpha + 1/\beta + 1/\gamma + 1/\delta + 1/\epsilon + 1/\zeta + 1/\eta + 1/\theta + 1/\iota + 1/\kappa + 1/\lambda + 1/\mu +$$
$$1/\nu + 1/\xi + 1/o + 1/\pi + 1/\rho + 1/\sigma + 1/\tau + 1/\upsilon + 1/\phi + 1/\chi + 1/\psi + 1/\omega +$$
$$1/\varepsilon + 1/\vartheta + 1/\varpi + 1/\varrho + 1/\varsigma + 1/\varphi +$$

$$A/2 + B/2 + C/2 + D/2 + E/2 + F/2 + G/2 + H/2 + I/2 + J/2 + K/2 + L/2 + M/2 +$$
$$N/2 + O/2 + P/2 + Q/2 + R/2 + S/2 + T/2 + U/2 + V/2 + W/2 + X/2 + Y/2 + Z/2 +$$
$$a/2 + b/2 + c/2 + d/2 + e/2 + f/2 + g/2 + h/2 + i/2 + j/2 + k/2 + l/2 + m/2 +$$
$$n/2 + o/2 + p/2 + q/2 + r/2 + s/2 + t/2 + u/2 + v/2 + w/2 + x/2 + y/2 + z/2 +$$
$$A/2 + B/2 + \Gamma/2 + \Delta/2 + E/2 + Z/2 + H/2 + \Theta/2 + I/2 + K/2 + \Lambda/2 + M/2 +$$
$$N/2 + \Xi/2 + O/2 + \Pi/2 + P/2 + \Sigma/2 + T/2 + \Upsilon/2 + \Phi/2 + X/2 + \Psi/2 + \Omega/2 +$$
$$\alpha/2 + \beta/2 + \gamma/2 + \delta/2 + \epsilon/2 + \zeta/2 + \eta/2 + \theta/2 + \iota/2 + \kappa/2 + \lambda/2 + \mu/2 +$$
$$\nu/2 + \xi/2 + o/2 + \pi/2 + \rho/2 + \sigma/2 + \tau/2 + \upsilon/2 + \phi/2 + \chi/2 + \psi/2 + \omega/2 +$$
$$\varepsilon/2 + \vartheta/2 + \varpi/2 + \varrho/2 + \varsigma/2 + \varphi/2 +$$

3.4.10 (Big) Operators Sans Serif Bold

$\sum_{i=1}^n x^n$ $\prod_{i=1}^n x^n$ $\coprod_{i=1}^n x^n$ $\int_{i=1}^n x^n$ $\oint_{i=1}^n x^n$
$\bigotimes_{i=1}^n x^n$ $\bigoplus_{i=1}^n x^n$ $\bigodot_{i=1}^n x^n$ $\bigwedge_{i=1}^n x^n$ $\bigvee_{i=1}^n x^n$ $\biguplus_{i=1}^n x^n$ $\bigcup_{i=1}^n x^n$ $\bigcap_{i=1}^n x^n$ $\bigsqcup_{i=1}^n x^n$

$$\sum_{i=1}^n x^n \quad \prod_{i=1}^n x^n \quad \coprod_{i=1}^n x^n \quad \int_{i=1}^n x^n \quad \oint_{i=1}^n x^n$$

$$\bigotimes_{i=1}^n x^n \quad \bigoplus_{i=1}^n x^n \quad \bigodot_{i=1}^n x^n \quad \bigwedge_{i=1}^n x^n \quad \bigvee_{i=1}^n x^n \quad \biguplus_{i=1}^n x^n \quad \bigcup_{i=1}^n x^n \quad \bigcap_{i=1}^n x^n \quad \bigsqcup_{i=1}^n x^n$$

3.4.11 Radicals Sans Serif Bold

$$\sqrt{x+y} \quad \sqrt{x^2+y^2} \quad \sqrt{x_i^2+y_j^2} \quad \sqrt{\left(\frac{\cos x}{2}\right)} \quad \sqrt{\left(\frac{\sin x}{2}\right)}$$

$$\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{x+y}}}}}}}$$

3.4.12 Over- and Underbraces Sans Serif Bold

$$\overbrace{x} \quad \overbrace{x+y} \quad \overbrace{x^2+y^2} \quad \overbrace{x_i^2+y_j^2} \quad \underbrace{x} \quad \underbrace{x+y} \quad \underbrace{x_i+y_j} \quad \underbrace{x_i^2+y_j^2}$$

3.4.13 Normal and Wide Accents Sans Serif Bold

$$\dot{x} \quad \ddot{x} \quad \vec{x} \quad \bar{x} \quad \overline{x} \quad \overline{xx} \quad \tilde{x} \quad \widetilde{x} \quad \widetilde{xx} \quad \widetilde{xxx} \quad \hat{x} \quad \widehat{x} \quad \widehat{xx} \quad \widehat{xxx}$$

$$\hat{x} \quad \check{x} \quad \tilde{x} \quad \acute{x} \quad \grave{x} \quad \dot{x} \quad \ddot{x} \quad \breve{x} \quad \bar{x} \quad \vec{x}$$

3.4.14 Long Arrows Sans Serif Bold

$$\leftarrow\!-\!-\!\rightarrow \quad \leftrightarrow \quad \longleftarrow \quad \longrightarrow \quad \longleftrightarrow \quad \Leftarrow\!=\!\Rightarrow \quad \Leftrightarrow \quad \Longleftarrow \quad \Longrightarrow \quad \Longleftrightarrow$$

3.4.15 Left and Right Delimiters Sans Serif Bold

$$-(f)--[f]--\lfloor f\rfloor--\lceil f\rceil--\langle f\rangle--\{f\}-$$

Using `\left` and `\right`.

$$-\left(f\right)--\left[f\right]--\left\lfloor f\right\rfloor--\left\lceil f\right\rceil--\left\langle f\right\rangle--\left\{f\right\}-$$

$$-\left)f\right(--\left]f\right[--\left/f\right/--\left\backslash f\right\backslash--\left/f\right\backslash--\left\backslash f\right/-$$

3.4.16 Big-g-g Delimiters Sans Serif Bold

$$-\Biggl\lfloor\biggl\lfloor\Bigl\lfloor\bigl\lfloor\lfloor-\rfloor\bigr\rfloor\Bigr\rfloor\biggr\rfloor\Biggr\rfloor- \qquad -\Biggl(\biggl(\Bigl(\bigl((-)\bigr)\Bigr)\biggr)\Biggr)-$$

$$-\Biggl\lceil\biggl\lceil\Bigl\lceil\bigl\lceil\lceil-\rceil\bigr\rceil\Bigr\rceil\biggr\rceil\Biggr\rceil- \qquad -\Biggl\{\biggl\{\Bigl\{\bigl\{\{-\}\bigr\}\Bigr\}\biggr\}\Biggr\}-$$

$$-\Biggl[\biggl[\Bigl[\bigl[[-]\bigr]\Bigr]\biggr]\Biggr]- \qquad -\Biggl(\biggl(\Bigl(\bigl((-)\bigr)\Bigr)\biggr)\Biggr)-$$

$$-\Biggl\langle\biggl\langle\Bigl\langle\bigl\langle\langle-\rangle\bigr\rangle\Bigr\rangle\biggr\rangle\Biggr\rangle- \qquad -\Bigg\int\bigg\int\Big\int\big\int\int-\Bigr)\biggr)\Biggr)-$$

$$-\Bigg\uparrow\bigg\uparrow\Big\uparrow\big\uparrow\uparrow-\downarrow\big\downarrow\Big\downarrow\bigg\downarrow\Bigg\downarrow- \qquad -\Bigg\Uparrow\bigg\Uparrow\Big\Uparrow\big\Uparrow\Uparrow-\Downarrow\big\Downarrow\Big\Downarrow\bigg\Downarrow\Bigg\Downarrow-$$

3.4.17 Binary Operators Sans Serif Bold

$x \pm y$	`\pm`	$x \cap y$	`\cap`	$x \diamond y$	`\diamond`	$x \oplus y$	`\oplus`
$x \mp y$	`\mp`	$x \cup y$	`\cup`	$x \bigtriangleup y$	`\bigtriangleup`	$x \ominus y$	`\ominus`
$x \times y$	`\times`	$x \uplus y$	`\uplus`	$x \bigtriangledown y$	`\bigtriangledown`	$x \otimes y$	`\otimes`
$x \div y$	`\div`	$x \sqcap y$	`\sqcap`	$x \triangleleft y$	`\triangleleft`	$x \oslash y$	`\oslash`
$x \ast y$	`\ast`	$x \sqcup y$	`\sqcup`	$x \triangleright y$	`\triangleright`	$x \odot y$	`\odot`
$x \star y$	`\star`	$x \vee y$	`\vee`	$x \lhd y$	`\lhd`	$x \bigcirc y$	`\bigcirc`
$x \circ y$	`\circ`	$x \wedge y$	`\wedge`	$x \rhd y$	`\rhd`	$x \dagger y$	`\dagger`
$x \bullet y$	`\bullet`	$x \setminus y$	`\setminus`	$x \unlhd y$	`\unlhd`	$x \ddagger y$	`\ddagger`
$x \cdot y$	`\cdot`	$x \wr y$	`\wr`	$x \unrhd y$	`\unrhd`	$x \S y$	`\S`
$x + y$	`+`	$x - y$	`-`	$x \amalg y$	`\amalg`	$x \P y$	`\P`

3.4.18 Relations Sans Serif Bold

$x \leq y$	`\leq`	$x \geq y$	`\geq`	$x \equiv y$	`\equiv`	$x \models y$	`\models`
$x \prec y$	`\prec`	$x \succ y$	`\succ`	$x \sim y$	`\sim`	$x \perp y$	`\perp`
$x \preceq y$	`\preceq`	$x \succeq y$	`\succeq`	$x \simeq y$	`\simeq`	$x \mid y$	`\mid`
$x \ll y$	`\ll`	$x \gg y$	`\gg`	$x \asymp y$	`\asymp`	$x \parallel y$	`\parallel`
$x \subset y$	`\subset`	$x \supset y$	`\supset`	$x \approx y$	`\approx`	$x \bowtie y$	`\bowtie`
$x \subseteq y$	`\subseteq`	$x \supseteq y$	`\supseteq`	$x \cong y$	`\cong`	$x \Join y$	`\Join`
$x \sqsubset y$	`\sqsubset`	$x \sqsupset y$	`\sqsupset`	$x \neq y$	`\neq`	$x \smile y$	`\smile`
$x \sqsubseteq y$	`\sqsubseteq`	$x \sqsupseteq y$	`\sqsupseteq`	$x \doteq y$	`\doteq`	$x \frown y$	`\frown`
$x \in y$	`\in`	$x \ni y$	`\ni`	$x \propto y$	`\propto`	$x = y$	`=`
$x \vdash y$	`\vdash`	$x \dashv y$	`\dashv`	$x < y$	`<`	$x > y$	`>`
$x : y$	`:`						

3.4.19 Punctuation Sans Serif Bold

x, y	`,`	$x; y$	`;`	$x \colon y$	`\colon`	$x \ldotp y$	`\ldotp`	$x \cdotp y$	`\cdotp`

3.4.20 Arrows Sans Serif Bold

$x \leftarrow y$	`\leftarrow`	$x \longleftarrow y$	`\longleftarrow`	$x \uparrow y$	`\uparrow`
$x \Leftarrow y$	`\Leftarrow`	$x \Longleftarrow y$	`\Longleftarrow`	$x \Uparrow y$	`\Uparrow`
$x \rightarrow y$	`\rightarrow`	$x \longrightarrow y$	`\longrightarrow`	$x \downarrow y$	`\downarrow`
$x \Rightarrow y$	`\Rightarrow`	$x \Longrightarrow y$	`\Longrightarrow`	$x \Downarrow y$	`\Downarrow`
$x \leftrightarrow y$	`\leftrightarrow`	$x \longleftrightarrow y$	`\longleftrightarrow`	$x \updownarrow y$	`\updownarrow`
$x \Leftrightarrow y$	`\Leftrightarrow`	$x \Longleftrightarrow y$	`\Longleftrightarrow`	$x \Updownarrow y$	`\Updownarrow`
$x \mapsto y$	`\mapsto`	$x \longmapsto y$	`\longmapsto`	$x \nearrow y$	`\nearrow`
$x \hookleftarrow y$	`\hookleftarrow`	$x \hookrightarrow y$	`\hookrightarrow`	$x \searrow y$	`\searrow`
$x \leftharpoonup y$	`\leftharpoonup`	$x \rightharpoonup y$	`\rightharpoonup`	$x \swarrow y$	`\swarrow`
$x \leftharpoondown y$	`\leftharpoondown`	$x \rightharpoondown y$	`\rightharpoondown`	$x \nwarrow y$	`\nwarrow`
$x \rightleftharpoons y$	`\rightleftharpoons`	$x \leadsto y$	`\leadsto`		

3.4.21 Miscellaneous Symbols Sans Serif Bold

$x \ldots y$	`\ldots`	$x \cdots y$	`\cdots`	$x \vdots y$	`\vdots`	$x \ddots y$	`\ddots`
$x \aleph y$	`\aleph`	$x \prime y$	`\prime`	$x \forall y$	`\forall`	$x \infty y$	`\infty`
$x \hbar y$	`\hbar`	$x \emptyset y$	`\emptyset`	$x \exists y$	`\exists`	$x \Box y$	`\Box`
$x \imath y$	`\imath`	$x \nabla y$	`\nabla`	$x \neg y$	`\neg`	$x \Diamond y$	`\Diamond`
$x \jmath y$	`\jmath`	$x \surd y$	`\surd`	$x \flat y$	`\flat`	$x \triangle y$	`\triangle`
$x \ell y$	`\ell`	$x \top y$	`\top`	$x \natural y$	`\natural`	$x \clubsuit y$	`\clubsuit`
$x \wp y$	`\wp`	$x \bot y$	`\bot`	$x \sharp y$	`\sharp`	$x \diamondsuit y$	`\diamondsuit`
$x \Re y$	`\Re`	$x \| y$	`|`	$x \backslash y$	`\backslash`	$x \heartsuit y$	`\heartsuit`
$x \Im y$	`\Im`	$x \angle y$	`\angle`	$x \partial y$	`\partial`	$x \spadesuit y$	`\spadesuit`
$x \mho y$	`\mho`	$x.y$	`.`	$x \vert y$	`|`	$x!y$	`!`

3.4.22 Variable-Sized Operators Sans Serif Bold

$x \sum y$	`\sum`	$x \bigcap y$	`\bigcap`	$x \bigodot y$	`\bigodot`
$x \prod y$	`\prod`	$x \bigcup y$	`\bigcup`	$x \bigotimes y$	`\bigotimes`
$x \coprod y$	`\coprod`	$x \bigsqcup y$	`\bigsqcup`	$x \bigoplus y$	`\bigoplus`
$x \int y$	`\int`	$x \bigvee y$	`\bigvee`	$x \biguplus y$	`\biguplus`
$x \oint y$	`\oint`	$x \bigwedge y$	`\bigwedge`		

3.4.23 Log-Like Operators Sans Serif Bold

$x \arccos y$	$x \cos y$	$x \csc y$	$x \exp y$	$x \ker y$	$x \limsup y$	$x \min y$	$x \sinh y$
$x \arcsin y$	$x \cosh y$	$x \deg y$	$x \gcd y$	$x \lg y$	$x \ln y$	$x \Pr y$	$x \sup y$
$x \arctan y$	$x \cot y$	$x \det y$	$x \hom y$	$x \lim y$	$x \log y$	$x \sec y$	$x \tan y$
$x \arg y$	$x \coth y$	$x \dim y$	$x \inf y$	$x \liminf y$	$x \max y$	$x \sin y$	$x \tanh y$

3.4.24 Delimiters Sans Serif Bold

$\boldsymbol{x(y}$	`(`	$\boldsymbol{x)y}$	`)`	$\boldsymbol{x \uparrow y}$	`\uparrow`	$\boldsymbol{x \Uparrow y}$	`\Uparrow`
$\boldsymbol{x[y}$	`[`	$\boldsymbol{x]y}$	`]`	$\boldsymbol{x \downarrow y}$	`\downarrow`	$\boldsymbol{x \Downarrow y}$	`\Downarrow`
$\boldsymbol{x\{y}$	`\{`	$\boldsymbol{x\}y}$	`\}`	$\boldsymbol{x \updownarrow y}$	`\updownarrow`	$\boldsymbol{x \Updownarrow y}$	`\Updownarrow`
$\boldsymbol{x\lfloor y}$	`\lfloor`	$\boldsymbol{x\rfloor y}$	`\rfloor`	$\boldsymbol{x\lceil y}$	`\lceil`	$\boldsymbol{x\rceil y}$	`\rceil`
$\boldsymbol{x\langle y}$	`\langle`	$\boldsymbol{x\rangle y}$	`\rangle`	$\boldsymbol{x/y}$	`/`	$\boldsymbol{x\backslash y}$	`\backslash`
$\boldsymbol{x\vert y}$	`|`	$\boldsymbol{x\Vert y}$	`\\|`				

3.4.25 Large Delimiters Sans Serif Bold

$\rmoustache$	`\rmoustache`	$\lmoustache$	`\lmoustache`	$\rgroup$	`\rgroup`	$\lgroup$	`\lgroup`
$\arrowvert$	`\arrowvert`	$\Arrowvert$	`\Arrowvert`	$\bracevert$	`\bracevert`		

3.4.26 Math Mode Accents Sans Serif Bold

$\hat{\boldsymbol{a}}$	`\hat{a}`	$\acute{\boldsymbol{a}}$	`\acute{a}`	$\bar{\boldsymbol{a}}$	`\bar{a}`	$\dot{\boldsymbol{a}}$	`\dot{a}`	$\breve{\boldsymbol{a}}$	`\breve{a}`
$\check{\boldsymbol{a}}$	`\check{a}`	$\grave{\boldsymbol{a}}$	`\grave{a}`	$\vec{\boldsymbol{a}}$	`\vec{a}`	$\ddot{\boldsymbol{a}}$	`\ddot{a}`	$\tilde{\boldsymbol{a}}$	`\tilde{a}`

3.4.27 Miscellaneous Constructions Sans Serif Bold

$\widetilde{\boldsymbol{abc}}$	`\widetilde{abc}`	$\widehat{\boldsymbol{abc}}$	`\widehat{abc}`
$\overleftarrow{\boldsymbol{abc}}$	`\overleftarrow{abc}`	$\overrightarrow{\boldsymbol{abc}}$	`\overrightarrow{abc}`
$\overline{\boldsymbol{abc}}$	`\overline{abc}`	$\underline{\boldsymbol{abc}}$	`\underline{abc}`
$\overbrace{\boldsymbol{abc}}$	`\overbrace{abc}`	$\underbrace{\boldsymbol{abc}}$	`\underbrace{abc}`
$\sqrt{\boldsymbol{abc}}$	`\sqrt{abc}`	$\sqrt[n]{\boldsymbol{abc}}$	`\sqrt[n]{abc}`
$\boldsymbol{f'}$	`f'`	$\frac{\boldsymbol{abc}}{\boldsymbol{xyz}}$	`\frac{abc}{xyz}`

3.4.28 AMS Delimiters Sans Serif Bold

$\boldsymbol{x\ulcorner y}$	`\ulcorner`	$\boldsymbol{x\urcorner y}$	`\urcorner`	$\boldsymbol{x\llcorner y}$	`\llcorner`	$\boldsymbol{x\lrcorner y}$	`\lrcorner`

3.4.29 AMS Arrows Sans Serif Bold

$x \dashrightarrow y$	\dashrightarrow	$x \dashleftarrow y$	\dashleftarrow
$x \leftleftarrows y$	\leftleftarrows	$x \leftrightarrows y$	\leftrightarrows
$x \Lleftarrow y$	\Lleftarrow	$x \twoheadleftarrow y$	\twoheadleftarrow
$x \leftarrowtail y$	\leftarrowtail	$x \looparrowleft y$	\looparrowleft
$x \leftrightharpoons y$	\leftrightharpoons	$x \curvearrowleft y$	\curvearrowleft
$x \circlearrowleft y$	\circlearrowleft	$x \Lsh y$	\Lsh
$x \upuparrows y$	\upuparrows	$x \upharpoonleft y$	\upharpoonleft
$x \downharpoonleft y$	\downharpoonleft	$x \multimap y$	\multimap
$x \leftrightsquigarrow y$	\leftrightsquigarrow	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \rightrightarrows y$	\rightrightarrows
$x \rightleftarrows y$	\rightleftarrows	$x \twoheadrightarrow y$	\twoheadrightarrow
$x \rightarrowtail y$	\rightarrowtail	$x \looparrowright y$	\looparrowright
$x \rightleftharpoons y$	\rightleftharpoons	$x \curvearrowright y$	\curvearrowright
$x \circlearrowright y$	\circlearrowright	$x \Rsh y$	\Rsh
$x \downdownarrows y$	\downdownarrows	$x \upharpoonright y$	\upharpoonright
$x \downharpoonright y$	\downharpoonright	$x \rightsquigarrow y$	\rightsquigarrow

3.4.30 AMS Negated Arrows Sans Serif Bold

$x \nleftarrow y$	\nleftarrow	$x \nrightarrow y$	\nrightarrow
$x \nLeftarrow y$	\nLeftarrow	$x \nRightarrow y$	\nRightarrow
$x \nleftrightarrow y$	\nleftrightarrow	$x \nLeftrightarrow y$	\nLeftrightarrow

3.4.31 AMS Greek Sans Serif Bold

$x\digamma y$ \digamma $x\varkappa y$ \varkappa

3.4.32 AMS Hebrew Sans Serif Bold

$x\beth y$ \beth $x\daleth y$ \daleth $x\gimel y$ \gimel

3.4.33 AMS Miscellaneous Sans Serif Bold

$x\hbar y$	`\hbar`	$x\hslash y$	`\hslash`
$x \vartriangle y$	`\vartriangle`	$x\triangledown y$	`\triangledown`
$x\square y$	`\square`	$x\lozenge y$	`\lozenge`
$x\circledS y$	`\circledS`	$x\angle y$	`\angle`
$x\measuredangle y$	`\measuredangle`	$x\nexists y$	`\nexists`
$x\mho y$	`\mho`	$x\Finv y$	`\Finv`^{u}
$x\Game y$	`\Game`^{u}	$x\Bbbk y$	`\Bbbk`^{u}
$x\backprime y$	`\backprime`	$x\varnothing y$	`\varnothing`
$x\blacktriangle y$	`\blacktriangle`	$x\blacktriangledown y$	`\blacktriangledown`
$x\blacksquare y$	`\blacksquare`	$x\blacklozenge y$	`\blacklozenge`
$x\bigstar y$	`\bigstar`	$x\sphericalangle y$	`\sphericalangle`
$x\complement y$	`\complement`	$x\eth y$	`\eth`
$x\diagup y$	`\diagup`^{u}	$x\diagdown y$	`\diagdown`^{u}

^{u} **Not defined in** `amssymb.sty`**, define using the** `\newsymbol` **command.**

3.4.34 AMS Binary Operators Sans Serif Bold

$x \dotplus y$	`\dotplus`	$x \smallsetminus y$	`\smallsetminus`
$x \Cap y$	`\Cap`	$x \Cup y$	`\Cup`
$x \barwedge y$	`\barwedge`	$x \veebar y$	`\veebar`
$x \doublebarwedge y$	`\doublebarwedge`	$x \boxminus y$	`\boxminus`
$x \boxtimes y$	`\boxtimes`	$x \boxdot y$	`\boxdot`
$x \boxplus y$	`\boxplus`	$x \divideontimes y$	`\divideontimes`
$x \ltimes y$	`\ltimes`	$x \rtimes y$	`\rtimes`
$x \leftthreetimes y$	`\leftthreetimes`	$x \rightthreetimes y$	`\rightthreetimes`
$x \curlywedge y$	`\curlywedge`	$x \curlyvee y$	`\curlyvee`
$x \circleddash y$	`\circleddash`	$x \circledast y$	`\circledast`
$x \circledcirc y$	`\circledcirc`	$x \centerdot y$	`\centerdot`
$x \intercal y$	`\intercal`		

3.4.35 AMS Relations Sans Serif Bold

$x \leqslant y$	`\leqslant`
$x \lesssim y$	`\lesssim`
$x \approxeq y$	`\approxeq`
$x \lll y$	`\lll`
$x \lesseqgtr y$	`\lesseqgtr`
$x \doteqdot y$	`\doteqdot`
$x \fallingdotseq y$	`\fallingdotseq`
$x \backsimeq y$	`\backsimeq`
$x \Subset y$	`\Subset`
$x \preccurlyeq y$	`\preccurlyeq`
$x \precsim y$	`\precsim`
$x \vartriangleleft y$	`\vartriangleleft`
$x \vDash y$	`\vDash`
$x \smallsmile y$	`\smallsmile`
$x \bumpeq y$	`\bumpeq`
$x \geqq y$	`\geqq`
$x \eqslantgtr y$	`\eqslantgtr`
$x \gtrapprox y$	`\gtrapprox`
$x \ggg y$	`\ggg`
$x \gtreqless y$	`\gtreqless`
$x \eqcirc y$	`\eqcirc`
$x \triangleq y$	`\triangleq`
$x \thickapprox y$	`\thickapprox`
$x \Supset y$	`\Supset`
$x \succcurlyeq y$	`\succcurlyeq`
$x \succsim y$	`\succsim`
$x \vartriangleright y$	`\vartriangleright`
$x \Vdash y$	`\Vdash`
$x \shortparallel y$	`\shortparallel`
$x \pitchfork y$	`\pitchfork`
$x \blacktriangleleft y$	`\blacktriangleleft`
$x \backepsilon y$	`\backepsilon`
$x \because y$	`\because`

3.4.36 AMS Negated Relations Sans Serif Bold

$x \nless y$	`\nless`	$x \nleq y$	`\nleq`
$x \nleqslant y$	`\nleqslant`	$x \nleqq y$	`\nleqq`
$x \lneq y$	`\lneq`	$x \lneqq y$	`\lneqq`
$x \lvertneqq y$	`\lvertneqq`	$x \lnsim y$	`\lnsim`
$x \lnapprox y$	`\lnapprox`	$x \nprec y$	`\nprec`
$x \npreceq y$	`\npreceq`	$x \precnsim y$	`\precnsim`
$x \precnapprox y$	`\precnapprox`	$x \nsim y$	`\nsim`
$x \nshortmid y$	`\nshortmid`	$x \nmid y$	`\nmid`
$x \nvdash y$	`\nvdash`	$x \nvDash y$	`\nvDash`
$x \ntriangleleft y$	`\ntriangleleft`	$x \ntrianglelefteq y$	`\ntrianglelefteq`
$x \nsubseteq y$	`\nsubseteq`	$x \subsetneq y$	`\subsetneq`
$x \varsubsetneq y$	`\varsubsetneq`	$x \subsetneqq y$	`\subsetneqq`
$x \varsubsetneqq y$	`\varsubsetneqq`	$x \ngtr y$	`\ngtr`
$x \ngeq y$	`\ngeq`	$x \ngeqslant y$	`\ngeqslant`
$x \ngeqq y$	`\ngeqq`	$x \gneq y$	`\gneq`
$x \gneqq y$	`\gneqq`	$x \gvertneqq y$	`\gvertneqq`
$x \gnsim y$	`\gnsim`	$x \gnapprox y$	`\gnapprox`
$x \nsucc y$	`\nsucc`	$x \nsucceq y$	`\nsucceq`
$x \nsucceqq y$	`\nsucceqq`	$x \succnsim y$	`\succnsim`
$x \succnapprox y$	`\succnapprox`	$x \ncong y$	`\ncong`
$x \nshortparallel y$	`\nshortparallel`	$x \nparallel y$	`\nparallel`
$x \nvDash y$	`\nvDash`	$x \nVDash y$	`\nVDash`
$x \ntriangleright y$	`\ntriangleright`	$x \ntrianglerighteq y$	`\ntrianglerighteq`
$x \nsupseteq y$	`\nsupseteq`	$x \nsupseteqq y$	`\nsupseteqq`
$x \supsetneq y$	`\supsetneq`	$x \varsupsetneq y$	`\varsupsetneq`
$x \supsetneqq y$	`\supsetneqq`	$x \varsupsetneqq y$	`\varsupsetneqq`

3.4.37 Math “Torture” Test Sans Serif Bold

Most of the following examples are taken from *The TEXbook* (Knuth 1984, see https://ctan.org/pkg/texbook) and were adapted for LATEX from Karl Berry’s torture test for plain TEX math fonts.

$x+y-z,\quad x+y*z,\quad z*y/z,\quad (x+y)(x-y)=x^2-y^2,$

$x\times y\cdot z=[x\,y\,z],\quad x\circ y\bullet z,\quad x\cup y\cap z,\quad x\sqcup y\sqcap z,$

$x\vee y\wedge z,\quad x\pm y\mp z,\quad x=y/z,\quad x:=y,\quad x\le y\ne z,\quad x\sim y\simeq z\,x\equiv y\not\equiv z,\quad x\subset y\subseteq z$

$\sin 2\theta=2\sin\theta\cos\theta,\quad O(n\log n\log n),\quad \Pr(X>x)=\exp(-x/\mu),$

$\bigl(x\in A(n)\bigm| x\in B(n)\bigr),\quad \bigcup_n X_n \bigm\| \bigcap_n Y_n$

In-text matrices $\binom{1\,1}{0\,1}$ and $\binom{a\ b\ c}{l\ m\ n}$.

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x}\frac{1}{1-x^2} = \frac{a+1}{b} \bigg/ \frac{c+1}{d}.$$

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \left|\phi(x+\mathrm{i}y)\right|^2 = 0$$

$$\pi(n) = \sum_{m=2}^{n} \left\lfloor \left(\sum_{k=1}^{m-1} \lfloor (m/k) / \lceil m/k \rceil \rfloor \right)^{-1} \right\rfloor.$$

$$\int_0^\infty \frac{t - \mathrm{i}b}{t^2 + b^2} e^{\mathrm{i}at}\,\mathrm{d}t = e^{ab} E_1(ab), \quad a, b > 0.$$

$$A := \begin{pmatrix} x-\lambda & 1 & 0 \\ 0 & x-\lambda & 1 \\ 0 & 0 & x-\lambda \end{pmatrix}.$$

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \begin{pmatrix} u & x \\ v & y \\ w & z \end{pmatrix}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

$$M = \begin{array}{c} \\ C \\ I \\ C' \end{array} \begin{array}{c} \begin{array}{ccc} C & I & C' \end{array} \\ \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{array}$$

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{converges if} \quad |z| < \left(\limsup_{n\to\infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x+\Delta x)-f(x)}{\Delta x}\to f'(x) \qquad \text{as } \Delta x\to 0.$$

$$\|u_i\|=1, \qquad u_i\cdot u_j=0 \quad \text{if } i\neq j.$$

$$\text{The confluent image of} \quad \begin{Bmatrix}\text{an arc}\\ \text{a circle}\\ \text{a fan}\end{Bmatrix} \quad \text{is} \quad \begin{Bmatrix}\text{an arc}\\ \text{an arc or a circle}\\ \text{a fan or an arc}\end{Bmatrix}.$$

$$\begin{aligned} T(n) \;\le\; T(2^{\lceil \lg n\rceil}) &\le c(3^{\lceil \lg n\rceil}-2^{\lceil \lg n\rceil})\\ &< 3c\cdot 3^{\lg n}\\ &= 3c\,n^{\lg 3}. \end{aligned}$$

$$\begin{aligned} (x+y)(x-y) &= x^2-xy+yx-y^2\\ &= x^2-y^2\\ (x+y)^2 &= x^2+2xy+y^2. \end{aligned}$$

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)^2 &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy\\ &= \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,dr\,d\theta\\ &= \int_0^{2\pi}\left(e^{-\frac{r^2}{2}}\Big|_{r=0}^{r=\infty}\right)d\theta\\ &= \pi. \end{aligned}$$

$$\prod_{k\ge 0}\frac{1}{(1-q^kz)}=\sum_{n\ge 0}z^n\Bigg/\prod_{1\le k\le n}(1-q^k).$$

$$\sum_{\substack{0<i\le m\\ 0<j\le n}} p(i,j) \neq \sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r} a_{ij}b_{jk}c_{ki} \neq \sum_{\substack{1\le i\le p\\ 1\le j\le q\\ 1\le k\le r}} a_{ij}b_{jk}c_{ki}$$

$$\max_{1\le n\le m}\log_2 P_n \quad \text{and} \quad \lim_{x\to 0}\frac{\sin x}{x}=1$$

Inline math: $\max_{1\le n\le m}\log_2 P_n$ and $\lim_{x\to 0}\frac{\sin x}{x}=1$

$$p_1(n)=\lim_{m\to\infty}\sum_{\nu=0}^{\infty}\bigl(1-\cos^{2m}(\nu!^n\pi/n)\bigr)$$

Inline math: $p_1(n)=\lim_{m\to\infty}\sum_{\nu=0}^{\infty}\bigl(1-\cos^{2m}(\nu!^n\pi/n)\bigr)$

Lebenslauf

Geboren am 24. Januar 1995 in Summacumlaudeville, wuchs ich in Neustadt (Nordrhein-Westfalen) sowie in Newcastle (Nova Landia, Neufundland) auf. Im Jahr 2013 erlangte ich am Gymnasium Neustadt die allgemeine Hochschulreife. Im Wintersemester 2013/2014 habe ich zunächst das Studium der Kunstgeschichte an der Rheinischen Friedrich-Wilhelms-Universität Bonn begonnen. Im Sommersemester 2014 nahm ich dann das Studium der Volkswirtschaftslehre auf, das ich im August 2018 mit dem Abschluss Master of Science (M. Sc.) beendete (Gesamtnote: 1,3). Meine Masterarbeit „The Influence of Stress on the Performance of BGSE Graduate Students" wurde von Prof. Dr. Lorem Ipsum betreut. Während des Masterstudiums besuchte ich im Herbst 2016 die Universität Tel Aviv in Israel als Austauschstudent. Im Oktober 2018 habe ich das Promotionsstudium an der Bonn Graduate School of Economics aufgenommen.